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FLIGHT TRAINING INSTRUCTION ADVANCED NFO T-45C/VMTS

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1. CNATRA P-825 (Rev. 02-17) PAT, "ALL WEATHER INTERCEPT (AWI), FLIGHT TRAINING INSTRUCTION, ADVANCED NFO T-45C/VMTS" is issued for information, standardization of instruction and guidance for all flight instructors and Student Naval Flight Officers within the Naval Air Training Command.

2. This publication shall be used as an explanatory aid to the Advanced Strike Fighter Naval Flight Officer (NFO) Training System curriculum. It will be the authority for the execution of all flight procedures and maneuvers therein contained.

3. Recommendations for changes shall be submitted via the electronic TCR form located on the CNATRA website.

4. CNATRA P-825 (Rev. 07-14) PAT is hereby cancelled and superseded.

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CHAPTER ONE FIGHTER MISSIONS AND HISTORY

100. INTRODUCTION

The primary purpose of the air combat role of the strike fighter aircraft is to obtain and sustain air superiority over contested airspace. The fighter's purpose is to prevent the enemy from using their air assets effectively (Figure 1-1). To successfully accomplish this, the strike fighter crew must execute an intercept in such a manner as to:

- Attain a position in space between the attacking aircraft and the defended force (positional advantage).
- Control the intercept geometry to achieve an optimal weapons Launch Acceptability Region (LAR) and deny an enemy Weapon Engagement Zone (WEZ). This is done to maximize the probability of kill (P_k), while simultaneously preventing the enemy from achieving weapons release.
- Arrive at the merge (if needed) with an advantage, either in position and/or in energy, and use basic fighter maneuvering to deter or destroy the enemy.

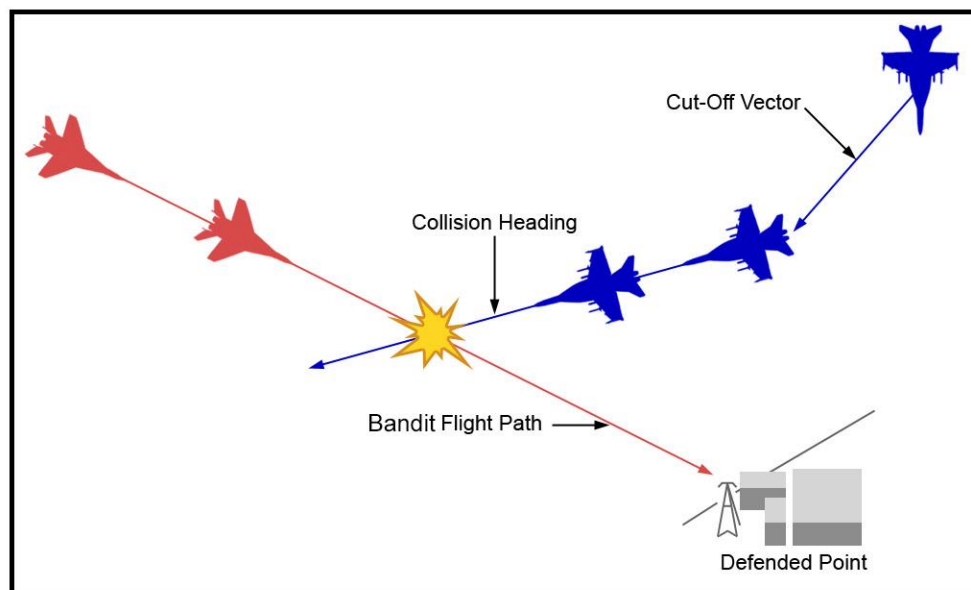


Figure 1-1 Fighter Purpose

101. AIR SUPREMACY AND SUPERIORITY

Air supremacy is defined as the complete dominance of the air power of an opposing force resulting in the complete control of the skies.

Air superiority is defined as the control of a contested airspace without prohibitive interference by an opposing force. Just by these two definitions, strike fighter aircrew can see that Air Supremacy cannot be achieved without Air Superiority.

– Factors to Obtain Air Superiority

Each battle space will have unique factors that must be considered to achieve mission success. Since World War I, the necessity for Air Supremacy to win major conflicts has been undisputed. Some of the requirements to gain air superiority include, but are not limited to:

- a. Enemy and own force capabilities
- b. Rules of Engagement
- c. Geographic area
- d. Fuel considerations and logistics
- e. Terrain
- f. Weather
- g. Geopolitical environment

102. METHODS OF INTERCEPTING

Normally, there are two primary methods that a strike fighter may use to close on a bandit:

- Establish a cutoff vector in order to obtain positional advantage.
- Establish a collision course.

A cutoff vector is one that will place the interceptor in a position between the bandit and the defended force. A collision course is a vector that will allow the strike fighter to close on the bandit in the fastest possible manner.

A collision course is a straight line course where the angle at which the strike fighter sees the bandit (angle-off) and the angle at which the bandit sees the strike fighter (target aspect or aspect angle) remains constant with decreasing range. In a co-speed situation, the two angles will be equal, but in opposite directions. This is also known as constant bearing, decreasing range (CBDR). When a speed differential exists, the angles will not be equal.

1-2 FIGHTER MISSIONS AND HISTORY

Remember, the strike fighter aircrew seeks to achieve optimum LAR on the enemy while simultaneously denying the enemy WEZ parameters on the strike fighter.

103. TYPES OF INTERCEPTS

There are three types of pursuit curves flown during the intercept: Pure Pursuit, Lead Pursuit, and Lag Pursuit.

1. Pure Pursuit

Pure pursuit is the primary means for achieving position in the rear quarter of the bandit. Essentially, pure pursuit occurs when the strike fighter constantly turns to keep the nose of his aircraft pointed directly at the bandit. Depending on the geometry initially generated, increasingly harder turns may be required to achieve a position at the bandit's six o'clock.

Pure pursuit is a form of collision when the fighter has a speed advantage or if both the fighter and bandit are in a pure pursuit scenario (fighter dead ahead of bandit).

Pure pursuit (Figure 1-2) may be used to give the bandit a difficult target to visually acquire.

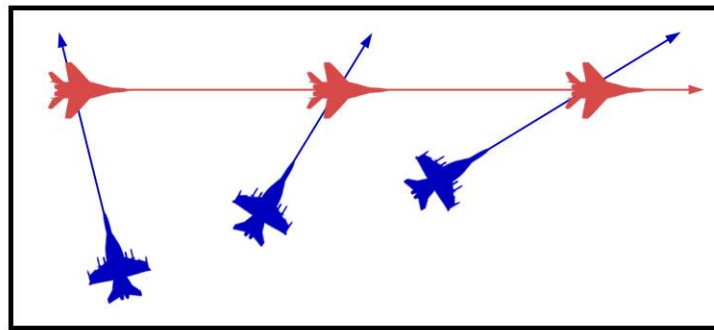


Figure 1-2 Pure Pursuit

2. Lead Pursuit

Lead pursuit is a situation where the strike fighter places his nose in front of the bandit. The amount of lead is how far in front of the bandit's nose the strike fighter is pointing. A collision intercept is a form of lead pursuit.

Lead pursuit (Figure 1-3) may be used to:

- a. Fire a forward quarter missile.
- b. Increase closure and decrease range to the target.
- c. Allow for a gun solution in the rear quarter.

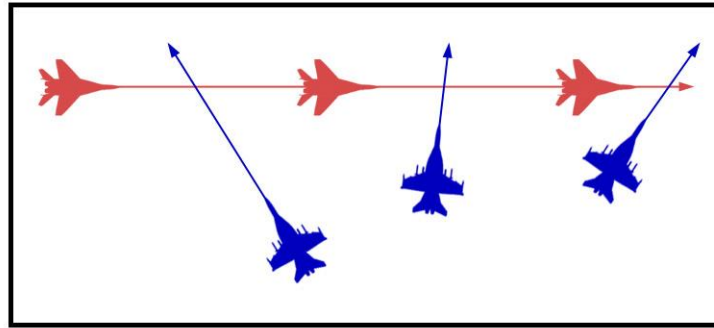


Figure 1-3 Lead Pursuit

3. Lag Pursuit

Lag pursuit is a situation where the strike fighter places his nose behind the bandit. The amount of lag is how far behind the bandit the fighter places his nose.

Lag pursuit (Figure 1-4) may be used to:

- a. Extend range in the rear quarter to complete a sidewinder missile solution.
- b. Decrease closure and increase range.

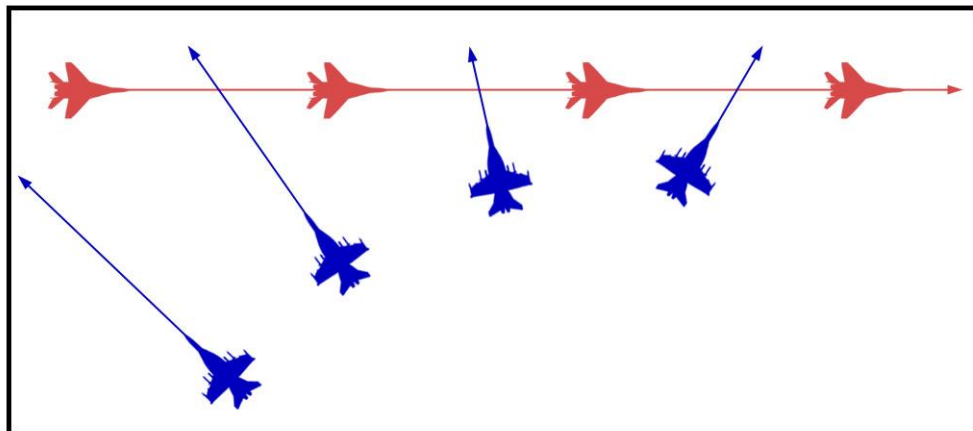


Figure 1-4 Lag Pursuit

104. NAVAL STRIKE FIGHTER MISSIONS

The F/A-18 Hornet was the nation's first purpose-built multi-role aircraft and the first to carry a dual Fighter/Attack designation. It represented a significant departure from previous designs, such as the F-4 Phantom and F-14 Tomcat which were purpose-built, air-to-air missile delivery platforms.

The F/A-18 series began with the F/A-18A and its two-seat trainer the F/A-18B. These aircraft were designed to be able to perform both air-to-air and air-to-surface activities in the same mission. They were also to replace the F-4 and the A-7 Corsair II by performing missions previously performed by both aircraft. Hornets also began to replace the F-4 in Marine Corps units.

In 1987, production began of the improved F/A-18C, and its associated trainer, the F/A-18D. The Marine Corps needed a replacement for the RF-4B, OA-4M, OV-10D and A-6E, and turned to a modified version of the F/A-18D that replaced the rear flight controls with sensor controllers and modified the cockpit for compatibility with night vision devices (NVD)s. Additionally, two aircraft in each Marine D-model squadron were modified for reconnaissance by addition of an electro-optic and infrared (EO/IR) sensor package that replaced the gun in the nose. The last F/A-18D was delivered to the Marine Corps in 2000.

In the late 1980s, the Navy needed a replacement for the A-6E and F-14. They chose a redesign of the F/A-18 and designated it the F/A-18E/F Super Hornet (Figure 1-5).

The E/F model Super Hornets have:

- 25% larger airframe
- Square intakes and larger leading edge extension (LEX)
- Additional store stations
- More powerful engines
- Increased fuel capacity
- Increased range and endurance
- Improved APG-73 radar, later upgraded to the APG-79 Active Electronically Scanned Array (AESA) radar

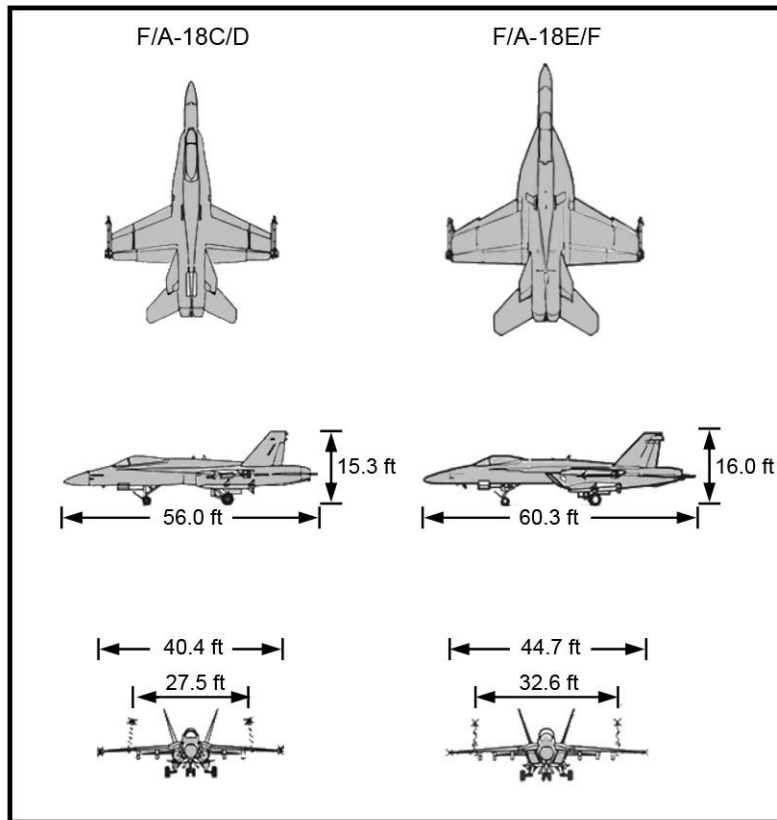


Figure 1-5 F/A-18C/D Hornet vs. F/A-18E/F Super Hornet

1. VFA Mission

The VFA Strike Fighter Mission is to provide combat ready strike fighter assets to conduct carrier based, all-weather, attack, fighter, and support missions as required by the Carrier Air Wing Commander, Strike Group Commander, or higher authority. The missions include Precision Strike, Close Air Support (CAS), Forward Air Control (Airborne) (FAC(A)), Offensive Counter Air (OCA), Defensive Counter Air (DCA), Reconnaissance (Recce), Surface Surveillance Coordination (SSC), Aerial Refueling, and Suppression of Enemy Air Defenses (SEAD).

Descriptions of these missions are as follows:

- a. Precision Strike consists of the employment and delivery of Laser, TV, IR or GPS guided weapons to destroy enemy targets with a high probability of destruction and minimal collateral damage.
- b. Close Air Support is air action by fixed- and rotary-wing aircraft against hostile targets that are in close proximity to friendly forces and which require detailed integration of each air mission with the fire and movement of those forces.

1-6 FIGHTER MISSIONS AND HISTORY

- c. Forward Air Control (Airborne) is a qualified Joint Terminal Attack Controller (JTAC) who performs terminal attack control from the aircraft in a forward area.
- d. Offensive Counter Air (OCA) consists of attacking enemy air defenses and their support systems. Defensive Counter Air (DCA) consists of defending a high value asset or point of interest (e.g., the carrier strike group).
- e. Airborne Reconnaissance missions include gathering imagery of points of interest for intelligence purposes.
- f. Surface Surveillance Coordination (SSC) is used for early warning to the battle group as well as engaging enemy vessels.
- g. Aerial refueling includes mission tanking towards the objective and recovery tanking overhead the aircraft carrier.
- h. Suppression of Enemy Air Defenses (SEAD) includes the temporary suppression via destructive (i.e., missiles or bombs) or electronic means (i.e., jamming) of enemy air defense systems. This is the primary mission of the E/A-18G Growler.

With such a wide range of missions, the F/A-18 E/F must be able to carry multiple types of weapons to achieve mission success. The variety of weapons the aircraft is capable of employing are displayed in Figure 1-6.

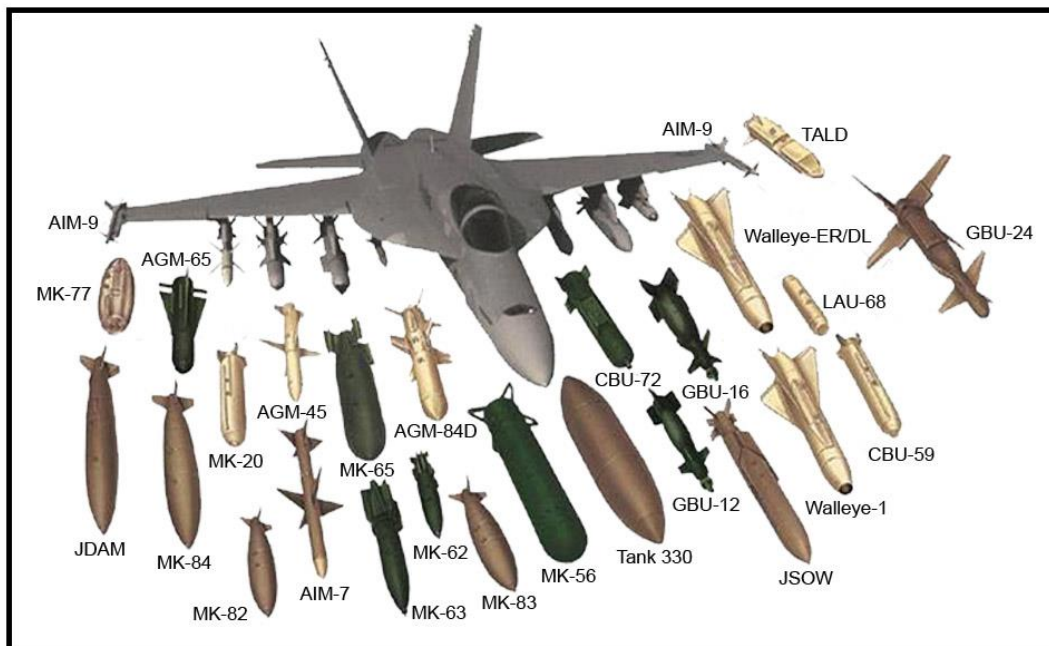


Figure 1-6 F/A-18E/F Weapons

2. Six Functions of Marine Corps Aviation

The six functions of Marine Corps Aviation are: assault support, anti-air warfare, offensive air support, electronic warfare (VMAQ), aircraft and missile control, and aerial reconnaissance. Each function is intended to support the air-ground task force (MAGTF) Commander.

- a. Control of Aircraft and Missiles integrates the other five functions by providing means for MAGTF Commander to exercise C2 authority over Marine aviation assets.
- b. Anti-Air Warfare is the action required to destroy or reduce to an acceptable level the enemy air and missile threat.
- c. Offensive Air Support includes any air operations conducted against enemy installations, facilities, and personnel to directly assist in the attainment of MAGTF objectives through the destruction of enemy resources or by the isolation of the enemy's military forces.
- d. Electronic Warfare is any military action involving the use of electromagnetic and directed energy to control the electromagnetic spectrum or to attack the enemy.
- e. Assault Support is the use of aircraft to provide tactical mobility and logistic support for the MAGTF, the movement of high-priority cargo and personnel within the immediate area of operations, inflight refueling, and the evacuation of personnel and cargo.
- f. Aerial Reconnaissance is used to obtain information concerning terrain, weather, and the disposition, composition, movement, installations, Lines of Communication (LOCs), electronic and communication emissions of enemy forces.

3. VMFA Mission

The VMFA mission is to support the MAGTF Commander by providing supporting arms coordination, conducting multi-sensor imagery reconnaissance, and destroying surface-targets and enemy aircraft day or night under all weather conditions during expeditionary, joint or combined operations.

Marine Corps F/A-18 A+/C squadrons do not have supporting arms coordination or multi-sensor imagery reconnaissance missions. These two missions are what set VMFA(AW) squadrons apart. There are currently two F/A-18D squadrons each at MCAS Miramar, California and MCAS Beaufort, South Carolina. Additionally, one VMFA(AW) squadron is based at MCAS Iwakuni, Japan.

4. VAQ and VMAQ Mission

The VAQ mission is to provide combat-ready Electronic Attack (EA) assets to conduct carrier based, all weather, EA missions as required by the Carrier Air Wing Commander, Strike Group Commander, or higher authority.

With the EA-6B Prowler nearing the end of its airframe life, the Navy also needed a replacement for this venerable Electronic Warfare asset. They turned again to the F/A-18 Super Hornet. The E/A-18G Growler is designed to perform electronic warfare and defense suppression missions. It is a highly modified F/A-18F. Although it retains some air-to-air capability, the Growler is specifically designed to fill the role of carrier-based electronic warfare.

The Growler saw its first use in combat in 2011 as part of Operation ODYSSEY DAWN, the establishment of a no-fly zone over Libya.

The Marine Corps still utilizes the EA-6B for an electronic attack platform in the VMAQ squadrons.

105. COUNTER AIR MISSIONS

1. Offensive Counter Air

OCA is defined as any mission designed to destroy, disrupt, or neutralize enemy aircraft, missiles, launch platforms, and their supporting structures and systems both before and after launch (Figure 1-7). OCA missions include attack operations, fighter sweep, escort, and SEAD.



Figure 1-7 F/A-18F Performing OCA

OCA mission planning considerations include, but are not limited to: Rules of Engagement, type and number of targets, collateral damage potential, integrated air defenses, friendly assets, time of day, weather, terrain, and fuel availability.

2. Defensive Counter Air

DCA is defined as all defensive measures designed to detect, identify, intercept, and destroy or negate enemy forces attempting to attack or penetrate the friendly air environment. DCA is often referred to as a point or area defense. Point defense is for a specific asset, such as the White House or the aircraft carrier. Area defense is for a broader geographic protection, such as a country or border.

DCA mission planning considerations include, but are not limited to: Rules of Engagement, point or area defense, mobility of the defended asset (e.g., aircraft carrier), type and number of enemy forces, friendly asset availability, time of day, and weather.

106. FIGHTER HISTORY

The history and evolution of fighter aircraft dates from the early years of World War I and continues to the present day strike fighters. While the machines and weapons have undergone evolutionary and revolutionary changes demanding ever increasing technical expertise, the aggressive and professional spirit of strike fighter aircrew has remained constant.

Mankind's first powered flight took place on a cold winter morning in Kill Devil Hills, North Carolina, on 17 December 1903. Orville and Wilbur Wright tossed coins, and Orville won the toss and thus became the first person on Earth to fly in a powered aircraft. His first flight lasted 12 seconds and travelled 120 feet. Today, the first flight could take place inside the cargo bay of the C-5 Galaxy (Figure 1-8). Nonetheless, the era of powered flight had begun.



Figure 1-8 From Kitty Hawk to the C-5 Galaxy

1. World War I

When WWI began in 1914, aircraft were primarily considered an extension of an army's observation capability. Airplanes flew with unarmed scouts locating targets and reporting enemy movements for their artillery, infantry, or cavalry forces to attack. The aircraft were slow with

1-10 FIGHTER MISSIONS AND HISTORY

limited maneuverability and included observation blimps. Eventually, scouts began carrying side arms to attack the enemy aerial observers.

As design challenges emerged for firing weapons and flying the aircraft, one solution was the two-place crew: a pilot and gunner. The significant advantages of the rotating machine gun seat design and an additional set of eyes were accompanied by the disadvantages of additional weight and limited maneuverability.

Two-seat fighter designs were fielded in WWI and WWII and continue to the present day. One only has to see the flight demonstration of an F/A-18F to recognize that speed, weight, and maneuverability are no longer limitations for two-seat strike fighters.

2. Famous WWI Aces and Aircraft Flown:

- a. SE-5a pilots:
 - i. Billy Bishop, RAF, Canada, (72)
 - ii. Edward Mannock, RAF, UK, (61)
 - iii. James McCudden, RAF, UK, (57)
- b. Sopwith Camel
 - William Barker, RAF, Canada (50)
- c. Nieuport 28/SPAD 13
 - i. Rene Fonck, AM (75, top scorer)
 - ii. George Guynmeyer (53)
 - iii. Charles Nungesser (43)
 - iv. Eddie Rickenbacker (26)
- d. German Aces of Note:
 - i. Baron Von Richthofen (80, top scorer overall)
 - ii. Ernst Udet (62).

3. World War II – Pacific

On December 7, 1941, six Japanese carriers 200 miles north of Oahu launched over 400 aircraft to support the attack on Pearl Harbor. Forty-eight fighters remained on patrol over the carriers. The remaining aircraft were divided into two attack waves of 183 and 171 planes.

The U.S. military was not prepared for the Japanese attack, which dealt a significant blow to the Pacific Fleet, especially the battleships moored at Ford Island, and the aviation units stationed around the island of Oahu. However, the Japanese primary targets were the aircraft carriers, which were not in port. LEXINGTON was delivering planes to Midway and ENTERPRISE was returning to Hawaii after delivering Marine Corps planes to Wake Island.

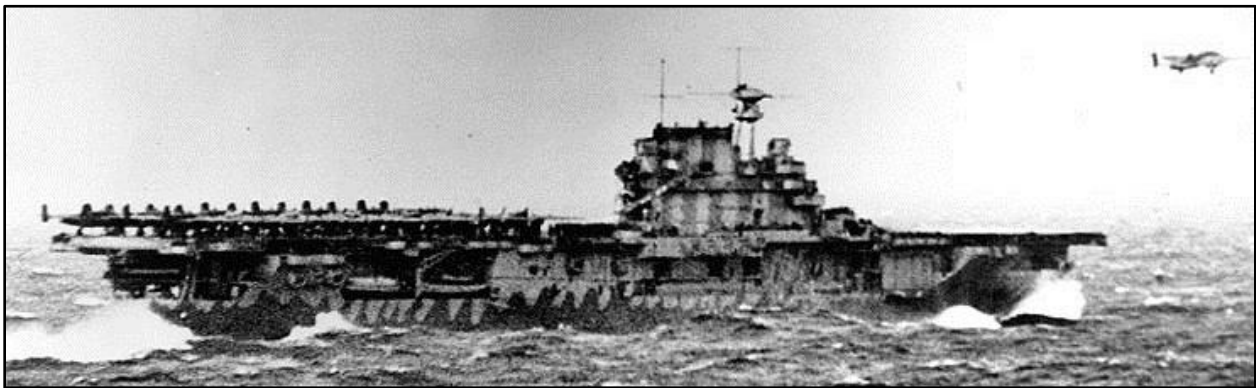


Figure 1-9 USS HORNET Launches Doolittle Raid Against Tokyo

America's first strike back came on 18 April 1942, when US Army LTC James H Doolittle led a raid of 16 B-25B bombers from the flight deck of the USS HORNET (Figure 1-9) to bomb Tokyo and other Japanese cities. Although the damage they did was minimal, it was a great public relations victory and provided a much needed morale boost in the early days of the war.

4. The Battle of the Coral Sea

In May 1942, the first naval battle in which the combatants struck at each other only with carrier-based aircraft was fought at the Battle of the Coral Sea. From 4-8 May 1942, the US Navy acted to prevent the Japanese from invading the island of New Guinea at Port Moresby, which would have severed the supply lines to northern Australia. The battle was a tactical defeat for the Americans, but a strategic victory. The U.S. sunk the light Japanese carrier SHOHO at the loss of the USS LEXINGTON and heavy damage to the USS YORKTOWN. YORKTOWN would return to Pearl Harbor to be rapidly repaired and returned to service, but the Japanese fleet carriers had to return to Japan to replenish their air groups. As a result, the ZUIKAKU and SHOKAKU were not available to assist in operations the next month. Although losses had been severe, the Japanese had been repulsed by Naval Aviation. The age of the battleship was over. From then on, carrier aviation would reign supreme in naval warfare in the Pacific.

5. Midway - The Turning Point

On 4 June 1942, the Japanese attacked Midway with a large carrier force consisting of four aircraft carriers. After initial success in destroying most of the Marine defenders' obsolete fighters (in fact the worst defeat ever suffered by Marine Corps fighter pilots), the Japanese were rearming and refueling for another attack on Midway. Then, word came that their scouts had discovered at least one American carrier. The Japanese commander halted rearming of aircraft for land attack and ordered that aircraft be reconfigured for sea action. This delay would prove deadly.

Based on information gathered by Naval Intelligence, Admiral Nimitz, the commander of the U.S. Pacific Fleet, had sent his three remaining carriers, HORNET, ENTERPRISE and the recently repaired YORKTOWN to Midway to surprise the Japanese and repel the invasion. The Japanese intended to use the invasion to draw the American fleet into a decisive battle, but did not realize the Americans would be waiting for them.

At around 10:00 on 4 June 1942, U.S. carrier torpedo planes made multiple unescorted attacks on the carriers and were slaughtered (37 of 42 lost). However, their sacrifice unknowingly drew Japanese fighter patrols to low level. By chance, between 10:20 and 10:30, three squadrons of Dauntless SBDs from USS ENTERPRISE and YORKTOWN arrived overhead. In what is probably the luckiest few minutes in the U.S. Navy's history, the SBDs inflicted fatal damage to three of the four Japanese carriers in what appeared to the Japanese to be a well-coordinated attack. The attack was, in fact, extreme luck on the part of the Americans. The effects of their bombs were amplified by the ordnance and fuel lines strung about their target's decks. The next day, they returned to sink the fourth carrier. YORKTOWN was sunk, but Japanese naval supremacy had ended. After Midway, the Japanese would never be able to replace the hundreds of experienced aircrew they lost.

For the remainder of the war, Japanese Naval Aviation would be operating at a disadvantage, unwilling to remove experienced pilots from front line roles to train new pilots. The American Navy, on the other hand, rotated their pilots to training commands so that the most recent experience with the enemy could benefit the newest aviators.

By the time the war ended in 1945, America's Pacific Fleet had grown from 3 carriers in 1942 to 18 fleet carriers, and over 50 escort carriers. The once mighty Imperial Japanese Navy, which entered the war with 10 carriers and the 2 largest battleships ever built, was decimated. There are many excellent resources available on the subject of Naval Aviation during WWII. Many modern squadrons trace their lineage to the squadrons of WWII. Many ships are named for WWII heroes. As a professional military officer, SNFOs are encouraged to learn more about the history of the Navy and Marine Corps operations in the Pacific in WWII, since those operations laid the foundation for how the Navy-Marine Corps team works today.

6. WWII – Europe

At the outbreak of WWII, and in keeping with their pre-war doctrine, the *Luftwaffe* fully supported the German army. The years of fighting in the Spanish Civil War, along with

disciplined training and development, allowed the Germans a tremendous advantage across all of mainland Europe. After rapidly over running the Allied Expeditionary Forces in France and Belgium, the remaining allied forces very narrowly avoided disaster as they evacuated to Great Britain from Dunkirk. Pearl Harbor was still over a year away. The Royal Air Force stood alone against the onslaught of German bombers.

7. Battle of Britain

After defeating the French in six weeks during the spring and early summer of 1940, the stage was set for Germany's invasion of England. The Battle of Britain began in July 1940. The advantages of early warning radar on the coasts of England and the Hurricane and Spitfire aircraft and crews surprised the Germans (Figure 1-10). The British Spitfire and Hurricane were the only fighter aircraft holding their own against the German Me-109 at the onset of the war. The Spitfire, with better speed and maneuverability, engaged the Me-109s while the Hurricanes, slower and greater in number, engaged the German bombers. At night, the airborne radar equipped Blenhiems and Beaufighters, engaged the German night bombers. England defeated the Germans, but just barely. The Battle of Britain demonstrated the need for close teamwork between intercept controllers and the fighters they control.

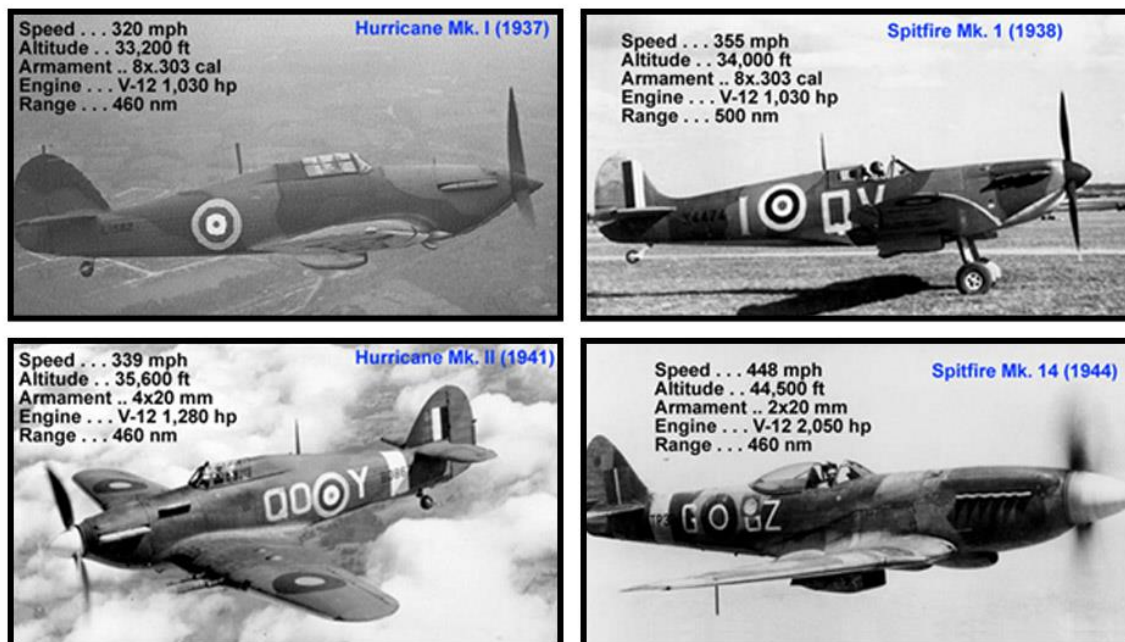


Figure 1-10 RAF's Best Fighters in 1940

8. The Allies Come Back

After Germany called off the invasion of Britain, the focus of the war in Europe shifted to North Africa. U.S. Naval Aviation operated in support of anti-submarine operations and convoy escort, as well as transporting aircraft to allies in North Africa and Malta.

Prior to the U.S. entering the war, British carrier aviation was partially responsible for sinking the German battleship BISMARCK. After Pearl Harbor, the U.S. Navy entered the war in force. U.S. Naval Aviation supported operations in Europe and transported entire Army Air Corps fighter groups into theater. Although there were no great carrier battles in the European Theater, the role of carriers in transporting replacement aircraft in the early days of the war allowed the allies to carry the fight to the enemy.

9. Transition to Jets - Korea

Following WWII, jet engine technology was recognized as the future of Naval Aviation. Although it took a decade to mature, by the end of the Korean War, carrier flight decks were filled with a mix of jets and prop fighters (Figure 1-11). At the end of the war, three of the five fighter squadrons in a carrier air wing were F9F Panthers. Also in the mid-1950s, angled flight decks made their first appearance, allowing for simultaneous launches and recoveries.



Figure 1-11 Korean War Fighters

10. Korea and the First Jet Aces

The Korean War produced the first jet aces. Most of these pilots had some WWII experience, and were extremely effective despite the MiG-15 being considered superior to U.S. designs until the F-86. On its arrival, the F-86 proved a formidable match to the MiG.

One of the U.S. aces, USAF COL Francis Gabreski, became the first “two war” ace, with 31 kills in WWII and 6.5 in Korea. His story is remarkable in that he was also one of the few pilots to get airborne from Wheeler Field, HI, on December 7, 1941.

Due to the Navy and Marine Corps missions in Korea, each service produced only one ace. Major John Bolt, USMC, became an ace on exchange duty with an F-86 squadron. LT Guy Bordelon became an ace flying F4U-5N night fighter variants of the Corsair.

At the end of the Korean War, the carrier air group had evolved into a mix of jet fighters and propeller driven attack aircraft (Figure 1-12).



Figure 1-12 Composite Air Group with Prop and Jet Aircraft

11. Vietnam - From Guns to Missiles and Back

By the 1960s, the advancements in technology lead aircraft designers to believe that guns were no longer required on fighter aircraft. This philosophy lead to designs like the F-4 Phantom II with no internally mounted cannon.

However, with rules of engagement requiring visual identification of hostile fighters, and missile technology still being rather primitive, U.S. pilots found themselves at a disadvantage in the skies over Southeast Asia. Simpler gunfighters, like the MiG-17 and MiG-19, could close with and gun down U.S. aircraft as they maneuvered to missile launch parameters.

Fortunately for the U.S. Navy, the F-8 Crusader was a true gunfighter and experienced considerable success against MiG-17s and MiG-19s. As a result, guns were fitted to future versions of all fighters. Even the F-35 Lightning II has provisions for a gun.

12. Post-Vietnam

The post-Vietnam era marked the beginning of the Fourth-Generation fighters. Examples of these include the F/A-18, F-16, F-14, and F-15. Emphasis on speed, sophisticated weapons systems, reduced detection capability, counter measures, infra-red, night vision, and stealth, are just a few of the technologies incorporated with the Fourth Generation fighters.



Figure 1-13 F-35B Lightning II - Marine Corps STOVL Version

Another important aspect of the Fourth-Generation of combat aircraft is the emphasis on multi-mission capability. Not only is the ability to have multi-mission, real time, flexibility a tremendous asset, it also reduces the production and maintenance costs to operate these aircraft. The tradeoffs for this flexibility and cost savings are some degradation in certain missions and the enemy's ability to dedicate intelligence to fewer allied platforms.

The F-35 Lightning (Figure 1-13) and F-22 Raptor marked the beginning of the Fifth-Generation strike fighters in the U.S. inventory. These aircraft differ from Fourth-Generation fighters in their extreme maneuverability and unprecedented integration of mission systems. These new systems allow the aircrew to avoid or engage threats as soon as they are detected, often without the threat knowing the fighter is present.

107. MODERN THREAT CAPABILITIES AND RECOGNITION

Following WWII, the advent of beyond visual range (BVR) missiles meant that strike fighter aircrew would no longer have to identify aircraft they were engaging. This approach proved to be fatally flawed. By the Vietnam conflict, Navy and Marine F-4 aircrews found themselves having to visually identify all aircraft prior to engagement. This requirement negated their BVR advantage and resulted in the rebirth of dog fighting.

Aircraft recognition is a required aircrew skill. Being able to visually identify aircraft is a starting point. In follow-on training you will be required to associate sensors and weapons systems with particular aircraft and sub-models in order to better prepare and brief your mission. The farther out an aircraft can be detected, recognized, and identified, the more time aircrew have available to make an engagement decision. However, in the air, identification can be difficult as many aircraft share design features (Figure 1-14).

Aircraft size also plays a role in how far away you can see or identify an aircraft.

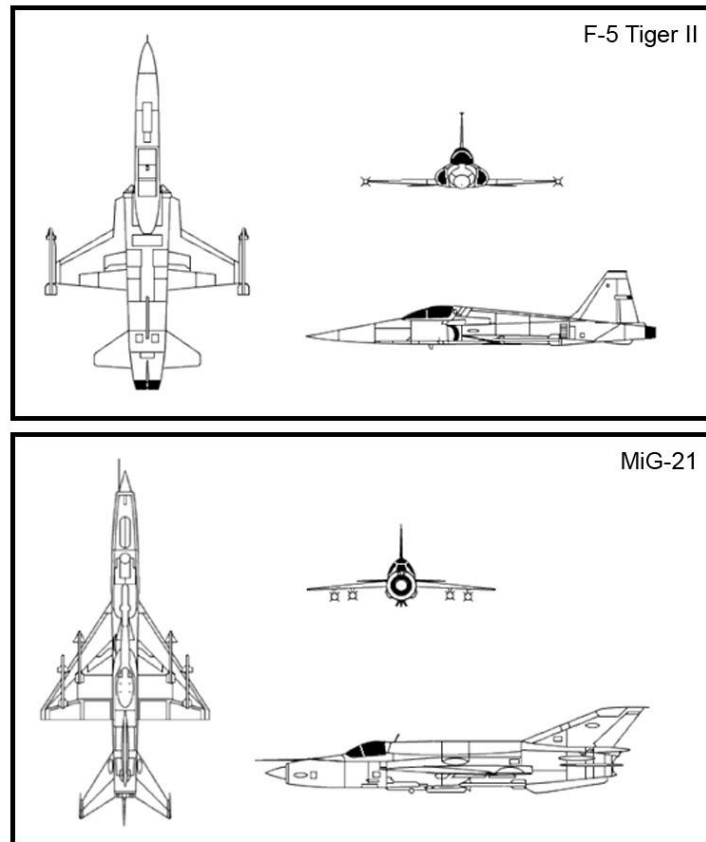


Figure 1-14 F-5E Tiger II and MiG-21

As an example, although they bear a passing resemblance to each other, the F-5 and the MiG-21 are remarkably similar in size. Their small size makes detection in the visual arena difficult, even with radar cueing.

1. North Atlantic Treaty Organization (NATO) Identifiers

Just as the F/A-18 have different variants, so do Russian-built aircraft. NATO identifiers add a letter to the name, which is different from the actual manufacturer's designation.

Care should be taken to ensure that you are referring to the aircraft by one name or the other. Bias your name recognition toward the NATO identifier. Subtle changes in manufacturing designation may not mean a significant change in capability.

For example there are distinct differences in capability between a FULCRUM-A and a FULCRUM-C. Similarly, F/A-18D aircraft from Lot XIV and Lot XV are both F/A-18D models, have different, non-compatible engines but very similar employment and sensor capabilities.

Russian and Chinese license made versions of Russian aircraft and missiles are referred to by their NATO identifier. As a convention, this identifier begins with a letter that designates the type or function of the system. As strike fighter aircrew, you should be familiar with this convention below:

- a. A: Air-to-Air Missiles (Archer, Apex)
- b. B: Bomber Aircraft (Badger, Bear)
- c. C: Cargo Aircraft (Condor, Coke)
- d. F: Fighter Aircraft (Fulcrum, Foxbat)
- e. G: Surface-to-Air Missiles (Goa)
- f. H: Helicopters (Hind, Havoc)

2. Aircraft Identification Fundamentals

All aircraft are built with the same basic elements, recalled as Wings, Engines, Fuselage, and Tail (WEFT) (Figure 1-15):

WEFT Features			
Wings	Engines	Fuselage	Tail
Type Position Slant Shape Taper	Type Number Location Intakes Exhausts	Shape Nose Mid Rear Cockpit Special	Location Slant Number Shape

Figure 1-15 WEFT Identification Factors

The WEFT features are unique to each aircraft, and will assist strike fighter aircrews with positive identification. The key to using these features is studying aircraft and knowing what the NATO identifications are for the aircraft being identified. Aircrew need to invest ample amounts of time on-deck studying aircraft from all corners of the globe. The Strike Group is mobile and strike fighter aircrew will operate anywhere in the world.

Friendly aircraft that strike fighter aircrews should be able to identify include, but are not limited to:

- a. F-15E Strike Eagle
- b. F-16C Fighting Falcon
- c. F/A-18A+/C/D Hornet
- d. F/A-18E/F Super Hornet and E/A-18G Growler
- e. F-35 Lightning II
- f. E-2C/D Hawkeye
- g. E-3 Sentry
- h. KC-10 Extender

Conversely, strike fighter aircrew should be able to identify potential enemy aircraft that include, but are not limited to:

- a. F-4 Phantom II
- b. F-5 Tiger
- c. F-14A
- d. Mirage 2000
- e. MiG-21
- f. MiG-29
- g. Su-27/30

All the above aircraft are good starting places to begin to build a library of knowledge for aircraft identification. When studying, use the WEFT features as the means to discriminate between aircraft and properly identify them.

3. Threat Air-to-Air Missiles

The proliferation of aircraft and missiles resulted in aircraft modifications to carry any missile compatible with their air frame or to allow for the carriage of missiles from multiple suppliers. Many “threat” aircraft may be carrying Russian, French, British or American made missiles or a combination. Being able to neutralize and defeat enemy weapons, and their supportive systems,

begins with strike fighter aircrew being able to identify and understand these weapons and their capabilities.

These missiles fall into three categories:

- a. Semi-Active Radar (SAR) Guided BVR Missiles. SAR missiles require continuous radar support information from the host aircraft radar system for terminal guidance. Examples include Vypel R-23/24 (AA-7 “APEX”), Matra Super 530 and Super 530D, Vypel R-27 (AA-10 “ALAMO”) (Figure 1-16).



Figure 1-16 ALAMO (left) and Super-530

- b. Active Radar (AR) Guided BVR Missiles. AR missiles are able to support themselves with internal radar and do not require information from the host aircraft for terminal guidance. Examples include the Vypel R-77 (AA-12 “ADDER”), MBDA MICA, and MBDA Meteor (Figure 1-17).



Figure 1-17 MBDA Meteor with Ramjet Intakes

- c. Infra-Red Guided WVR Missiles. IR missiles use heat signature from targeted aircraft for guidance. These missiles are widely proliferated and are generally considered shorter-range and/or Within Visual Range (WVR) weapons. Examples include the Matra R550 Magic, Rafael Python III, IV and V, Vympel R-60 (AA-8 “APHID”), and the Vympel R-73 (AA-11 “ARCHER”) (Figure 1-18).



Figure 1-18 MiG-29 with A/A Missiles

4. Threat Surface-to-Air Missiles

In addition to recognition and understanding of airborne threats, strike fighter aircrew need to be familiar with and understand the types of surface-to-air threats that may potentially impact mission accomplishment.

The following surface-to-air missiles (SAMs) and Anti-Aircraft Artillery (AAA) are lethal and require aircrew knowledge to neutralize and defeat.

- a. Man Portable Air Defense Systems (MANPADS). MANPADS are typically shoulder fired rockets with IR guidance. These weapons are easy to carry and becoming increasingly capable with technology. Examples include the FIM-92 Stinger, SA-7 “GRAIL” (Figure 1-19), SA-14 “GREMLIN,” SA-16 “GIMLET,” and SA-18 “GROUSE.”



Figure 1-19 SA-7 "GRAIL "

- b. Command Guided Radar Surface to Air Missiles (SAM). Command guidance is equivalent to radio control of the missile. The missile itself has no seeker and is told where to fly to intercept the target by signals from the launching system. Examples include the SA-2 "GUIDELINE," SA-3 "GOA," SA-8 "GECKO," and SA-15 "GAUNTLET."
- c. Semi Active Radar Homing SAMs. Semi-Active Radar Homing missiles use an internal seeker to detect and guide on the reflected energy from a tracking radar or target illuminator. Some systems combine illuminators and tracking radars on the same chassis, such as the SA-6 where they are collectively known as "Straight Flush." Examples include SA-6 "GAINFUL" (Figure 1-20), SA-11 "GADFLY," and the MIM-23 HAWK.



Figure 1-20 "STRAIGHT FLUSH" Radar System with SA-6 SAM

- d. Anti-Aircraft Artillery (AAA). Anti-Aircraft Artillery, or AAA (triple-A) is any weapon system above 12.7 mm (.50 caliber) specifically designed to engage airborne targets. Generally, these systems fire explosive shells which have contact, time-delay or altitude fuses. “Light” AAA is up to 30 mm in caliber. “Medium” AAA is from 30 mm to 80 mm (Figure 1-21). “Heavy” AAA is anything above 80 mm. Examples include the ZU-23 23 mm and S-60 57 mm.
- e. Mobile and Combination Systems. Mobile systems are designed to accompany troop formations and provide for local air defense, mobile AAA systems are also fairly well deployed worldwide. Most incorporate a fire-control radar to aim the guns, increasing their accuracy. Combination systems utilize short range IR or laser guided SAMs with light AAA guns on one chassis. Coupled with a radar system for target detection and tracking, these systems are especially effective against attack helicopters, but are also effective against low flying aircraft. The SA-19 “GRISON” depicted below is an example of a modern combination system active in today’s combat environment.



Figure 1-21 2S6 Combination Air Defense System with SA-19 Missile

This chapter covered an extensive amount of information beginning with the strike fighter purpose in the air-to-air environment. Basic intercept pursuit curves and geometry were emphasized. Remember that Air Supremacy is the ultimate goal in the air combat environment. The history of military aviation was broadly covered and the major technological developments that lead to Fourth and Fifth Generation Strike Fighters in today’s combat environment. The last sections focused on threat recognition and identification of airborne and ground systems.

CHAPTER TWO

AIR INTERCEPT CONTROL

200. INTRODUCTION

The strike fighter aircrew does not operate in a vacuum. Rather, the fighters are part of an integrated team that includes the systems and system operators that work together to create a complete air picture. This complete picture is communicated to all the members of the team to ensure the proper management of airborne threats. Strike fighter aircrew must understand the capabilities, limitations, and roles of all the components of the intercept control system.

201. BASIC RADAR

An Air Intercept Controller (AIC) is the person trained to provide communications, radar direction, and cueing (which they term control) to fighter aircraft during an intercept. Controllers are thoroughly versed on established control procedures, combat performance of aircraft, fuel consumption data, requirements to affect a landing in adverse weather, aircraft ranges with respect to remaining fuel, vectoring aircraft, and passing information between command and control authorities and intercepting aircraft.

Both the Air Intercept Controller and the function or Air Intercept Control use the acronym AIC. When aircrew discuss AIC, they are referring to both the systems and the people in place to provide the AIC function.

AIC provides an operational picture of the airspace of concern as well as a communication conduit to other command and control nodes throughout the battle space. The controllers are the conduit by which this information is passed.

AIC training defines three types of control, from which fighters define the types of intercept control that will be discussed later in this chapter:

- Close control: intercept of aircraft is the controller's responsibility.
- Tactical control: intercept of aircraft is pilot's responsibility, but the controller will continue to pass information and instructions.
- Broadcast: the pilot plans and executes intercept with AIC providing information as needed.

Unlike the fire control radars used by the fighters, AIC typically use long range radars, in some cases even ATC radar, with detection ranges of up to 250 NM. These radars can also resolve IFF signals in excess of 200 NM. These systems usually rotate through 360 degrees in their scan pattern at a low rate. This combination of factors means that the long range acquisition is at the cost of resolution. Because of this, AIC is not able to provide close control to fighters at long range using only their available radars.

Strike fighter aircrew must recognize the importance of complete, accurate, and thorough briefs and debriefs with controllers. In training, working area constraints are always briefed and monitored by controllers and aircrew.

202. E-2 HAWKEYE, E-3 AWACS AND SURFACE-BASED SYSTEMS

AIC radars may be surface or air based. The U.S. military has built two aircraft to provide Airborne Warning and Control System (AWACS) mission. These two aircraft are the E-2 Hawkeye and the E-3 Sentry (Figure 2-1).

1. E-2C/D Hawkeye

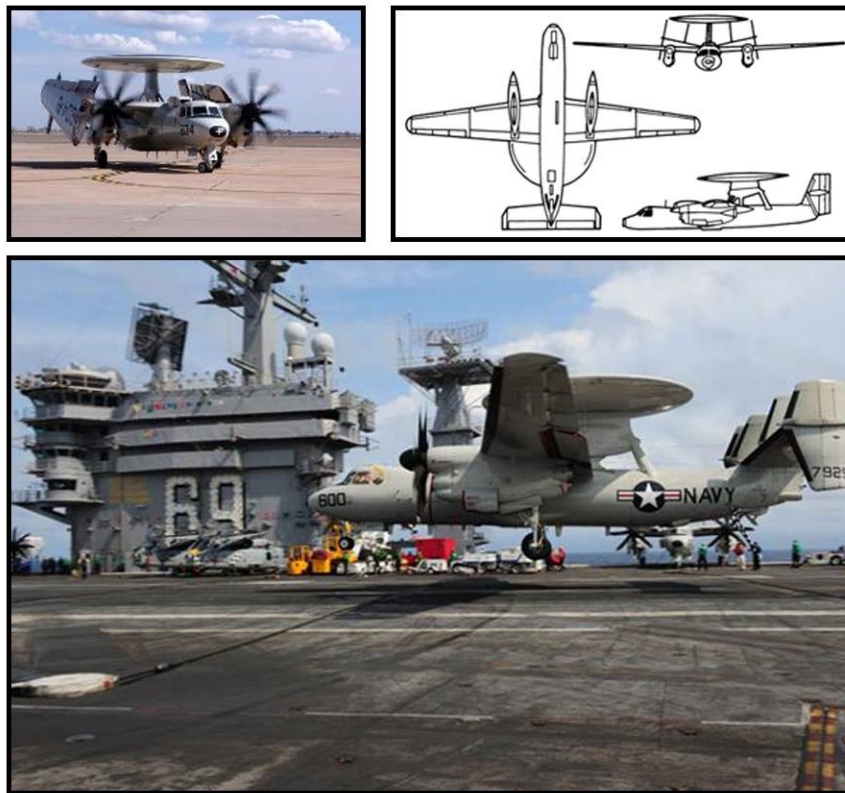


Figure 2-1 E-2C/D Hawkeye

The E-2 Hawkeye is the carrier-based Airborne Early Warning (AEW) platform with a 24-foot diameter saucer-shaped radome housing the APS-145 or APY-9 radar. The E-2C/D Hawkeye can detect targets more than 230 miles away, track over 250 targets, and provide control for more than 30 separate airborne intercepts simultaneously. The Hawkeye crew is also responsible for the “big picture” coordination of surface and air warfare assets. Hawkeyes are assigned to each carrier air wing (CVW) in the VAW squadron.

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2. E-3 Sentry

The E-3 Airborne Warning and Control System (AWACS) is the Air Force's specially modified Boeing 707 that houses a sophisticated airborne command and control center. The operating altitude of the E-3 (Figure 2-2), combined with the capability of the radar, allows it to detect targets from 200 to 300 nautical miles, depending on target size. Like the E-2, the E-3 AWACS is resistant to electronic jamming by enemy forces. The E-3, like all long-range radar platforms, is limited in its ability to provide close control to the fighters at the merge. The E-3 airframes are being updated with the latest battlefield data link and management systems. E-3s are found in Air Command and Control Squadrons in various wings in the United States Air Force and other friendly air forces. Their very long on station times make them a national asset that aircrew will encounter in any large, joint or coalition operation.

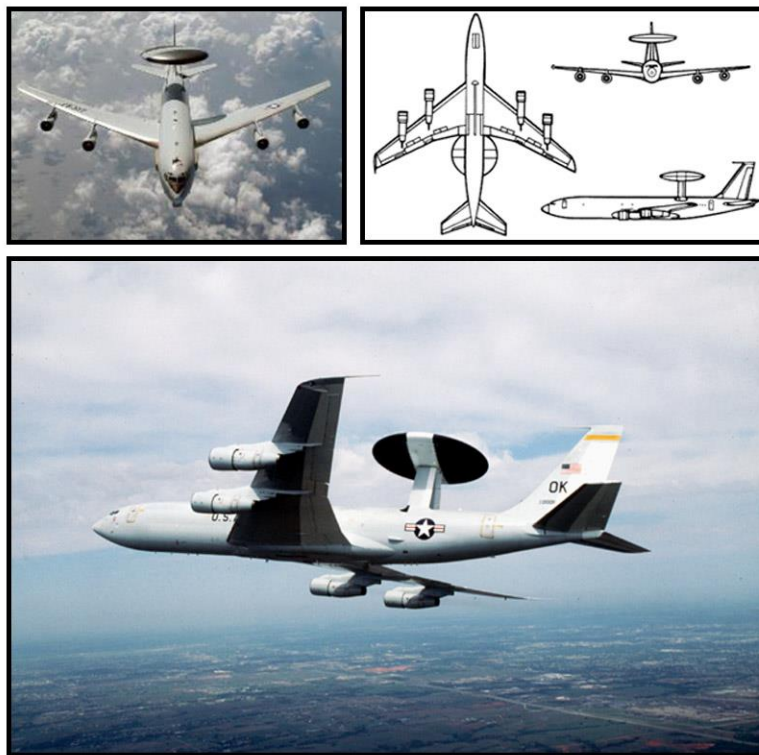


Figure 2-2 E-3D Sentry

3. Surface-Based Systems

In addition to E-2 and E-3 aircraft, there are a number of surface based radars that are used for AIC functions. These include the AN/SPS-48, a shipboard, 3-dimensional radar found onboard large combatants like carriers and amphibious assault ships, the AN/SPS-43, a shorter ranged system found on carriers which provide ASR and carrier controller approach functions as well as doubling as an AIC asset, and the AN/SPS-49, a two-dimensional search radar that provides range and bearing information to airborne contacts. The AN/SPS-49 is a complement to the SPS-48 system.

The premier ship-based system is the Aegis/SPY-1 radar system (Figure 2-3). The SPY-1 radar is a phased-array radar tied into the Aegis fire control system. The Aegis system is a state of the art system that manages the entire battle space around the strike group and not just the air picture. SPS-49 radars are also found on any Aegis equipped ships. These ships are just as capable as a carrier to provide AIC support. Onboard ships, whether Aegis or SPS radar equipped, intercept controllers will most often be enlisted sailors in the Operations Specialist (OS) rating. These sailors will vary greatly in experience or skill: from young sailors fresh out of training, to a veteran Senior Chief.



Figure 2-3 Aegis-Guided Missile Destroyer (Left) and Carrier Superstructure (Right)

4. Marine Corps Air Command and Control System (MACCS)

In a Marine Air Wing (MAW), AIC functions are performed by the Tactical Air Operations Center (TAOC). These personnel are organic to the Marine Air Control Squadron (MACS) as part of the Marine Air Control Group (MACG). The TAOC performs these functions for the MAGTF commander:

- a. Detects, identifies and classifies aircraft and missiles within the assigned sector
- b. Selects and assigns appropriate weapons to engage and destroy air threats
- c. Recommends air defense sectors, subsectors and engagement zones
- d. Deploys sensors and supporting systems to provide air surveillance
- e. Displays and disseminates appropriate air/ground information to higher, adjacent and subordinate units

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To accomplish this, the Marine Corps deploys two radar systems for the AIC role. These are the AN/TPS-53, a three-dimensional, phased-array system similar to the SPS-48, and the AN/TPS-63, a two-dimensional system very similar to ATC radar (Figure 2-4). The TPS-53 is used for intercept control and battle management, while the TPS-63 provides both AIC and ATC functionality. Although intercepts controlled from the ground were once termed Ground Controlled Intercept (GCI) much in the same way AIC was used, AIC has become the standard term for any intercept control.

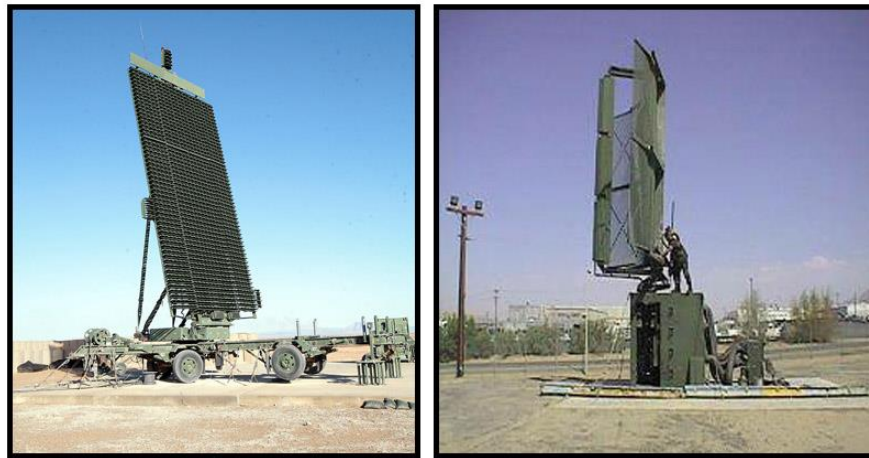


Figure 2-4 TAOC Radars

203. PLANNED POSITION INDICATOR (PPI) DISPLAY

AIC radars use a type of display called a planned position indicator, or PPI, display. A PPI display gives controllers a “God’s eye view” of the area. Most PPI displays are circular and traditionally have the radar at the center of the display. However, modern PPI displays may display only a sector or a portion of the radar’s entire scan. This allows the controllers to provide precise control in their sector of responsibility. AIC PPI displays, such as those on the E-2, E-3, or Aegis systems, give the controller off-centering ability, range scaling, and cursors, allowing the controller to screen out sectors, focus on a particular area, or quickly provide ranges and bearings to contacts. The AIC display can show both raw radar returns, called “primary targets” and the associated IFF return, called a “secondary target.” In fleet operations, this information can be combined with information from data link systems to provide an enormous amount of information to controllers including fuel state, weapon load and even radar status.

Figure 2-5 shows the actual PPI display used by SEABREEZE controllers to control aircraft in the W-155. Note how the display is circular, but offset to the south with the W-155 areas highlighted. The radar is located at NAS Pensacola, Label A on Figure 2-5. W-155 and its subdivisions are part of the sector lines annotated by Label B also on Figure 2-5.

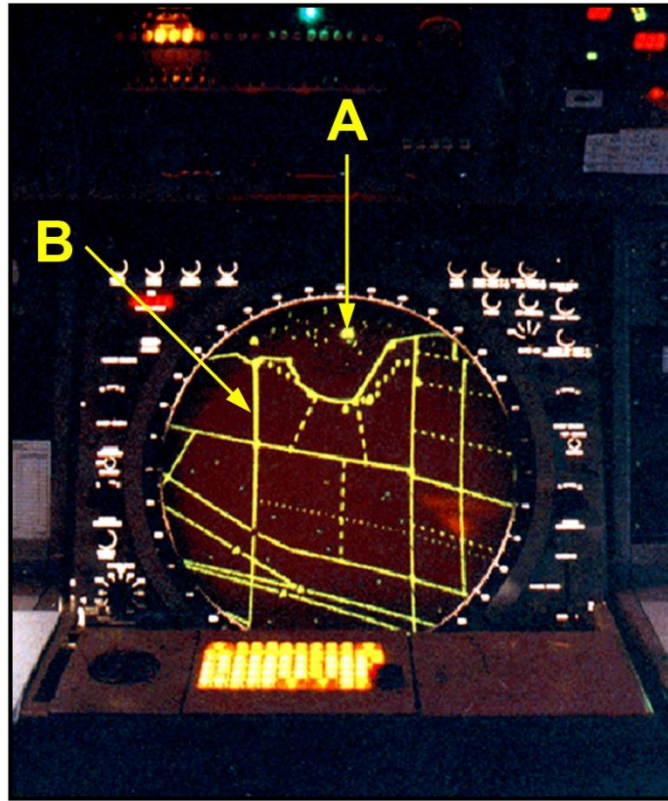


Figure 2-5 SEABREEZE's AIC Display for W-155

204. AIC – ATC TRANSITION AND SAFETY OF FLIGHT

Upon departure from NAS Pensacola, the aircraft will be under positive control of Pensacola Approach ATC. Once clear of traffic and NAS airspace, the aircraft are transferred to the Fleet Air Control and Surveillance Facility, or FACSFAC (pronounced “faks-fak”), for the warning area. This agency’s call sign is “SEABREEZE” in Pensacola. FACSFAC provide positive control into and out of the special use airspace (SUA) and will assign operating areas to the fighters.

Once established in the warning area, the fighters will be directed to contact AIC for their mission. At mission completion, the fighters will contact FACSFAC, which will coordinate their transition back into the ATC system.

When operating in SUA with AIC control, AIC controllers will assist fighters in managing their location to ensure they do not trespass into threats in combat, or stray out of their assigned areas in training. While in a restricted or warning area, AIC does not provide positive control of fighters, but will provide advisories of other aircraft (also called “interlopers”) that may be traveling through the area. In a MOA, the fighters are operating VFR and the controllers will usually be in contact with ATC to provide information about any aircraft that may affect the fighter’s maneuvering.

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Ultimately, however, when operating in special use airspace, the location and safety of the flight is the flight Lead's responsibility.

205. CONTROL TYPES AND FORMATS

1. Control Types

Broadcast control is communication from the controller to all aircraft on the net/control frequency and provides information in a way that is useful to all aircraft. This communication format does not reference any specific friendly unit.

Tactical control is communication from a controller to a specific strike fighter, CAP station, section, or division. Tactical control is tactically and specifically useful to that specific strike fighter element only. However, all other blue forces may be able to use some of the transmitted information to build SA (like altitude).

2. Control Formats

The format of information in either type of control may use either of two formats, bullseye or BRAA (Bearing, Range, Altitude, Aspect).

In bullseye format, contact information is passed by the controller, referenced to a specific geographic reference with latitude and longitude coordinates used by all friendly aircraft. Thus, if you do not know where bullseye is, the information is useless to you. Bullseye format can be used with either broadcast control or tactical control.

BRAA is an acronym for Bearing, Range, Altitude, and Aspect referenced from the strike fighter to the called contact. The purpose is to give a specific strike fighter group quicker situational awareness to a called contact.

- a. Bearing - magnetic bearing from the strike fighter to the group
- b. Range - slant range in nautical miles from the strike/fighter to the group
- c. Altitude - in thousands of feet
- d. Aspect - As shown in Figure 2-6

Broadcast control will be in bullseye format while tactical control may be in either bullseye or BRAA format.

3. AIC Information

During any intercept scenario, any AIC transmission pertaining to a detected group will include the following information

- a. Location of the group – In broadcast control, the group's location will be defined from a common reference point. In tactical control, the group's location may use a reference point (bullseye format) or the fighter's nose (BRAA format).
- b. Altitude to the nearest thousand feet.
- c. Direction of movement – In broadcast control, this will be a cardinal or sub-cardinal direction (i.e., "North" or "Northwest"). In tactical control, this will be a reference to the target aspect the group is presenting to the referenced fighter flight. These aspects are shown in Figure 2-6: (A) 0-30 TA "HOT;" (B) 31-60 TA "Flank;" (C) 61-120 TA "Beam;" (D) 121+ TA "Drag." Flank, beam and drag descriptors will be accompanied by a sub-cardinal direction to denote aspect and track direction.

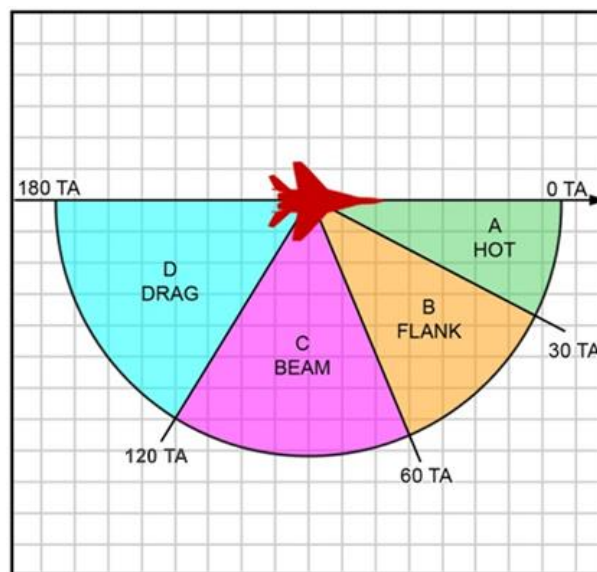


Figure 2-6 Target Aspect

- d. Declaration – The declaration determines the level of permission the fighter has to destroy the group. There are four declarations:
 - i. Hostile – The group meets rules of engagement criteria (ROE) criteria for engagement and weapons release is authorized.
 - ii. Friendly – Group is friendly.
 - iii. Bandit – The group is positively identified as enemy, but does not meet ROE criteria to engage; weapons release is not authorized.

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- iv. Bogey – The group’s identity is unknown; it may be the fighter’s responsibility to obtain a positive identification through sensor or visual means.
- v. Standby/Unable – Although not a declaration, AIC is attempting, or is unable to determine the identity of the group. Standby implies group identity is pending; unable implies AIC cannot make the ID.

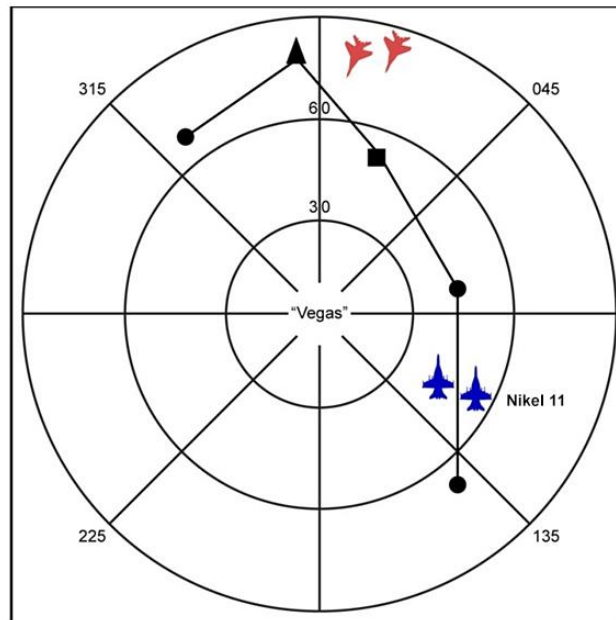


Figure 2-7 AIC Information in Bullseye or BRAA Format

In Figure 2-7, fighters are flying a route with a hostile group to their North. The same situation can be communicated in three different ways with the definitive elements of each in **bold**:

- a. Broadcast control, Bullseye format - “SABRE, Single Group, Vegas 015, 80, 22 thousand, **track southwest, hostile**”
- b. Tactical Control, Bullseye format - “Nikel 11, SABRE, Single Group, Vegas 015, 80, 22 thousand, **hot, hostile**”
- c. Tactical Control, BRAA format - “Nikel 11, SABRE, Single Group **BRAA 345, 90, 22 thousand, hot, hostile**”

At range, before the fighters commit to the intercept, AIC will normally use broadcast control in a bullseye format. Once the fighters are committed to intercepting a group, AIC transitions to tactical control in a bullseye format. Outside of 30 NM, the fighters will brief AIC to use bullseye format. Inside 30 NM, AIC will use BRAA format (Figure 2-8).

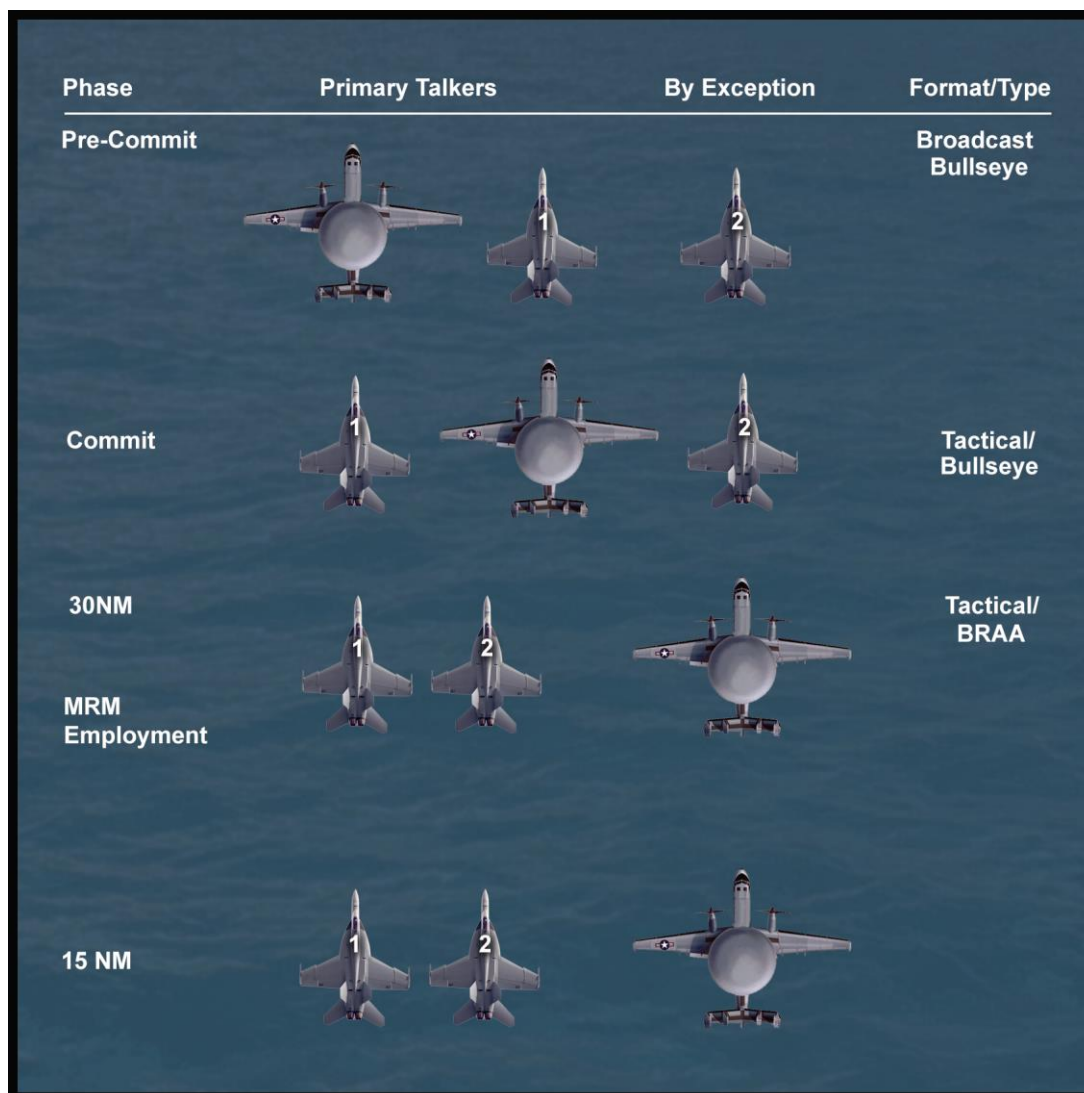


Figure 2-8 Communication Priority

206. CONTROLLER/FIGHTER TEAM WORK

Effective intercepts require that the fighter and AIC controller work as a team to establish a common understanding of the situation, also called the air “picture.” A thorough briefing with AIC prior to the mission will ensure that the controller understands the fighter’s intent for employment and the situation prior to the on station time. Although many of the issues of coordination will be SOP, fighter aircrew should always brief and debrief their controllers to ensure communications are open and effective.

When the fighter requests a picture from AIC, AIC will respond with the location of all known hostile, bandit, or bogey groups. If the fighter has contact with a group that is not identified by AIC, then the fighter should identify that group, using the format in use at that time, and request a declaration. To avoid confusion, AIC will repeat the location of the group to the fighter and add the group’s name, if known, and a declaration. For example:

Fighter - *“SABRE, Nikel 11, contact BRAA 315, 40, 33 thousand, declare”*

AIC - *“Nikel 11, SABRE, single group BRAA 315, 40, 33 thousand, hot, hostile”*

The fighter has a number of ways of determining group position from the radar display. These include:

- Cursor BRAA position (OFT and VMTS)
- Cursor bullseye position (VMTS)
- Kneeboard spider card

Placing the cursor over the contact will provide BRAA to the contact in the OFT and both BRAA and bullseye position of the group in VMTS.

During a mission, a bullseye reference card, also called a spider card, can be used to assist in building situational awareness. The card is normally a blank bullseye grid, in which the fighter can assess whatever information is relevant to the mission. This information may include:

- CAP positions
- Routes
- Locations of defended assets
- Locations of known hostile surface threats

A blank spider card example is referenced in Figure 2-9 for SNFO to copy and use.

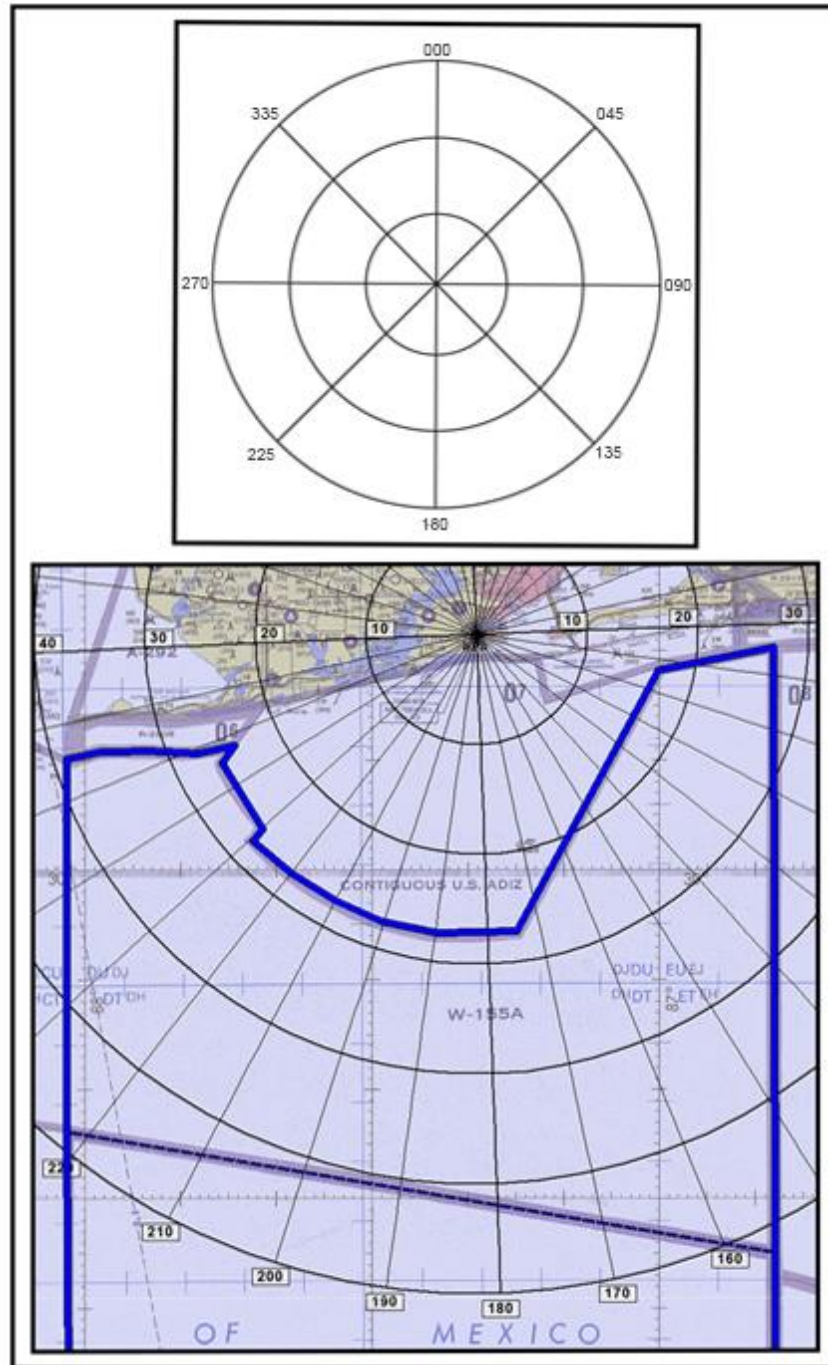


Figure 2-9 Spider Card

207. CONCLUSION

The fighter/AIC controller relationship is very important. The fighter should include AIC controllers in planning, briefing and debriefing of any mission in which AIC controllers are involved. Fighter aircrew must be thoroughly familiar with control methods and formats in order to be an effective air-to-air participant.

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CHAPTER THREE

DIRECTIVE AND DESCRIPTIVE COMMENTARY

300. INTRODUCTION

Directive and descriptive communications provide the foundation for effective intercept communications. The use of applicable and appropriate communication brevity terms can facilitate easier understanding of the tactical situation and allow for simple, universally understood tasking to be accomplished with minimal communications. With the limited time available for transmission of information in the air, standard terminology allows complex ideas to be expressed with minimal verbiage. These terms must be concise, comprehensive, and free of any confusing qualities.

Communications brevity, or comm brevity, standardizes tactical communication terms so that all listeners on a communications net understand the situation in the fewest or shortest possible transmissions. Comm brevity is used across the U.S. military and NATO. Intimate familiarization with multi-service brevity terms is required for mission success. The aircrew at VT-86 will be responsible for knowing and applying the comm brevity words in Appendix A. These terms are taken from joint publications and are standard across the U.S. military and other friendly forces.

301. TYPES OF COMMENTARY

There are two types of intra-cockpit aircrew commentary: directive and descriptive:

- Directive commentary is specific instructions to members of the aircrew. At VT-86, directive commentary is used to direct the pilot to change the aircraft's altitude, heading, or airspeed in order to attain the optimal position for intercept control and employment of weapons.
- Descriptive commentary provides the aircrew information concerning the progress of the intercept and the relative position of a bandit from the fighter. At VT-86, it is used to "paint a verbal picture" of the intercept for the pilot.

Effective communication is critical to mission accomplishment. There can be no misunderstandings between the front and rear cockpits, or between the fighter and intercept control. The aircrew must be able to utilize both directive and descriptive communications in order to ensure a common air picture is understood and that tasks are executed in a timely and correct manner. A successful intercept and combat engagement is dependent on the appropriate and timely flow of both types of information.

302. VOICE INFLECTION AND MODULATION

Voice control is just as important to effective communication as proper terminology. Intercept communications, both inside and outside the cockpit, must be calm and void of any tendency to become loud and high pitched with excitement. However, increasing one's volume and pitch is certainly one way to express the importance of the words being spoken. Avoid long pauses, periods of indecision, expletives, and unnecessary talking. During tactical events, the cockpit should be a sterile environment so that every communication is meaningful to the other crew and flight members. Enunciation must be clear and concise. Words should be clipped and sharp so that syllables are not drawn out.

303. DIRECTIVE COMMUNICATIONS

Directive communications are sent with the intent that the receiver will perform an action without question upon receipt. Directive communications are distinguished from descriptive communications by the use of voice inflection, forcefulness, or tone. Without an audible change in vocal tone, important directive communication may get lost in the "background noise" of the intercept.

Because WSOs are not in physical control of the aircraft, they must learn to use directive communication effectively in the cockpit. Directive comm instructs the pilot to take specific positive action necessary to execute the mission. In effect, the weapons officer controls the fighter via commands to the pilot. A fighter may be maneuvered by changing its bank, pitch, and/or airspeed. In addition, the WSO issues directive radio communications to control the intercept outside the cockpit as well. Directive communications in the cockpit fall into five categories:

- Turn Commands
- Turn Modification Commands
- Altitude Commands
- Airspeed Commands
- Intercept Control Communications

1. Turn Commands

Turn commands control the heading of the fighter. With the exception of an easy turn, direction of turn is stated first so that the pilot will begin the roll and then capture the appropriate rate.

- a. *"Easy left/right"* - Half standard rate turn at approximately 15 degrees angle of bank (AOB).
- b. *"Left/Right standard"* - 30 degrees AOB standard rate turn.

3-2 DIRECTIVE AND DESCRIPTIVE COMMENTARY

- c. “*Left/Right hard*” - A Mil Powered turn, using G to maintain airspeed.
- d. “*Break Left/Right*” - Max performance, tactical maneuver (same as in BFM).

2. Managing Intercept Geometry Commands

The SNFO may utilize the following comm on ICS to direct the intercept.

- a. “*Go pure (pursuit)*” - Pilot will place the contact/aircraft on the nose.
- b. “*Put ‘em XX Left/Right*” - Pilot will place the contact at the commanded antenna train angle (ATA).
- c. “*Lead ‘em*” - Pilot will place and hold the contact/aircraft 30 degrees in lead.
- d. “*Lag ‘em*” - Pilot will place and hold the contact/aircraft 30 degrees in lag.

3. Turn Modification Commands

These commands are used once established in a turn to modify the turn rate.

- a. “*Harder*” - Pilot will increase turn performance to the next highest (easy to standard, standard to hard, etc.).
- b. “*Ease*” - Pilot will begin a smooth roll toward wings level.
- c. “*Hold*” - Pilot will maintain current AOB.
- d. “*Steady up*” - Roll out on the current heading.
- e. “*Steady (Heading)*” - Turn at the commanded rate and rollout at the commanded heading.
- f. “*Reverse*” - Immediate roll to capture the same AOB in the opposite direction. Can also specify the turn rate to capture in the opposite direction “*Reverse Hard.*”

4. Altitude Commands

Directive commands to adjust altitude include the following:

- a. “*Descend ____*” - The pilot will descend to specified altitude or designated number of feet while maintaining constant airspeed.

- b. “*Climb ____*” - The pilot will climb to specified altitude or designated number of feet while maintaining constant airspeed.
- c. “*Level off*” - The pilot will attempt to capture current altitude. Also used as a correction if pilot has missed designated altitude “*Level off, 15 thousand*” if descent or climb rate passing 15K was not under control.

5. Airspeed Commands

Airspeed commands are used to change airspeeds or configurations that affect airspeed. Some directive airspeed commands are comm brevity, which mostly apply to flight management.

- a. “*Set ____*” - Accelerate to the IAS or IMN designated. If no speed is specified speed is allowed to increase to the maximum allowed for the aircraft. “*Buster*” is comm brevity to set military power.
- b. “*Hold speed*” - Given when aircraft reaches a desired or intermediate speed, which was not otherwise specified.
- c. “*Throttle back, set speed ____*” - Pilot decreases speed until the aircraft reaches the IMN commanded.
- d. “*Idle/boards*” - Pilot will go immediately to idle and extend the speed brakes.

6. Intercept Control Commands

Comm brevity terms that are directive and commonly used in an intercept include:

- a. “*Commit*” - Call to commit on the indicated AIC picture
- b. “*Cold*” or “*Cold Left/Right*” - Turn away from threat; not to be confused with cold aspect, which is descriptive information as part of AIC communications
- c. “*Hot*” or “*Hot Left/Right*” - Turn toward the threat; not to be confused with hot aspect, which is descriptive information as part of AIC communications
- d. “*Target*” - Assigns targeting responsibilities to flights or flight members

Directive comm always takes precedence over descriptive comm. The SNFO should never hesitate to interrupt descriptive communication to give a necessary directive command.

Directive comm should be given with a sharp authoritative voice and with a different inflection to preclude any possibility of misinterpretation as descriptive comm.

Any time a maneuver is directed which exceeds the fighter’s performance capabilities, the pilot will comply with the direction up to the fighter’s limits.

304. DESCRIPTIVE COMMENTARY

Descriptive communications provide amplifying information to increase the situational awareness of the receivers. It should be clear, concise, and convey the most accurate information available. Descriptive communications can also rely heavily on comm brevity. However, if brevity does not fit the situation, clear concise plain language can be used as a last resort.

Descriptive communications can be divided into two subsets:

- Intra-cockpit descriptive commentary
- Globally applicable descriptive communications

1. Intra-Cockpit Descriptive Commentary: The AREO Report

The AREO report states, in order, the bandit's azimuth, range, elevation, and overtake (rate of closure (V_C)). AREO informs the pilot where to look for a contact during the transition from BVR to WVR. AREO reports will be made on the ICS within the aircraft. As with all descriptive comm, AREO reports should be given smoothly in a natural tone of voice. AREO reports should begin no earlier than 10 NM, since visual acquisition outside of 10 NM is not likely. The frequency of AREO reports should not be less than every 2 NM, and should increase as range decreases to the merge until the pilot calls "*Tally*." Once a tally by the pilot is achieved, AREO calls are no longer required.

Information in the AREO comes directly from the radar attack display as shown below:

- a. Azimuth - reported in degrees left or right of the nose. Degrees are implied and do not need to be stated. If less than 5 degrees, report as "*On the nose*"
- b. Range - reported in nautical miles as interpreted from the radar
- c. Elevation - reported in degrees high or low. If no elevation difference, report "*Level*"
- d. Overtake - Rate of Closure (V_C). (This is an optional item)

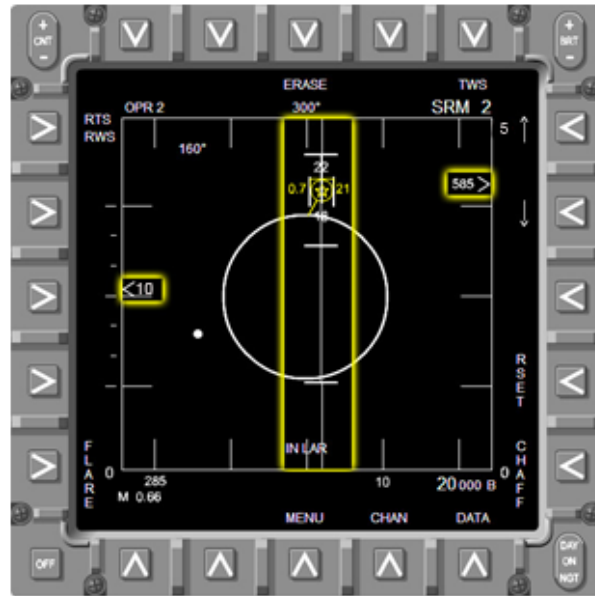


Figure 3-1 AREO Report Information from Radar Attack Display

For Figure 3-1, the correct call should be:

SNFO (ICS): *“On the nose, 4 miles, 3 high, 585 over”*

AREO reports should not be given when the intercept situation requires directive commentary to control intercept progression.

2. Globally Applicable Descriptive Communications

The initial contact communications made by the fighter or AIC are descriptive. These include calls made in both bullseye and BRAA format. These communications must be clear, concise, and accurate as possible. When more than one group or more than one fighter is present, clear, concise descriptive comm will build SA for the other fighters and for AIC, while poor communication can destroy everyone’s SA. These intercept specific descriptive communications include:

- a. Declarations
 - i. “Bogey” - unknown contact, proceed with intercept, weapons release not authorized
 - ii. “Bandit” - Confirmed enemy, weapons release not authorized
 - iii. “Hostile” - Confirmed enemy, weapon release is authorized
 - iv. “Unable” - ID capability not available, proceed with intercept

3-6 DIRECTIVE AND DESCRIPTIVE COMMENTARY

- b. Altitudes will be given in thousands of feet as whole numbers
 - 15,000 is said as “fifteen thousand”.
- c. Other globally applicable descriptive intercept communications brevity
 - i. “Cons/conning” - Non-friendly aircraft leaving contrail
 - ii. “Marking” - Friendly aircraft leaving contrail
 - iii. “Hot/flank/beam/drag” - Contact’s aspect relative to the fighter in tactical AIC control
 - iv. “Track (direction)” - Contact’s direction of travel; used in broadcast AIC control
 - v. “Heavy” - Group contains more than two contacts.
 - vi. “Echelon” - Two or more groups separated in both azimuth and range

305. COMM BREVITY GLOSSARY

Aircrew should be familiar with communication brevity terms. An incomplete glossary of these terms is included as Appendix A.

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CHAPTER FOUR

INTERCEPT DISPLAYS AND FLIGHT PATH VISUALIZATION

400. INTRODUCTION

The aircrew must correctly position the fighter's weapons system in preparation for weapons release. The SNFO must understand the orientation of the basic geometry of the intercept. The first step in understanding the intercept's geometry is a thorough understanding of intercept displays and the orientation of both the bandit and fighter flight paths. Correctly orienting the fighter and bandit flight paths will allow aircrew to advantageously position the fighter and employ weapons systems.

401. INTERCEPT PERSPECTIVES

There are three ways the intercept may be viewed:

1. AIC or "God's Eye View" Perspective

This is a two dimensional, top down orientation to the intercept as presented by the PPI display to the intercept controller. Regardless of the source of intercept control (AIC/GCI/E-2/AWACS/AEGIS), the air intercept controller's radar system and displays provide the earliest and most encompassing, "God's eye" view with 360 degree situational awareness. The large area covered, and the perspective by which it is shown, mean that the AIC will have a much better idea of what is going on around the fighter than the fighter aircrew will. Although, with the advent of data-links and network centric warfare systems, this information can be shared with a modern fighter and even displayed on the fighter's sensors.

2. Bandit's Perspective

The bandit's perspective pertinent to the fighter is called target aspect. Target aspect (TA) is the position of the strike fighter from the bandit's perspective. TA can be a visual assessment and/or displayed on the radar. TA is always expressed in degrees with a left or right. Degrees are assumed and not stated; meaning that "30 Right TA" is written and spoken as "thirty right" and not "thirty degrees right."

3. Fighter's View

The most important view is the fighter's view. This is determined by Antenna Train Angle (ATA). Antenna Train Angle is the position left or right of the fighter's nose on the radar attack display. Like TA, ATA is always expressed in degrees with a left or right, with degrees implied and not written/spoken.

While ATA and TA represent a first person perspective to the intercept from the fighter and bandit's viewpoints, respectively, the AIC's "God's eye" view allows the most complete view of the geometry of the intercept. Fighter aircrew need to learn to build the three dimensional view from the information presented on their displays and through effective intercept communication.

4. Relationship Between TA and ATA

TA is how the bandit sees the strike fighter. ATA is how the strike fighter sees the bandit. When both ATA and TA are equal in magnitude and opposite in direction, a collision course is established.

Collision course is when TA and ATA are equal in magnitude, but opposite in direction, for two co-speed, co-altitude aircraft (Figure 4-1). If two aircraft are co-altitude and on collision course, there is a high likelihood they will collide if neither makes a change to heading, altitude, or airspeed.

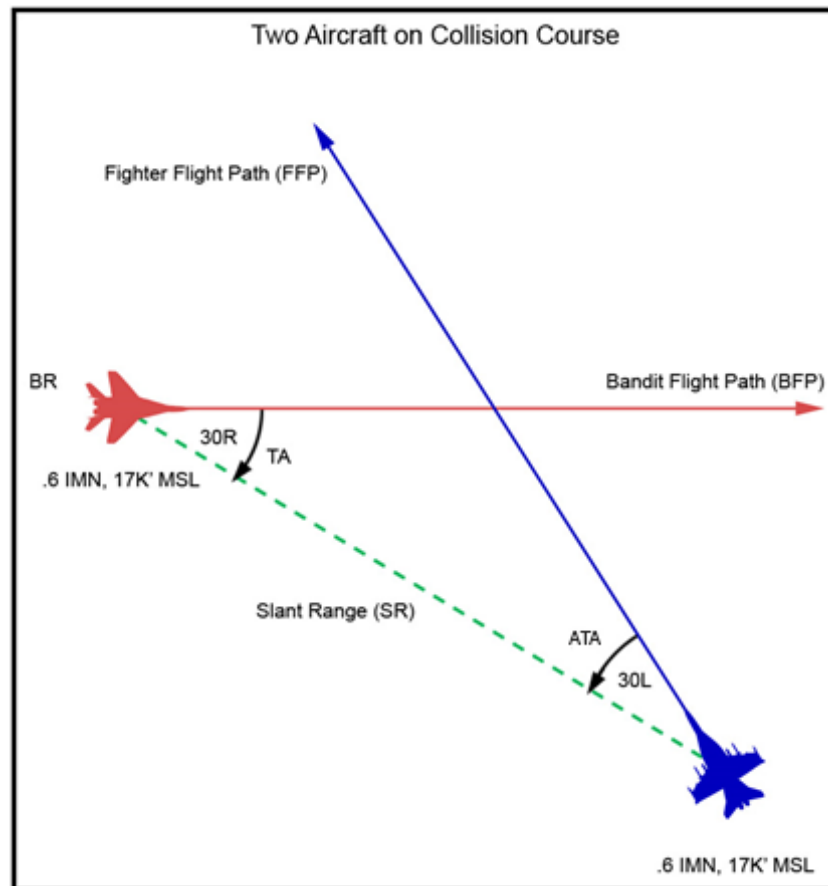


Figure 4-1 Two Aircraft on Collision Course

402. INTERCEPT DISPLAYS

There are a number of displays available to aid the fighter in determining the spatial relationships in an intercept. These include

- Radar attack display
- HSI (VMTS and OFT)

4-2 INTERCEPT DISPLAYS AND FLIGHT PATH VISUALIZATION

- EW page (OFT)
- Situational Awareness (SA) display (VMTS)
- Kneeboard spider card

1. Radar Attack Display

The radar attack display is the primary intercept display in the cockpit. The radar attack display shows information about the fighter's current position, the bandit's position and flight information, and weapon employment information.

Ownship information is called out from the radar attack display in Figure 4-2.

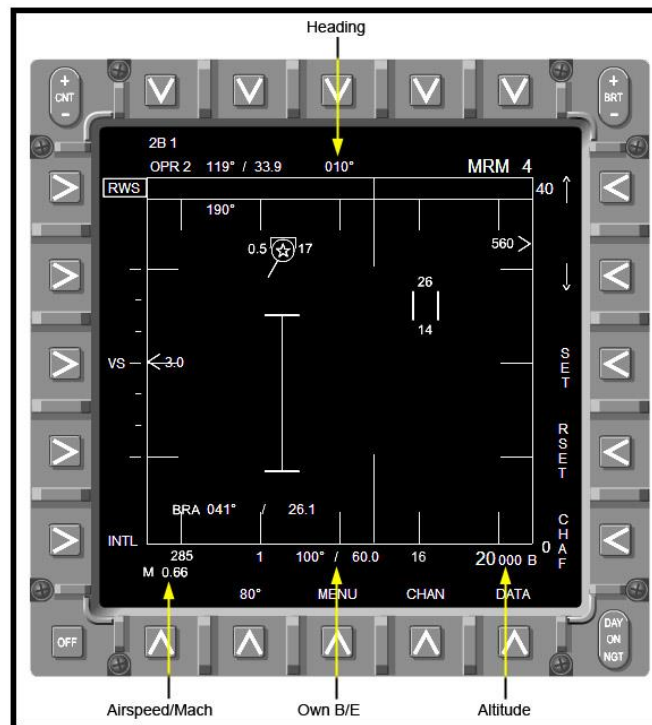


Figure 4-2 Ownship Information on VMTS Radar Attack Display

Ownship information displayed on the attack display includes:

- Heading
- Airspeed and IMN
- Ownship bullseye position
- Altitude

Note that ownship bullseye position is not displayed in the OFT, but is available in the VMTS aircraft.

In addition to ownship information, information about the launch and steering (L&S) trackfile is presented. This information on the radar attack display about the L&S includes:

- a. Heading as computed by the radar
- b. Airspeed in IMN
- c. Altitude rounded up to the nearest thousand feet
- d. Target Aspect vector providing TA with 0 degrees being straight down the display
- e. Range as indicated as a caret on the range scale at the right of the display
- f. Rate of closure (V_C) is indicated next to the range caret
- g. Elevation caret indicating degrees above or below the aircraft
- h. Differential altitude next to the elevation caret
- i. Bearing and range to the cursor
- j. Bullseye location of the cursor (VMTS only)

This information provides the fighter aircrew with every piece of information about the contact needed to affect a successful intercept. All of these indications, with the exception of ownship bullseye position and cursor bullseye position, are information from the fighter's perspective. Bullseye information provides an immediate connection to the AIC perspective by tying the attack display information to a top down view (Figure 4-3).

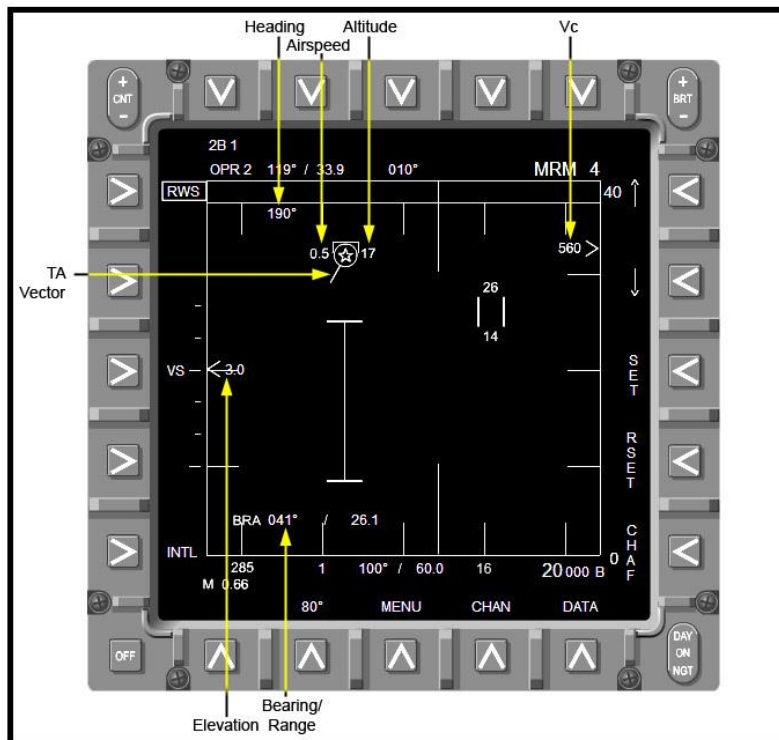


Figure 4-3 Contact Information on Radar Attack Display

2. Horizontal Situation Indicator (HSI) Display (OFT and VMTS)

The HSI is used in PLAN mode to help visualize the intercept situation by translating intercept information into a recognizable navigation instrument (Figure 4-4). By recognizing that the fighter is at the center of the display, the bearing to the bandit can be easily identified on the compass rose. Using the range scale as a reference, the fighter can then establish the relative position of the bandit to geographical reference points, waypoints or route/area features as depicted on HSI symbology; however, the contact itself is not displayed on the HSI and the intercept is too dynamic for bandit location to be entered as a waypoint or offset. This limits the usefulness of the HSI to providing coarse intercept awareness.

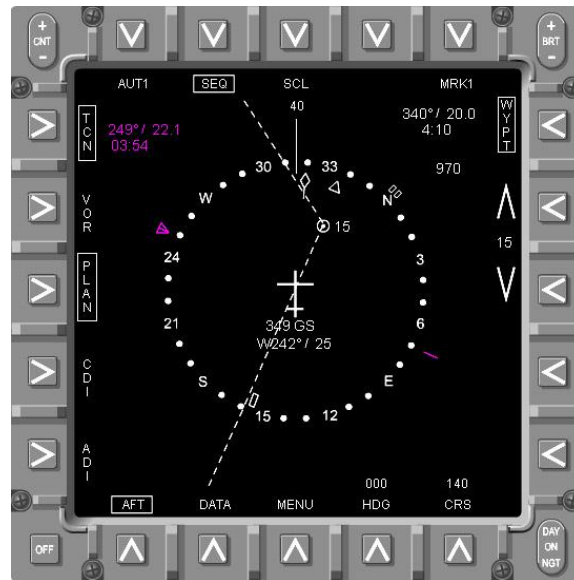


Figure 4-4 VMTS and OFT HSI Displays

3. Electronic Warfare (EW) Display

In the OFT, the EW display (Figure 4-5) provides an indication of the direction of arrival of detected threats, threat identification by type of emitter and threat status. The EW display also provides a quick reference to place the threat emitter in the beam by the marks at the aircraft's 12, 9, 6, and 3 o'clock position. No heading indication is given which limits the usefulness of the display on its own. Direction of arrival may not be accurate enough for defense; SNFOs should not rely on the EW page for calculating defense headings.

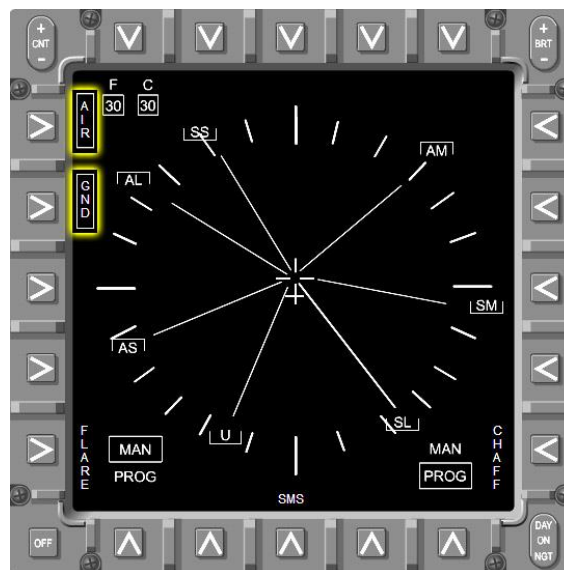


Figure 4-5 OFT EW Display

The EW display provides counters for chaff and flares, as well as dispense capability for both. It is accessed through the left MFCD EW option on the SMS display. The SMS option on the EW display returns the display to the SMS format. The OFT displays the threat symbols (Figure 4-6).

Symbol	Meaning
	Air Threat
	Ground Threat

Symbol	Meaning
AL	Air (>25NM <40NM)
AM	Air Medium (>10NM <25NM)
AS	Air Short (<10NM)
SL	Surface Long (>25NM <40NM)
SM	Surface Medium (>10NM <25NM)
SS	Surface Short (<10NM)
U	Unknown Emitter

Figure 4-6 OFT EW Page Symbols

Status change of the emitter is simulated by changes in the aural cues associated with the emitter. Different tones indicate the threat has locked the fighter and employed weapons.

4. Situational Awareness (SA) Display (VMTS)

The VMTS SA display combines the functionality of EW display and the HSI into a common display. The VMTS SA display (Figure 4-7) is very similar to those in use in the F/A-18. As shown, the fighter has waypoint steering information, sequence information, auto sequence option and range selection options as in the HSI. Additionally AIR and GND options allow the selection of display of airborne or surface threats, respectively. VMTS EW symbols are simplified and shown below in Figure 4-8.

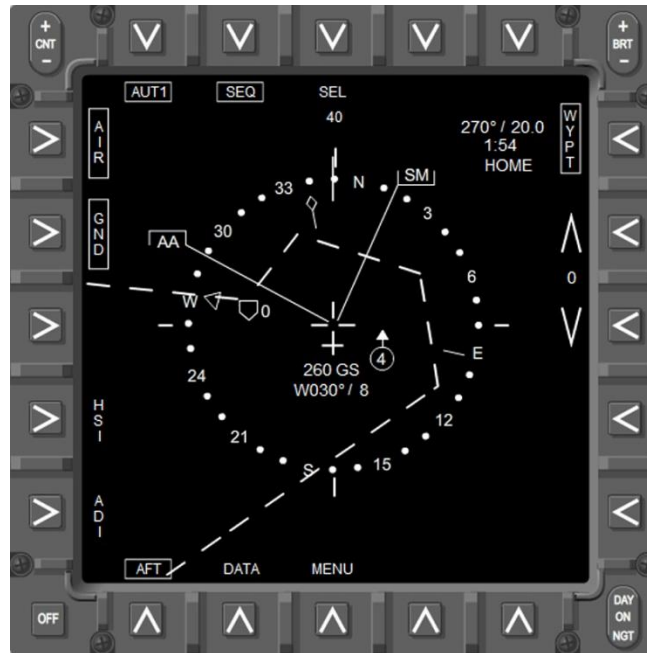


Figure 4-7 VMTS SA Display

Symbol	Meaning
	Air Threat
	Ground Threat
Symbol	Meaning
AA	Air Threat (any range)
SL	Surface Long (>25NM <40NM)
SM	Surface Medium (>10NM <25NM)
SS	Surface Short (<10NM)
U	Unknown Emitter

Figure 4-8 VMTS SA Page Symbolology

The SA display allows the fighter to reference one display for route/area management, waypoint steering/geographic reference (GeoRef) and EW information, making it a much better display than the HSI or EW display during an intercept. The only disadvantage of the SA page is that chaff cannot be dispensed from the SA page. This must be done from the attack display.

5. Kneeboard Spider Card

The spider card on the kneeboard is also an intercept display in that it allows for near real-time plotting of group and fighter positions to build situational awareness. The kneeboard spider card is only as good as the information aircrew put on it and does not take the place of the attack display. Rather, the spider card should be used to enhance SA to the group positions relative to

the route, area or target. The spider card will be used in intercepts where broadcast control in a bullseye format is used.

When flying a route or designating GeoRefs, it is highly recommended that these also be plotted on the spider card to allow for quick reference to the tactical situation.

6. Intercept Displays, Active Listening and the Mind's Eye View

The most important intercept situational awareness tool is the mind's eye view of what the situation is. This represents a fusion of the available information into a coherent mental image of the intercept situation. It combines the information presented on all cockpit displays to form a mental picture of what is happening in three dimensional space. This is a skill that must be developed.

The first step to being able to visualize an intercept begins with mission preparation. The fighter aircrew must be thoroughly familiar with the briefed mission area, bullseye position, CAP or route placement, and expected threat presentations. Once airborne, the strike fighter aircrew can determine their aircraft position and the group contacts as communicated by AIC. A spider card can be very helpful for visualizing the intercept at this point in the mission but the most important skill to develop is active listening.

Active listening means the receiver, in this case the fighter aircrew, are able to feed back to the communicator exactly the information sent, not similar, paraphrased or their own interpretation. This ensures accurate and effective communications are taking place. In order to establish a correct mind's eye view, the fighter must do the following:

- a. Actively listen to AIC communication to understand the bandit presentation/location.
- b. Use available information, especially the radar attack display, to confirm the location of groups as transmitted by AIC.
- c. Correlate with properly formatted communications that the information presented by AIC is what the aircrew see on their sensors.
- d. If the aircrew has information not presented by AIC, they must communicate that information to AIC and request a declaration to that group to ensure a common air picture is realized.

403. CORRELATING AIC INFORMATION

Once the aircrew have established their position and have received bandit location information from AIC, they can begin to build the spatial picture. In VMTS, correlation of this information is done through BRAA to cursor and Bullseye to cursor information on the radar attack display. In the OFT, this correlation is done via BRAA to cursor information and projecting that information from the mind's eye view to the HSI and/or spider card.

In Figure 4-9, the cursor is not on the contact, this means that the cursor position information will not reflect bandit position. For example, AIC communicates:

“Hammer 31, SABRE, Single group BRAA 350, 35, 17 thousand, hot, hostile”

The fighter correlates this by placing the cursor over the contact and noting the BRAA position. Since this matched what the fighter is seeing on the radar attack display, the fighter can ask for a declaration.

SNFO (PRI) - *“SABRE, Hammer 31, contact BRAA 350, 33, 17 thousand, declare”*

AIC will respond with the entire BRAA location again and add a declaration, if not previously given.

AIC (PRI) - *“Hammer 31, Single group BRAA 350, 30, 17 thousand hot, hostile”*

From this information the fighter knows the picture is a single group, and that the group is correlated to what the fighter sees. From the information on the radar attack display, the fighter can create the spatial image presented in Figure 4-10.

The information on the radar attack display provides the fighter heading, bandit heading, angle-off and target aspect. In addition, the range, rate of closure (V_C), and elevation are known. The fighter can now take the steps required to affect an intercept on the bandit. From the presentation here, it is obvious that the fighter must turn significantly to the left in order to intercept the bandit. This turn to intercept will change the intercept geometry into the intercept triangle. A thorough discussion of intercept geometry is the purpose of the next chapter.

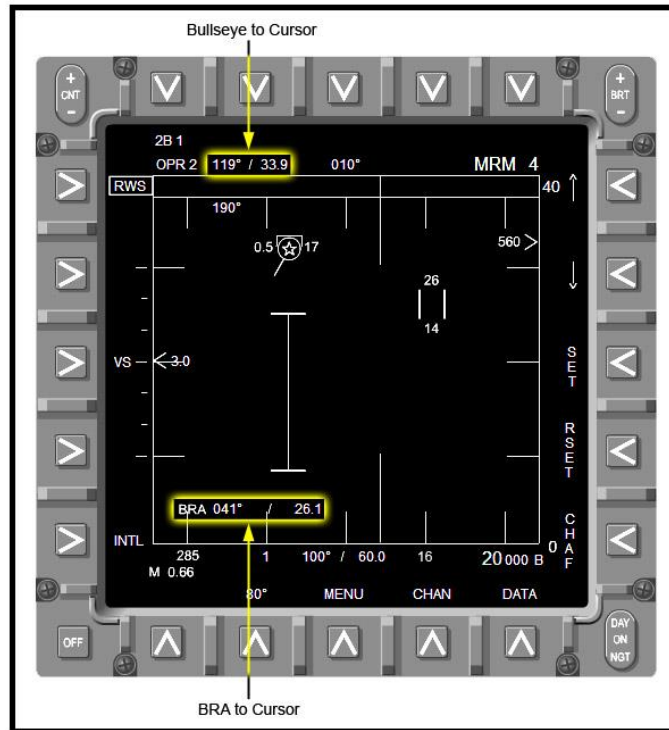


Figure 4-9 Bullseye and BRAA Information on VMTS Radar Attack Display

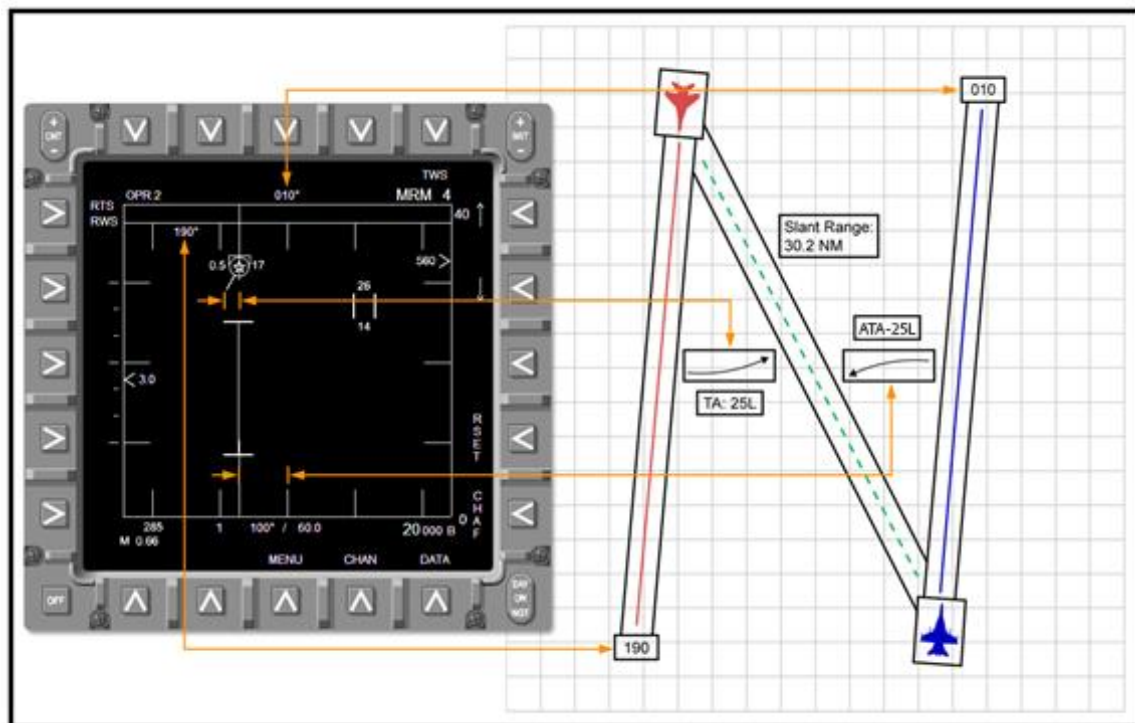


Figure 4-10 Intercept Visualization

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CHAPTER FIVE

FUNDAMENTALS OF INTERCEPT GEOMETRY I

500. INTRODUCTION

Armed with the information presented through AIC communications and intercept displays, the fighter aircrew can recognize initial intercept geometry. Once the initial geometry is recognized, the fighter can take the necessary steps to modify that geometry to affect an intercept with the targeted group.

501. INTERCEPT TERMS

In order to discuss intercept geometry a vocabulary of intercept terminology must be established. Terms such as ATA and TA have been discussed previously, but are included here for reference as they relate to other intercept terms. These terms include:

- Fighter Heading (FH) – Heading of the fighter which, if extended through space, defines the fighter's flight path.
- Fighter Flight Path (FFP) – The logical extension of the direction of the fighter's travel through space on its current FH.
- Bandit Heading (BH) – Heading of the intercepted aircraft.
- Bandit Flight Path (BFP) – The logical extension of the bandit's travel through space on the current BH.
- Bandit Reciprocal Heading (BR) – Bandit reciprocal heading, also called bandit recip (BR) is 180 degrees in the opposite direction from BH.
- Bandit Bearing (BB) – Line of sight (LOS) bearing between the fighter's position and the bandit's position. BB is independent of both FH and BH.
- Antenna Train Angle (ATA) – Angle between FH and BB; also called Angle Off (AO).
- Target Aspect (TA) – Angle between BH and the bearing from the bandit to the fighter; or from BR to BB.
- Aspect Angle (AA) – Number of degrees from the BB to BH; defines the number of degrees to the bandit's six o'clock. AA plus TA always equals 180 degrees. "High Aspect" BFM refers to high AA, not TA, as they are complementary and measure from opposite sides of the bandit. For example, 45 R TA implies 135 R AA, and vice versa.

- Elevation Angle – Number of degrees high or low the contact is detected; can be difficult to determine in a multi-bar scan.
- Heading Crossing Angle (HCA) – The number of degrees between FH and BH at the point of intercept. Also called Degrees to go (DTG).
- Cut – Measure of angle from FH to BR which is equal to 180 degrees minus HCA. Although not commonly used in operating forces, discussions using cut are easier than annotating “180 degrees-HCA” in intercept geometry discussions.
- Rate of Closure (V_C) – Sum of the components of fighter and bandit velocities that contribute to downrange travel.
- Vertical Displacement – The amount of distance between FFP and BFP based on current fighter and bandit altitude.
- Slant Range (SR) – Direct LOS distance between fighter and bandit.
- Lateral Separation (LS) – Also called lat sep, LS is the horizontal distance between the fighter’s current position and BFP.

In addition to these terms, there are some important relationships between these terms that should be understood

The relationships between BH, TA, FH and ATA are the most important in the intercept and need to be understood such that fighter aircrew immediately recognize the impact changing any of these parameters has on the intercept.

502. BANDIT HEADING, TARGET ASPECT AND LATERAL SEPARATION

Bandit heading determines target aspect. Target aspect will change instantaneously with a change in bandit heading. A bandit that turns toward the fighter will decrease the target aspect. Likewise, a turn away from the fighter’s nose can instantaneously create large target aspect, changing the geometry of the intercept.

Because TA represents the difference between the bandit’s bearing and the bandit’s flight path, a bandit that maneuvers by changing heading changes not only TA but the fighter’s location to the bandit’s flight path.

– Lateral Separation – Target Aspect Relationship

Lateral separation is the horizontal distance from the fighter to the BFP. LS is turning room. LS (in thousands of feet) is computed by: $LS = TA \times SR \times 100$. Since SR will only change over time, but TA can change instantaneously, LS will change instantaneously with changes in TA and very slowly with changes in SR.

For example, a bandit at 20 NM with 20 TA will have 40,000 feet (written as 40K) of LS. If the bandit turns to put the fighter on the nose, the LS equation yields $LS = 0$, meaning there is no separation between the fighter and the bandit's flight path at that moment.

In Figure 5-1, the bandit begins with 30 L TA and 60K LS at 20 NM. By turning 30 degrees into the fighter, that is placing the fighter on its nose, the bandit has reoriented its flight path such that the fighter now has 0 LS and is seeing 0 TA. If the bandit maintains a nose-on position, there is nothing the fighter can do to create lateral separation.

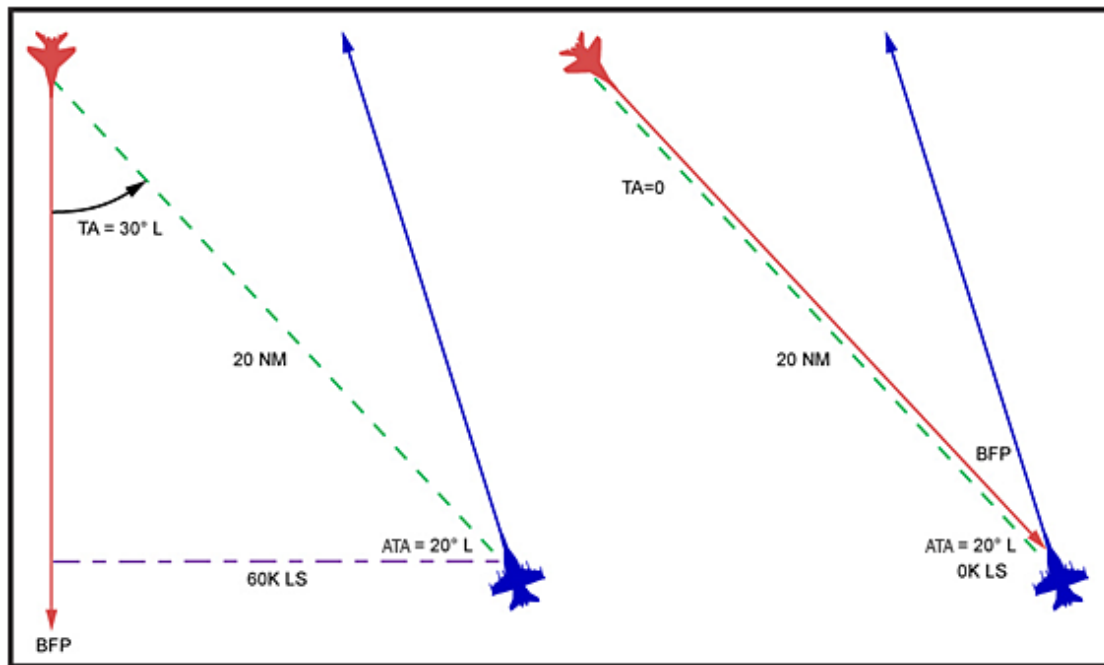


Figure 5-1 Bandit Heading Change

Lateral separation is an instantaneous measurement and will change as both slant range and target aspect change. In Figure 5-2, at 20 NM, the fighter has:

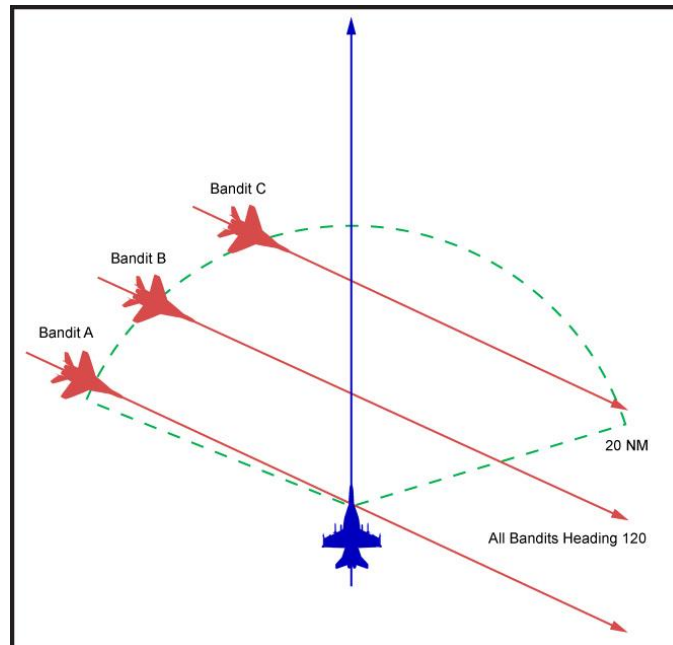


Figure 5-2 Fighter Location from BFP

- a. 0 LS from Bandit A; the fighter is on Bandit As flight path (20 NM x 0 TA)
- b. 40K LS on the right from Bandit B (20 NM x 20 TA)
- c. 80K LS on the right from Bandit C (20 NM x 40 TA)

TA varies even though all of the bandit headings are the same. Also note that whenever the bandit is nose-on, the fighter has 0 LS. This means any bandit maneuver in heading will instantaneously affect LS, while the fighter movement will change LS over time.

In summary, bandit maneuvers always change LS and TA. It is up to the fighter to determine what has happened and how to react to the maneuver. In stern conversions, the bandit will not maneuver, so the fighter must be able to recognize their location in relation to BFP based on TA determination and LS calculations

503. TARGET ASPECT AND V_C

Since TA is directly related to BH, and BH determines the amount of the bandit's velocity that contributes to V_C , the lower TA is, the higher closure will be. V_C can be easily computed, theoretically, by determining the component of the fighter's and bandit's velocities that are contributing to their mutual downrange travel (Figure 5-3).

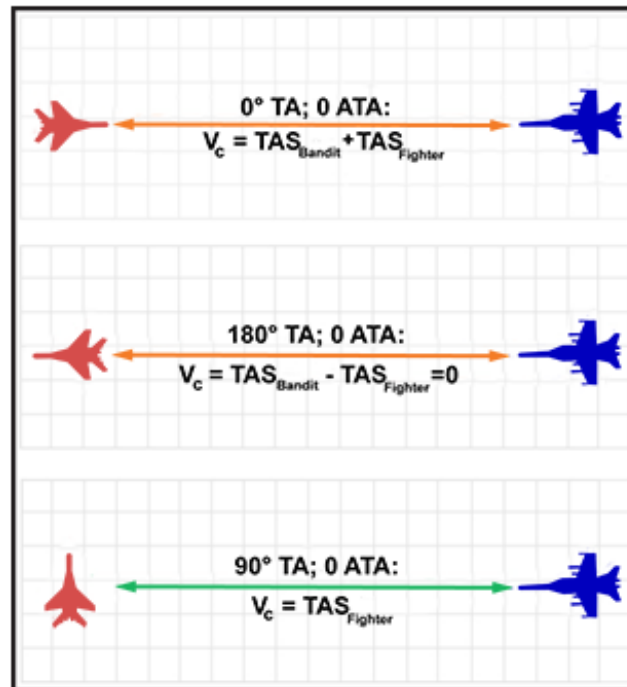


Figure 5-3 V_C and TA

With a co-speed bandit, maximum V_C will be double the fighter's speed (300 KGS + 300 KGS in AWI). With 90 TA, V_C will be equal to the fighter's speed (300 KGS + 0 KGS component to downrange travel) and the bandit may be hidden by the radar's clutter filter. With 180 TA, closure will be 0 (300 KGS + (- 300KGS of receding bandit).

This means the only time a 600 V_C will be seen is with 0 TA and the bandit on the nose. Normal V_C numbers for AWI intercepts are between 420 and 580 KTS.

504. TARGET ASPECT, FIGHTER HEADING AND ANTENNA TRAIN ANGLE

Unlike changes in bandit heading, TA will not change immediately with a change in fighter heading. Rather, with a bandit on a constant heading, the movement of the fighter through space in relation to BFP is required for the fighter to affect TA.

Antenna Train Angle (ATA) will change instantaneously with fighter heading, as any turn by the fighter changes the number of degrees off the nose the fighter sees the bandit.

Against a non-maneuvering bandit, such as those seen in stern conversions, TA can only be changed by the fighter maneuvering to change its position in relation to BFP, thereby changing the fundamental geometry of the intercept. This concept is shown in Figure 5-4.

The bandit begins with 30 L TA and 60K LS at 20 NM and the fighter has 20 L ATA. When the fighter turns 20 degrees into the bandit, that is placing the bandit on its nose, see that the bandit's flight path remains where it was and that there is no change to the target aspect or to the lateral separation.

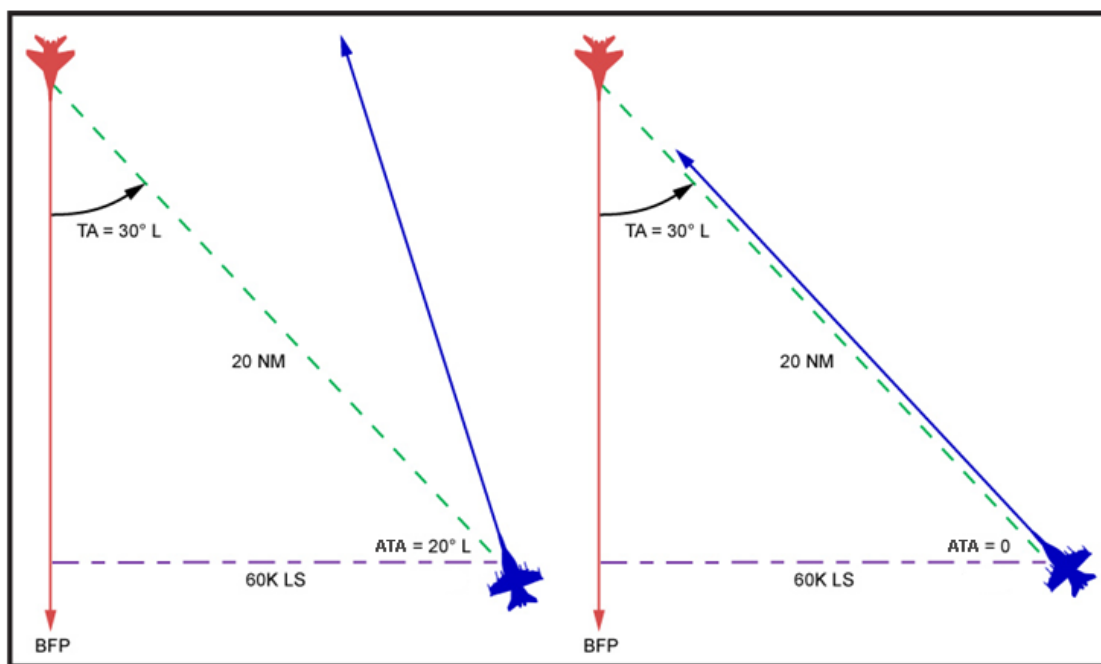


Figure 5-4 Effect of Fighter Heading Changes Only ATA

505. VERTICAL DISPLACEMENT

Vertical displacement (Figure 5-5) is the perpendicular distance the fighter is located above or below the bandit's flight path. This represents the altitude difference between the fighter and bandit and should be computed in feet. Note that the formula for vertical displacement is the same as the one for LS with the vertical displacement (feet) = Elevation X SR X 100.

5-6 FUNDAMENTALS OF INTERCEPT GEOMETRY I

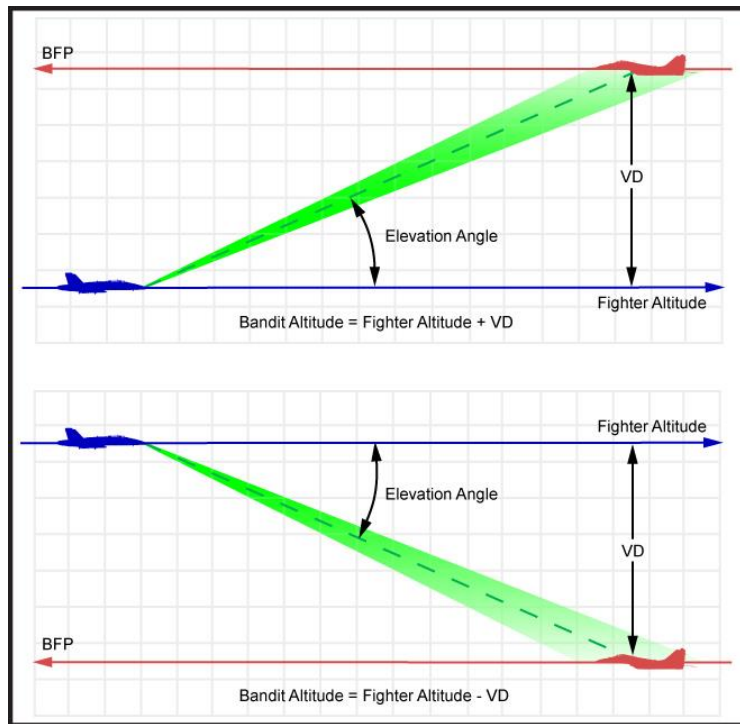


Figure 5-5 Vertical Displacement

If the bandit is 4 degree high at 20 miles, $4 \times 20 \times 100$ means that the bandit is 8,000 feet above the fighter.

During an intercept, AIC will inform the fighters of the altitude of the bandit and the fighters must then compute the elevation angle in order to determine where to place the radar scan volume. Aircrew must be able to work the vertical displacement problem in reverse. For example, if the fighter is at 20,000 feet and AIC calls bandit altitude at 30,000 feet at 20 miles, the fighter should ensure that the scan volume indicates that the bars are covering an elevation angle of at least 5 degree high from the following derivation:

- $30,000 \text{ feet} - 20,000 \text{ feet} = 10,000 \text{ feet of vertical displacement}$
- $10,000 \text{ feet} = 20 \text{ (NM)} \times \text{Elevation angle} \times 100$
- $100 = 20 \times \text{elevation angle or elevation angle} = 5 \text{ degree of elevation}$

Obviously if the fighter is above the bandit, the elevation angle is negative (i.e., depression).

506. INTERCEPT RELATIONSHIPS

Once intercept terms, their relationships, and the information on the radar attack display is understood, the fighter should be able to build a mind's eye view of the intercept geometry based on the information presented on the attack display. The fighter aircrew can also interpret the

information presented to recognize whether the contact(s) they have is (are) the one(s) being correlated by AIC information.

In Figure 5-6, the fighter can determine the following directly from the attack display:

- BH is 190 degrees
- SR is 33 NM
- V_C is 560
- Vertical displacement is 3,000 feet
- Elevation angle is -1 degree
- Bandit altitude is 17 thousand
- Bandit airspeed is 0.5 IMN
- The fighter sees the bandit at an ATA of 20L
- The bandit TA vector is approximately 20L

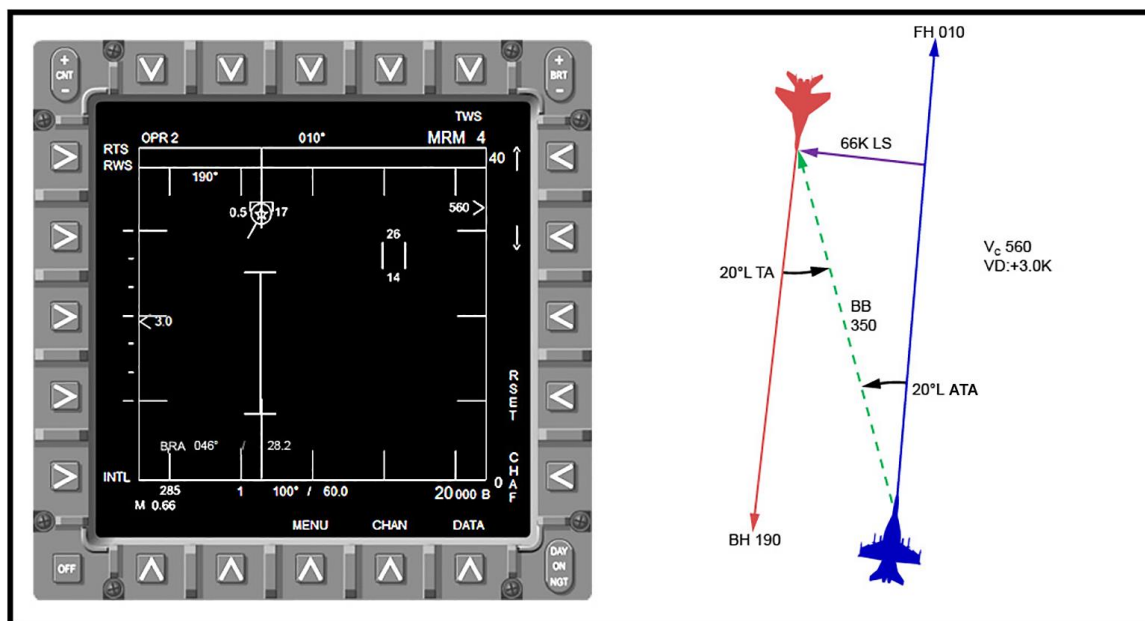


Figure 5-6 Interpretation of Radar Attack Display Information into Mind's Eye View

From this information, the fighter can determine:

- BR is 010 degree
- LS is 66,000 feet (66K)

With this information, the fighter should recognize that FH is on BR, which has a significant impact on future intercept conditions and conduct.

507. FIGHTER POSITIONS AND POSITIONAL ADVANTAGE

The concept of intercept quadrants was touched on in Chapter 5 with the introduction of TA quadrants referenced by AIC. These are shown in Figure 5-7: (A) Head-On Quadrant, 0-30 TA "HOT," (B) Forward Quarter, 31-60 TA "Flank" Aspect; (C) 61-120 TA Beam Quarter "Beam" Aspect; (D) Rear Quarter 121+ TA "Drag" Aspect.

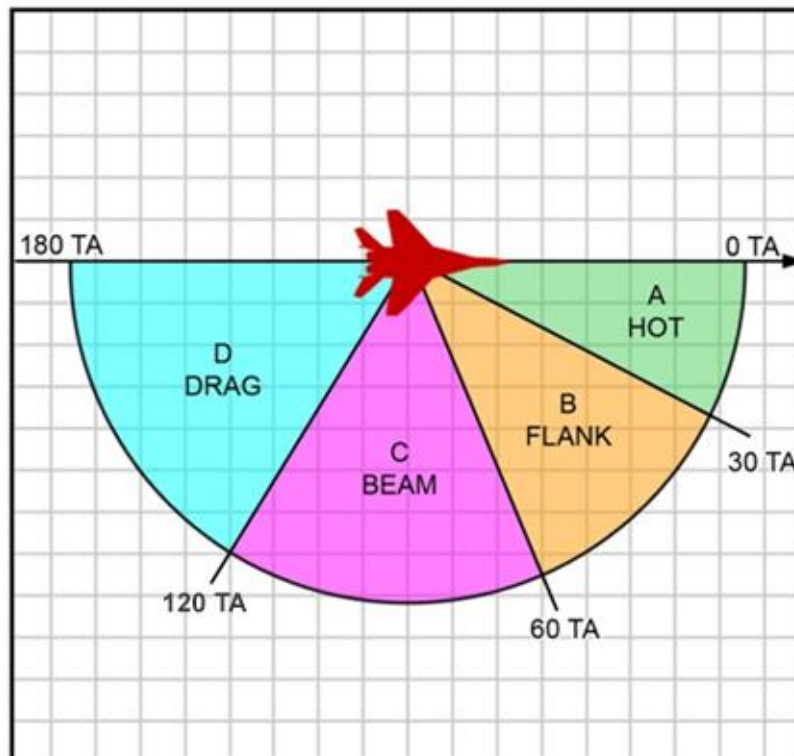


Figure 5-7 Target Aspect

These TA regions also correspond to initial approach quadrants in an intercept and determine whether or not the fighter is beginning the intercept with a positional advantage.

Positional advantage is a relative term used to describe the fighter's ability to attack a bandit or to defend a target in a beyond visual range (BVR) engagement. A fighter with a positional advantage can have an enormous effect on the outcome of the bandit's mission. A fighter beginning with a positional disadvantage will have a difficult time influencing the bandit's actions. In other words, with a positional advantage comes the tactical initiative; without positional advantage, the tactical initiative is not held by the fighter. In Figure 5-8 there is a bandit approaching a friendly ship. Only one of the three fighters has a positional advantage.

Fighter A has a positional advantage since the aircrew can maneuver to place themselves between the bandit and the target. Fighter B is at a positional disadvantage because the distance it must cover to achieve a Forward Quarter (FQ) shot may mean the bandit has attacked the target and is engaged by the fighter on the egress. Fighter C is in a distinct positional disadvantage because it cannot affect the bandit's actions until bandit egresses.

From this, it should be obvious that the fighter in the head-on or forward quarter, that is hot or flank aspect, has a positional advantage. A fighter in the beam quarter, with beam aspect, is in a neutral or disadvantageous position. A fighter seeing drag aspect is in the rear quarter and is at a positional disadvantage. Recognizing whether or not the fighter has a positional advantage at the beginning of the intercept is possible once the initial geometry is recognized. Once an advantage or disadvantage is recognized, the fighter can take action to change the geometry to create an advantage, mitigate a disadvantage or coordinate with AIC for other assets to engage a group that the fighters are out of position to engage.

In the example above, positional advantage is demonstrated from an AIC or "God's eye" view of the situation with the bandit threatening a near stationary target. However, positional advantage may also include factors like a large vertical displacement, where the fighter has an advantage due to its great altitude differential over the bandit, or the placement of the fighter's planned route or target. Fighter aircrew must recognize the impact of their location, both current and planned, on the tactical situation when evaluating positional advantage and take the appropriate steps to create positional and tactical advantages whenever possible.

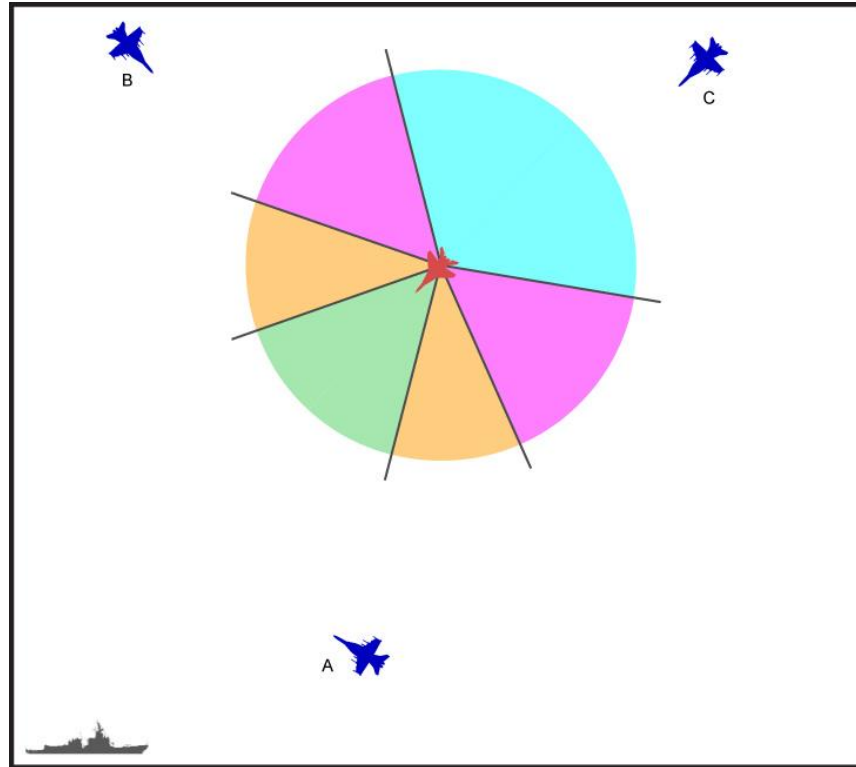


Figure 5-8 Intercept Quadrants and Positional Advantage

Actions used to create or capitalize on a positional advantage include changing heading, altitude, or airspeed or a combination of these. At this stage, with a non-maneuvering, co-speed bandit, geometry changes will be made primarily through heading and speed changes.

508. THE INTERCEPT TRIANGLE

In order for an intercept to take place, the fighter flight path must intersect with the bandit flight path. The intersection point will be the point of intercept. Once aircrew establish the bandit and fighter positions on their respective flight paths, the intercept triangle is formed by the flight paths of the aircraft and the bearing line between aircraft.

The intercept triangle (Figure 5-9) is a useful tool to visualize angles such as TA, ATA, HCA and cut. When combined with the fact that the three interior angles a triangle always add up to 180 degrees, a complete spatial picture of the intercept can be obtained. TA and consequently AA can always be calculated if either BB or ATA and HCA are known.

Manipulation of fighter and bandit heading, thereby changing ATA and TA, respectively, change the geometry of the intercept triangle. Additionally, changes in bandit or fighter airspeed will also change the geometry of the intercept triangle.

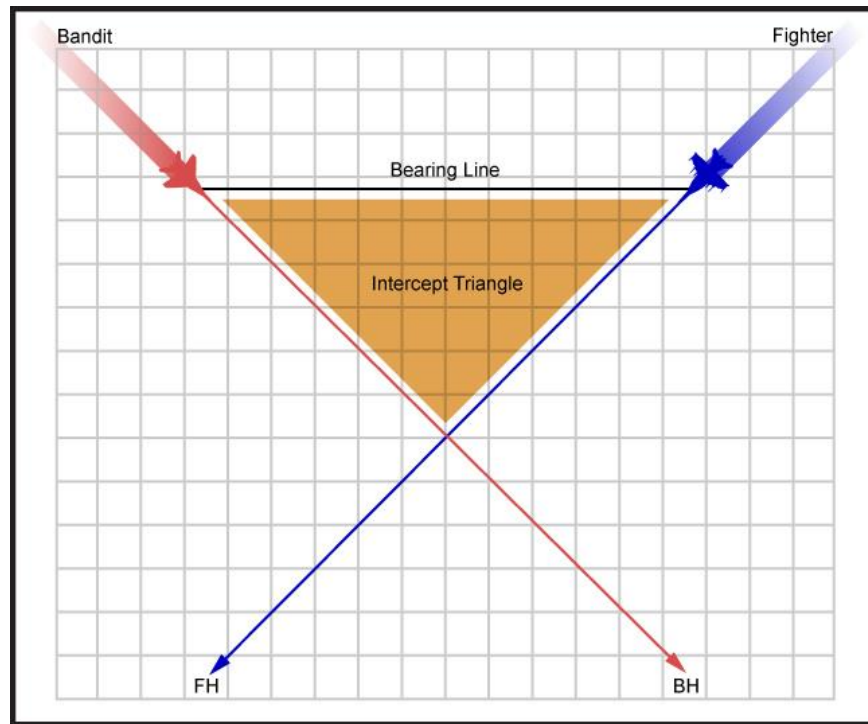


Figure 5-9 Intercept Triangle

509. CONCLUSION

Knowing intercept terminology allows the aircrew to discuss the intercept using a common vocabulary with specific definitions. Fighter aircrew must be well versed in this vocabulary to perform their mission.

The intercept triangle will be explored in much greater detail in the rest of this manual. The recognition that the intercept is a triangle allows the fighter to manipulate the geometry of the intercept to affect a rendezvous to identify and employ weapons on the bandit.

The fighter must use geometry and the intercept triangle to obtain a positional advantage in order to accomplish the briefed mission. An in depth understanding of the relationships between intercept parameters is key in understanding the actions to take to create a positional advantage in any intercept.

CHAPTER SIX

FUNDAMENTALS OF INTERCEPT GEOMETRY II

600. INTRODUCTION

With the information from chapter 5, aircrew can examine the intercept triangle in greater detail to discuss what constitutes optimal intercept geometry in any given situation. For stern conversions, it is assumed that:

- The use of geometry is preferable to burning fuel to affect an intercept. This is not to say a fighter speed advantage will never be used; rather, geometry will be used as the primary means of affecting an intercept.
- The geometry game plan for intercepts should minimize in cockpit calculations.
- The geometry game plan should affect geometry in a logical and predictable manner.
- With these fundamentals in mind, intercept geometry can be discussed in great detail.

601. CONTACT DRIFT

During conduct of an intercept, the contact will move on the radar attack display. This movement is called drift. There are several types of drift and often more than one type is occurring at the same time. Understanding these different types of drift is crucial to understanding what is happening on the attack display and why. The effect of drift will be revisited more than once during the discussion of intercepts at VT-86. If the aircrew knows what kind of drift to expect, then they will be able to detect abnormal conditions, such as bandit maneuvering, much more easily. The types of drift discussed are:

- **Turn Drift:** Movement of the contact on the radar attack display as a result of fighter maneuvering in heading. Occurs strictly as a result of turning the fighter. To illustrate, if the fighter makes a turn 30° to the right, the target will appear to drift to the left by the same amount and rate as the fighter's turn.
- **Intercept Drift:** Movement of the contact due to the movement of the fighter and bandit through space. This is the result of changes in geometry during the intercept. For example, as two aircraft approach the merge, the bearing to the other aircraft changes as range decreases. At range, this same effect is seen, but is less pronounced. Controlling intercept drift is a primary goal during any intercept.
- **Displayed Drift:** The sum of intercept and turn drift, displayed drift is what the aircrew perceives on the radar attack display. It is very important for aircrew to recognize what is causing the resultant displayed drift and what this drift means.

Recognition of the type of drift that is occurring and why is a fundamental fighter aircrew skill.

602. COLLISION COURSE AND THE ISOSCELES INTERCEPT TRIANGLE

An air-to-air intercept is a relative motion problem in which the fighter is constantly striving to control its position in relation to the bandit. Changes in the bandit and fighter's position will result in intercept drift. The ultimate manner of controlling intercept drift is establishing a collision course. On a collision course, the fighter has established a constant bearing, decreasing range situation that, if not changed, may result in a midair collision. A contact which is on collision will not drift. A contact which is not on collision must drift and it will always drift away from collision bearing (CB). CB has a direct impact on intercept control.

Recall from Chapter 5 that the intercept triangle is made from the fighter and bandit flight paths and the bearing between the two. A collision course will exist when geometry is such that the bandit remains on a constant bearing with no changes in heading by the fighter. When this occurs, the following conditions will be present:

- ATA will be equal in magnitude, but opposite in direction to TA
- The distance to the intercept point for both the fighter and bandit will be equal
- V_C for the intercept will be higher but not at the maximum (with both aircraft at the same airspeed)

This is illustrated in Figure 6-1.

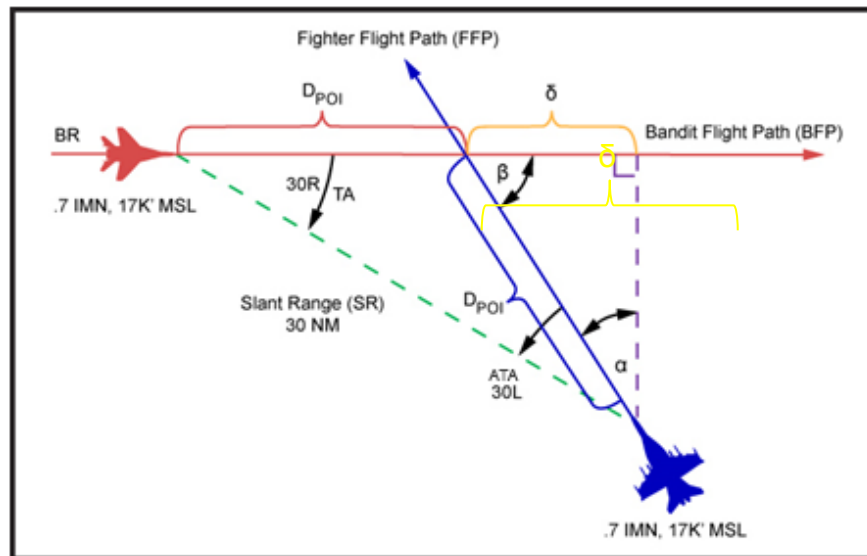


Figure 6-1 Co-Speed Fighter and Bandit On Collision

Using basic trigonometry and knowing TA, ATA and SR, one can show that, for the distance to the point of intercept (D_{POI}) the fighter must travel:

- $\sin TA = LS/SR$; therefore $LS = SR \times \sin TA$; in this case $0.5 = LS/30$ or $LS = 15 \text{ NM}$
- Angle $\alpha = 180 \text{ degrees} - 30 \text{ degrees} - 90 \text{ degrees} - 30 \text{ degrees} = 30 \text{ degrees}$
- $\cos \alpha = LS/D_{POIFighter}$; therefore $D_{POIFighter} = LS/\cos \alpha$;

Therefore $D_{POIFighter} = 15/\cos 30 \text{ degrees} = \mathbf{17.32 \text{ NM}}$

For the distance the bandit must travel:

- $\cos TA = (D_{POIBandit} + \delta)/SR$; therefore $D_{POIBandit} + \delta = SR \times \cos TA = 25.981 \text{ NM}$
- $\tan \alpha = \delta/LS$; therefore $\delta = \tan 30 \text{ degrees} \times 15\text{NM} = 8.660 \text{ NM}$
- $D_{POIBandit} = SR \times \cos TA - \tan \alpha \times LS = 30 \cos 30 \text{ degrees} - 15 \tan 30 \text{ degrees} = 25.981 \text{ NM} - 8.660 \text{ NM} = \mathbf{17.32 \text{ NM}}$

To minimize the time-to-kill the fighter should establish a heading that maximizes V_c and minimizes the distance traveled to the intercept point. If a fighter initially maximizes only V_c by maintaining a pure pursuit path, the distance traveled will be much further than that of establishing a collision course. The mathematical proof of this is too long to discuss in this text, however, it is a logical inference for anyone who has ever played football, soccer or basketball that you don't run AFTER the person with the ball, you position yourself to intercept that person at the place you estimate they will be. This is essentially the same fundamental principle used in an intercept, where establishing a collision course is preferable to remaining in pure pursuit. When established on collision:

- TA will not change
- ATA will not change
- The contact will not drift

Knowing that TA will always equal ATA when a collision intercept has been established in a co-speed situation, the fighter can establish a collision intercept by adjusting its heading to match the ATA with the previously calculated TA. Collision course is an effective tool to rapidly close with the bandit.

603. TA AND ATA WITH A BANDIT OFF COLLISION COURSE

If a CB intercept is not established, then one of two things can happen:

- Either the bandit will pass in front of the fighter (a good thing at close range).
- Or the fighter will pass in front of the bandit (a bad thing).

A contact not on a collision course will be subject to intercept drift. This intercept drift will always be away from collision bearing.

For example, if the fighter and bandit are established on opposite headings, that is FH equals BR, and there is LS between their flight paths, then ATA and TA will increase with a decrease in range.

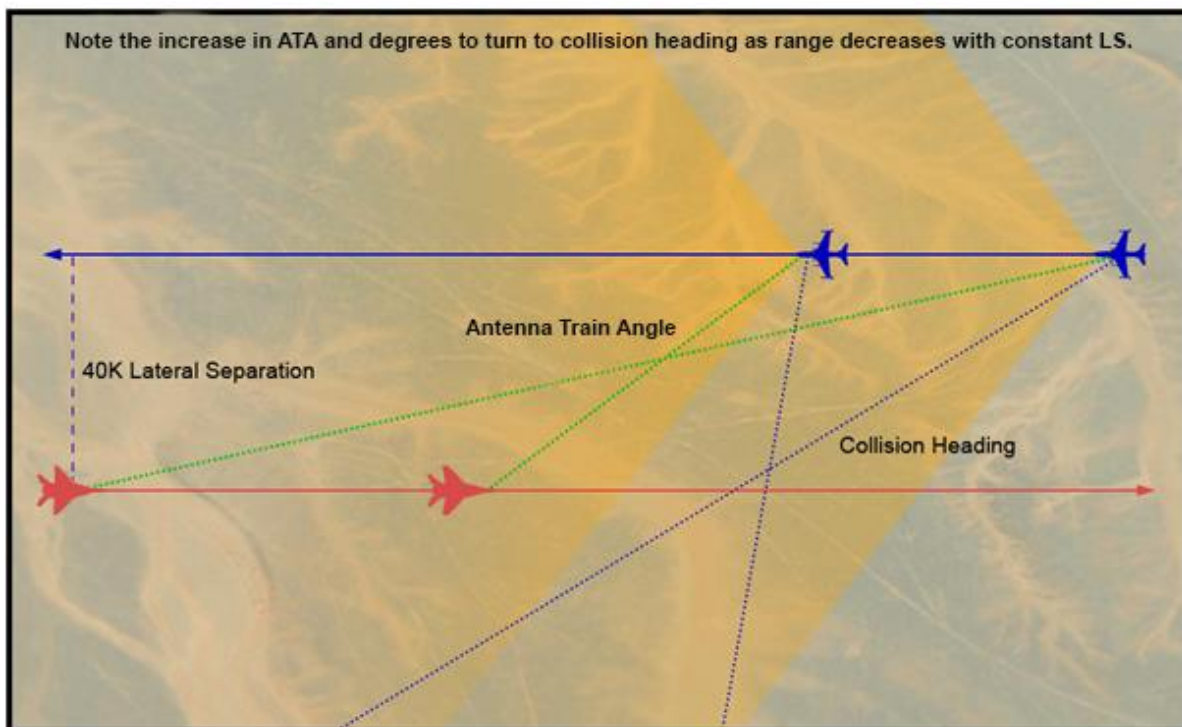


Figure 6-2 Relationship between ATA, TA and Slant Range

From Figure 6-2, it is easy to see how TA and ATA both increase as slant range decreases. This will occur whenever the bandit is not on a collision course.

604. PREDICTING ATA AND THE CHANGE IN TA

Knowing that the contact will drift when not on collision, and that the drift that occurs will be away from collision bearing, the drift of the contact on the radar attack display becomes very predictable. The most simplistic example of this is when fighter heading is bandit recip (FH=BR). Since TA is also measured from BR to BB, the drift of the contact away from BR represents a change in TA. For every degree the contact drifts, TA increases by one degree. This is shown in Figure 6-3.

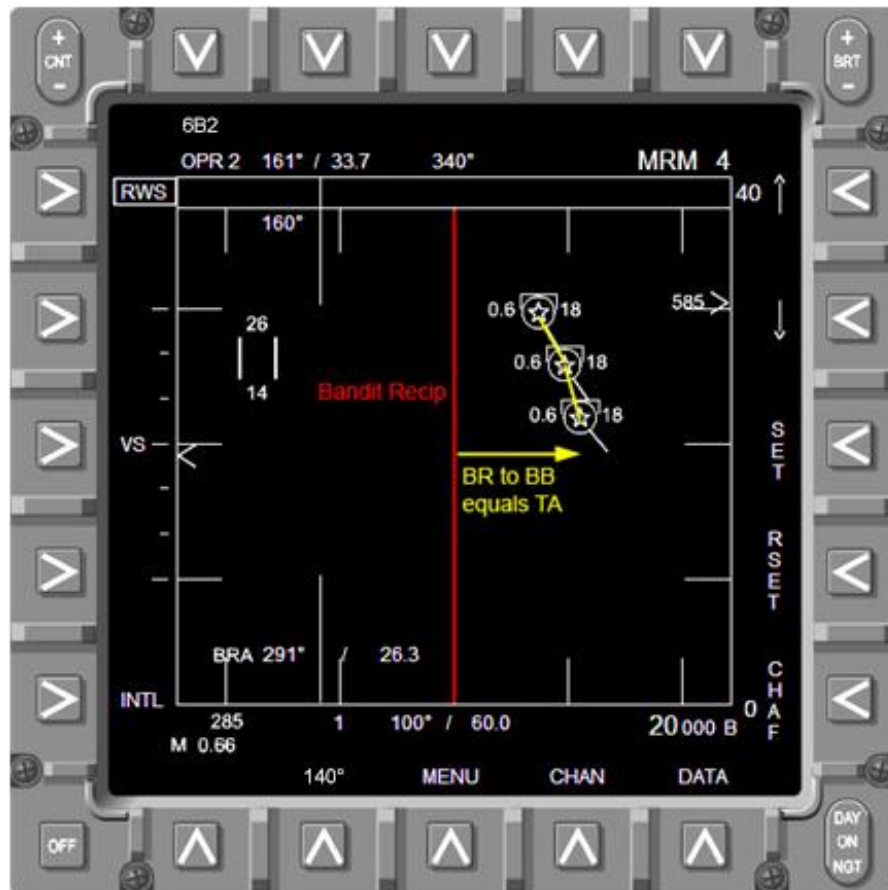


Figure 6-3 ATA Drift

Similarly, a contact that is not on collision will drift away from collision. For example, if the fighter turned to place a contact on collision bearing at 30 degrees left ATA, and had misread the target aspect vector as 30 degrees right when it was actually 40 degrees right, the contact will drift away from the CB, which will be an increase in TA. This is shown in Figure 6-4.

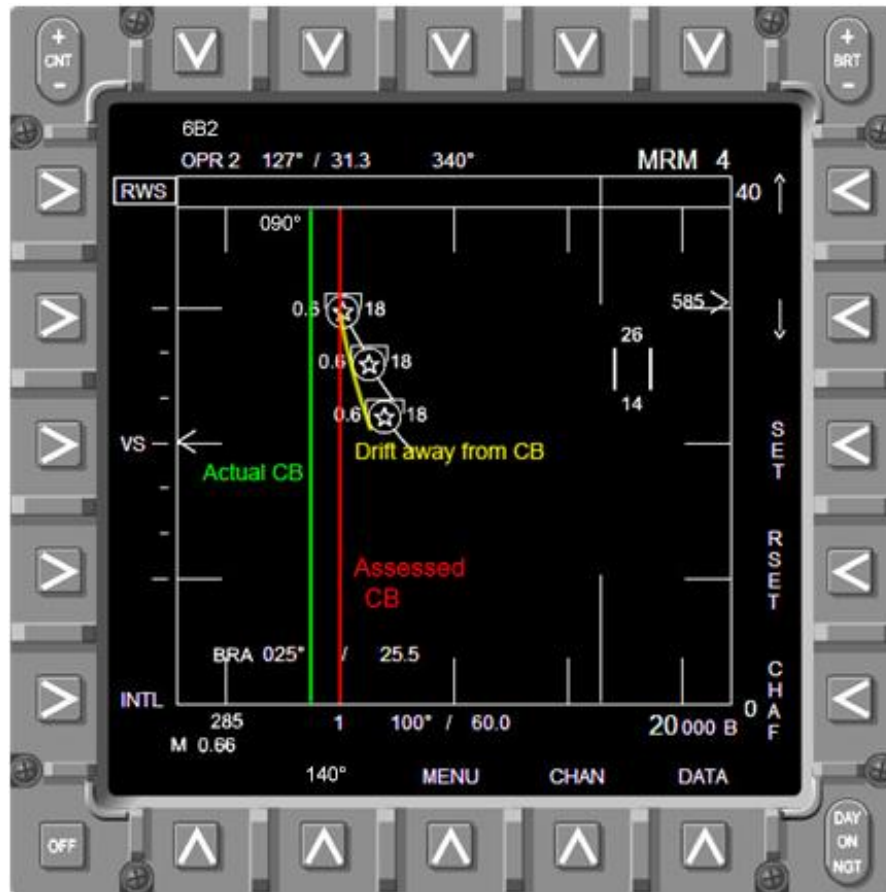


Figure 6-4 Drift with Contact Not on Collision

In another example (Figure 6-5), the fighter turns to try to place the contact on collision and forces a drift that reduces TA. In this case, the contact is placed at 40 degrees right ATA with 30 L TA. The result is that the contact will drift to the right, away from the actual CB at 30 R. Since this drift is toward BR, this drift is a decrease in TA. In order to arrest this change in TA, the fighter must make a change in the geometry of the intercept.

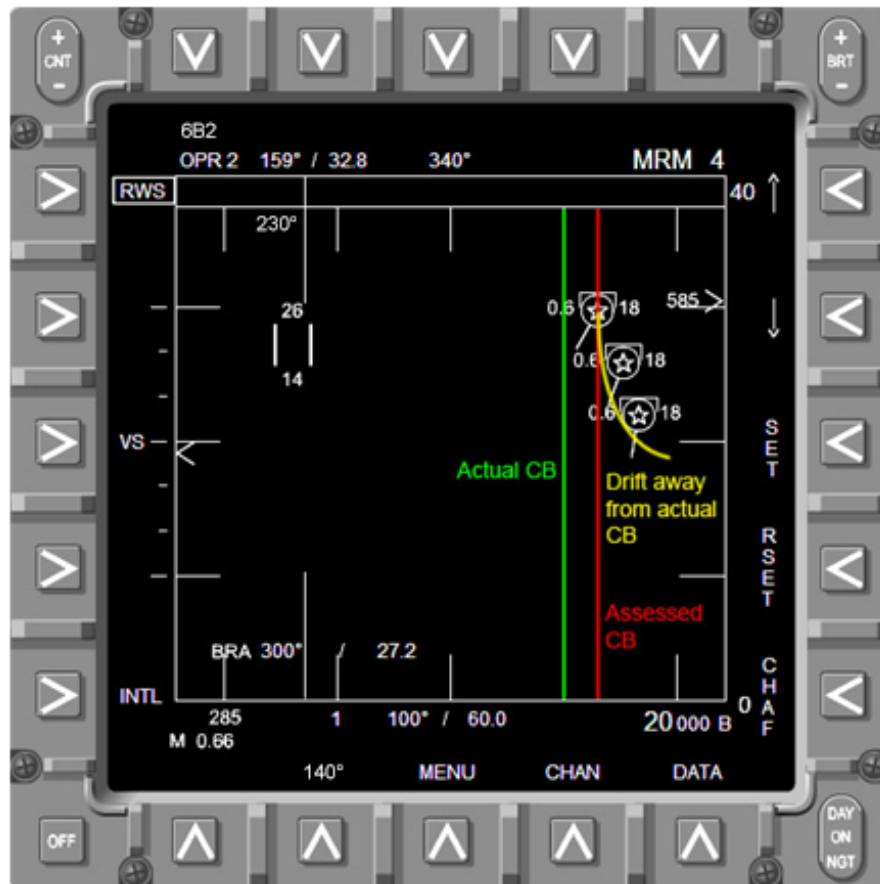


Figure 6-5 Bandit Off Collision

While collision course is useful, there are many occasions where the fighter wants to either increase or decrease TA, and the associated LS, for various reasons. This requires that the fighter understand the geometry created by changing fighter heading, thereby changing the HCA and the associated cut.

When discussing intercepts, cut is used instead of HCA because the cut is easily referenced on the radar attack display. The cut corresponds to BR. Once the fighter has found the bearing on the display that represents BR, the cut is known. This is further aided by the BRAA to cursor information being displayed on the radar. Knowing where the cut and BR are let the fighter immediately determine how TA will change and how the contact will drift.

605. HEADING CROSSING ANGLE, CUT, ATA AND LS RELATIONSHIPS

Heading crossing angle (HCA) is the angle between fighter heading and bandit heading the remaining angle between fighter heading and bandit reciprocal heading (BR or “recip”) is called the cut (Figure 6-6). The term “cut” has fallen out of common use in operating forces, but it is a useful learning tool in intercept geometry. Cut is used because the math is easier than using “ $180 - \text{HCA}$ ” every time the relationship is used. A 90 cut is equal to a 90 HCA, as the fighter then has 90 degrees to turn to match bandit heading. Changing the cut changes HCA and vice-versa and can be changed by either fighter or bandit heading changes.

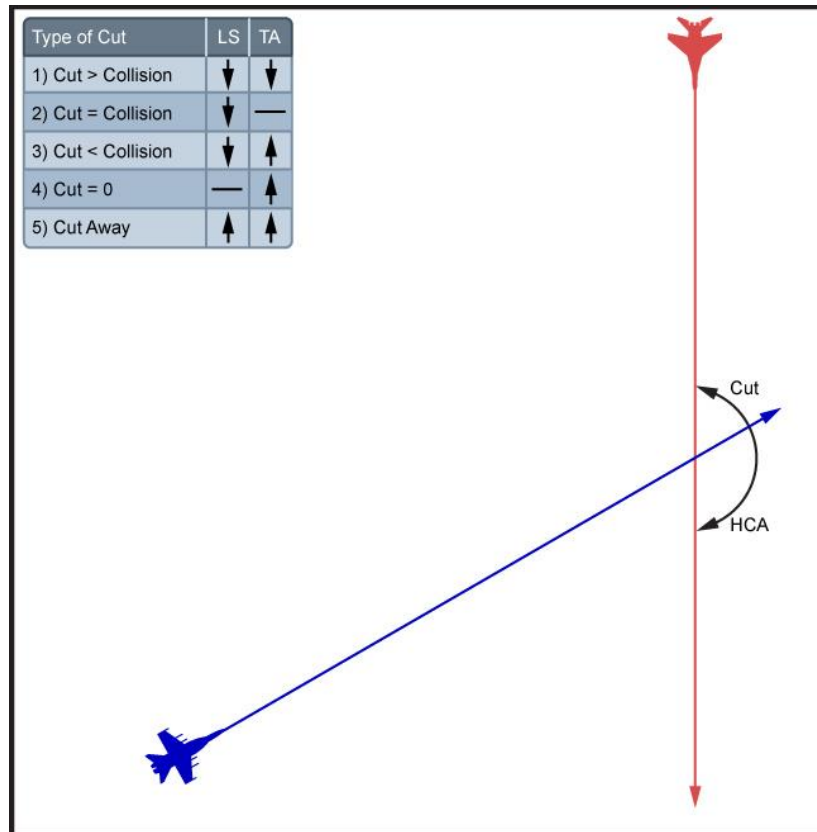


Figure 6-6 Cut and HCA Relationship

There are five types of cuts which fall into three categories. These categories are:

- **Cut into bandit flight path** – an HCA creates a cut into bandit flight path whenever the flight paths of the fighter and bandit cross if neither changes heading. A cut into will decrease Lat Sep. There are three types of cuts into based on how they are related to collision.
- **Cut away from bandit flight path** – an HCA creates a cut away from bandit flight path when the heading places the fighter’s flight path such that the fighter will move away from the bandit’s flight path over time. A cut away will increase Lat Sep.

- **Cut equal to bandit reciprocal** – this is also called a zero cut. A zero cut will maintain Lat Sep.

Each of these types of cuts has specific properties that make it useful in controlling intercept geometry.

1. Cuts Into Bandit Flight Path

Cuts into bandit flight path will always reduce lateral separation (LS). However their effect on TA depends on their relation to the cut required for collision. These are detailed below:

- Cut into equal to collision (Figure 6-7) – Cut into equal to collision places the contact on collision. TA will remain unchanged while LS will decrease. ATA will be equal in magnitude and opposite in direction from TA. If co-altitude, the aircraft may collide and will, at a minimum, be in very close proximity at flight path fly through. This cut is a type of lead pursuit and represents the most efficient geometry to close with the bandit.

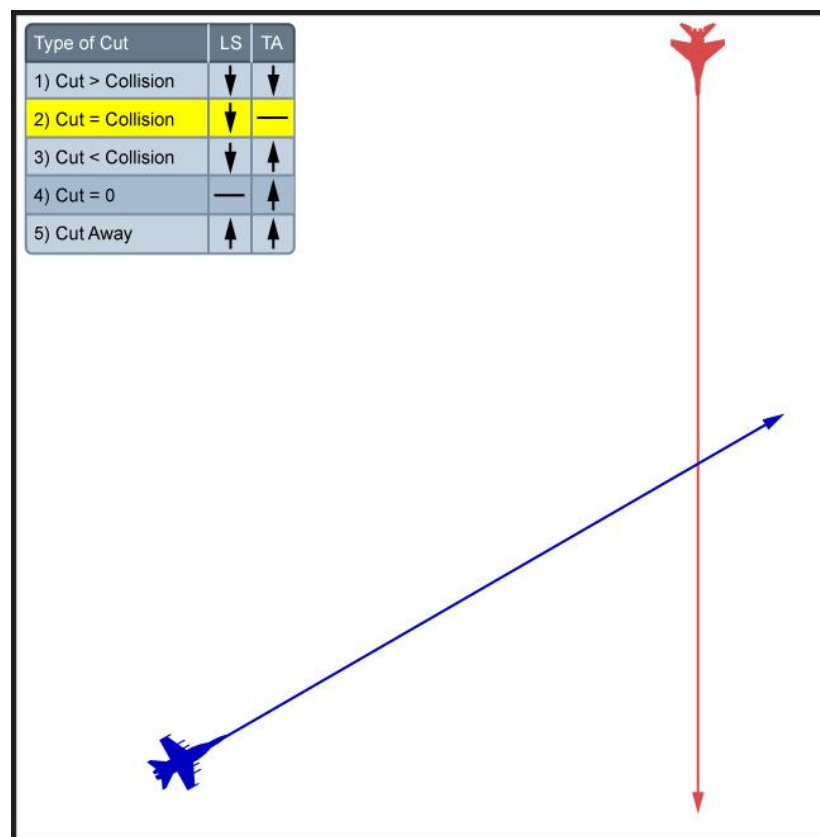


Figure 6-7 Cut Into Equal To Collision

- b. Cut into greater than collision (Figure 6-8) – A cut into greater than collision will reduce LS and TA over time. The fighter's flight path is well in front of the bandit with the fighter in a lead pursuit. The fighter can reduce both TA and LS with this type of cut. In fact, this is the ONLY cut which will reduce TA. To see this kind of cut on the radar attack display the fighter needs to see the TA vector pointing across the display with the contact at an ATA greater than that required for collision. The contact will drift toward the edge of the display, which will indicate a reduction in TA. Drift will be away from the TA vector. To establish a cut into greater than collision on the radar display below, the fighter would have to turn left in order to place the bandit in the shaded area of the display.

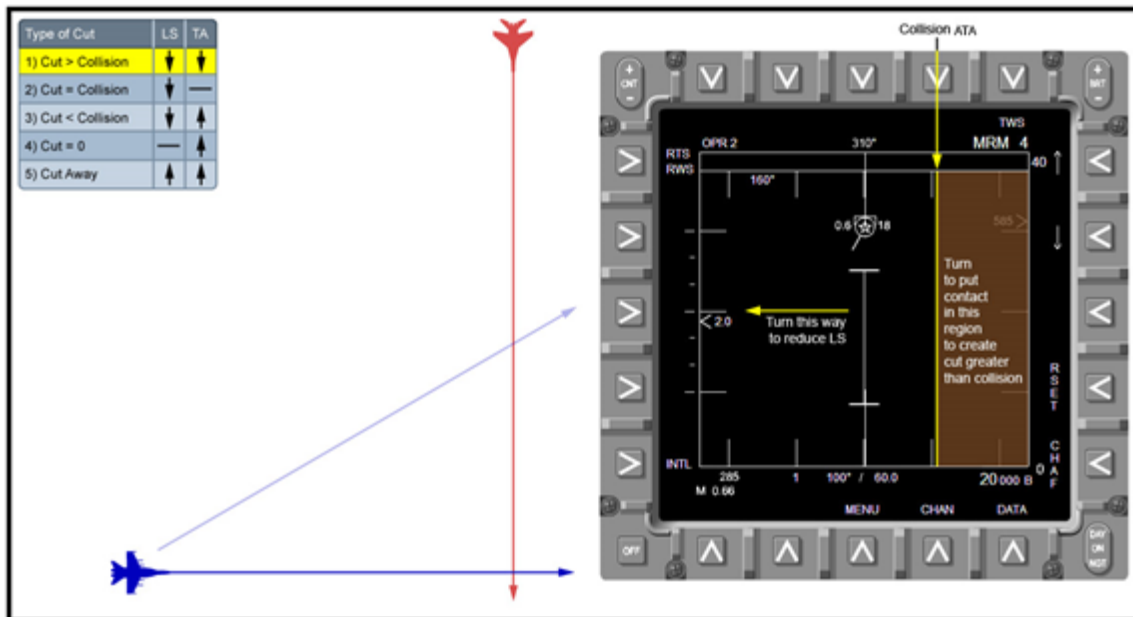


Figure 6-8 Cut Into Greater Than Collision

- c. Cut into less than collision (Figure 6-9) – A cut into less than collision places the fighter in either lead, pure or lag pursuit with an HCA less than that required for collision. Pure and lag pursuit are always a cut into less than collision. Fighter headings between pure and collision are lead pursuit with a cut less than collision. This type of cut will reduce LS while allowing TA to increase, which is useful when the bandit is approaching or is within visual range, but unaware of the fighter. On the attack display, the contact's TA vector will be pointed toward the edge of the display with the fighter's nose in pure or behind the contact. If in lead, the contact's TA will be more than the contact's ATA and drift will be in the direction of the TA vector.

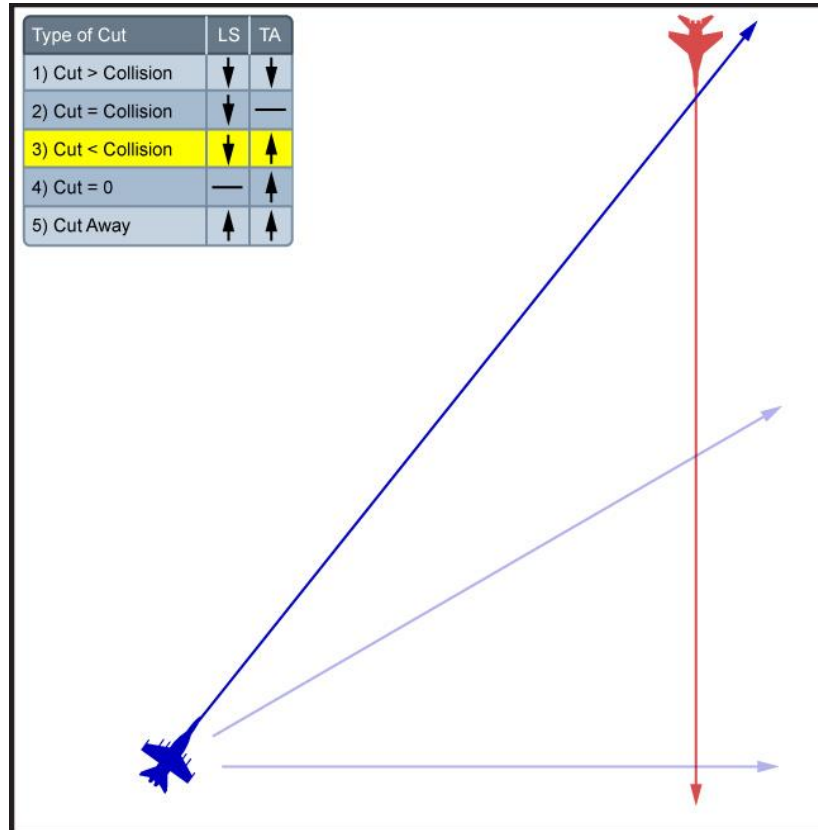


Figure 6-9 Cut Into Less Than Collision

2. Zero Cut – Fighter Heading On Bandit Recip

When fighter heading is equal to the bandit reciprocal, the fighter is on a zero cut (Figure 6-10). HCA is 180 and the fighter has 180 degrees to turn to bandit heading. With 0 TA, the bandit is on collision and will merge in close proximity with fighter nose-on if they do not collide, when range reaches zero. With $TA > 0$, the fighter is on a parallel, opposite direction, flight path. TA will double as range halves. For example, on a zero cut with 15 TA at 30 NM, there will be 30 TA at 15 NM. This is the only cut that does not change LS, TA will grow. Since no LS is being removed, this is not a type of pursuit.

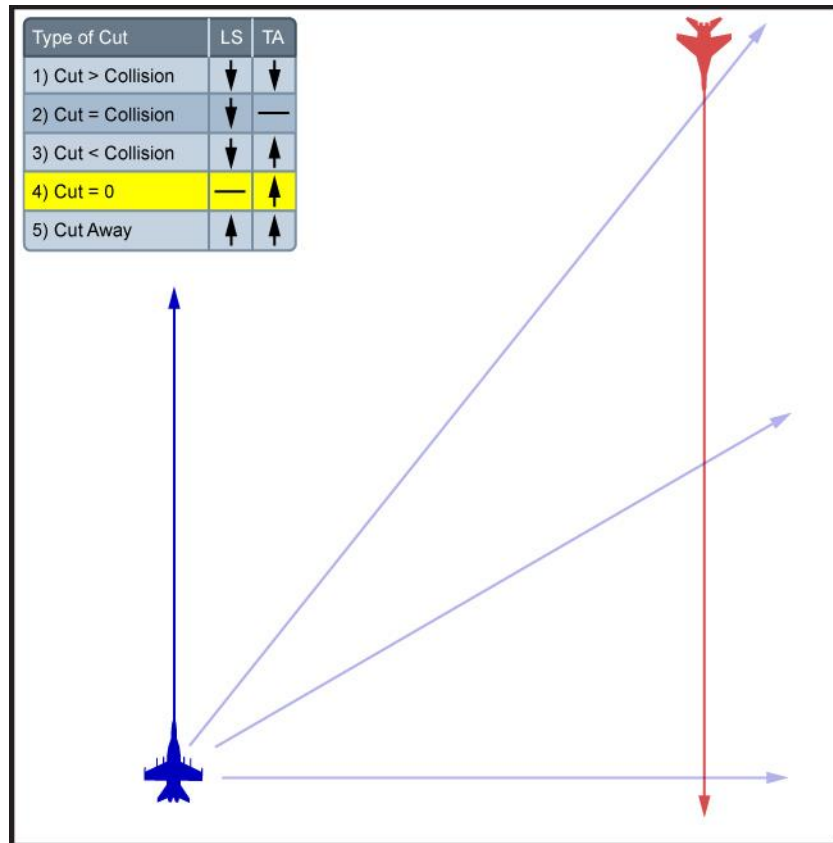


Figure 6-10 Zero Cut

These properties of a zero cut are very important in that this is the only cut which preserves the current LS.

3. Cut Away from Bandit Flight Path

A cut away is the only way to increase LS (Figure 6-11). At high TA, a cut away will mean the contact is off the attack display, while a cut away for low TA may keep the contact on the radar attack display. A cut away will be used by the fighter any time LS needs to be increased. Since LS is being generated, it is not a type of pursuit.

Understanding the five types of cuts will allow the fighter to manipulate intercept geometry to accomplish whatever goal is required.

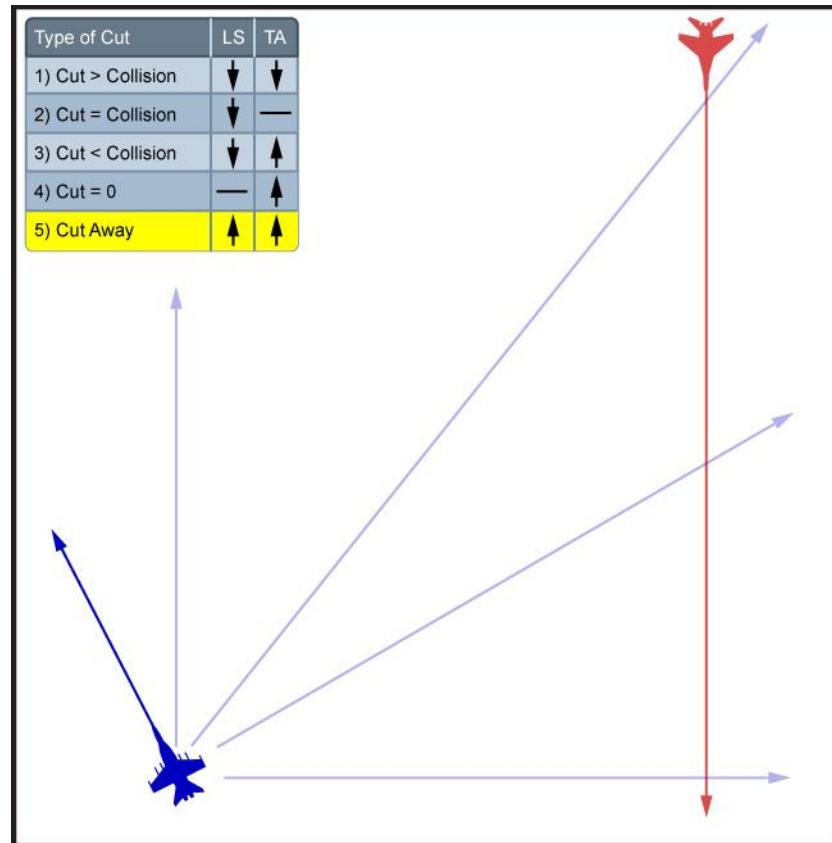


Figure 6-11 Cut Away from Bandit Flight Path

606. MAXIMIZING LS/TA CHANGES IN THE CO-SPEED INTERCEPT

The effect of HCA on intercept geometry should be readily apparent. While an instantaneous snapshot of the intercept can be made by evaluating TA, ATA and slant range and determining LS, it is the cut that determines how this geometry will change over time. With a non-maneuvering contact, the fighter's changes in heading will create or remove lateral separation and build or remove TA over time. How the fighter can manipulate intercept geometry is the focus of stern conversion intercepts.

In a co-speed intercept with a non-maneuvering bandit, the fighter has control over LS and TA. The limiting factor for the ability to impact these is the range at which the intercept begins. A couple of examples that help highlight this:

1. Maximum LS Generation from a 0 TA, 0 LS Start

Assuming a 0 TA, 0 LS start against a co-speed, non-maneuvering contact, the fighter can immediately turn to a 90 cut (90 HCA) to build the maximum amount of LS in the minimum amount of time. Assuming 20 NM start 300 KGS (approximately 0.5 IMN), or 5 NM per minute, and that the fighter must turn hard as possible to turn to a zero cut when LS equals 40,000 feet (desired turning room in stern conversions) or nose-on when SR reaches 10 NM (nominal visual range) the following can be demonstrated:

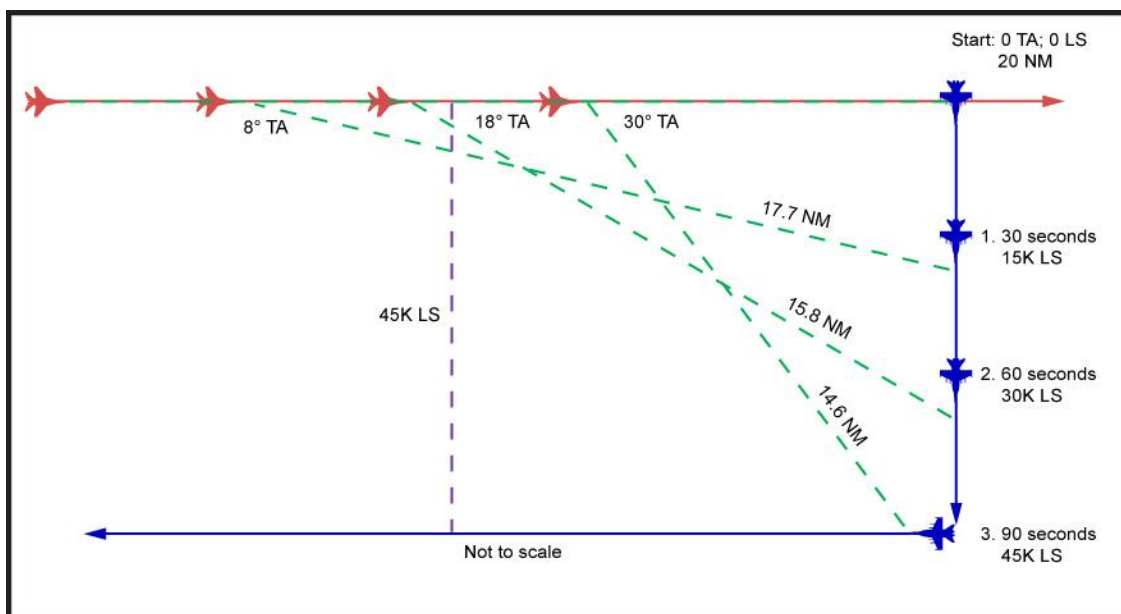


Figure 6-12 Fighter using 90 Cut to Generate Turning Room

From Figure 6-12:

- a. After thirty seconds, the fighter has generated 2.5 NM, or 15,000 feet of LS
 - i. SR has decreased from 20 NM to 17.7 NM
 - ii. TA has increased to 8.1 degrees ($8.1 \times 17.7 = 14,390$ feet LS per the formula)
- b. After one minute, the fighter has generated 5 NM, or 30,000 feet LS
 - i. SR is now 15.8 NM
 - ii. TA is now 18.1 degrees (LS formula yields 28,662 feet LS)

- c. At ninety seconds into the intercept, the fighter has generated 7.5 NM or 45,000 feet of LS
 - i. SR is 14.6 NM
 - ii. TA is 29.5 degrees (LS formula yields 42,980 feet LS)

At this point the fighter should turn hard as possible to BR to capture and hold 40K of LS. When the fighter turns to a zero cut, the bandit will be about 30 ATA at a slant range of about 14 NM. TA will continue to grow and, if no further heading changes are made, the fighter and bandit will pass with 45K LS about one minute later. The fighter would not have radar contact to the bandit until the turn to a zero cut. Once on the zero cut, the contact will be at 30 degrees right ATA and will drift toward the edge of the radar attack display as TA increases.

2. Maximum LS Removal Example

A 90 cut into can be used to remove the maximum amount of TA and can be used to remove the maximum amount of LS. If the fighter began with 30 TA start at 20 NM slant range (120,000 feet), the fighter would begin with 60,000 feet LS (60,000 feet LS by the formula as well). In this example, the goal of the fighter is to establish a position as close to the bandit's flight path as possible, but the fighter must be nose-on at 10 NM. The fighter would expect to see the following:

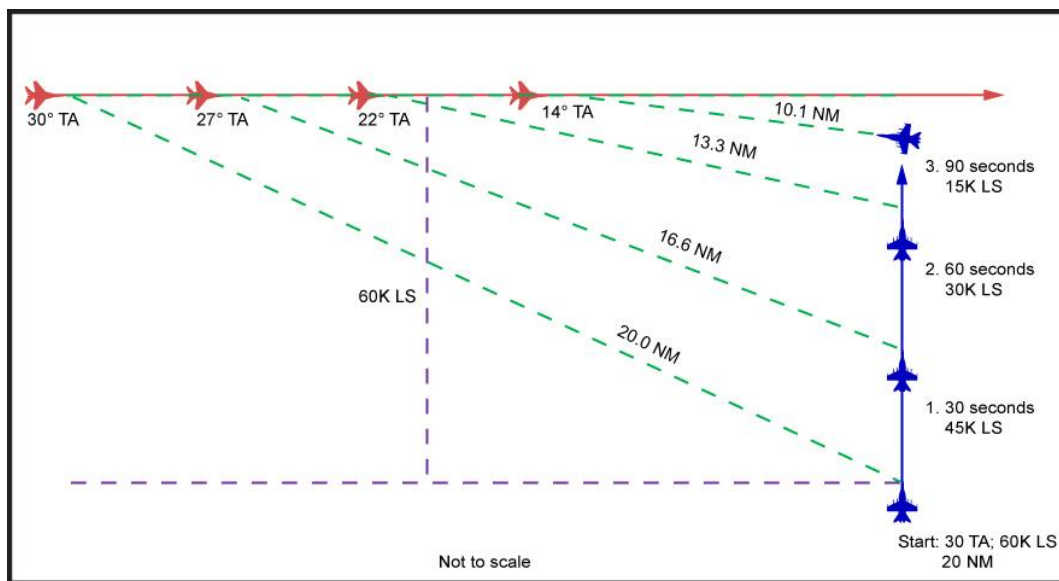


Figure 6-13 LS Removal Example

From Figure 6-13:

- a. Initially, the fighter has 60K LS
 - i. TA is 30 degrees on the right

- ii. ATA is 60 degrees on the left
- b. At 30 seconds, the fighter has removed 15K of LS
 - i. TA has been reduced to 26.8 degrees
 - ii. SR is 16.6 NM
 - iii. ATA is 63.2 degrees on the left
- c. At 60 seconds, the fighter has removed 30K LS
 - i. TA is reduced to 22.1 degrees
 - ii. SR is 13.3 NM
 - iii. ATA is 67.9 degrees (recall that radar attack display limitation is 70 degrees left or right in azimuth)
- d. At 90 seconds, the fighter has removed 45K LS
 - i. TA has been reduced to 14.3 degrees
 - ii. SR is 10.1 NM
 - iii. ATA is 75.8 degrees

What is important to note here is that although the change in LS from this example is the same as in the previous example, the removal of LS changes TA more slowly. The reason for this is that the rate of change of TA increases the further off collision bearing the contact is. In other words, the further a contact is from collision bearing, the faster the contact will drift. This drift will always be away from collision bearing.

Also note that the drift away from the fighter's nose with the cut into greater than collision will be in the opposite direction of the contact's TA vector. ANY time a contact drifts the opposite direction from the aspect vector, the fighter is removing LS.

Additionally, when using a 90 cut away to create LS, the contact was not on the attack display. This is not preferable. Likewise, using a 90 cut into remove LS made the contact drift off the radar attack display. This is also not preferable.

607. HOT AND COLD ON THE ATTACK DISPLAY

Based on TA, ATA, and fighter heading (which determines the HCA and, therefore, the cut), the fighter can identify the “hot” and “cold” sides of the display (Figure 6-14). These are defined as follows:

- The hot side of the display is the portion of the attack display where the fighter should place the contact to increase the rate at which LS is being removed. This represents a turn toward bandit flight path.
- The cold side is the direction the fighter should turn to place the contact to slow down, stop or reverse the removal of LS. In other words, this is a turn away from bandit flight path.

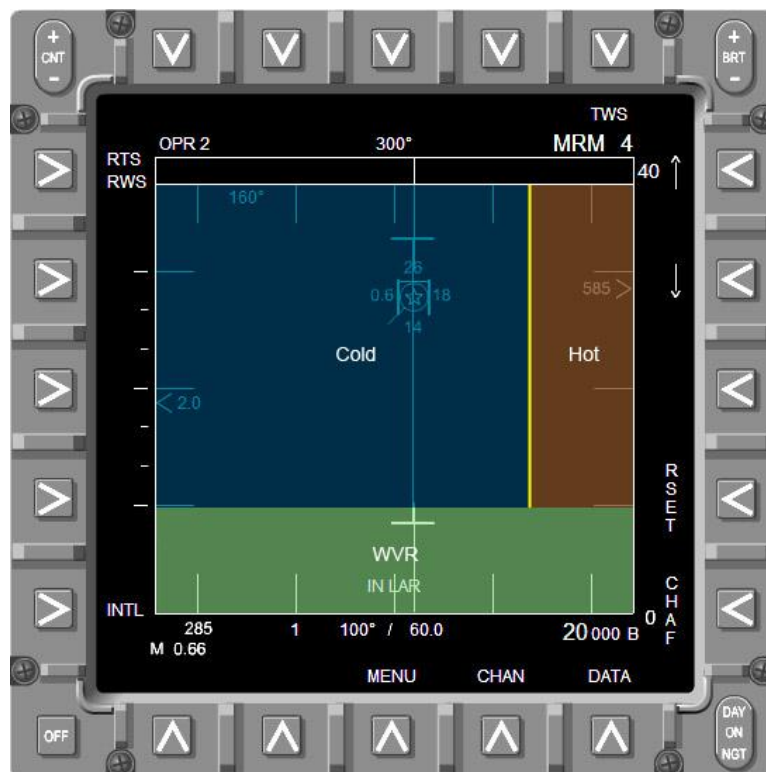


Figure 6-14 Hot and Cold Sides of the Attack Display

The dividing line between the hot and cold sides of the display is the ideal intercept curve, or where the contact is on collision.

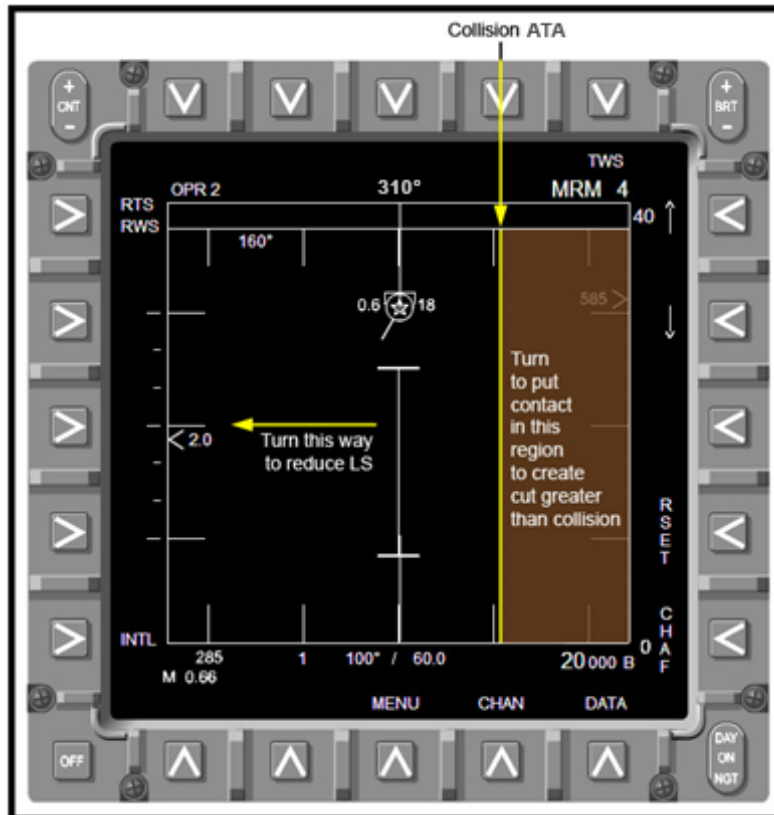


Figure 6-15 Hot Side of Attack Display

Because a TA vector is used instead of a heading vector on the L&S symbology, where it points can be used to denote to the fighter where hot and cold are on the attack display. Simply put, the TA vector stick always points in the direction to turn to move the contact toward the hot side of the attack display (Figure 6-15). Turning toward the aspect vector will heat up the intercept. Turning away will cool off the intercept. Although interpretation of the TA vector may not seem intuitive, having a TA vector provides better SA to the positional relationship between aircraft than a heading vector does. The dividing line between hot and cold on the contact is collision bearing (CB). As TA changes the amount of turn to place a contact hot or cold will change with the corresponding change in CB. The SNFO should continue to assess TA to identify collision bearing and the subsequent hot/cold sides of the attack display.

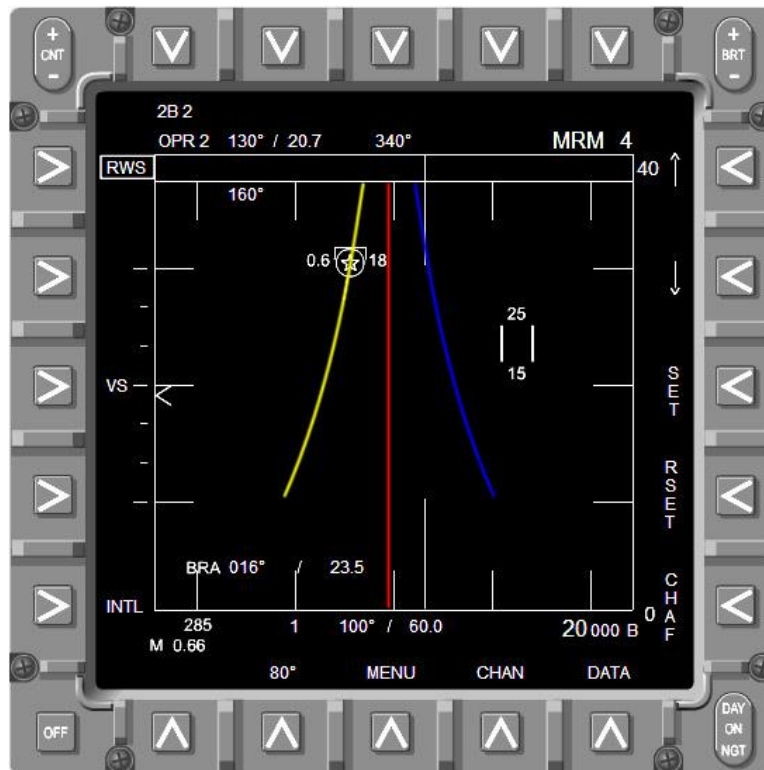


Figure 6-16 Change in Collision ATA and TA on a Zero Cut

In the example above (Figure 6-16), the red line represents BR. The fighter is on a zero cut which will maintain LS with an increase in TA. The yellow line equates to 40K LS. The blue line is collision bearing. From a zero cut, turning to place contact right of collision bearing at any range (the blue line) will create a cut into greater than collision. TA and LS will decrease.

Turning to place the contact left of the yellow line will be a cut away. LS and TA will increase and may need to be reduced later. To capture TA, the fighter must turn to place the contact on the blue line (ATA equal and opposite to TA) and then hold that heading. LS will decrease, but TA will remain constant.

Intercept geometry can be changed by either fighter or bandit changes in heading. Contact drift is predictable in these intercepts. Fighter aircrew must understand how and why the contact will drift and the relationship that drift has to TA, ATA and LS. In the next chapter, methods of using different types of cuts and fighter heading changes to manage LS and TA will be explored.

608. INTERCEPT FORMULA QUICK REFERENCE

The following formulas are used for quick reference in the cockpit and should be committed to memory:

- Aspect Angle (AA) = 180 - Target Aspect (TA) : AA = 180 - TA (same direction as TA)
- Cut = Fighter Heading (FH) to Bandit Reciprocal (BR) : Cut = FH to BR
- Target Aspect (TA) = Bandit Reciprocal to BB : TA = BR to BB
- Collision Bearing (CB) = $\frac{1}{2}$ Cut : CB = $\frac{1}{2}$ Cut
- Lateral Separation = Target Aspect x Slant Range x 100 : LAT SEP = TA x SR x 100
- Altitude Differential = Elevation Angle x Slant Range x 100 : ALT DIFF = Elev x SR x 100
- Antenna Train Angle = Angle Off (AO): ATA = AO
- Degrees to Go (DTG) = Fighter Heading to Bandit Heading
 - i. DTG = FH to BH
 - ii. DTG = 180 - Cut

CHAPTER SEVEN

TARGET ASPECT CONTROL

700. INTRODUCTION

The stern conversion intercept is the introductory intercept at VT-86. Stern conversions teach introductory air-to-air radar employment skills, intercept geometry and timeline awareness. Stern conversion intercept procedures are used to join other friendly aircraft or arrive in an offensive position on non-friendly aircraft. It is a critical skill needed by fighter aircrew to achieve mission success. This chapter will describe aircrew actions from the commencement of the intercept at beyond visual range (BVR) to the transition to within visual range (WVR) at 10 NM.

701. TARGET ASPECT VISUALIZATION

During an intercept, the fighter must accomplish the following tasks:

- Pre-commit sanitization and picture building
- Correlate the picture with AIC information
- Get declaration on all groups
- Commit to the highest priority group
- Correlate the group's position
- Set intercept geometry to affect the bandit
- Employ weapons
- Transition to visual range
- Perform a stern conversion counterturn to arrive in the bandit's rear quarter

In a co-speed intercept with the fighter and bandit at 0.5 IMN, the V_C for a head on aspect will be in the vicinity of 600 KTS (Figure 7-1). This means that the fighter and bandit will close with each other at 10 NM per minute. At this speed a 30 NM intercept will take 3 minutes from commit to the merge. This rate translates to 30 seconds every five miles. Based on this, most intercept timeline tasks will be allowed 30 seconds to complete.

This allows the fighter to build a timeline on which intercept tasks are placed. The intercept timeline should be committed to memory to ensure that no matter what the range, the fighters can step into the timeline and execute proper intercept mechanics.

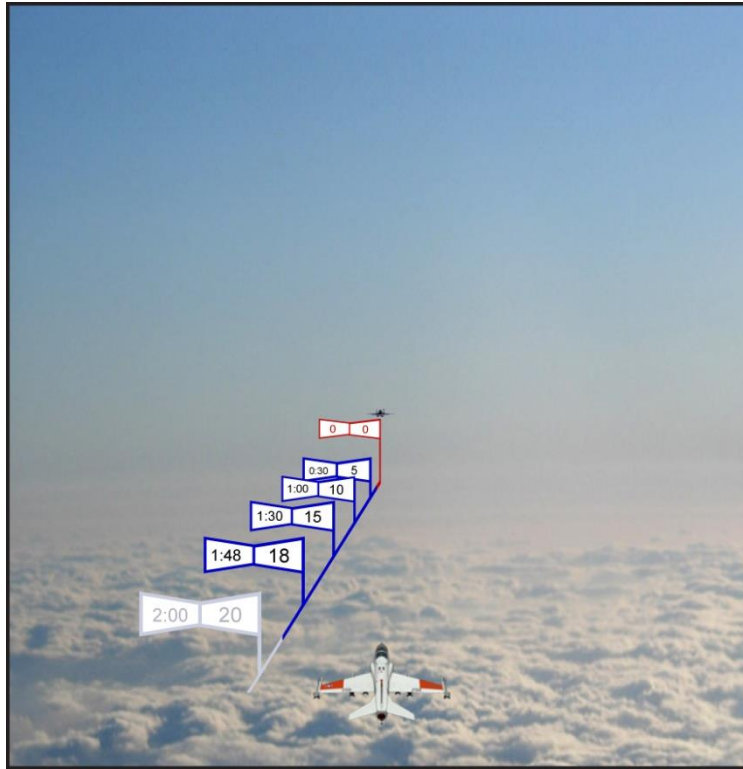


Figure 7-1 Time to Accomplish Actions with 600 KTS V_C

The timeline is a guide. All ranges on the timeline are no later than (NLT) ranges. The fighter must be able to enter the timeline and execute tasks from that point forward; however, the first task will always be correlation and declaration. The timeline is normally briefed line by line to address formation management, sensor employment/usage, communication format with examples and crew coordination. This helps ensure completeness in the brief for each important range or task on the timeline.

In the stern conversion intercept, the fighter will accomplish the following tasks outside of 10 NM:

- Commit, correlate and declare the single group
- Determine target aspect and LS of the contact
- Set intercept geometry to achieve/maintain 40K LS by 10 NM

This chapter focuses on the fighter's actions to 10 NM.

7-2 TARGET ASPECT CONTROL

702. PRE-COMMIT, COMMIT AND CORRELATION

Once established in the area and ready to begin the intercept, the fighter will start the flight with a “fight’s on” call. After the “fight’s on” call has been echoed by all players, the fighter will call AIC for the picture. AIC will respond with the picture in broadcast control in bullseye (BCBE) format. Once the SNFO has determined the picture meets commit criteria and commits, the comm format shall transition to Tactical Control in BRAA format (TCBRAA) from the fighter.

1. Group Definition

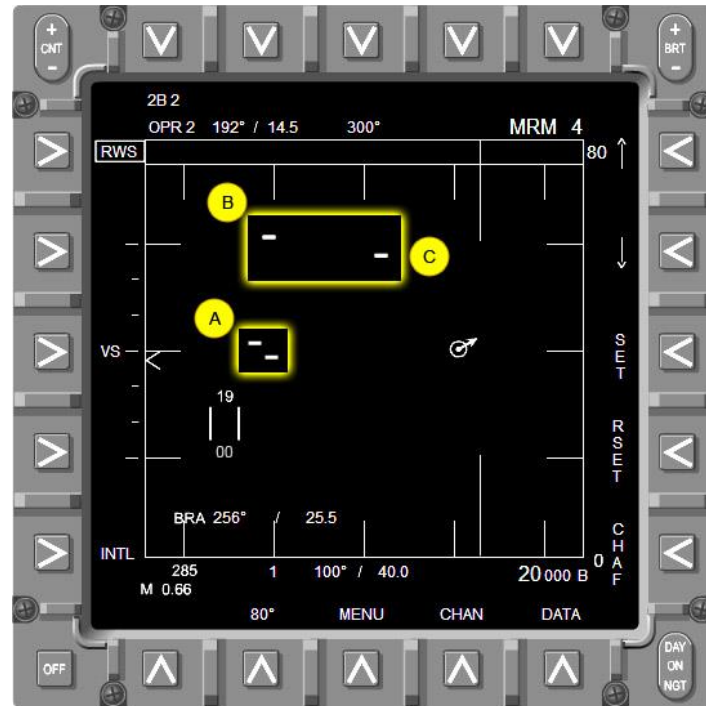


Figure 7-2 Group Definition

A group by definition in VT-86, is assumed to be two aircraft within 5 NM and 5,000 feet of altitude. Thus, in Figure 7-2, the two contacts in group A are a group while the single contacts in group B and C are two separate groups. Scenarios with multiple groups, or groups with more than one aircraft, will be addressed in Section Radar Attacks and beyond. In stern conversion intercepts, pictures will be a single group and will consist of a single contact only.

2. Commit

The first fighter action on the timeline is commit. When the fighters commit they are leaving their assigned CAP or route to prosecute an intercept to a logical conclusion on a group that triggered the commit. Fighters commit to a picture, not an individual group. In future stages, options after the commit will be discussed in more detail.

The fighter will commit on any contact that has 60 or less TA, no later than 30 NM. Radar contact with the group that triggers commit is not required. Thus, the fighter may commit once AIC has called a group with hot/flank aspect approaching 30 NM.

3. Correlation

The fighter can correlate the location of the group in bearing, range and altitude using the BRAA information on the attack display. If no contacts are detected on the attack display, the fighter should place the acquisition cursor (called the cursor hereafter in this manual) at the location of the BRAA called by AIC, and then adjust the elevation of the scan volume to bracket AICs called altitude, if given. If AIC does not pass altitude information, the fighter should manually adjust the scan volume only after two complete frames. Recall from radar theory that the bar scan of the radar means the antenna will move in elevation, as evidenced by the movement of the elevation caret. If in a 6-bar scan, the radar operator should ensure that any elevation adjustments made occur after sanitization of the complete frame at that elevation setting.

Once the contact is detected, the fighter should place the cursor over the contact and reference the displayed altitude and BRAA to cursor position. For the fighter to be properly correlated, the bearing and range being called by AIC must match within 3 degrees and 3 NM. This correlation is made by either having cursors over the contact while AIC is talking or by making a contact call to AIC using the BRAA format.

- a. Lead SNFO (PRI) “*SABRE, Hammer-11, contact BRAA 177, 28, 22 thousand, declare*”

AIC will repeat the call to correlate the group and provide the declaration.

- b. AIC (PRI): “*Hammer-11, single group BRAA 177, 27, 22 thousand, hot, hostile*”

The fighter can then begin to prosecute the intercept.

703. TARGET RADAR INTERPRETATION

Once the fighter has committed, the fighter should turn to place the AIC BRAA call on the nose to gain radar SA and to “point and assess” the geometry of the intercept. The fighter can expect to see 0-60 TA on either side during this stage. Once intercept geometry has been assessed, the fighter will turn to execute one of the game plans presented in this chapter to create, preserve or maintain 40K of LS.

1. Target Aspect and the TA Vector

The TA vector on the attack display shows the current computed TA (Figure 7-3).

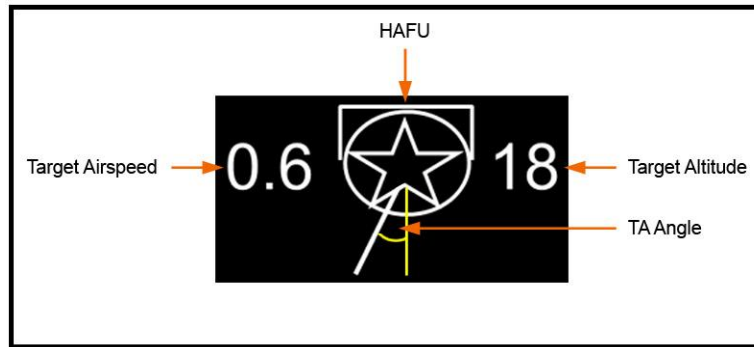


Figure 7-3 TA Vector on the Trackfile Symbology

SNFOs at VT-86 are expected to interpret TA to the nearest 10 degrees. However, TA is whatever the SNFO calls it. Proper execution of the correct game plan against the stated TA, TA management, and control of the intercept are more important than precise TA interpretation. The SNFO should make a logical and supportable TA assessment based on the information presentation by AIC and on the radar attack display. That being said, the information is present on the display, and in this manual, to interpret TA to the nearest degree.

2. Low TA

Low TA (Figure 7-4) setups are those with TA of 0 to 15 degrees and are characterized by initial rates of closure equal to the sum of the fighter and the contact's airspeeds. For example, at 0 TA, with both aircraft doing 0.5 IMN, the aircrew should expect to see closure rates around 600 KTS.

TA of 10 or less is considered very low. V_C will be near 600 KTS and LS at the beginning of the intercept will be very small.

With TA between 10 and 15, the fighter is beginning the intercept with close to the 40K goal. For a 30 NM setup, the 40K goal is met with 13.333 degrees TA. Since this is an unrealistic and irrelevant level of precision, it should be specifically noted that 15 degrees TA at 30 NM is 45,000 feet of LS.



Figure 7-4 Low TA

3. Medium TA

Medium TA (Figure 7-5) setups are those where the TA is between 15 and 35. These intercepts are characterized by initial V_C in the region of 450-500 KTS.



Figure 7-5 Medium TA

7-6 TARGET ASPECT CONTROL

In an intercept that begins with 30 NM of separation and medium TA, the fighter has between 45,000 and 105,000 feet of LS. The fighter will need to reduce LS to meet the 40K LS goal by 10 NM.

4. High TA

High TA (Figure 7-6) intercepts begin with TA of 35-45 degrees. In these intercepts, the fighter has at least 120,000 feet of LS. The fighter has to remove a large amount of lateral separation prior to 10 NM. Typical V_C for high TA intercepts is around 480 KTS.



Figure 7-6 High TA

5. Very High TA

With very high TA, defined as more than 45 degrees, the fighter will have to change the geometry at a greater rate than is available in a co-speed intercept. These intercepts will be discussed in a later chapter dedicated to speed variations.

6. TA Interpretation

Interpretation of target aspect is an important skill (Figure 7-7). However, there are some indications that should make TA interpretation much easier.

- a. At 0 TA the vector points at the bottom of the display.
- b. At 10 TA, the aspect vector is just off of vertical.
- c. At 20 TA, the aspect vector is noticeably off of vertical, but not to the point of the L&S star.
- d. At 30 TA, the aspect vector is just below the lower point of the L&S star.
- e. At 36 TA, the aspect vector is lined up with the point on the L&S star.
- f. Around 45 TA, the aspect vector is just past the bottom point of the star.

- g. At 60 TA, the aspect vector is well beyond the point of the star.

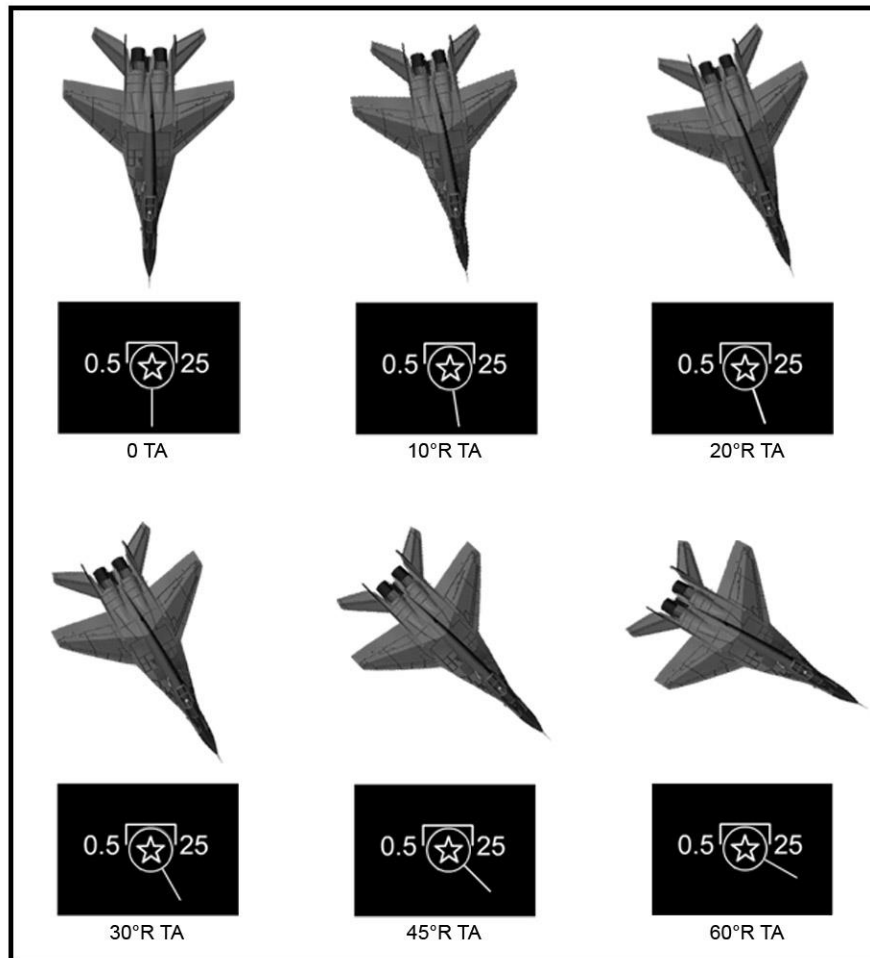


Figure 7-7 TA and Associated Symbology

If these reference points are too hard to see, recall that at 90 degrees the TA vector will point directly into the airspeed or altitude numbers of the L&S symbology.

704. THE 40K LATERAL SEPARATION GOAL

Arriving IN LAR in the rear quarter is dependent on the aircrew's ability to manage LS outside of 10 NM. This is the same technique used to rendezvous on a tanker or friendly formation.

The 40K of LS is not an arbitrary number. Lateral separation equates to turning room (Figure 7-8). With 40K of LS the fighter has enough turning room to turn nose-on at 10 NM, maintain pure pursuit and rollout in a rear quarter (RQ) LAR for an SRM. The lateral separation (LS) goal in this stage of intercepts is 40,000 feet (40K) at 10 NM. At 10 NM, the fighter will bring the contact to the nose to perform a stern conversion. LS in the radar environment equates to turning room in the visual arena. The fighter needs to be able to determine the amount of LS at any point in the intercept and recognize increasing or decreasing LS trends. With too little LS at

7-8 TARGET ASPECT CONTROL

10 NM the fighter may have to make an energy depleting break turn to roll out in the bandit's rear quarter. This deceleration will result in a fuel critical fighter arriving in the rear quarter, but at a longer range than desired. This distance must also be closed by accelerating. In fuel critical fighters, burning extra fuel due to poor geometry management is not acceptable.

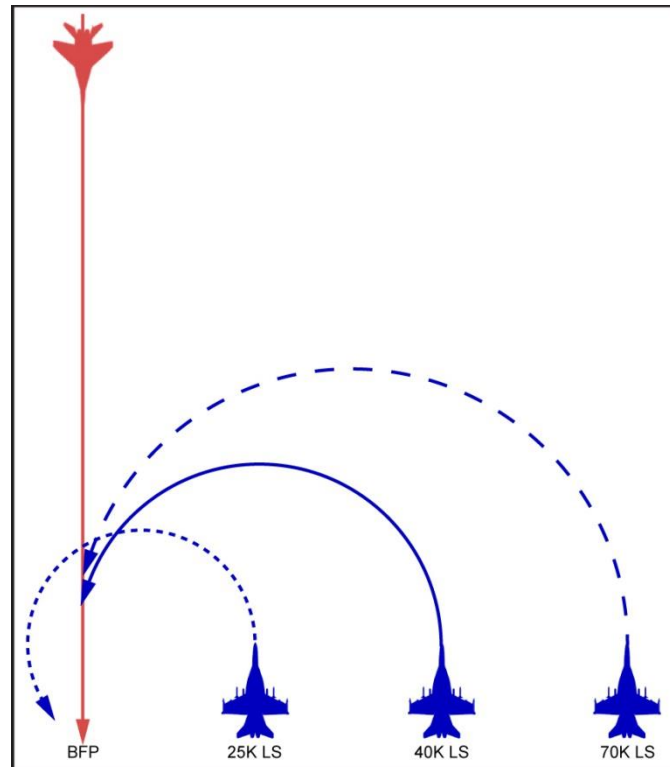


Figure 7-8 40K LS and Turning Room

705. LS GATES AND THE 40K GOAL

In order to meet this goal, the fighter will have to do one of the following:

- Turn to create LS.
- Turn to preserve LS.
- Turn to remove LS.

To do this, the fighter must have a fundamental understanding of the calculation of LS and how fighter heading changes geometry to affect LS. Recall the LS equation:

$$LS = TA \times SR \times 100$$

40 NM		LS
TA	10	40,000
	20	80,000
	30	120,000
	40	160,000
	50	200,000
	60	240,000

30 NM		LS
TA	10	30,000
	20	60,000
	30	90,000
	40	120,000
	50	150,000
	60	180,000

25 NM		LS
TA	10	25,000
	20	50,000
	30	75,000
	40	100,000
	50	125,000
	60	150,000

20 NM		LS
TA	10	20,000
	20	40,000
	30	60,000
	40	80,000
	50	100,000
	60	120,000

15 NM		LS
TA	10	15,000
	20	30,000
	30	45,000
	40	60,000
	50	75,000
	60	90,000

10 NM		LS
TA	10	10,000
	20	20,000
	30	30,000
	40	40,000
	50	50,000
	60	60,000

Figure 7-9 LS Computation

Using the LS equation, it is easy to generate a LS table in Figure 7-9 and find corresponding TA/Range combinations that correspond to 40K LS. From the formula and Figure 7-9, it is easy to see that at 30 NM, 15TA equates to 45K of LS. At 20 NM, 20 TA is 40K of LS. Thus, key relationships to commit to memory are

- 15 TA at 30 NM
- 20 TA at 20 NM
- 30 TA at 15 NM

Each of these mean the LS goal has been reached.

7-10 TARGET ASPECT CONTROL

The fighter's goal is 40K LS, but air-to-air intercepts are as much art as they are science. Given the dynamic nature of the intercept environment any LS determination made by the fighter will be a snapshot in time. The fighter aircrew should not focus on the details of precise calculation of LS, but rather the aircrew need to focus on trends and correcting those trends back toward the goal.

So while it is important to know what LS is at any point in the intercept, this is only the first piece of information the fighter needs to control the intercept. A thorough understanding of intercept geometry and how geometry will be used to create, remove or preserve LS is required.

706. CAPTURING 40K LS BY 10 NM

In order to have 40K LS at 10 NM the fighter has two choices. First, the fighter can turn to the bandit reciprocal with 40K of LS and maintain this to 10 NM. Second, the fighter can capture 40 TA and place this on collision to 10 NM. From Figure 7-10, it is easy to see that once the fighter maneuvers to 40K LS, to maintain this, fighter heading must equal bandit reciprocal. Recalling from Chapter 6 that a fighter heading equal to bandit recip creates a 180 degree HCA, or a zero cut, and that on a zero cut LS does not change, the importance of identifying bandit recip becomes readily apparent.

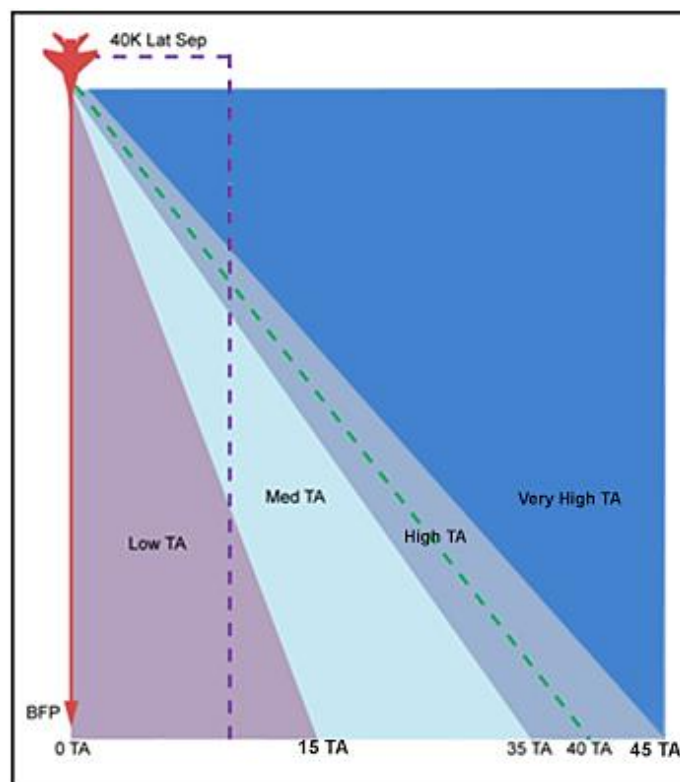


Figure 7-10 Potential Intercept Areas

1. Identifying Bandit Recip

As previously discussed, when the fighter's heading is on bandit recip, LS remains unchanged as range decreases and TA increases. If the fighter turns to a zero cut, the current Lat Sep is captured and will not change. It is very important to be able to rapidly identify bandit recip. There are three techniques to do this. The first is to simply memorize the recip table in Figure 7-11. The second is to look at the HSI. The third is to compute it via the "plus two, minus two method." In this technique, for headings from 020 degrees to 180 degrees adding 2 to the hundreds and subtracting 2 from the tens provides the reciprocal. For headings greater than 190 degrees, subtracting 2 from the hundreds and adding 2 to the tens provides the recip. The recip of 300 degrees is 120 degrees; the recip of 020 degrees is 200 degrees. This works for all headings except 190 degrees and 010 degrees, which are reciprocals of each other.

Hdg → Recip	Recip ← Hdg
360	180
010	190
020	200
030	210
040	220
050	230
060	240
070	250
080	260
090	270
100	280
110	290
120	300
130	310
140	320
150	330
160	340
170	350

Figure 7-11 Recip Headings

Once bandit recip is computed, the fighter can turn to the bandit's recip to preserve the current LS.

2. Turning to Collision with 40 TA

Recalling that the bandit is on collision when TA is equal in magnitude but opposite in direction from ATA, it is self-evident that to capture 40 L/R TA the contact must be placed at 40 R/L ATA.

Recognition of 40 TA can be achieved by the previously discussed interpretation of the TA vector or the distance from BR to ATA when BR is computed.

SNFOs are expected to use the TA vector and interpret this to within 10 degrees; however, BR to ATA is a direct and exact measurement of TA and is especially easy when the fighter is on a zero cut. SNFOs are encouraged to calculate exact TA when time permits.

707. STERN CONVERSION INTERCEPT TIMELINE

Based on the 600 KTS V_C worst case scenario, the fighter can establish timeline to accomplish the required intercept tasks to reach a 40K LS goal by 10 NM. With this goal in mind, the fighter can create a visual representation of the timeline that depicts the expected actions, formation and geometry management highlights, sensor employment considerations and intercept communication formats/examples.

The stern conversion intercept timeline is shown below in Figure 7-12.

Range	Action	Form/Geometry	Sensor	Comm
	Pre-Commit	Cap = Alt As Briefed/250 KIAS	MRM/RWS/	"Rage-11, Tapes on, Fights on" PRI BCBE
30	Commit	Point and assess		Tactical – BRAA "Rage-11-Commit" PRI "Check tapes M/A" AUX
		Set Geometry – K&B, BR, Collision, 50 ATA	STT	
25	Assess Geometry	Climb/Descend 1K below Set A/S match IMN		
20	20 TA Gate			
15	30 TA Gate			
10	Employ – Hostile	Nose-on/CT begins		"Rage-11 Fox-3" PRI
5	Timeout Fox-3	60 TA/450 Vc to 500 Vc	STT/SRM WACQ/SRM - clean	"Rage-11 timeout, single only, ZZ thousand, single group" PRI
2.0	CT Check Point	90 TA/300 Vc		
1.0	VID Employ – Hostile	Trail – 1K Look up		"Rage-11, standby single group" "Shoot, Shoot, Fulcrum" or "Friendlies, Friendlies, skip it" PRI "Rage-11 Fox-2" PRI "Rage-11 Kill, single only, ZZ thousand, single group" PRI
	Post Engagement - KIO			"KIO, KIO, Fighters KIO" PRI "Bandits KIO" PRI "AIC, KIO" PRI "Rage-11 KIO" AUX "Rage-12 KIO" AUX
	Separation	Swap Caps		"Rage-11 flow west as fighter, Reload, 1.4 good G" AUX "Rage-12 flow east as bandit, Reload, 1.3, good G" AUX

Figure 7-12 Stern Conversion Timeline

The timeline begins with pre-commit setup with CAP at 250 KIAS and the radar RWS/140/80NM/6B range with MRM selected. Target aging should be discussed during the brief and the option should be set to longer than frame time (normally 16 seconds for these parameters). Communications are separated into examples and control/format changes. The depiction then walks through each event during the intercept and actions to take.

This format makes the brief easier to talk to because all of the required actions are on the board. The fighter aircrew can address each action item and expand where needed.

The timeline for each stage of training should be committed to memory.

708. TARGET ASPECT CONTROL PROCEDURE

For the initial turn into the threat sector, the fighter has the ultimate responsibility of picking the correct turn direction to preserve target aspect. This means at times, the shortest direction may not be the correct option to prevent target aspect from growing out of control while follow-on steps are being conducted. To pick the correct turn direction, the SNFO needs to either reference track direction (pre-commit) or bandit aspect (post-commit) during the picture call.

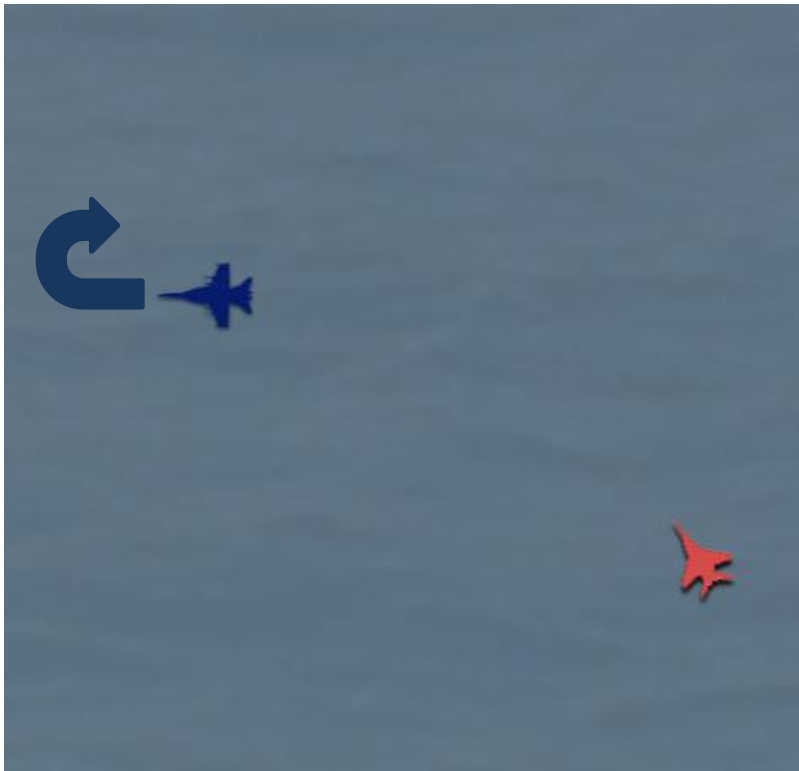


Figure 7-13 Fighter Turn In

In Figure 7-13, the fighter is headed west and the bandit is tracking northwest. After the commit, the fighter hears “Ripper 11, Sabre, single group, BRAA 120, 35, twenty thousand, flank northwest, hostile.”

7-14 TARGET ASPECT CONTROL

Although a left-hand turn to 120 would be the shortest direction of turn, a right-hand turn to 120 would preserve target aspect and would be the correct choice.

On the initial turn in, the fighter should stop the turn when the AIC called bandit bearing is on the nose. This allows the fighter to point its nose at the bandit while assessing what steps to take next. This is called a “point and assess” strategy (as opposed to blindly setting geometry based on AIC information). The point and assess strategy allows the fighter to do the following:

- Place the expected contact location in the center of the radar’s scan volume, thereby increasing the probability of detection
- Ensures that the fighter is closing with the contact while sanitization is taking place
- Keeps the fight in front of the fighter to make geometry more apparent

Once the fighter has detected the contact, the fighter should use the BRAA information on the attack display to correlate the contact with AIC information. As previously discussed, AIC will provide a declaration on the group. Once the declaration is obtained, the fighter must make an assessment of target aspect and execute the target aspect management plan. The procedure for the intercept will therefore be:

- Detect and correlate the contact; command STT upon correlation
- Identify TA and LS (point and asses)
- Determine if current LS is greater or less than the 40K goal
- Depending on TA:
 - i. Turn to generate required LS; OR
 - ii. Turn to a zero cut to preserve LS; OR
 - iii. Turn to collision to remove unwanted LS
- Capture 40K LS or 40 TA

Depending on the initial target aspect, the fighter may execute more than one turn to manage geometry; however, the goal will always be to reach 40 TA and hold that on collision to 10 NM or turn to BR with 40K LS and hold that to 10 NM.

709. TARGET ASPECT MANAGEMENT PLANS

There are four possible situations the fighter can be in at the beginning of the intercept. These are:

- Low TA without enough LS (0-15 TA)
- Medium TA with enough LS (15-35 TA)
- High TA ($35 < TA < 45$)
- Very High TA ($45 < TA$)

How these scenarios are handled is discussed below.

1. **Low TA Management; Kick and Build**

With TA at 0, the fighter must build LS to meet the 40K requirement. Because of the range, the fighter will reach a 40K LS gate before reaching 40 TA. This can be done rapidly at range by placing the contact at 50 ATA cold which is also called the “kick and build” game plan (Figure 7-14).

With the bandit at 50 degrees ATA cold, the fighter rapidly creates LS while maintaining radar contact with a low possibility of gimbaling the contact. Due to intercept drift, the contact will drift away from the fighter’s nose toward the edge of the attack display. The fighter will be on a cut away to build LS and TA. If at any time during the creation of LS, gimbaling becomes a possibility due to contact drift, this can be corrected by turning 10 degrees toward the contact to return it to 50 ATA cold.

When the 40K LS goal is met, the fighter captures this LS by turning to BR and maintaining this heading until 10 NM. TA will increase to 40 at 10 NM. The fighter must recognize the 20 NM, 20 TA goal, or be ready to turn to BR at 15 NM with around 30 TA.

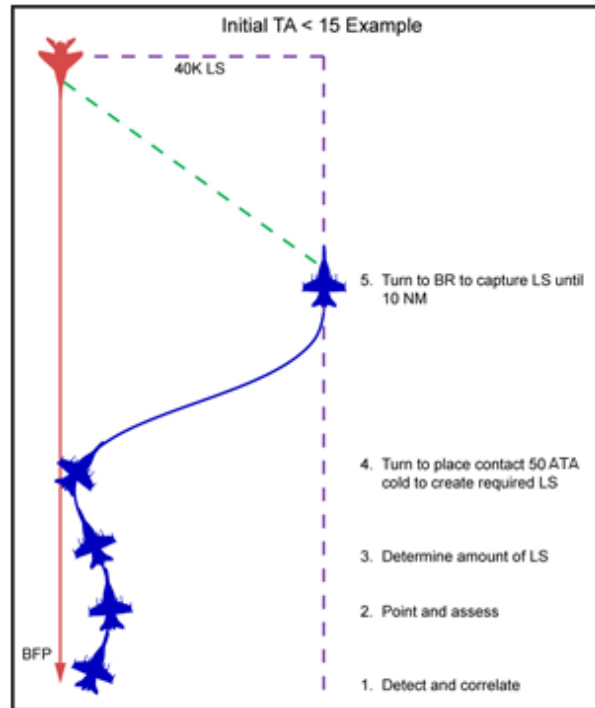


Figure 7-14 Kick and Build Game Plan

Example 1: Figure 7-14 provides a top down view of this example. During this run, the fighter has correlated and has turned to point and assess. Fighter heading is 035, Bandit heading is 220 degrees, BR is 040 degrees.

The fighter assesses TA = 5 degrees L and LS about 15,000 feet on the left. The fighter must generate LS. The fighter turns right hard to place the contact 50 ATA cold for a fighter heading of 085 degrees. The contact will drift out, so the fighter will have to make 10 degree turns into the bandit to maintain contact. As TA and LS build, the fighter holds the contact at 50 ATA cold until 40K LS is achieved as indicated by 30 degrees L TA at about 15 NM. At this point, the fighter turns to BR to capture 40K LS.

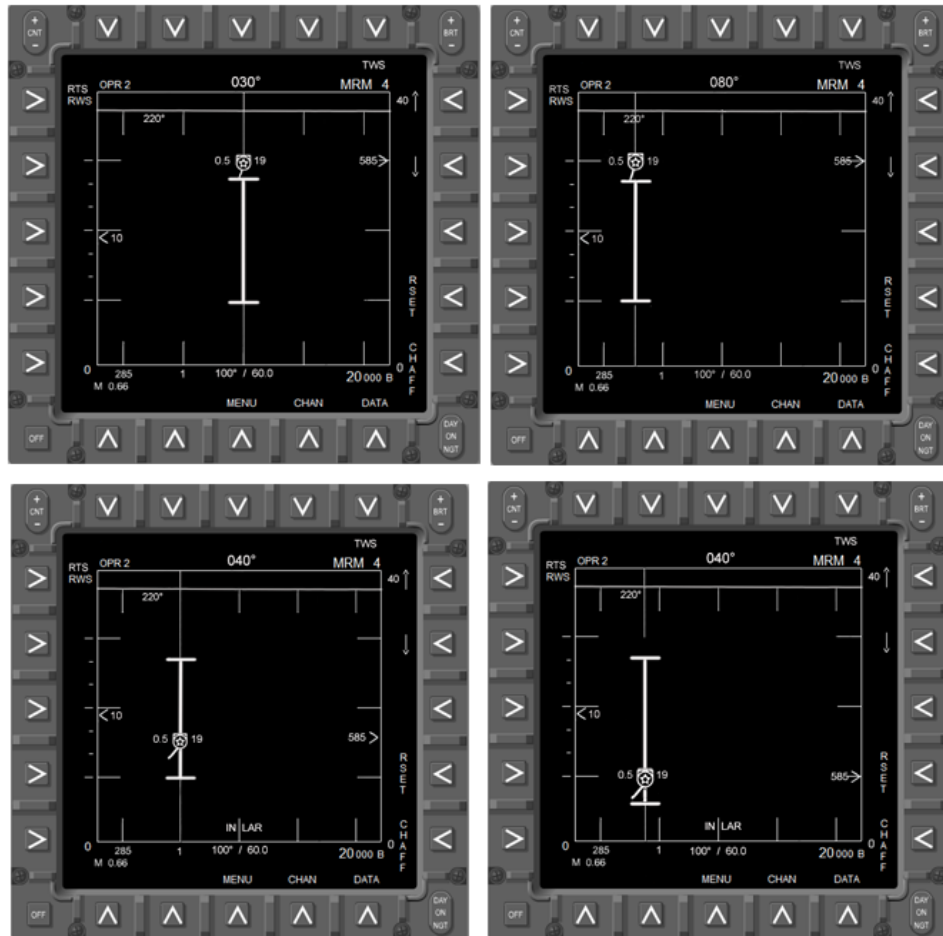


Figure 7-15 Low TA Drift

Example 2: Figure 7-15 provides a radar depiction of this intercept. During the point and assess step, the fighter has the contact on the nose and shows about 10L TA. (Pane 1)

Since the fighter has 30K of LS, the fighter turns right to place the contact cold at 50 L ATA to increase both LS and TA. (Pane 2)

While performing this cut away the fighter continues to assess TA and calculate LS. As the contact drifts the fighter will need to perform check turns into the bandit to keep the contact at 50 ATA cold. When the contact drifts to 40K LS, the fighter turns to a zero cut (FH=BR) and captures 40K LS. (Pane 3)

After the fighter turns to the zero cut LS will remain constant at about 40K. When on a zero cut ATA=TA and TA will continue to increase. As the contact approaches 10 NM, the fighter has met the goal of 40K LS and 40 TA at 10 NM.). (Pane 4)

2. Medium TA (15 < TA <35) Management

Medium TA is from 15 TA to 35 TA. From the LS equation, it is easy to determine that initial setups at 30 NM will result in between 45K and 105K of LS. Therefore, the fighter must remove between 5K and 65K of LS.

After achieving correlation, the fighter should assess current TA and LS. With a TA between 15 and 35 at 30 NM, the fighter has more than enough LS. Rather than chase the 40K LS goal, the fighter will turn to bandit recip and let TA build to 40. Once TA approaches 40, the fighter turns to capture 40 TA on collision until 10 NM. During the turn, the bandit is not yet on collision so the contact will continue to drift. The rule to use to account for this drift is that the bandit will have 1 degree of drift during every 20 degrees of fighter heading change. The turn to capture 40 TA should be 80 degrees of heading change and should be started when the contact is at 36 TA, which normally corresponds to 36 ATA.

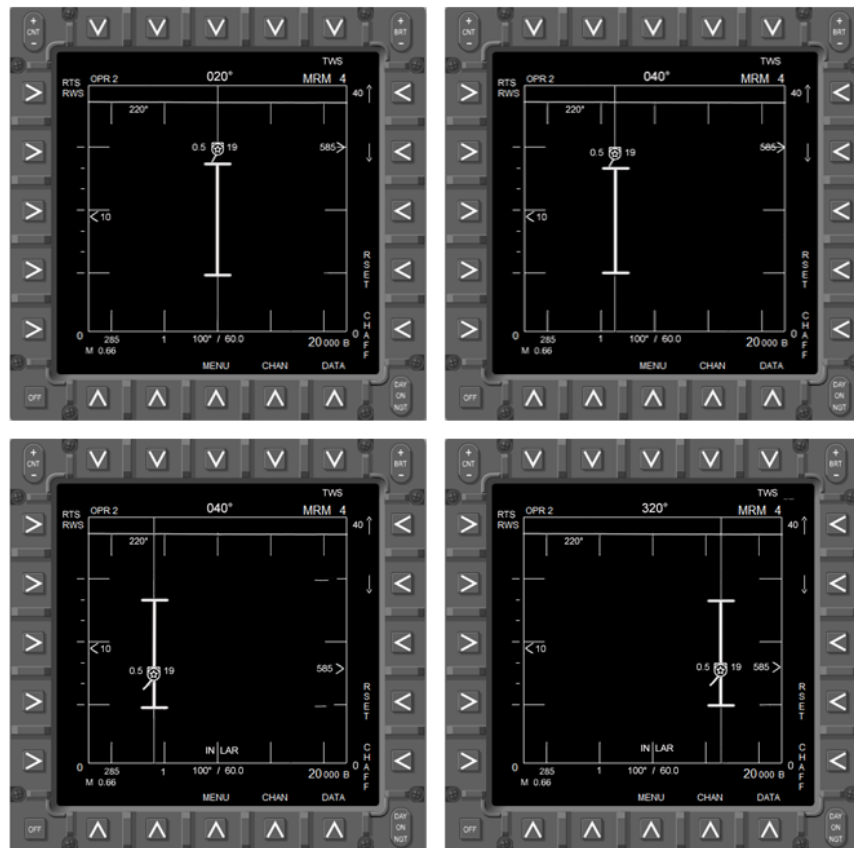


Figure 7-16 Medium TA Drift

For example: Figure 7-16 shows radar depiction of Med TA gameplan execution. The fighter has correlated and has turned to point and assess. Bandit heading is 220 degrees, BR is 040 degrees. Fighter heading is 020. (Pane 1)

The fighter assesses TA = 20 degrees L and LS about 60K on the left. Rather than chase the 40K LS goal, the fighter turns to a zero cut to allow TA to build. (Pane 2)

Once 36 L TA is recognized (by the fact that ATA=TA on a zero cut) (Pane 3) the fighter turns left hard to place 40L TA on collision at 40R ATA and holds this geometry to 10 NM. (Pane 4)

The turn to place the contact on collision will be 80 degrees since the fighter must turn 40 degrees left to bring the contact to the nose and another 40 degrees to place the contact 40R ATA. The turn should begin when the contact has 36L TA to allow for drift in the turn.

3. High TA ($35 < TA < 45$)

With a bandit that initially has 35-45 TA, immediately turn to place this near collision bearing at 40 degrees L/R ATA as appropriate and keep the contact there until 10 NM. The SNFO should attempt to assess and act on this situation as rapidly as possible. (Figure 7-17)

The contact should be placed on collision at 40 ATA and kept there until 10 NM. This is an excellent time for the SNFO to do an accurate TA calculation leading up to 10 NM. .

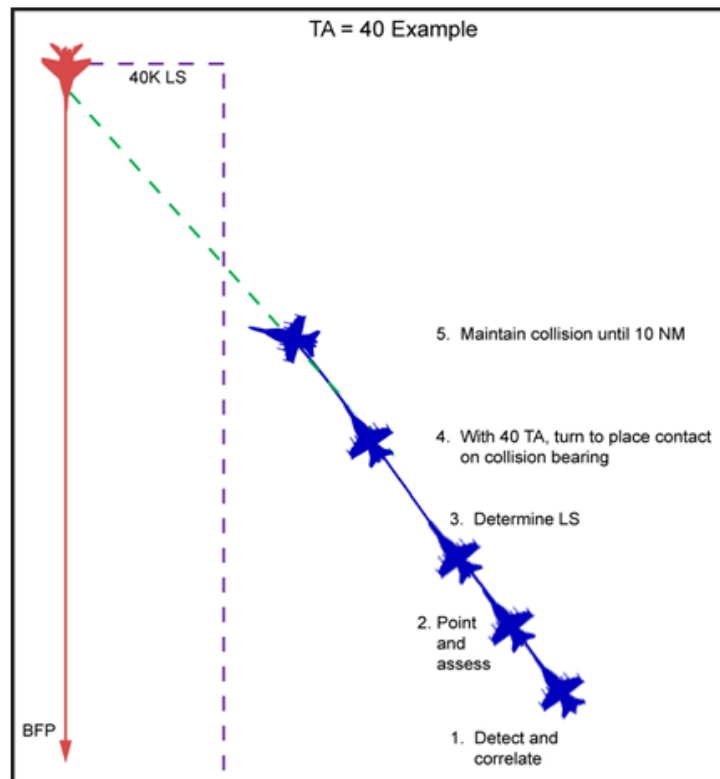


Figure 7-17 Execution with a 40 TA Start

4. Very High TA ($TA > 45$)

With TA greater than 45 the contact should be placed at 50-60 ATA hot with a 0.1 IMN speed advantage. For any aspect less than 50, this will reduce both LS and TA. If the fighter assesses

that 40 TA has been reached, turn immediately to capture 40 TA on collision, remove the speed advantage and continue to 10 NM (Figure 7-18).

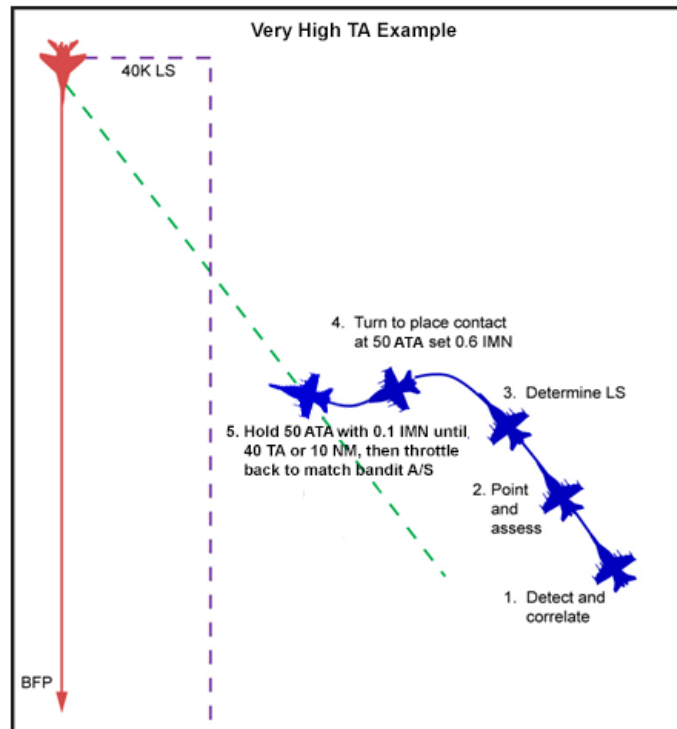


Figure 7-18 Very High TA Example

With TA greater than 50, the fighter is close to losing positional advantage. To maintain a positional advantage the fighter must go beyond the normal isosceles, co-speed intercept triangle (Figure 7-18). Using geometry, it can be shown that the distances A, B, C, and D are:

- a. $A = 30 \text{ NM} \times \sin 60 \text{ degrees} = 25.98 \approx 26$
- b. $B = 30 \text{ NM} \times \sin 55 \text{ degrees} = 24.57 \approx 24.6$
- c. $C = A / \sin 80 \text{ degrees} = 26.4 \text{ NM}$
- d. $D = B / \sin 80 \text{ degrees} = 24.98 \text{ NM}$

Since $C > D$, the fighter must cover a greater distance than the bandit in order to arrive at the intercept point at the same time. To achieve this, the fighter must increase its airspeed to cover more distance in the same amount of time.

Recall that at 300 KGS, the bandit will cover the 25 NM to the intercept point in 5 minutes. At 360 KGS, the fighter can cover the 26.4 NM in 4.4 minutes (4 minutes 24 seconds). This means the fighter, has re-established a positional advantage while maintaining the contact at 50 ATA. This can be achieved by the fighter adding a 0.1 IMN speed advantage.

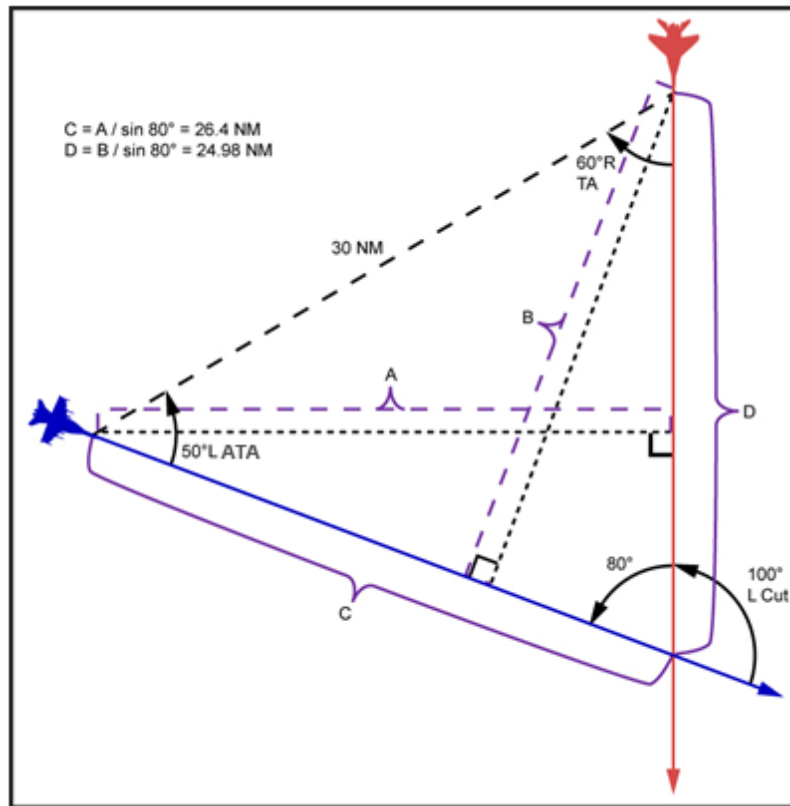


Figure 7-19 Intercept Triangle Geometry for Intercepts with TA > 50

The SNFO should command “*Set point 6*” to instruct the pilot to increase airspeed.

When 40 TA has been achieved but no later than 10 NM, the SNFO should direct “*Throttle back, set point 5*” to return to 0.5 IMN. This applies to all contacts with TA of 45 or greater. The effect of a fighter 0.1 IMN speed advantage is to move collision bearing toward the fighter’s nose by 5 degrees. The fighter can then use a predictable geometry to manage LS, and even reduce it toward the 40K goal. Since the speed advantage will move collision bearing in by 5 degrees, a contact with 50 TA at 50 ATA with a speed advantage will slowly drift away from the nose, reducing target aspect.

When faced with a non-isosceles intercept, the fighter is not likely to reach a 40K LS situation. However, anytime the fighter has the bandit on collision, TA will not change. Procedures for addressing too much LS at 10 NM will be covered in the next chapter.

710. TA MANAGEMENT SUMMARY

The TA management plan at range is summarized in Figure 7-20.

TA Assessment at 30 NM	Initial LS	First Turn	Second Turn	Third Turn
0 – 10	0 – 30K	50 ATA Cold	Bandit Recip	Collision at 40 ATA Hot
20	60K	Bandit Recip	Collision at 40 ATA Hot	Nose-on at 10 NM
30	90K	Bandit Recip	Collision at 40 ATA Hot	Nose-on at 10 NM
40	120K	Collision at 40 ATA Hot	Nose-on at 10 NM	
>=45	135K+	50-60 ATA Hot with 0.1 IMN speed advantage	Collision at 40 ATA Hot and remove speed advantage	Nose-on at 10 NM

Figure 7-20 TA Management Summary

Evaluate TA and LS, then turn to place the contact per the game plans to meet the goal of 40K LS at 10 NM. In all cases with the goal met, the fighter will turn nose-on at 10 NM.

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CHAPTER EIGHT

DISPLACEMENT TURN FUNDAMENTALS

800. INTRODUCTION

In the event the fighter arrives at 10 NM with significantly more or less than 40K of LS, a maneuver must be made to create turning room while maintaining radar contact with the bandit or other aircraft being joined on. This turn is called the displacement turn.

Although during an intercept the displacement turn is only executed if the fighter has arrived at 10 NM with more or less than the desired amount of turning room, the principle applies when joining on a formation of aircraft or on an aircraft on a predictable flight path, like a tanker.

801. THE NEED TO CREATE TURNING ROOM

As previously identified, lateral displacement in the BVR arena translates to turning room in the visual arena. Recall that AWI intercept procedures are designed to be just that, an all-weather procedure, meaning that the fighter can execute these procedures and arrive in a rear quarter LAR without ever obtaining a tally on the intercepted aircraft. Thus, the fighter should be comfortable with the concept that once the pilot achieves a tally, likely with the assistance of AREO calls from the SNFO, the intercept will proceed visually. However, if conditions are such that a tally is not made, the SNFO should prosecute the intercept.

The displacement turn, or DT, is essentially a contingency scenario in that, if the fighter has executed intercept procedures correctly and achieved 40K of LS at 10 NM, no DT is required. It is addressed here because there are many various reasons that the fighter may find itself at 10 NM not having 40K of LS and the DT procedures will apply, even beyond stern conversion intercepts.

Often when a fighter arrives at 10 NM with 0 TA, uninformed crews will see two solutions to the turning room problem. The first is to accept that no turning room exists and set up for high aspect BFM entry.

The problem with this is that at 10 NM and 0 TA, there is a high likelihood that the bandit is aware of the fighter's presence. 10 NM is outside SRM employment range, and the fighter is then spending almost one third of the intercept managing a head on charge at the bandit. In IMC or poor visibility, there is a significant chance that the fighter will not see the bandit and will be unable to engage or make an ID if one is required. This effectively means the fighter has failed in its mission to engage the bandit or ID the unknown aircraft and the bandit can blow through the merge to continue on its mission. Obviously, this is not desirable.

The second action aircrew often perform is to place the contact at the maximum available ATA in an attempt to rapidly build TA and LS. The problem with this is that at short ranges, any bandit maneuver will result in a rapid line of sight, or ATA change. If the bandit is at greater than 60 ATA and maneuvers away from the fighter even slightly, contact will be lost as the bandit drifts rapidly off the edge of the fighter's attack display. Once contact is lost inside of 10

NM, the fighter must rely on AIC information and lost contact procedures to reacquire the bandit. Again, this is not desirable.

The fighter must execute a planned turn to create LS in a controlled fashion. This is what the DT aims to achieve.

802. OBJECTIVES OF THE DISPLACEMENT TURN

The DT is required when the fighter arrives at 10 NM without making the goal of 40K of LS. This situation could arise from improper execution of the target aspect management plan or bandit maneuvers. This chapter will focus on the former, with reactions to bandit maneuvers covered in advanced 1v1 intercepts.

There are three goals of a displacement turn.

- To create close to 40K of LS to allow the fighter to execute a standard counter turn
- To ensure that the fighter maintains radar contact on the bandit through the maneuver in preparation for the counterturn (CT) and VID/SRM employment
- To ensure that the fighter remains in control of bandit TA and the intercept

In previous versions of intercept training displacement was an integral part of the intercept. With the advent of better radars and more realistic training scenarios, the DT has become a secondary training objective, but is still a skill that fighter aircrew need to understand.

803. ACCOMPLISHING THE DT OBJECTIVES: THE RULE OF 40

The fundamental rule for displacement is the Rule of 40. This rule states that the proper displacement point for a contact at 10 NM is 40 ATA cold minus the current TA. From this, a quick table can be built to illustrate proper displacement points (Figure 8-1).

Target Aspect	LS @ 10 NM (in feet)	Displacement ATA (40 ATA Cold - TA)
0	0K	40 ATA, either side
10 L/R	10K	30 ATA L/R
20 L/R	20K	20 ATA L/R
30 L/R	30K	10 ATA L/R
40 L/R	40K	0 ATA
50 L/R	50K	10 ATA R/L

Figure 8-1 Displacement Turn Chart

8-2 DISPLACEMENT TURN FUNDAMENTALS

It should be noted that intercept procedures will state the contact should be on the nose at 10 NM with 40K LS. From the table, the displacement point for 40 TA is on the nose.

As an example, with 30 TA the fighter has 30K of LS, rather than the goal of 40K of LS. This situation is manageable through the displacement turn. Because we are closer to the bandit flight path than we want to be, we need to generate more turning room, and we do this by putting ourselves in lag pursuit initially. In this example, using the rule of 40, 40 minus 30 is 10 so the contact should be placed at 10 ATA on the cold side. The opposite is true for TA greater than 40 and fighters will need to make the initial turn to get into lead pursuit with an ATA on the hot side.

Anytime the goal of 40TA and 40K of LS has not been met at 10 NM the fighter will apply the rule of 40 to displace the contact to the proper displacement point and begin the counter turn.

The displacement turn is complete once the contact has been displaced to the displacement point. Any maneuvering after displacement is counter turn management, discussed in the next chapter.

804. HOW MUCH LS CAN A DT GENERATE?

Based on the requirement to maintain radar contact to the bandit, the SNFO can expect to generate approximately 25,000 feet of LS from a 0 TA start and coming nose-on at 5 NM. The fighter should prepare for a hot counterturn.

805. MINIMUM DISPLACEMENT

The SNFO should announce the intent to turn for displacement by stating:

SNFO (ICS) - *“Left Hard, put him 20 right”*

Ten degrees prior the SNFO can announce:

SNFO (ICS) - *“Ten to go”*

At the proper displacement ATA the SNFO can command:

SNFO (ICS) - *“Steady Up”*

The turn to displacement is not based on a specific heading, but the desired displacement ATA as assessed at 10 NM by the SNFO. The purpose of these calls is to ensure the pilot understands what the intent of the SNFO is and that the SNFO is thinking about the geometry of the intercept.

806. REVIEW OF DT PRINCIPLES

Simply put, a displacement turn is required if the goal of 40 TA was not reached at 10 NM. If this is the case, turn to place the contact at 40 ATA for 0 TA, 30 ATA for 10 TA, 20 ATA for 20 TA, or 10 ATA for 30 TA. Once the contact has been displaced, the counterturn begins.

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CHAPTER NINE COUNTERTURN FUNDAMENTALS

900. INTRODUCTION

The stern conversion intercept gets its name from the position the fighter arrives in at the conclusion of the intercept, i.e., the bandit's stern. The stern conversion means that the fighter converts front hemisphere target aspect into the stern, or rear quarter, aspect.

The 40K LS goal sets up the fighter for this stern conversion turn. Due to the rapidly changing geometry at this stage of the intercept, the fighter must perform a counterturn in order to compensate for intercept drift, maintain radar contact and arrive within SRM parameters in the rear quarter. This chapter discusses the stern conversion counterturn.

901. DRIFT

Previously, the concepts of intercept and turn drift were discussed as they applied at relatively long BVR. Although the CT begins within visual range (WVR), the same concepts of intercept and turn drift apply. In fact, the effects of both are magnified by the short range and high line of sight rate changes that occur.

The drift management goal of the fighter in the CT is to use turn drift, adjusted by varying the fighter turn rate, to counter intercept drift to keep the bandit on the fighter's nose from 10 NM until SRM parameters are satisfied.

902. COUNTERTURN OBJECTIVES

The goal of the rear quarter counterturn (CT) is to arrive in the rear quarter within SRM parameters. It counters outward drift from the contact. Ideally, the CT will begin at 10 NM with 40K feet of lateral separation. Just like in the BVR portion of the intercept, the WVR portion of the intercept has range and TA gates to assist with geometry.

903. NORMAL COUNTERTURN FROM 40K LS AT 10 NM

The counterturn is a dynamic environment in that without the fighter maintaining a continuous turn to keep the contact on the inside gimbal limits of the radar, the contact would rapidly drift off of the attack display. If the fighter has achieved the goal of 40K LS at 10 NM and brought the contact to the nose on timeline, then the counterturn is pretty easily accomplished.

1. 10 NM Gate

At 10 NM the fighter should have 40K LS with 40 TA and about 530 V_C . The fighter should go pure. With 40K LS, the fighter has the proper amount of turning room to keep the contact on the nose and roll out within SRM range (1.5-0.5 NM) at the contact's six o'clock position. If the fighter comes nose-on at 10 NM and maintains a wings level position, the contact will drift in the direction of the TA vector toward the edge of the attack display. The fighter should counter this intercept drift by entering a series of harder turn rates until in the rear quarter LAR. The initial turn rate will be easy and will increase to a standard by 7 NM.

2. 5 NM Gate

As the fighter remains nose-on, TA will continue to increase as range decreases. At 5 NM, the fighter should see approximately 60 TA and 500 V_C . The fighter should maintain pure pursuit. The fighter will be in a hard turn.

3. 4 NM Gate

As the fighter remains nose-on, TA will continue to increase as range decreases. At 4 NM, the fighter should see approximately 70 TA and 450 V_C . The fighter should maintain pure pursuit. The fighter will remain in a hard turn.

4. 3 NM Gate

As the fighter remains nose-on, TA will continue to increase as range decreases. At 3 NM, the fighter should see approximately 80 TA and 400 V_C . The fighter should maintain pure pursuit. The fighter will remain in a hard turn.

5. 2 NM 90 TA, 90 DTG Gate

At 2 NM, the fighter should see 90 TA. The fighter will have 90 DTG to bandit heading and V_C should be 300. The fighter should be in pure pursuit in a hard turn.

6. 1.5 NM 150 TA Gate

If the fighter has remained nose-on, at 1.5 NM the fighter should be IN LAR for an SRM. If a valid shot exists and the contact is hostile, then the fighter should employ. The fighter should begin to roll out of a hard turn toward wings level.

7. Wings Level, 1NM 180 TA

If the CT has been executed correctly, the fighter will roll out wings level at the bandit's six o'clock, on bandit heading with 1 NM of separation and 1,000 feet of lookup. This represents the culmination point of an ideal intercept.

904. CORRECTING THE COUNTERTURN

For any number of reasons, the fighter may not arrive at 40K at 10 NM or may deviate from the ideal gates listed in Figure 9-1. The fighter must have a good understanding of the relationship between range and V_C during the CT. These relationships are shown below and should be committed to memory. Additionally, if the SNFO has utilized the rule of 40 to initiate a DT at 10 NM, they will need to utilize V_C gates and sight picture to get back onto a standard CT profile.

Essentially, the CT assessment can be boiled down to:

- Assess range and TA during the CT and compare to checkpoints.
- Compare V_C to range and check TA.
- If V_C is higher than called for at range, the CT is hot.
- If V_C is lower than checkpoint at range, the CT is cold.
- Example: V_C is 450 kts at beyond 5 NM, the CT is COLD.
 - i. The corrective action is to pull 20-30 degrees of lead
 - ii. “Lead him” to pilot
 - iii. Conversely, if approaching 5 NM and V_C is much higher than 500 kts, the CT is HOT; “Lag him” to cool off.

Range	TA (Ideal)	V_C (Approximate)
10	40	580
7	50	530
5	60	500
4	70	450
3	80	400
2	90	300
1.5	150 (Fox-2)	150
1.0	180 (Ideal)	0

Figure 9-1 Counterturn Checkpoints

As the fighter maintains a nose-on position, the SNFO should evaluate the range and V_C against the table to determine if the CT is on track or needs correcting. A fighter that sees 500+ V_C at 5 NM is hot and needs to cool the CT off. Likewise a fighter that sees 450 V_C prior to 4 NM is cold and needs to heat up the CT.

1. Heating Up

If the fighter finds that they are cold, they need to be aggressive in their hot correction. To heat up a CT, the fighter must increase its turn rate to place the contact 20-30 degrees ATA hot. This is, in fact, placing the fighter in lead pursuit to close with the bandit's flight path.

Once the fighter recognizes the intercept has returned to the desired pattern, the turn rate should be eased to allow the contact to drift to the nose. Once on the nose, the turn rate should be increased again to hold pure pursuit into the rear quarter.

2. Cooling Off

If the fighter recognizes that V_C is too high for the current range, the fighter is hot and needs to cool off the CT. Fighters need to be cautious with their cold corrections because one of the worst things is to leave a cold correction in too long and end up too far in the RQ. To fix a hot CT, the fighter eases the turn rate to allow the bandit to drift no further than 20 degrees ATA cold, placing the fighter's nose in lag pursuit. The fighter does not want to lag by more than 20 degrees in order to avoid an overshoot. Once the fighter recognizes the correct gate is reached, the turn rate should be increased to return to pure pursuit. If the fighter has let the contact drift greater than 20 degrees, a hard turn may not generate a turn rate sufficient to get the fighter's nose out of lag.

3. The 2 NM, 90 DTG Gate

The 2 NM, 90 DTG, 300 V_C gate is important for the following reasons:

- a. Meeting this window means the fighter will roll out IN LAR for SRM.
- b. 300 V_C check will help remind aircrew to remove speed advantage.
- c. This is the last time for fighter to go to lead; past this point the fighter cannot generate an energy sustaining turn rate sufficient to pull lead.
- d. This is the final opportunity to use geometry to solve range and V_C issues; after this point, the intercept is a tail chase.

The fighter should never ease the turn rate at 2 NM and 90 DTG or a hot overshoot will be induced. The 2 NM gate is the only "must make" gate (Figure 9-2).

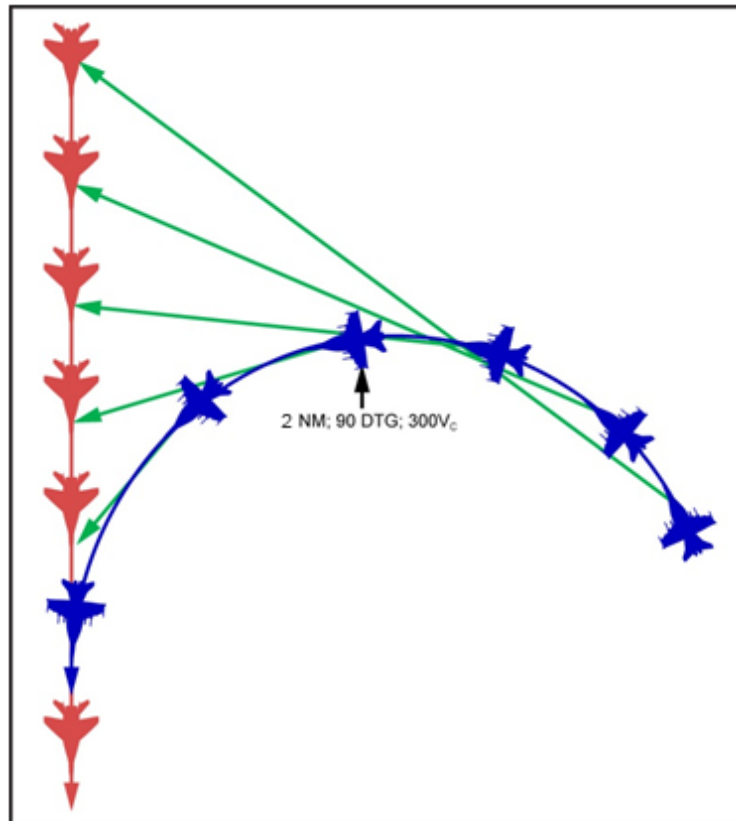


Figure 9-2 Counterturn

905. RECOGNIZING AND COUNTERING THE HOT OVERSHOOT

The fighter will induce a hot overshoot by easing at 2 NM or by not cooling off a hot CT. A hot overshoot is characterized by:

- Fighter is unable to keep bandit on the nose in a hard turn.
- Bandit line of sight rate increases as L&S drifts rapidly toward the edge of the display.

To counter a hot overshoot:

- Never ease the turn, this will exacerbate the overshoot.
- Maintain a hard turn.

If radar contact is lost:

- With no tally call “*Rage 11, clean single group.*”
- Select WACQ /SRM.
- Turn hard as possible 30 degrees past bandit heading (if known).
- After 5 seconds, turn to bandit heading in a standard turn; if AIC provides a BRAA to the contact, place that bearing on the nose.

This procedure will isolate the bandit on one side of the fighter. It is possible that the fighter may maintain radar contact through a hot overshoot. In this case, the fighter should maintain the hard as possible turn until the bandit is on the nose.

1. Fighter Induced Deviations – CT After Displacement

SNFOs should recognize that if the fighter begins the CT from a position other than 40K LS, the CT gates here, specifically those from 5 NM in, can be used as a gauge to control the CT and intercept into the rear quarter. Procedurally, the fighter is starting from a hot or cold position and should fix the intercept by changing the method of pursuit until the CT gates are being met.

2. The Phantom 60 Knots of Closure

SNFOs should also note that if 0.1 IMN speed advantage was applied earlier in the intercept to convert TA and not removed, an extra 60 KTS V_C will be seen in the counterturn. The speed advantage should be removed once turning to collision but no later than 10 NM.

906. COUNTERTURN COMMUNICATIONS

The SNFO should use standard turn management terminology to manage the counterturn. AREO calls should continue until the pilot calls tally. Directive communications should be prioritized over descriptive commentary.

1. Visual Identification (VID)

If during the intercept the group was given a declaration of “Bogey” a visual identification of the group may be required. Prior to reaching visual identification ranges (typically inside of 5nm) broadcast on AIC frequency “C/S, standby (group name)” to clear the frequency.

Aircraft identification should be communicated to AIC using the NATO identifier, such as: “*Shoot, shoot, Fulcrum.*” If identified as a friendly aircraft the call would be “*Friendlylies, friendlylies, skip it.*” Aircrew should be well prepared for threat and friendly aircraft ID and the ROE for weapons employment. The event brief should include the type of aircraft expected. Appendix B includes aircraft presented in the OFT. The SNFO is responsible for correct visual identification and terminology.

9-6 COUNTERTURN FUNDAMENTALS

907. COUNTERTURN SUMMARY

The counterturn can be summarized as follows:

- At 10 NM, evaluate TA.
- If 40 TA at 10 NM, put the contact on the nose and fly normal CT honoring the gates from the table.
- If TA is not 40 at 10 NM, initiate CT using the rule of 40 and monitor checkpoints; heat up or cool off as required to meet 2 NM gate.
- At 2 NM be at 90 DTG with 300 V_C in STT with SRM selected; prepare to employ/VID.
- Make the VID as early as possible; do not wait to rollout to employ the SRM against a hostile contact.

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CHAPTER TEN

INTERCEPT PROGRESSION

1000. INTRODUCTION

The previous chapters discussed identifying the spatial relationships at the beginning of the intercept, target aspect management, pursuit curves, and proper drift control from the initial setting of geometry through the counterturn and into the bandit's rear quarter. This chapter addresses the stern conversion intercept as distinct phases, giving the aircrew a logical framework from which to approach all intercepts.

The stern conversion is the basic radar intercept procedure used to join other aircraft. It is a critical skill needed by fighter aircrew to achieve mission success. This lesson will describe how aircrew manage intercept geometry beyond visual range (BVR) and transition to within visual range (WVR) to execute an expeditious rendezvous for a SRM or VID. These procedures will ensure aircrew success not only at VT-86, but also in the FRS and fleet.

1001. INITIAL INTERCEPT CONDITIONS AND ACTIONS

The initial conditions presented in an intercept determine what the fighter's course of action should be. The fighter needs to analyze initial geometry to determine TA, LS and what actions need to take place to manage the change in LS. The fighter is given information from AIC which defines the location of the bandit. From this information, the fighter can obtain radar contact and analyze the situation.

The fighter must perform the following actions in this order before setting geometry:

- Turn to place the contact call on the nose (point and assess).
- Sanitize airspace to detect the group (sanitization).
- Correlate the detected group with AIC information.
- Obtain a declaration.

1. Communication Formats

Pre-Commit Communications

The picture will be given in broadcast control, anchored to bullseye, until the fighter's commit. A typical call will be:

LEAD (PRI) - "*SABRE, Rage 11, picture*"

AIC (PRI) - “SABRE, three groups; group Vegas 215, 25, 19 thousand, track south, hostile; group Vegas 270, 50, 22 thousand, track east, hostile; group Vegas 250, 60, 22 thousand, track east, hostile.”

Post-Commit Communications

Once a group meets commit criteria, the fighters will commit and control will change to tactical control. Tactical control can be either anchored to bullseye or given in BRAA (bearing, range, altitude) format. BRAA format always references the fighter’s nose. Recall from Chapter 2 that broadcast control does not reference bandit aspect in its calls, where tactical control does. Initial picture calls for all AWI events will be in the broadcast control, anchored to bullseye format. During Stern Conversion intercepts AIC will use tactical control in the BRAA format to aid the fighter in acquiring the contact.

2. Sanitization

The fighter should ensure that airspace in the current scan volume is “clean” prior to adjusting the scan volume. Initially, the fighter should center the scan altitude volume around ownship altitude with the cursor at 25 NM. If the contact does not appear, the fighter can place the cursor at the AIC called BRAA and center the elevation scan coverage. Again, the fighter should be patient as a RWS/140 degrees/6B frame takes approximately 13 seconds to complete.

3. Commit Criteria

The following two requirements are the commit criteria for stern conversions at VT-86:

Flank or less TA ($TA \leq 60$ degrees)

Range NLT 30 NM

Once the fighter determines that the AIC information meets one of these two criteria, the fighter will commit with the following call:

SNFO (PRI) - “Showtime, Commit”

SNFO (AUX) - “Check tapes, master arm”

The bullseye card assists the aircrew’s understanding of intercept geometry, geographic threat location and threat relevance of the called contact, with respect to the fighter, defended position, route, commit line, etc.

4. Point and Assess

The point and assess game plan involves the fighter pointing their own aircraft towards threat sector and obtaining radar contact, then using radar information to assist with the intercept geometry decisions. As shown in Figure 10-1, the point and assess plan allows the fighter to

10-2 INTERCEPT PROGRESSION

make informed commit and geometry decisions. Following the commit, the fighter will point at the bandit to assess the initial intercept conditions.

- a. Advantages:
 - i. More detailed contact information from radar contribution to fighter aircrew (L&S or STT).
 - ii. Own aircraft weapon system is directed at the contact in the event of a Hostile declaration.
- b. Disadvantages:
 - i. Compresses the range between fighter element and contact element.
 - ii. Allows contact element to gain SA that they are targeted when radar energy is detected from fighter element.

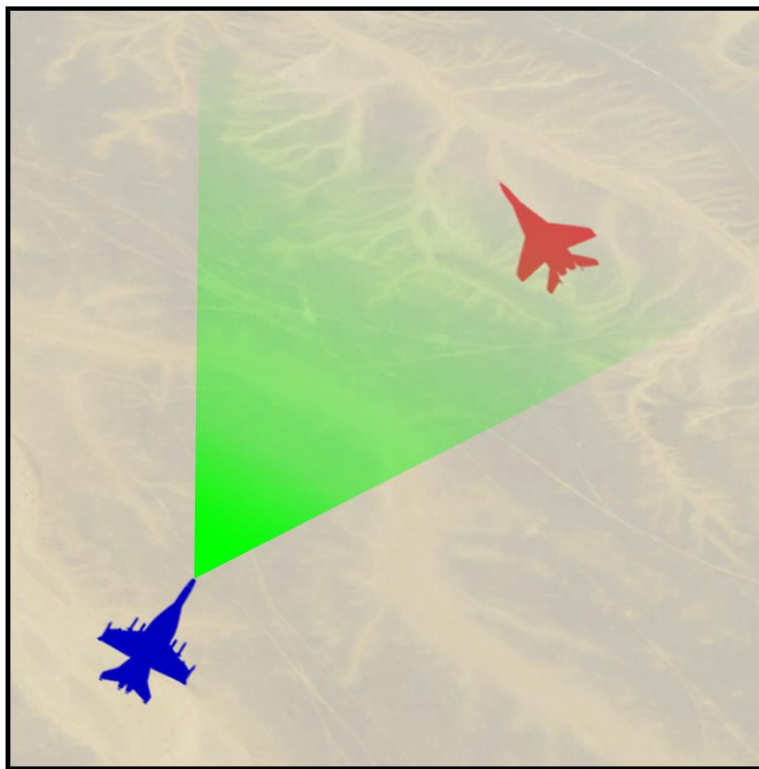


Figure 10-1 Point and Assess Plan

5. Correlation and Declaration

Once radar contact occurs, the SNFO will make a contact call to the AIC controller to correlate the radar contact to the AIC called group. An example of the communication format is as follows:

SNFO (PRI) - *“SABRE, Showtime 11, contact BRAA XXX, YY, ZZ thousand, declare.”*

AIC will respond with a correlated contact and declaration:

AIC (“SABRE”) - *“Showtime 11, single group BRAA XXX, YY, ZZ, Hot, Hostile.”*

Communication for the declaration call is in the tactical control format. AIC response will label and name the picture/group “single group.” The aircrew must remember the AIC called declaration in order to not violate ROE by employing weapons without authorization. If there is any confusion with the declaration, another declaration call should be made.

SNFO (PRI) - *“SABRE, Showtime 11, declare Single Group.”*

The fighter needs to analyze initial geometry to determine TA, LS and what actions need to take place to manage the change in TA and LS. The fighter is given information from AIC which defines the location of the bandit. From this information, the fighter can obtain radar contact and analyze the situation.

Once the fighter is coorelated, they should command an STT and set their initial geometry. This is done by initiating a gameplan per the TA analysis, and setting the correct airspeed and altitude for the intercept. For airspeed, the fighter should match the bandit’s airspeed but not slower than 0.5 IMN. For altitude, the fighter will climb or descend to be 1,000 feet below the bandit.

SNFO (ICS) – *“Left hard, put ‘em 50 right, set point six, climb twenty one thousand”*

6. TA and LS Analysis

The BVR geometry goal of the stern conversion at VT-86 is to achieve 40K LS and 40 TA by 10 NM. Lateral separation (LS) is the distance between the fighter and projected bandit flight path (Figure 10-2).

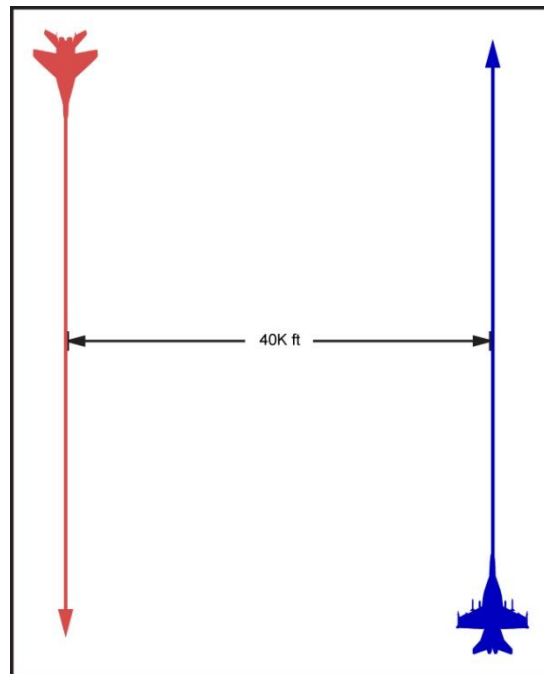


Figure 10-2 40K Lat Sep Goal

The formula for LS is:

$$LS = TA \times Range \times 100$$

To calculate LS, the SNFO is expected to identify TA to the nearest 10 degrees and act based on that determination. Once the initial geometry is understood, the intercept's initial phase is over.

1002. INTERCEPT MID-PHASE; TA MANAGEMENT PLAN EXECUTION

Initial conditions for Stern Conversion AWIs will be 0-60 TA at approximately 30 NM. However, SNFOs should understand geometry management to at least 65 TA since slow recognition of initial conditions or wrong application of a game plan may allow TA to build to 60 or greater. The SNFO should apply geometry and a speed advantage to stop the growth of TA out to 70 TA with timely application of the correct game plan and good headwork.

Once the initial situation is recognized, the fighter will apply the game plan for a low, medium, high, or very high TA contact.

1. Summary of Starting Positions

Figure 10-3 shows the fighter's potential starting positions and how they relate to the 40K LS goal.

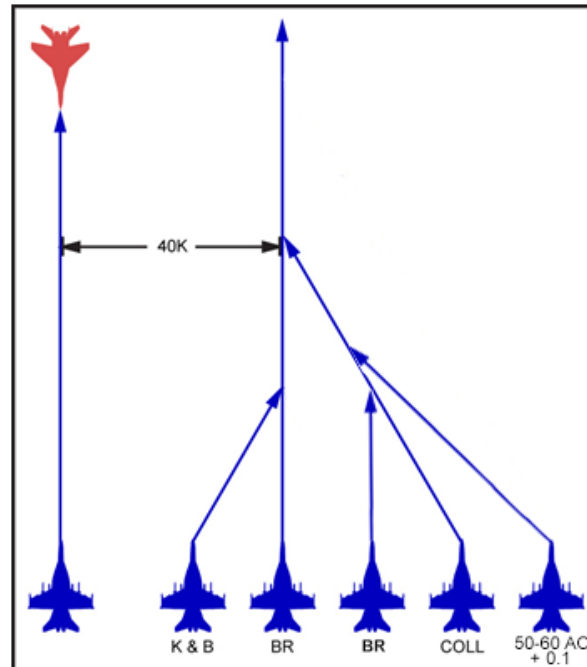


Figure 10-3 Summary of Starting Positions

Figure 10-4 provides radar display examples for various TAs. These TA checkpoints, used in combination with the Lat Sep Gates, assist aircrew in recognizing when 40K LS is achieved.

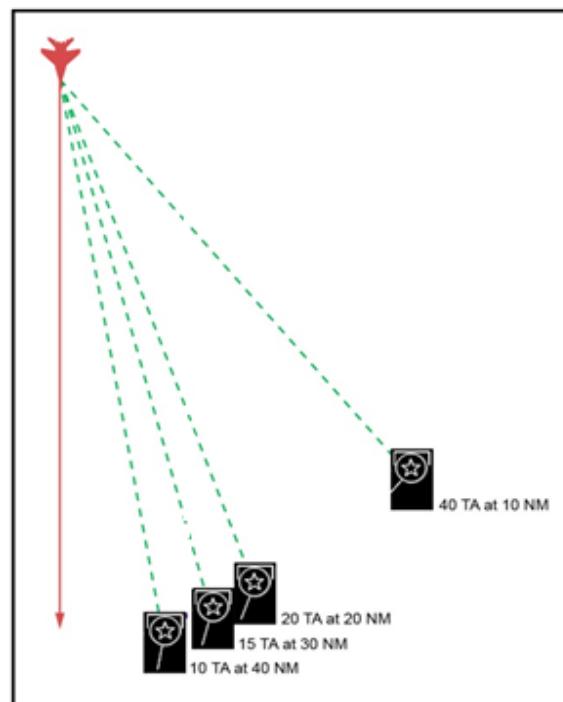


Figure 10-4 Radar TA and Intercept Starting Points

10-6 INTERCEPT PROGRESSION

1003. RECOGNIZING AND CAPTURING THE 40K LS GOAL**1. Recognizing 40K TA**

The fighter must recognize when the 40 LS goal has been reached. The table in Figure 10-5 shows the relation between TA and range for 40K of LS.

Target Aspect	Range
10	40
15	30 (45K LS)
20	20
30	15 (45K LS)
40	10

Figure 10-5 TA vs. Range for 40K LS

These gates shall be committed to memory for effective intercept execution.

In Figure 10-6, the red line is bandit reciprocal (BR, or a zero cut), the blue line represents the ATA to place the contact for collision, and the yellow line shows the growth of TA, as measured from BR to BB. This is the expected drift pattern for a zero cut execution that starts at about 13 TA at 30 NM. As the contact drifts left on the display, TA is increasing, but LS is remaining constant at 40K.

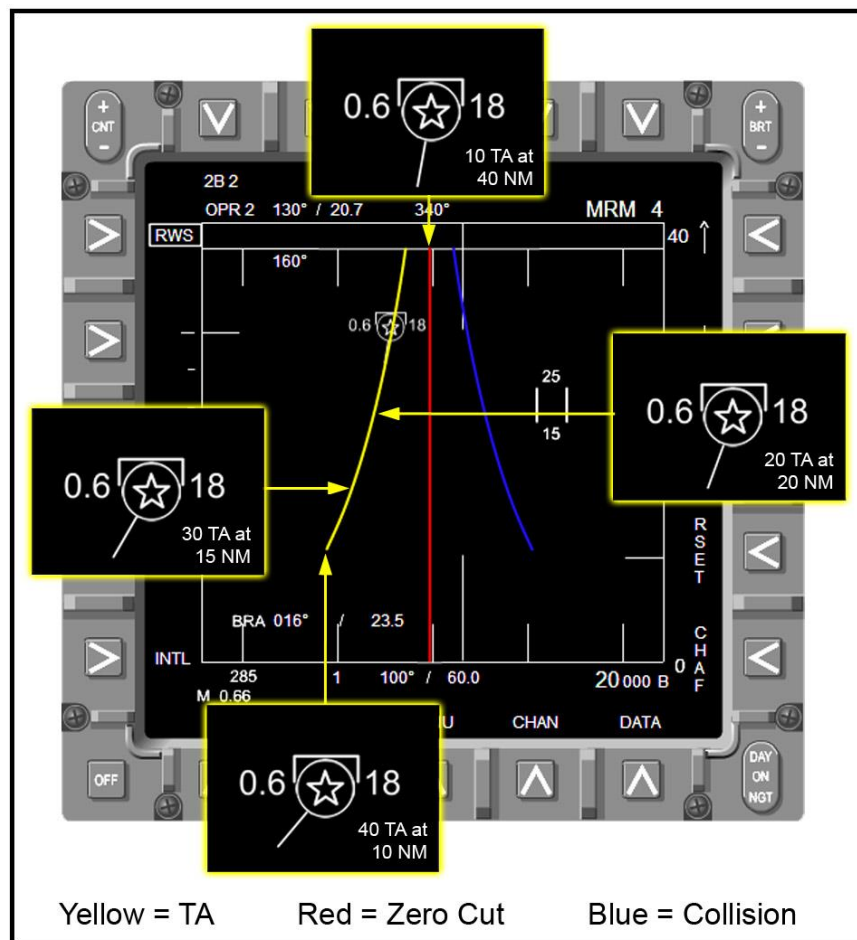


Figure 10-6 Attack Display Showing 40K LS Goal

The overlap between game plan execution and TA is shown below in Figure 10-7.

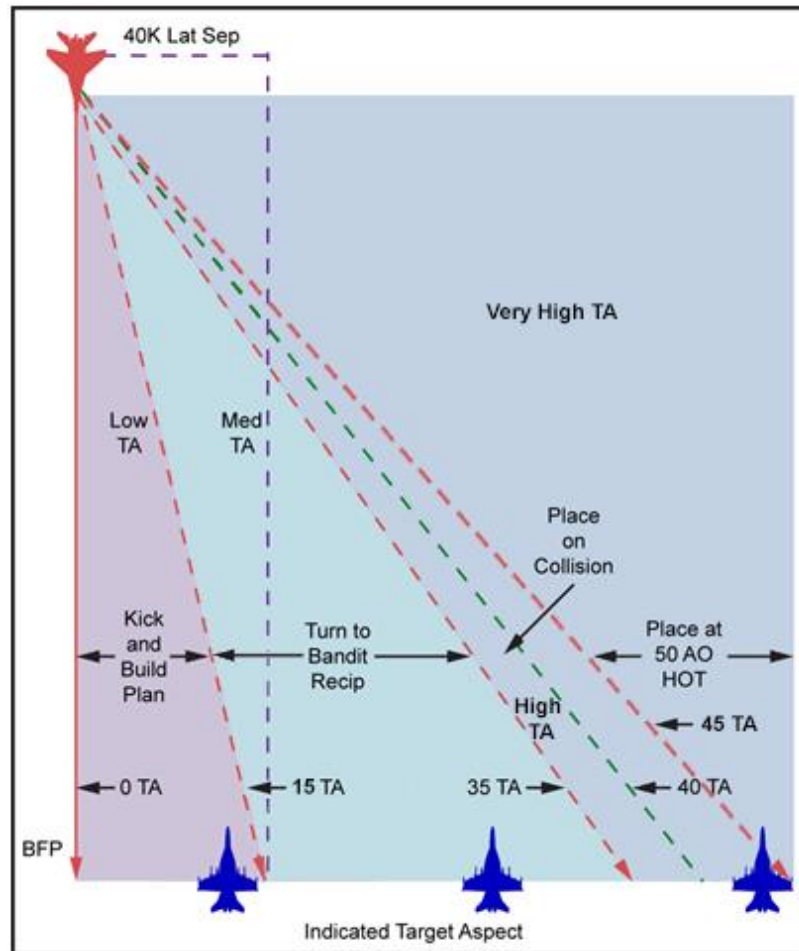


Figure 10-7 TA Regions in Relation 40K Goal

2. Capturing

Capturing 40K LS will occur one of three ways:

- The fighter will turn to a zero cut after building 40K LS.
- The fighter will turn to BR and maintain 40K LS to 10 NM.
- The fighter will continue to reduce LS to 40K by placing the bandit on collision once 40 TA has been achieved until 10 NM.

The key is for the fighter to recognize which of these scenarios applies and execute one of the four gameplans.

- a. Example 1 - 10 TA start: With 10 TA at 30 NM, the fighter has 30K LS. The “kick and build” plan applies, this is a cut away. The fighter should place the contact at 50 ATA cold and monitor TA gates. The most likely checkpoint for 40K recognition will be 20 TA at 20 NM. Turning to a zero cut at 25 NM with 15 TA equates to 37.5K of LS, which is too early because it does not meet the LS goal.
- b. Example 2 - 50 TA start: With 50 TA at 30 NM, the fighter has 150K of LS. The fighter must work immediately to reduce LS. The fighter should turn to place the bandit 50-60 ATA hot and add a 0.1 IMN speed advantage, this is a cut into greater than collision. By 20 NM, the fighter may have reduced TA to 40, at which time the fighter could turn to place the bandit on collision at 40 ATA hot and remove the speed advantage.
- c. Example 3 - 20 TA start: With 20 TA at 30 NM the fighter has 60K of LS and needs to stop LS from building. To preserve LS, the fighter should turn to a zero cut, and let the contact drift to 36 ATA. The fighter should then turn into the bandit 80 degrees to place the contact on collision with 40 TA.
- d. Example 4 - 40 TA start: With 40 TA at 30NM the fighter has 120K of LS. The fighter needs to decrease LS but keep the current TA, a cut into equal to collision is required. The contact should be placed at 40 ATA hot and kept there until 10 NM. Remember, if the contact is truly on collision it will not drift.

These four game plan execution examples show the logic the SNFO should use approaching the geometry management phase of the intercept.

3. Correcting TA Management

TA and LS should be continually assessed. The SNFO should ask, “What is happening to LS and TA with this geometry? Is the current gameplan I am executing producing the desired results?” If the answer is no to the second question, the SNFO should act, considering the following:

- a. Immediately assess TA
- b. Compute LS for the current TA
- c. Apply the gameplan that is appropriate for the current LS and TA

The SNFO should be proactive and work the current problem, not try to fix the previous problem that was not done correctly.

4. Actions at 10 NM

At 10 NM, just prior to the turn to place the bandit on the nose, the fighter should perform another TA assessment. The fighter should then determine for a final time if 40K LS has been achieved and update intercept geometry as required.

The fighter should also commence AREO calls to talk the pilot's eyes onto the bandit.

1004. STERN CONVERSION TURN

If 40K of LS has been achieved, the fighter begins the counterturn (CT). When performed correctly, this will result in the fighter arriving in the rear quarter with:

- 1,000 feet of look up
- 0.5 to 1.5 NM of range
- STT
- Tally (preferred)

The goal of the rear quarter counterturn (CT) is to arrive in the rear quarter within SRM parameters. It counters outward drift from the contact. Ideally, the CT will begin at 10 NM with 40K of lateral separation and the fighter will remain in ***pure pursuit*** all the way to SRM parameters.

1005. DISPLACEMENT TURN CONTINGENCY

If the fighter encounters a situation where 40K LS has not been achieved at 10 NM, a displacement turn will be required (Figure 10-8). The goal of the displacement turn is to create or remove LS for the counterturn. The DT will not be necessary if the conversion game plan discussed in this lesson is executed. This is a tool to use in the event of a quick geometric decision due to compressed range between the fighter and contact or low SA.

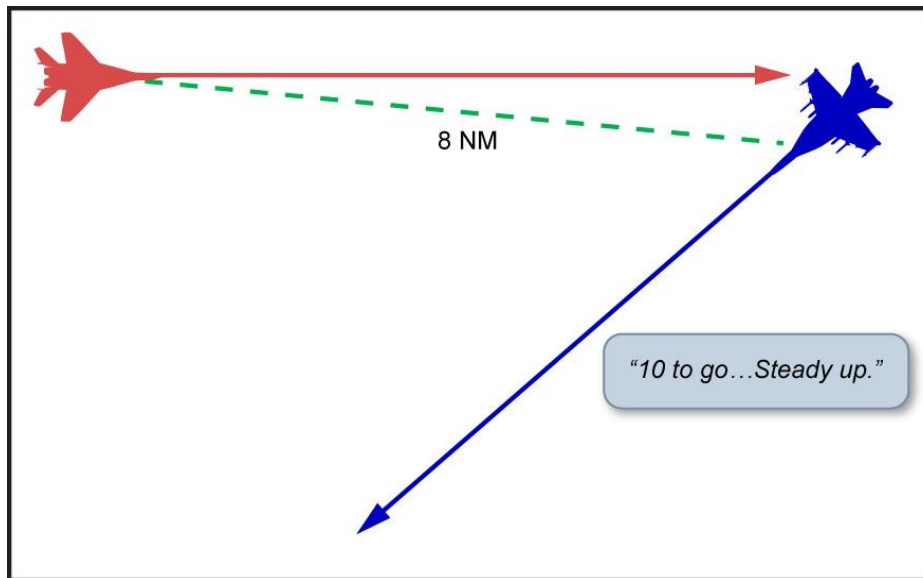


Figure 10-8 Displacement Turn

The two objectives of the displacement turn (DT) are:

- To orient the fighter's flight path (heading) so as to gain or remove lateral separation as required.
- Achieve an intercept geometry that ensures that the drift rate in the counturn is controlled and that the counturn does not require high AOB and G-Load (especially important at night or on NVGs to avoid disorientation).

Five steps are used to accomplish the DT:

- Determine TA.
- Determine distance from displacement range (10 NM).
- Determine the displacement point (ATA), left or right of the fighter's nose from the previous TA assessment.
- Direct the pilot to turn in the direction necessary to place the contact on the displacement point with *"Left/Right Hard, put him XX L/R."* Once within 10 degrees of the displacement point, describe *"10 to go."*
- Once the contact is displaced to the correct ATA, keep the contact there until it is assessed that the ROC gates will be met and give AREO calls to the pilot. Once the pilot is tally, AREO calls should stop but continue to QA the ROC gates as the pilot runs the counturn visually.

The displacement point chart below, Figure 10-9, assists aircrew with the proper displacement points.

Target Aspect	Displacement Point
0	L/R to 40° Lag
10	30 Lag
20	20 Lag
30	10 Lag
40	Go Pure
50+	10 Lead

Figure 10-9 Displacement Points

Once again, the DT is a contingency procedure that is not required if the TA management game plan is properly executed. However, these procedures should be used to correct for too little or too much LS at close range (< 10NM) on a stern conversion.

1006. CONCLUSION

Stern conversions are a critical skill required for tactical aircraft employment. Formation, radar, and communication are the critical skills for success in the air-to-air environment. Most missions will require a rendezvous on a Flight Lead, airborne tanker, or visual identification of an unknown aircraft. The information and procedures in this chapter will be utilized throughout the training at VT-86 and in the fleet.

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CHAPTER ELEVEN

AIR-TO-AIR MISSILES

1100. INTRODUCTION

As with most industries, technological innovation played a critical role in making today's air-to-air missiles extremely lethal. The technological evolution spans from unguided rockets to missiles with internal radar systems and guidance in the current combat environment. It is imperative strike fighter aircrew understand the capabilities of modern air-to-air weapons and how to employ them. Understanding the enemy's capabilities is also critical and those capabilities are highlighted in Chapter 1. This chapter will focus on the history and components of the U.S. air-to-air missiles; the AIM-9, AIM-7, and AIM-120.

1101. AIR-TO-AIR MISSILE HISTORY

The end of WWII and its aftermath marked a period of furious innovation in air-to-air missile development. The AIM-9 missile project was initiated in the early 1950s and the production models entered operational service in 1956. The Sparrow project began in 1946, with the first test firing of a missile in 1952. The missiles became operational in 1956 aboard F3H Demons and F7U Cutlass aircraft. The late 1960's witnessed the innovation of active missiles with the advent of the AIM-54 Phoenix missile and subsequently the AIM-120 AMRAAM, carried on today's U.S. tactical Strike Fighters.

The first guided missile kill occurred in September 1958 when employed by the Chinese Nationalists (Taiwan) against the Communist Chinese during the Formosa Straits conflict. It was also during this conflict that an unexploded AIM-9 was recovered by the Communists, and eventually turned over to the Soviets, who exploited the technology to develop the AA-2 Atoll.

The AIM-9 utilized Infrared (IR) guidance. IR guidance seeks the infrared emissions from a target in the electromagnetic spectrum. Missiles with these passive guidance systems are commonly referred to as "heat seekers."

The AIM-7 Sparrow variants have been utilized as air-to-air and surface-to-air weapons since the missile's inception in 1946. The weapon utilizes semi-active radar guidance. Once the strike fighter has illuminated the target with a radar lock, the Sparrow guides on the reflected energy of the target by using a forward-looking planar-array antenna.

During the 1960s, the growing Soviet bomber threat required a missile that utilized look-down/shoot-down capability with either conventional or nuclear warhead systems. In response, the U.S. Department of Defense initiated a program that would evolve into the AIM-54 Phoenix missile supported by the AWG-9 weapon system. This was a revolutionary technological advance in modern air-to-air weapons because it was the first "active" missile. Active missiles have the capability of guiding themselves with radar systems within the missile. No longer did the missile require radar support from the strike fighter until detonation. A critical added benefit to this technological break-through was the new ability to shoot multiple missiles. Now that a missile could support itself in flight, the strike fighter aircrew could locate another target and

shoot again! This capability was precisely what the U.S. DoD needed to counter the Soviet threat.

1102. AIM-9 SIDEWINDER HISTORY

The AIM-9 Sidewinder is a short-range, all-aspect, heat-seeking missile. The Sidewinder has proven itself to be the weapon of choice because of its simplicity, reliability, and high probability of kill. Since the missiles inception in the early 1950s, the Sidewinder has seen vast improvements in its basic design.

The early Sidewinders suffered from poor seeker head discrimination (i.e., could not distinguish one heat source from another) and a very limited firing envelope. During the Vietnam War, the Sidewinder received improved seeker head capabilities with Sidewinder Expanded Acquisition Mode (SEAM), which enabled the missile to have a wider field of view for acquiring targets.

During the 1970s, the Sidewinder received a new seeker head, fuse, and warhead combination that gave it all-aspect capability. Since then, additional improvements include better turn performance, flare rejection, helmet-mounted sight integration, and smokeless motors. The AIM-9X is the latest version of the Sidewinder family. This extremely capable weapon will be introduced in later training at the FRS.

1103. AIM-9 DESIGN

The AIM-9 is a small, lightweight, low-cost missile. Although it still carries a price tag of \$100,000, this is relatively economical when compared to its radar-guided big brothers. The Sidewinder has four major sections: guidance, fuse, warhead, and rocket motor.

1. Physical Characteristics

Refer to Figure 11-1 for physical characteristics.

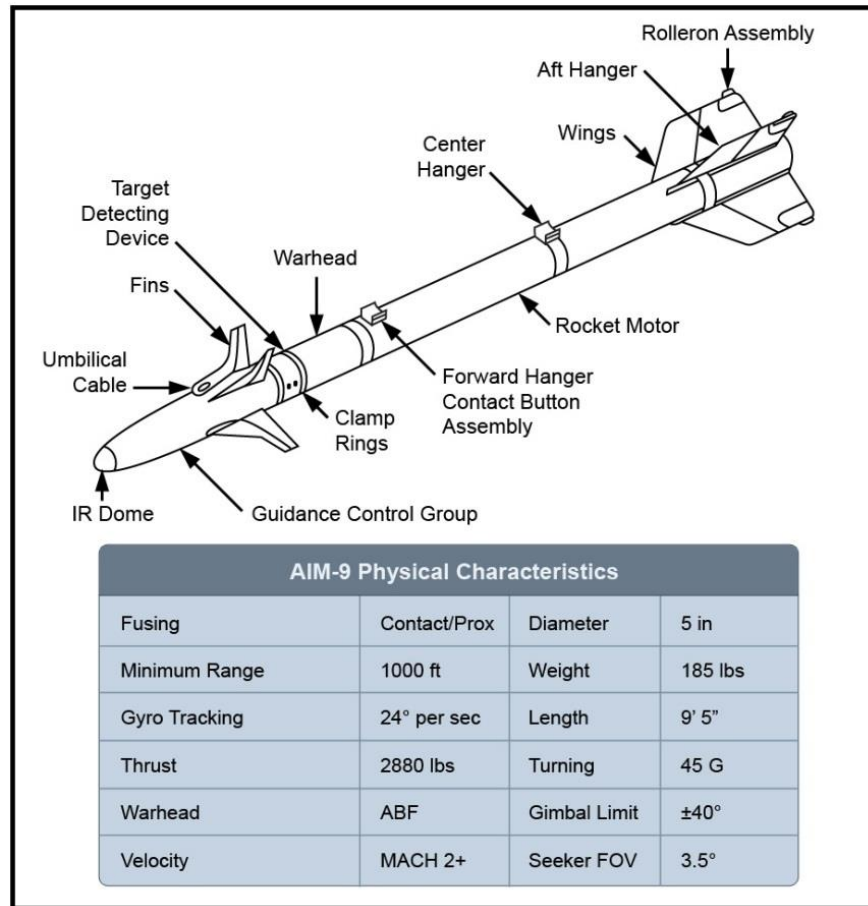


Figure 11-1 AIM-9 Sidewinder

2. Guidance and Control

The guidance and control section consists primarily of an IR seeker head that is cooled by compressed nitrogen and contains indium antimonide (InSb), which makes it very sensitive to IR emissions. A thermal battery provides power for 60 seconds time of flight.

Target acquisition is indicated by a sharp tone in the pilot's headset. The tone increases in intensity as the target approaches the center of the seeker head field of view. The missile is guided by the deflection of the 4 forward fins that are pneumatically controlled by a gas generator ignited at launch.

3. Fuse

The AIM-9 utilizes two types of fusing mechanisms, contact and proximity. The contact fuse feeds a firing pulse to the warhead upon missile impact. The proximity fuse is a laser-optical system that detects when the missile is within a lethal range, sending a firing pulse to the warhead.

The Sidewinder employs a safe and arm device that senses missile launch and will not allow the warhead to arm until the missile is safely ahead of the firing aircraft.

4. Warhead

The AIM-9L/M has an annular blast fragmentation (ABF) warhead consisting of a high explosive encased by titanium rods and a zirconium disk. The titanium rods penetrate an aircraft's skin and the zirconium disk fragments will set fire to any combustible material.

5. Rocket Motor

The Mk-36 rocket motor produces 2,880 pounds of thrust for a 5.0 second burn time that accelerates the AIM-9 to approximately 2.0 Mach above the launch aircraft's airspeed. The "smokeless" motor is actually 95% smoke free, leaving only a thin wisp of smoke in its trail, which would be imperceptible to the enemy.

1104. AIM-7 SPARROW HISTORY

The AIM-7 Sparrow is a medium range, all-aspect, all-weather, semi-active radar guided missile. Unlike many of its predecessors, the current Sparrow (AIM-7M) is a reliable and highly lethal weapon. In the late 1950s and 1960s, the AIM-7 evolved into versions using semi-active radar, or SAR, homing guidance. In this method, the missile "homes in" on the reflected energy provided by the launching aircraft, which is "illuminating" the target with its radar. The missile could then fly to an intercept point along the target's flight path. In 1963, production switched to the AIM-7E version. It used a new propulsion system, a solid-fueled rocket motor. The new motor significantly increased range and performance of the missile. Effective range of course depended greatly on firing parameters like launch speed, altitude, and relative velocity of the target.

Although a great improvement over beam riding, SAR still required the launching aircraft to provide radar illumination, via STT, until weapon impact. This limited the strike fighters' maneuverability, required the strike fighter to engage one target until it was destroyed, and often guaranteed a within visual range (WVR) encounter with the target's wingman. Development continued on semi-active seekers and the Sparrow III was deployed very widely with U.S. and allied forces.

1105. AIM-7 DESIGN

The AIM-7 is a 12 foot long, 500 pound missile. Although it is not a launch and leave missile like the Sidewinder, the high reliability and performance level of the Sparrow make it a viable weapon. Like the Sidewinder, the Sparrow can be broken down into four major sections: guidance, fuse, warhead, and rocket motor.

1. Physical Characteristics

Refer to Figure 11-2 for physical characteristics.

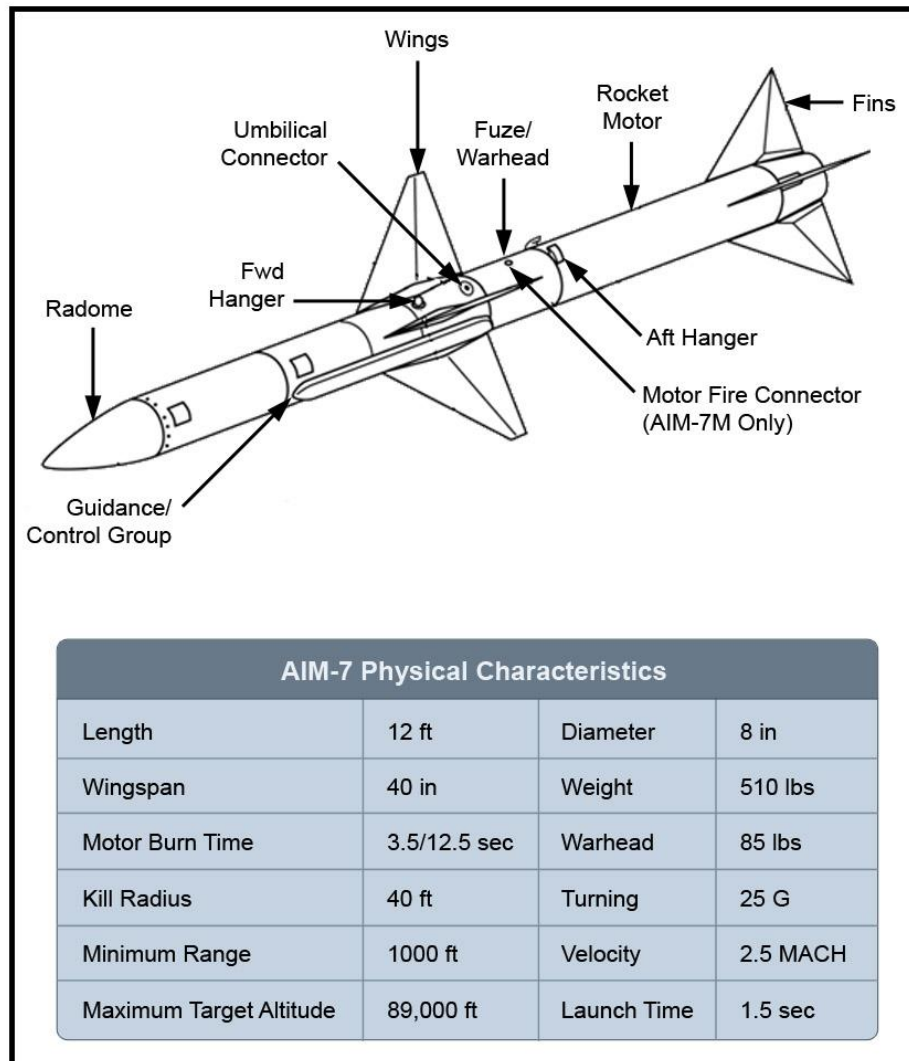


Figure 11-2 AIM-7 Sparrow

2. Guidance and Control

Once the strike fighter has illuminated the target with a radar lock (STT), the Sparrow guides based on the reflected energy from the target using a forward-looking planar-array antenna. The missile receives guidance information from the strike fighter's own radar via a rear facing antenna on the end of the missile. The forward wings are controlled by an open-loop hydraulic system that is pressurized upon trigger squeeze. Once the hydraulic fluid is used, it is vented out of the missile. When the fluid has been exhausted, the missile can no longer maneuver.

3. Fuse

The AIM-7 has a safe and arm device that delays arming of the missile until it has traveled a safe distance in front of the launch aircraft. The Sparrow has both a contact fuse and a proximity fuse, which will detonate the missile at the closest point of approach to the target.

4. Warhead

The AIM-7M employs an 85 pound annular blast fragmentation warhead that explodes into thousands of steel fragments. The hot gases that propel these fragments also serve to ignite all combustible materials.

5. Rocket Motor

The Sparrow has an Mk-56 boost-sustained, solid propellant rocket motor. The initial boost lasts for 3.5 seconds and propels the missile to its cruising speed of 2.5 Mach over the launch aircraft's speed. The motor then sustains the thrust for an additional 12.5 seconds to allow the missile to maintain its speed over a much greater range.

1106. AIM-120 AMRAAM HISTORY

The Advanced Medium-Range Air-to-Air Missile (AMRAAM) is the high priority replacement for the aging AIM-7 Sparrow and retired AIM-54 Phoenix missiles. Currently a joint program for the Navy and Air Force, the AMRAAM is designed to have higher performance and lethality, at a lower cost.

The AMRAAM program began in 1975 as a joint Air Force/Navy program. In 1981, Hughes was awarded a full scale development contract after a fly-off against Raytheon. After eight years of testing, the AIM-120 became operational in 1991. It was first used in combat in December of 1992 when a USAF F-16 used an AMRAAM to down an Iraqi MiG-25 Foxbat during a confrontation over Southern Iraq.

1107. AIM-120 DESIGN

The AMRAAM is slightly smaller in size and weight in comparison with the AIM-7, with many of the same features as the AIM-54. The AMRAAM has four major sections: guidance, fuse, warhead, and rocket motor. Its appearance is similar to that of an AIM-7 Sparrow.

1. Physical Characteristics

Refer to Figure 11-3 for physical characteristics.

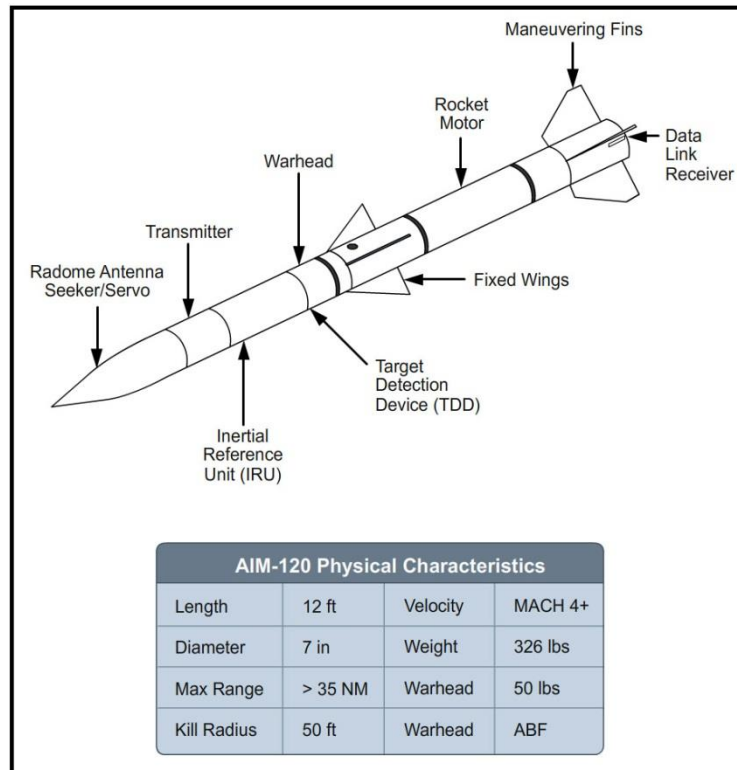


Figure 11-3 AIM-120 AMRAAM

2. Guidance and Control

At medium to short ranges, the strike fighter is not required to illuminate the target with a radar lock because the AMRAAM can guide via a Track While Scan system similar to the AIM-54. These advanced tracking techniques allow any fighter carrying the AMRAAM to have multi-target capability without a multi-target radar system.

3. Fuse

Like all previously mentioned missiles, the AIM-120 also has a safe and arm mechanism that delays arming of the missile for the safety of the firing aircraft. The AMRAAM also uses a laser scanning device to verify external shape, and thus confirm aircraft type, in its fusing mechanism.

4. Warhead

The AIM-120 employs a 50 pound annular blast fragmentation (ABF) warhead that explodes into thousands of tiny fragments. Although one third smaller than the warhead of the AIM-7M,

the AMRAAM utilizes higher density gases to propel fragments more explosively than the Sparrow.

5. Rocket Motor

The AMRAAM has a high-impulse motor giving rapid acceleration to a Mach number higher than 4.0 above aircraft airspeed. The rocket motor is designed to produce very little smoke to aid in reducing the launch signature.

1108. CONCLUSION

This chapter provided a historical perspective of the development of air-to-air missiles, as well as, an in-depth analysis of the designs of the AIM-9, AIM-7, and AIM-120. All these missiles possess the same basic design of guidance, fusing, warhead, and rocket motor. Strike fighter aircrew must know the characteristics of their own air-to-air weapons and their enemies. The studying continues as technology evolves in the world wide air-to-air missile inventory.

CHAPTER TWELVE

MRM AND SRM EMPLOYMENT

1200. INTRODUCTION

This chapter discusses the purpose of Beyond Visual Range (BVR) and Within Visual Range (WVR) employment training, the characteristics and display of the Launch Acceptability Region (LAR), the factors affecting successful BVR and WVR employment, the MRM and SRM radar modes, and the missile firing range display information (Max Range, Min Range, No Escape Range). The last segment of the chapter will focus on some rules of thumb and terminology used in BVR and WVR communication. Overall, the MRM and SRM employment in the OFT and VMTS provide an excellent initial training environment to prepare the student for follow-on AIM-120 AMRAAM and AIM-9X training in the F/A-18F Super Hornet.

1201. MISSILE EMPLOYMENT TRAINING

The purpose of missile employment training is to provide the SNFO with introductory BVR and WVR employment simulation. Any engagement that takes place beyond the normal range for visual acquisition of the bandit (generally accepted to be 10 NM) is considered to be BVR, anything inside of this range is considered WVR. To employ weapons with precision, the student must utilize the standard aviation skills of aviating, navigating, and communicating, as well as weapons systems knowledge, radar display interpretation, tactical crew coordination, and tactical communication. This training is extremely challenging and requires extensive study, simulation training, flight preparation and debriefing. A key point to understand during this segment of training is that the SNFO is learning *skills*, not tactics. Tactics will be taught at the FRS. Specific tactics related to the AMRAAM and AIM-9 are evolving and beyond the scope of entry level training. The training goal is to produce functional system operators, not tactically proficient aviators.

1202. LAUNCH ACCEPTABILITY REGION

The launch acceptability region, or LAR, is a three dimensional volume of space around a hostile aircraft into which the fighter must fly in order to have a chance to successfully employ its weapons. LARs are often depicted in a 2-dimensional (Figure 12-1), top down view that is only valid at the stated altitude and airspeeds.

The fighter will maneuver in altitude, airspeed, and heading in order to achieve the best weapon solution for his opponent. The LAR is largest (i.e., longest R_{MAX}) with 0 TA, at high airspeed and high altitude and is smallest (i.e., shortest R_{MAX}) in the rear quarter at low altitude and low airspeed. Missiles like altitude, airspeed, and closure to achieve maximum kinematics.

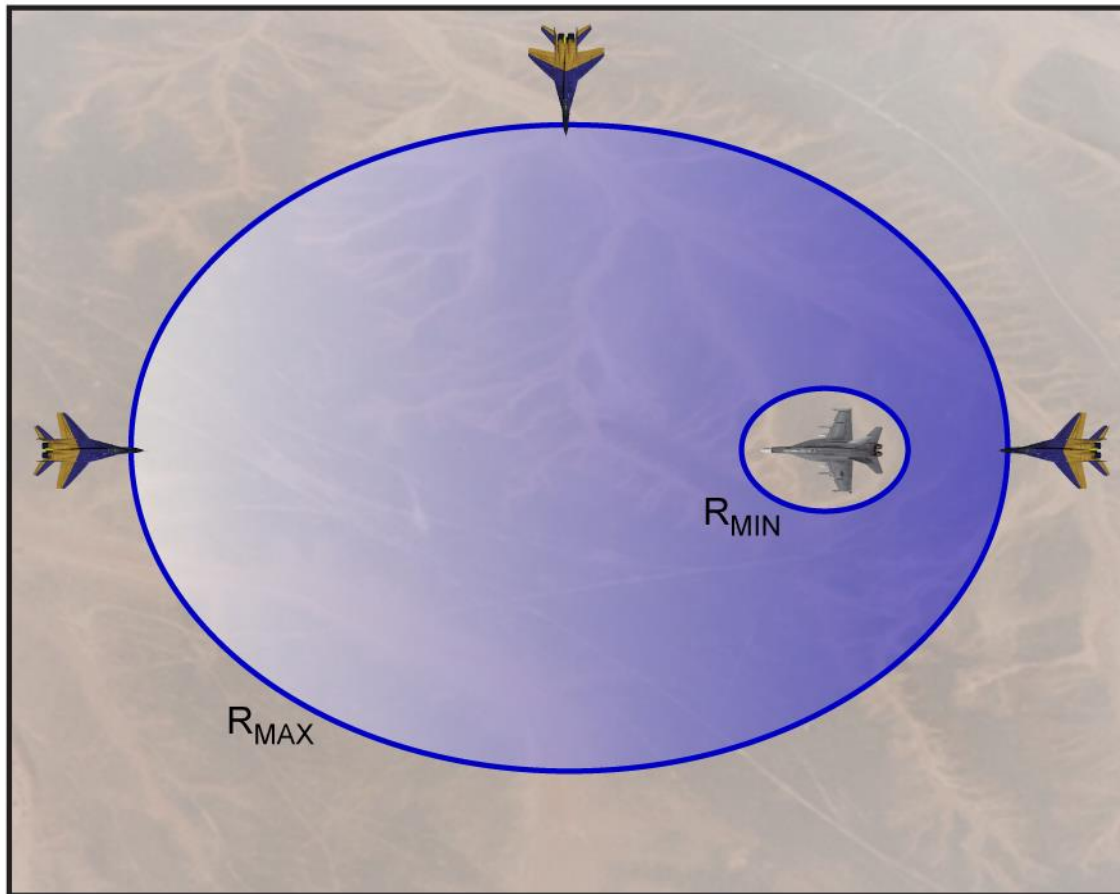


Figure 12-1 Two-Dimensional LAR Representation.

1. **Additional Terminology and Definitions**

- a. R_{MAX} : Maximum range of an air-to-air missile. The maximum range from which an A/A missile can be fired and reach its target given the current target and fighter conditions (V_c , aspect, altitude, etc.).
- b. R_{NE} : No escape range. The range at which a launched missile will have the energy and maneuverability to intercept the target regardless of the target's post launch maneuvering.
- c. R_{MIN} : Minimum range. The shortest range a missile can be launched, acquire the target, receive post launch guidance, and fuse its warhead to intercept its target.
- d. WEZ: Weapon Engagement Zone. The three-dimensional volume of airspace around a fighter into which the hostile aircraft must fly to employ weapons.

- e. ASE circle: The allowable steering error, or ASE, circle and steering dot represent deviation from optimal heading for collision course and deviation from optimal heading and elevation to have the missile on collision course at launch.
- f. Steering dot: The steering dot, used in conjunction with the ASE circle, is a “fly to” cue, meaning you should turn toward the dot to center it in the ASE circle to have the missile on collision course at launch.

2. LAR vs. WEZ

The strike fighter's LAR is dynamic based on parameters of airspeed, altitude, aspect and closure. The bandit's WEZ (Figure 12-2) expands and contracts, just like the Strike Fighter LAR, based on changing flight parameters. As one can see, the understanding of the enemy's weapons capability and how the various parameters affect the WEZ is critical for survival.

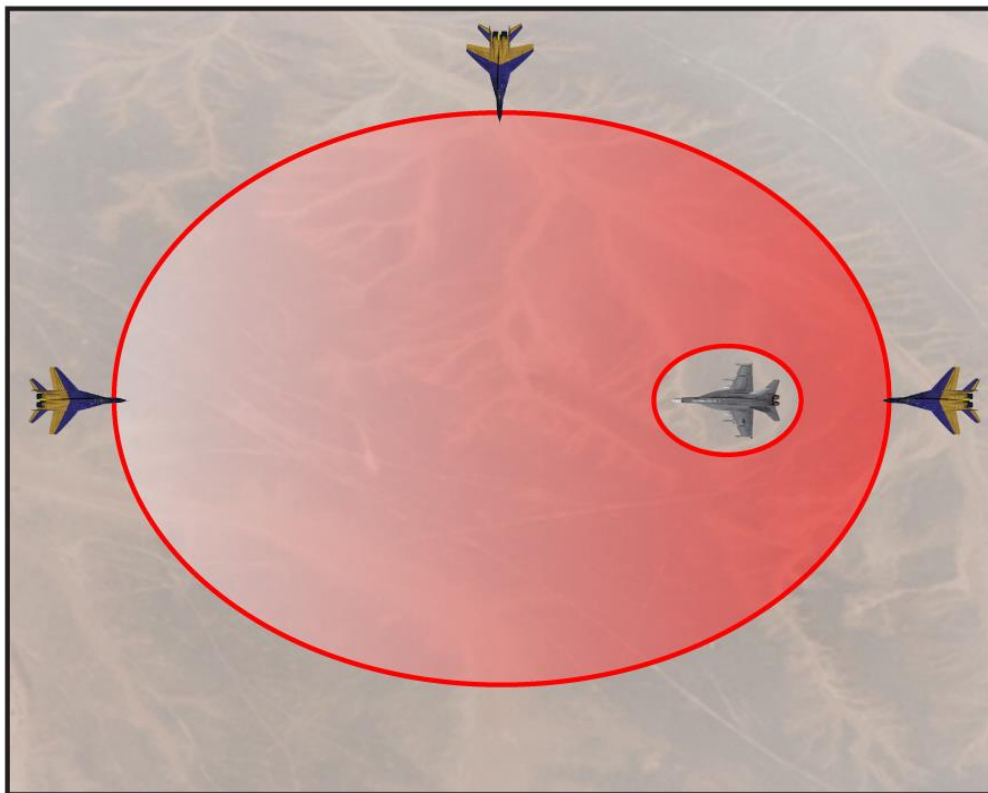


Figure 12-2 Representation of Bandit WEZ

1203. LAR DISPLAYS

The displayed LARs in the OFT and VMTS provide aircrew situational awareness for missile launch decisions. Maximum range for both systems is highlighted in Figure 12-3.

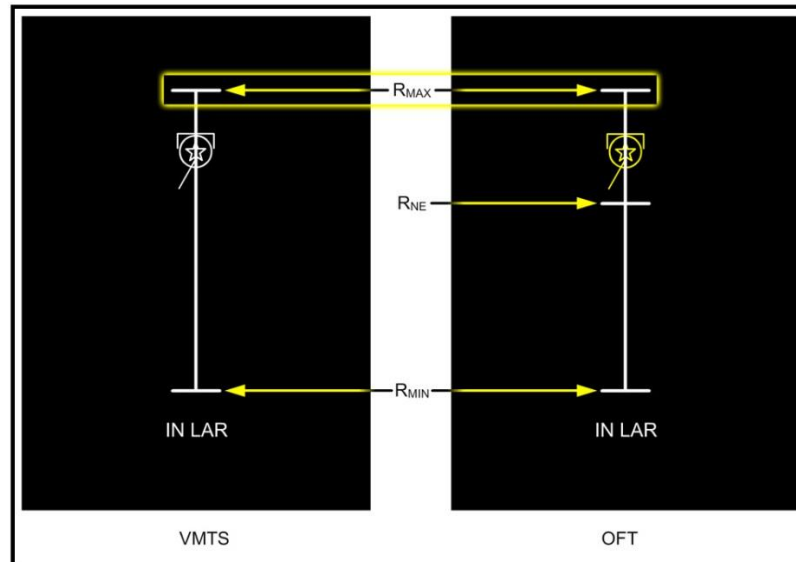


Figure 12-3 Maximum Range or R_{MAX}

No escape range is important because launching at a bandit at or inside R_{NE} guarantees the missile will have the kinematic energy to reach the bandit, regardless of maneuvers. R_{NE} is displayed in the OFT but not displayed in VMTS (Figure 12-4).

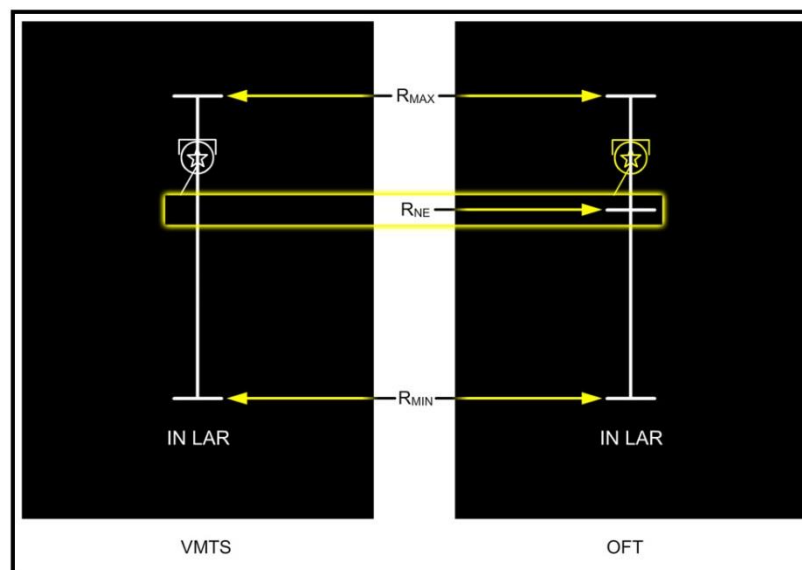


Figure 12-4 No Escape Range is Only Displayed in the OFT

Minimum range, the shortest range to the bandit a missile can be launched and fuse properly, is generally smaller for IR missiles than for radar missiles (Figure 12-5). This is due to IR missiles requiring less information from the host aircraft than radar missiles. The same symbology is used for MRM and SRM.

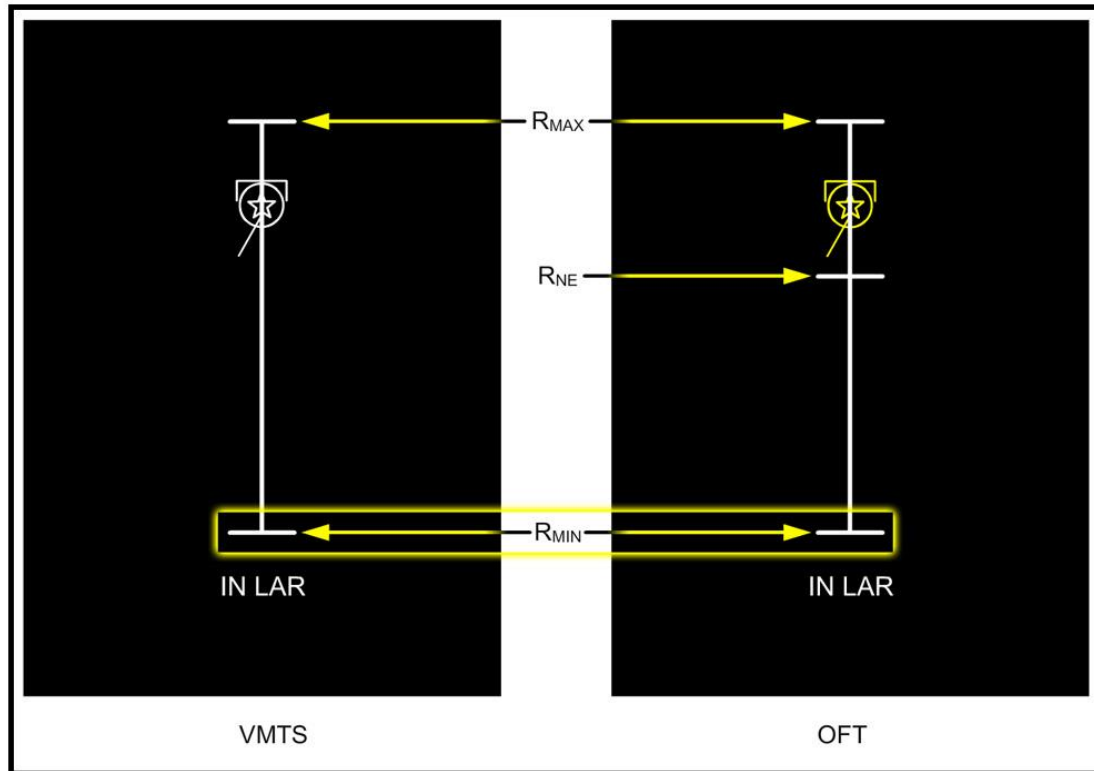


Figure 12-5 Minimum Range

The ASE circle and steering dot appear inside $1.2 R_{MAX}$ providing the optimal elevation and azimuth to shoot the MRM (Figure 12-6). This provides the missile with the best chance of reaching the bandit in the current situation. In the figure below, the bandit is in LAR with the steering cue (or “dot”) inside the ASE circle. These cues are only displayed in the OFT, and are not referenced for validity during MRM employment.

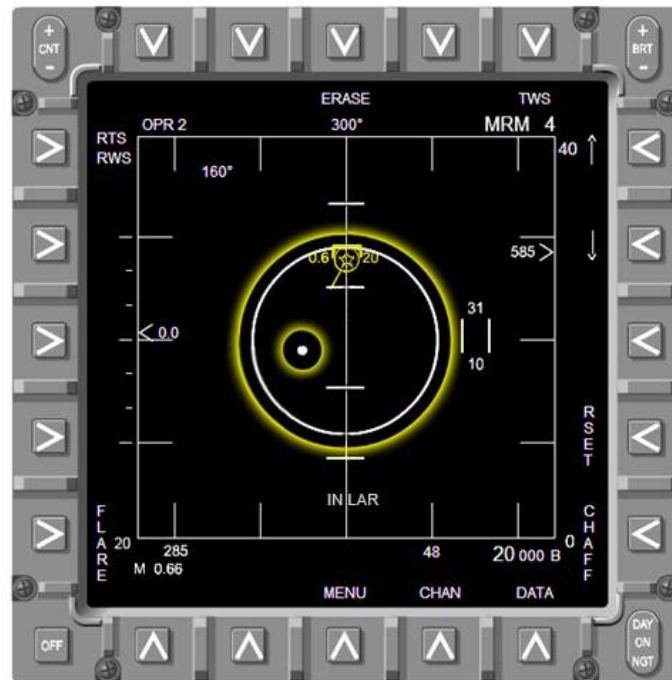


Figure 12-6 Allowable Steering Error Circle and Steering Dot

1204. CONTROLLABLE EMPLOYMENT FACTORS

There are many factors that can affect the true probability of kill (P_k) for a missile in any given situation. The strike fighter can control the following conditions in an intercept:

- **Employment range:** The range a strike fighter employs weapons can have a significant effect on the success of the missile. At long range, the bandit has the time to recognize he is being shot at and maneuver to defeat the missile. It is for this reason the R_{NE} is an important range to understand. Any shots taken at less than maximum range improve the missile's chance of success.
- **Radar mode:** MRM employment requires the radar mode to be STT or TWS. STT provides the most accurate bandit information and smallest uncertainty volume. If a mode other than STT is used, there is a greater uncertainty to the bandit's actual location. Additionally, if radar modes are changed during flyout, for example from STT to TWS, then uncertainty about the bandit's location will grow accordingly.
- **Track quality:** Closely associated with the concepts of radar mode effects on P_k are those related to the quality of the trackfile just prior to, and after launch. If a trackfile is consistently in and out of a MEM condition, then the quality of that trackfile is suspect. Weapons employment should be delayed until the trackfile quality improves, or an STT can be commanded on that track.

- Strike fighter conditions at launch: The strike fighter can have an impact on the success of a missile throughout its time of flight by establishing favorable flight parameters prior to launch. Generally, the faster and higher a fighter is, the larger the available LAR for any given bandit. This is because the missile can accelerate from an already higher airspeed in thinner air and use gravity to assist in end game maneuverability.
- Bandit ATA at launch (Lead): The missile is much faster than the fighter. If the missile was considered to be twice as fast as the fighter, in order to place the missile on collision at launch, ATA would have to be reduced by half. In other words, the correct lead ATA for a missile is $\frac{1}{2}$ collision ATA based on TA. With newer, faster missiles, better fire control systems and better radars, lead ATA control has become unnecessary. Rather these computations are taken into account with the IN LAR symbology. Always shoot with IN LAR displayed.

1205. EMPLOYMENT FACTORS CONTROLLED BY THE BANDIT

The bandit is able to control its flight parameters to challenge the P_k of the strike fighter's missiles. The bandit may also use electronic counter measures (ECM) to deny the strike fighter's radar system from attaining a reliable radar track.

If a bandit is aware that it is being engaged, it will most likely take steps to defend itself. These steps may include chaff, active ECM (jamming), aggressive maneuvering, or a combination of these. If an enemy pilot can get the strike fighter's radar to stop looking at his aircraft for only a short period of time, there is a high likelihood that the fighter's missile will be decoyed, especially at long ranges.

1. Max Range and P_k Enhancing Shot Considerations

Once a missile has been employed and is being supported through its TOF, the strike fighter must decide whether follow-on BVR employment is desired or required. If the strike fighter shoots at maximum range and bandit maneuvers aggressively, the missile is likely defeated. Another BVR LAR will likely present itself to the strike fighter when the bandit tries to reacquire the fighter by pointing its nose at the strike fighter. This follow-on employment will likely be inside of R_{NE} and have a high probability of success. The strike fighter may employ additional weapon(s) against a bandit prior to time out of the first weapon if dictated by the tactical situation. This second employment would be called a " P_k enhancing" shot (Figure 12-7).



Figure 12-7 Second Employment, Enhancing the Probability of Kill (P_K)

1206. VALID MRM SHOT CRITERIA AND EMPLOYMENT TERMINOLOGY

In AWIs the MRM will be employed from the Forward Quarter (FQ). Radar information is required in order to direct the missile initially. The range at which the MRM is employed is determined by the student's block of training.

1. MRM Employment Validity

- a. The following criteria must be met for a MRM valid shot:
 - i. Radar in a shot support mode (STT or TWS throughout missile TOF) and stable trackfile (Trackfile did not enter MEM through TOF, due to bandit maneuvers, jamming, or operator error).
 - ii. IN LAR indication and Correct System Setup (missile selected, Master Arm, A/A Master Mode).

- b. The MRM shall be considered defeated if, during the time of flight, any of the following occur:
 - i. Lost Cue appears
 - ii. Radar SA is lost (MEM cue appears for > 3 sec)
 - iii. RWS is entered

2. MRM Employment Terminology

The terms below are commonly communicated during an intercept where BVR missiles are launched. Students will use these terms in training with the MRM. There are certainly more terms to know, but these are an excellent starting point and utilized in almost every air-to-air training intercept.

Fox-3 is the radio call made to announce that a strike fighter has employed an active radar air-to-air missile, assumed against the targeted group.

SNFO (PRI) - "Rage 11, Fox-3"

If two missiles are employed on two separate contacts in a group, the comm will be:

SNFO (PRI) - "Rage 11, Fox-3, two ship"

If two missiles are employed on the same contact:

SNFO (PRI) - "Rage 11, second Fox-3"

Timeout is an informative call that announces a previously fired missile has reached its expected target. Timeout does not imply a kill. Rather it is used to let other aircraft know the status of your missile. If radar contact with the bandit is lost following timeout, a kill may have occurred. The timeout comm should be in the format of who the missile was shot at, the group they are a part of, and the altitude they are at.

SNFO (PRI) - "Rage 11, timeout, west man, single group"

SNFO (PRI) - "Rage 11, timeout, single only, twenty thousand, single group"

Kill is a comm brevity term that provides directive clearance to destroy a bandit. In training, a kill call is a *descriptive* call that kill criteria has been met. Simply saying "kill" in multi-aircraft scenarios adds confusion. Identify the contact specifically. In multi-aircraft environments, kill calls may also include group name or the target aircraft's current maneuver. In combat, the term "Splash" is used to indicate the downing of a hostile aircraft.

SNFO (PRI) - "Rage 11, kill, east man, west group"

Trashed is an informative comm brevity term that the current missile in flight has been rendered non-viable by ECM, bandit maneuvers, or radar problems. Trashed should be used when a LOST (OFT) or SL (VMTS) cue is displayed beneath the flyout symbol.

SNFO (PRI) - "Rage 12, shot trashed"

1207. VALID SRM SHOT CRITERIA AND EMPLOYMENT TERMINOLOGY

In AWIs the SRM will be employed from the Rear Quarter (RQ) and from the Forward Quarter (FQ). Though actual IR Missile and A/A gun employment do not require a radar lock, a radar lock greatly improves the ability of the aircraft's weapons systems to acquire the target. Aircrew should always strive for a radar lock. At VT-86, a radar lock is required to employ an SRM.

In the aircraft, radar indications of IN LAR will be visible when the target meets launch parameters, but there is no audible tone (Figure 12-8). Any shot taken with an IN LAR cue is considered valid, so with a radar lock, use the displayed LAR. In the OFT, an audible tone is provided to simulate the feedback from an SRM that is tracking a target along with a radar indication of IN LAR.

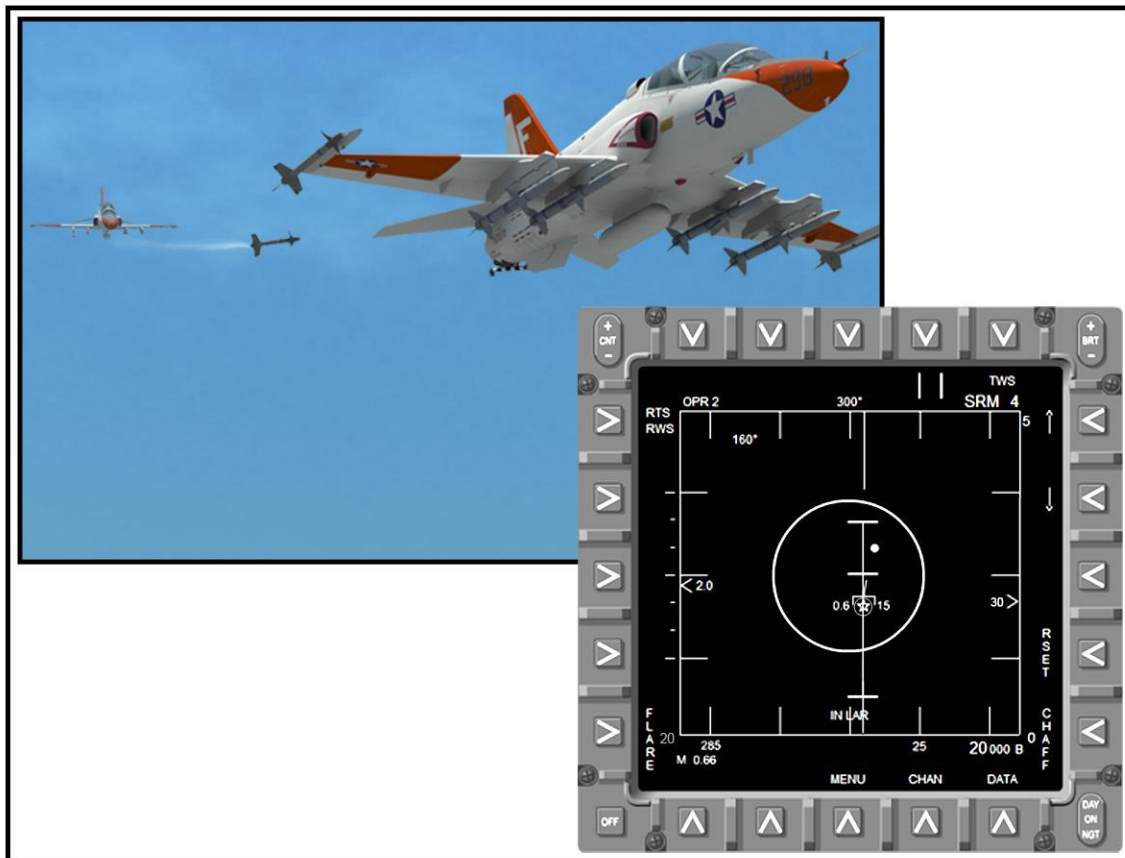


Figure 12-8 RQ SRM with a Radar Lock

12-10 MRM AND SRM EMPLOYMENT

With no radar lock, the SNFO should continue to attempt to acquire a radar lock in WACQ. However, with a tally, the SNFO should direct maneuvering to the valid RQ SRM LAR used in BFM training. For AWI purposes, visual shots (SRM without radar lock) are considered “not assessable” and therefore should not be taken.

1. SRM Employment Validity

- a. The following criteria must be met for a RQ (TA $\geq$ 90) SRM valid shot:
 - i. Radar mode in STT and stable trackfile (no MEM).
 - ii. IN LAR indication and Correct System Setup (missile selected, Master Arm, A/A Master Mode).
 - iii. Range: 1.5NM down to Break-X.
- b. The following criteria must be met for a FQ (TA < 90) SRM valid shot:
 - i. Radar mode in STT and stable trackfile (no MEM).
 - ii. IN LAR indication and Correct System Setup (missile selected, Master Arm, A/A Master Mode).

Note: FQ SRMs are allowed for 1v1 events and beyond.

2. SRM Employment Terminology

The terms below are commonly communicated during an intercept where WVR missiles are launched. Students will use these terms in training with the SRM.

Fox-2 is the radio call made to announce that a strike fighter has employed an IR air-to-air missile, assumed against the targeted group.

SNFO (PRI) - "Rage 11, Fox-2"

Timeout calls are not made when employing SRMs. In lieu of the timeout call, if either aircrew have a tally and assess missile impact, the kill call can be made.

IP (PRI) - "Rage 11, kill, lead man, north group"

IP (PRI) - "Rage 11, kill, locked, twenty thousand, lead group"

1208. MISSILE TIME OF FLIGHT RULE OF THUMB**1. Rule of Thumb (ROT)**

The MRM time of flight (TOF) rule of thumb depends on the quadrant from which the missile is employed (Figure 12-9).

- a. Forward quarter fired missiles close with the bandit at a rate of 1 NM every 2 seconds or will impact at half the shot range.
- b. Beam fired missiles close at 1 NM every 3 seconds.
- c. Rear quarter missiles will close 1 NM every 4 seconds for a co-speed bandit.

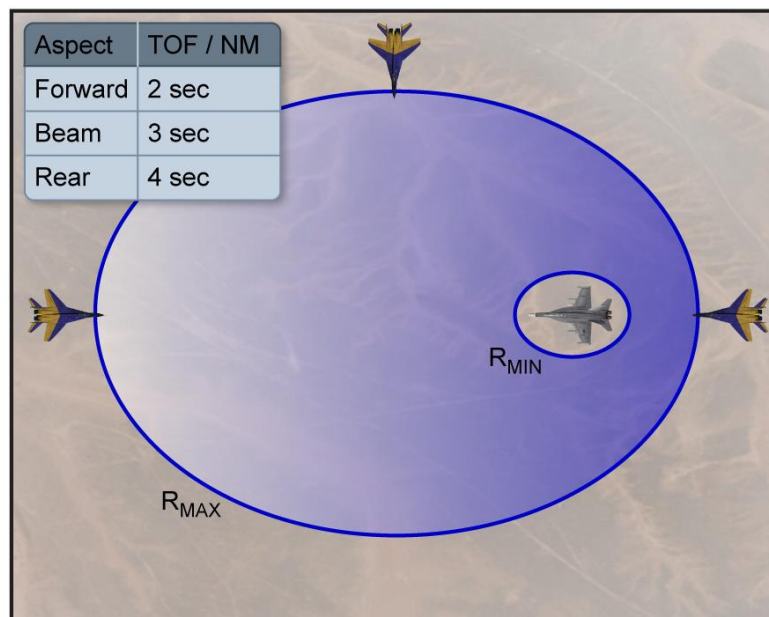


Figure 12-9 VT-86 Missile TOF ROT

1209. CONCLUSION

Successful employment of BVR and WVR weapons depends on a solid understanding of the capabilities and limitations of your radar and your weapon systems. Strike fighter aircrew must also know the enemies capabilities and tactics.

This chapter covered a number terms, concepts, radar displays, and definitions for employing the MRM and SRM weapons. The chapter also specified the valid shot requirements for the OFT and VMTS that students will be graded on at VT-86. Tactical considerations are addressed in this chapter but actual tactics will be taught at the FRS and fleet level. All of this information is critical for success at VT-86 and for building skills to successfully employ future tactics in the fleet.

12-12 MRM AND SRM EMPLOYMENT

CHAPTER THIRTEEN

ADVANCED INTERCEPTS

1300. INTRODUCTION

The following chapter details the fighter's decisions, timeline, radar mechanics, communication, recognition and corrective reactions to a maneuvering, and potentially threatening, bandit. Fighter aircrew must constantly work to enhance situational awareness of the entire battle space during a mission. The bandit's intentions and actions are no exception to this rule. At VT-86, the Advanced Intercepts phase of training introduces the SNFO to an uncooperative bandit. This chapter teaches the SNFO to recognize a potentially threatening situation and react in order to neutralize or destroy the threat.

1301. RULES AND RESTRICTIONS

Rules and restrictions exist in both the combat and training environments. In the combat environment, the Rules of Engagement (ROE) dictate the threat-specific rules for combat operations. In the training environment, Training Rules exist to attain maximum training proficiency while mitigating risk. During training events, fighter aircrews brief Training Rules with the bandits before every air-to-air training sortie.

Range and air space restrictions are typically referenced during the brief. For large force training exercises, such as Air Wing Fallon, the range and air space restrictions are usually briefed in a mass brief the first day of training. Whether a combat or training mission, the rules and restrictions must be understood and adhered to by fighter aircrew. The SNFO must refer to the most recent version of standardization notes (STAN Notes) for a complete description of the current rules and restrictions governing Advanced Intercept training.

1302. THREAT PICTURE WITH BULLSEYE CONTROL

Bullseye control allows multiple friendly forces in different locations, to be able to assess bandit locations with respect to a universally known reference point, generically known as Bullseye. The Bullseye location is usually chosen based on tactical or geographical significance. Once the location of a Bullseye is established among friendly forces, controllers can provide contact positions with bearing and range from Bullseye. In this way, anyone listening to the common AIC frequency can gain SA concerning the location of bogey, bandit, hostile, neutral, or friendly forces. Advanced intercepts at VT-86 utilize both broadcast and tactical control anchored to bullseye.

Broadcast control communicates useful information from the controller to all the fighters (and other friendly assets), and is not directed to one particular flight element. An example of broadcast control anchored to bullseye is below. In this comm example, "SABRE" is the controller and "Vegas" is bullseye.

Controller ("SABRE") - *"SABRE, single group Vegas two one five, thirty, 18 thousand, track south, bogey"*

Notice in the above example that the controller gives a cardinal direction of the contact (“track south”). This is the distinct difference from tactical control anchored to bullseye which gives aspect of the called contact to a specific fighter element. An example of TCBE is below.

Controller (“SABRE”) - “*SABRE, single group Vegas, 215, 30, 18 thousand, hot, bogey*”

In this example, the controller transmits the aspect of the group (“hot”) to Showtime 11.

– Labels and Names

Once the called group(s) meets the fighter’s commit criteria (discussed later in this chapter) and the fighter element commits, the controller will then label the picture and name the group(s). A label is the relationship between groups. Examples include a single group, two groups azimuth, two groups range, three groups vic, and three groups champagne.

A name is the title given to each individual group in the labeled picture. Examples include, single group, east group, north group, lead group, etc. The only time a label and name are the same is for a single group. The example is below. Remember, the controller labels and names the picture after the Fighters commit.

Controller (“SABRE”) - “*SABRE, two groups azimuth ten, east group Vegas 215, 30, 19 thousand, hot, hostile...west group Vegas 208, 27, 10 thousand, drag north, hostile*”

1303. BULLSEYE CARD

The bullseye card is an effective tool used by fighter aircrew to gain situational awareness to the air picture. After receiving the initial picture call from the controller, aircrew can plot the locations of the called bandits on the bullseye card. The bullseye card is a two dimensional representation and does not reflect altitude. Critical to the successful use of the Bullseye card is aircrew knowledge of their own position on the card. Only by comparing the fighter’s position to the bandit’s position can the aircrew assess the situation. An example of the bullseye card can be found in Figure 13-1.

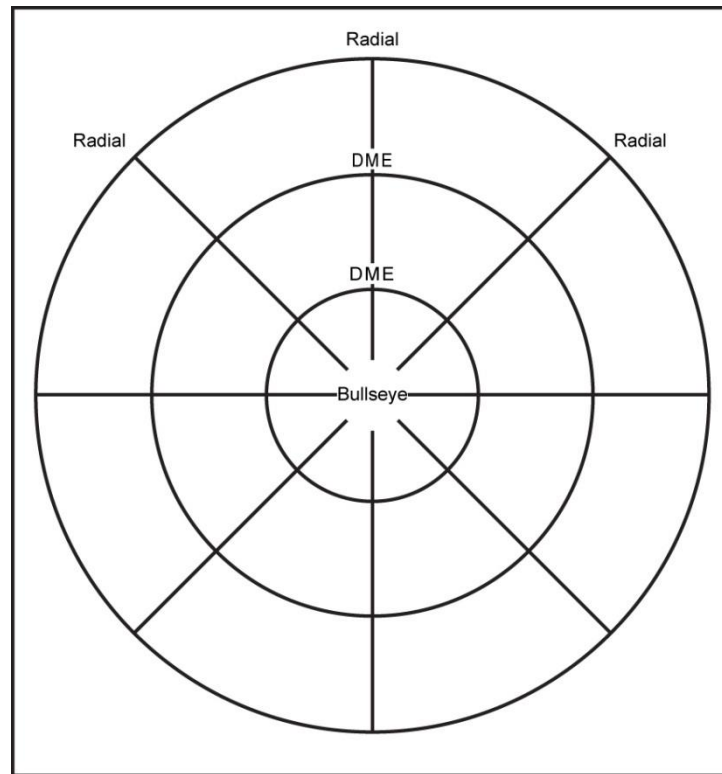


Figure 13-1 Bullseye Card Format

The bullseye center should have the mission assigned name of the geo-reference and the DME scale will be dependent on the battle space. With strike missions, the route often includes a bullseye card overlay.

1304. ADVANCED INTERCEPT PRINCIPLES

The Advanced Phase brings together all previous terminal objectives as expressed in the Master Curriculum Guide. In support of these objectives, the underlying learning objective is proper interpretation of the information provided by the weapon system officer's resources (radar, comms, instruments, visual cues, etc.).

The terminal objectives are accomplished by the demonstration of the student's ability to interpret available resources (with an emphasis on the radar) and act appropriately to achieve successful air-to-air radar operation, intercept, communication, and aircraft operation. Real time decisions and reactions to bandit maneuvers and timeline awareness will challenge SNFO's throughout their career. Advanced intercepts is the first opportunity to learn and test these skills in a dynamic air-to-air environment.

1305. RADAR SETUP AND SCAN

Radar game plans are designed to optimize the probability of detection. The initial radar setup (Figure 13-2) at VT-86 for “fencing-in” is as follows:

- RWS
- 6 Bar
- 140 Azimuth
- 80 NM scale
- Acquisition cursor at 25 NM
- Centered on own aircraft altitude

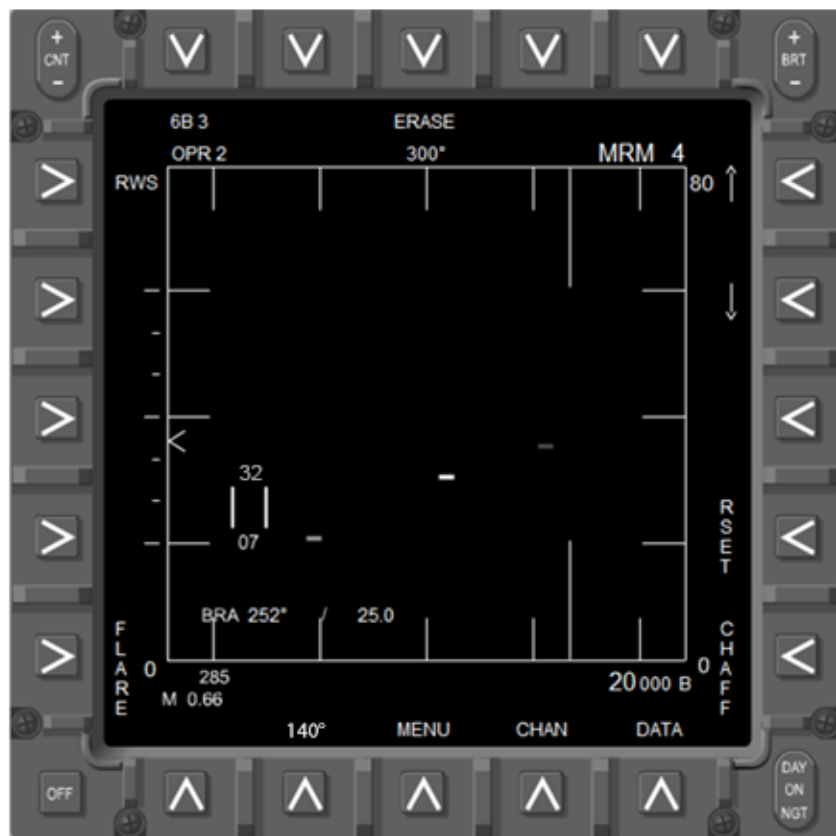


Figure 13-2 Initial Radar Setup

As always, aircrew will need to listen to the calls from the controller, particularly in altitude, to ensure their notional radar system is pointed in the correct threat sector and altitude volume.

13-4 ADVANCED INTERCEPTS

1306. COMMIT CRITERIA

Commit criteria is derived from the mission objectives. If the mission objective is to defend a high value asset, the fighter assets will have a more restrictive commit criteria in order to not be “dragged away” from the protected asset. Missions, such as Offensive Counter Air, will have less restrictive commit criteria as they are offensive in nature, vice defensive.

Since tactics are not taught at VT-86, the commit criteria for advanced intercepts is standardized. Commit criteria will be established in the brief. For 1v1 AWIs, commit criteria will be:

- Range NLT 35 NM
- Target Aspect ≤ 60 degrees (Flank or less)

After the commit, pilots will accelerate to tactical airspeeds (vice fuel conservation airspeeds on CAP) and work to establish initial intercept geometry. An example of a group meeting commit criteria and the fighter commit call is depicted in Figure 13-3.

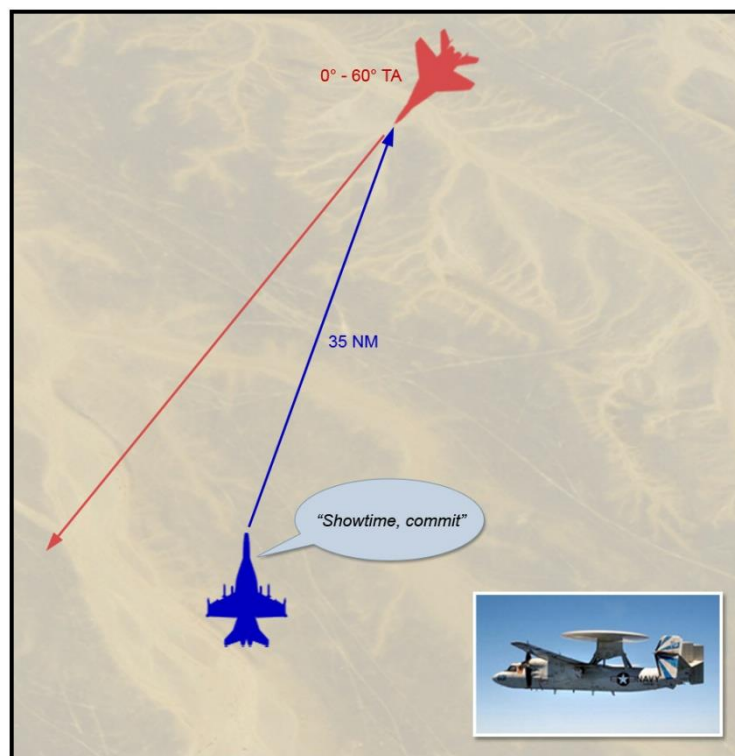


Figure 13-3 Commit Criteria

1307. INITIAL INTERCEPT PHASE

Once the fighter has pointed into the threat sector and gained radar SA, they will need to set the beginning of intercept geometry while executing their “point and assess” plan. The point and assess plan does not necessarily need to be done with the contact on the nose of the aircraft, but the fighter should ensure they have radar SA while making decisions on geometry. The pilot will also at this point accelerate to tactical airspeeds and assume tactical formation (for section and division).

The beginning intercept geometry will be accomplished by establishing a single side offset (SSO), which is defined as 20 degrees ATA. This offset can happen in either direction, left or right, regardless of hot or cold TA, and the fighter should turn in the shortest direction to achieve the SSO. However, if area is a limiting factor, the fighter can use this to determine SSO direction.

It is important to note that with a higher initial TA of the bandit (for example, 50-60 degrees TA), a 20 degree ATA offset in either direction may not suffice to control intercept geometry. In this case, the fighter should use their knowledge of collision bearing to achieve the best possible solution to preserve TA. With the example given of 50-60 degrees TA, a possible SSO solution would be to set the contact at 30-50 ATA hot.

1. Formation, Sensor, Comms

“Formation, Sensor, Comms” will be briefed and stressed on every air-to-air mission. In Advanced 1v1, formation will not be a tactical concern (since there is no Wingman). However, radar mechanics along with clear and concise tactical communications will be critical for mission success.

After the commit, perform the tasks of search and sanitization using the initial radar setup discussed earlier. Once a contact is detected, correlate the contact. For the fighters to be properly correlated, the bearing and range being called by AIC must match within 3 degrees and 3 NM. This correlation is made by either having cursors over the contact while AIC is talking or by making a contact call using the BRAA format. For example:

SNFO (PRI) - *“SABRE, Showtime 31, contact BRAA 190, 32, 25 thousand, declare”*

AIC will then provide a call so that the fighters can achieve correlation that will include a declaration to the detected contact in the format:

Controller (“SABRE”) - *“Showtime 31, single group BRAA 191, 31, 25 thousand, hot, hostile”*

Once correlation is complete, make the contact an L&S, if able, and continue with setting geometry. For 1v1 intercepts and beyond, the fighter should climb or descend in order to be as close as possible to the bandit while remaining in a fighter block. Set airspeed to match the bandit's, but not slower than 0.5 IMN.

SNFO (ICS) – *“Put ‘em 20 left, point six, nineteen thousand”*

2. Targeting

Normally, the next communication will be to target the correlated group. Targeting is used to inform all players that a certain group of fighters are intending on prosecuting that group. Depending on the scenario, targeting may come before or after correlation has taken place. Note that radar SA is not required for targeting. In early training, with only a single group, the targeting call is often forgotten by students. When multiple groups are part of the air picture, targeting becomes more complex and a critical decision in solving multi-group air-to-air engagements. Targeting should occur as close to tactical range as possible, but no later than 30nm. The targeting call for a single group is simple:

SNFO (PRI) - *“Showtime, target single group”*

3. Tactical Range

A tactical range call is given in order to ensure all players are at the same point on the timeline. Anyone with radar SA can make the tactical range call. At VT-86, the tactical range that will be referenced is 30 NM. The “30 Miles” call is transmitted either by AIC or one of the fighters to provide SA that they are 5 NM or 30 seconds (for a 0.5 IMN co-speed intercept) from melding.

SNFO (PRI) - *“Showtime 31, single group 30 miles”*

4. Meld

Meld is a directive call to cease radar sanitization and focus radar energies into the targeted group. The meld will ensure a specific radar mechanic procedure is used to provide radar SA to the targeted group. It is used in conjunction with a comm call to help the out-of-AOR aircraft adjust their radar to the correct threat sector (most often in elevation). The meld should be called by the fighter with radar SA no later than 25 NM. The meld mechanics at VT-86 are:

- a. Range to 40 NM
- b. Reduce to 80 degrees Azimuth
- c. Center azimuth scan volume on meld call bearing with full action trigger
- d. Place cursor 1-2 NM in front of meld called range
- e. Center elevation scan volume to bracket meld called altitude
- f. Command STT on contact

The meld comm call for 1v1 is practiced for later air-to-air events in a multi-plane formation. A meld call will include the bearing, range and altitude of the group. A complete meld call example is:

SNFO (AUX) - *“Showtime, meld BRAA 190, 25, 22 thousand”*

1308. MRM EMPLOYMENT

After committing, targeting and melding into the targeted group, the time has come for weapon employment.

The timeline calls for employment at 20 NM. Take note of the range at which MRMs are employed in order to make the correct follow-on flow decisions. Remember to have a Hostile declaration prior to employment. The timeline in Figure 13-4 depicts the game plan for Advanced Intercepts at VT-86.

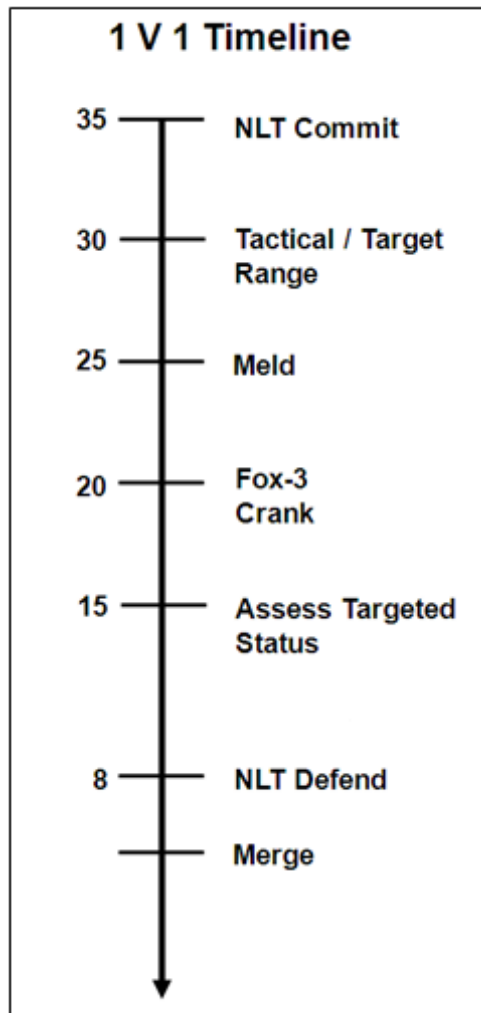


Figure 13-4 Timeline

In order to employ an MRM at the maximum possible range, two conditions must be satisfied:

- Radar in a proper shot support mode (TWS or STT) with a stable trackfile (no MEM)
- IN LAR indication with proper system setup (proper missile selected, Master Arm, A/A Master Mode)

Employment comms are as follows:

Fighter (PRI) – “*Showtime 11, Fox-3*”

Fighter (AUX) – “*Showtime, Crank Right*”

Crank is an informative/directive call that informs the section/division which direction the fighter element is turning. In 1v1 AWIs, even though there are no wingmen, still call for the crank on AUX. The crank is performed by placing the contact 50 degrees ATA in the shortest direction. Just like with the SSO, crank also should be done in the shortest direction, regardless of hot or cold TA.

The crank maneuver:

- Minimizes threat WEZ and slows the downrange travel.
- Must be at least 50 degrees ATA to be effective. Fighter velocity vector must have more of a cross-range component than a downrange component.
- Should be done as to not gimbal the contact, lose track and trash the shot.

While in the crank, evaluate the follow-on flow for the intercept.

- Should the fighter continue to prosecute the intercept by employing follow-on weapons, going to the merge if necessary, or should the fighter time out the current missile and exit the fight?
 - i. Continuing the intercept can be very risky and has a much higher chance of resulting in fighter losses, but may be required for mission success.
 - ii. Going “out” has less risk, but may not result in mission accomplishment. Likewise, fighter losses may be unacceptable and going out may be the only option.
 - iii. If the fighter is spiked or targeted by the bandit, the crew must decide whether to defend or flow to the merge.

This decision is not arbitrary and should be based on the best information available.

1309. DECISION RANGE

In the crank, consider two things to determine if “winning” or “losing” the engagement. These two factors are:

- Timeline adherence: Did the fighters employ weapons on their contracted sort on timeline?
- Bandit awareness: Does the bandit show indications of targeting the fighters? This is defined as 30 degrees or less TA or 60 degrees or less TA and the fighter is spiked.

The combination of these two factors yields the matrix (Figure 13-5) and defines the actions taken upon missile timeout. This decision matrix should be committed to memory.

		TA ≤ 30° or Spiked?	
		Yes	No
On Timeline	Yes	Winning	Winning
	No	Losing	Winning

Figure 13-5 Winning vs. Losing

There may be situations when it is not tactically smart to continue the fight, i.e., the fighters are in a “losing” situation or not in an advantageous position to kill and survive. Based on the determination of winning or losing, and the mission objectives, aircrew will determine if they will “plan banzai” or “plan skate.”

“Banzai” is a directive call to execute launch and defend tactics. Banzai means the fighter element will support any missile in flight to timeout and then proceed to the merge with the targeted group.

“Skate” is a descriptive call to timeout missiles currently in flight and go out. “Abort” is a directive call to execute the abort mechanics. “Plan skate” means the fighter element will still support any missile in flight to timeout, but will not merge with the targeted group. After missile timeout, the fighter element will execute an abort which is a maneuver to aggressively get away from the targeted group (i.e., turn “Out”).

- 1. Unaware Bandit Scenario**

TA > 30 degrees and naked or TA > 60 degrees will constitute that the bandit does not have SA to the fighter. NLT 15 NM, assess if the bandit is aware or unaware. If unaware, the fighter should attempt to force the merge with the bandit using the following procedures:

- a. **Unaware group declared hostile** – It is imperative to control the merge by placing the contact on the hot side of the scope once the decision has been made to force the merge. This ensures that TA does not balloon and the merge can be made. If IN LAR at 20 NM, employ an MRM. Once the fighter recognizes unaware criteria has been met, put the contact at 50-60 ATA hot. If you are unable to reduce TA, maintain the contact at 60 ATA, set a 0.1 IMN speed advantage, and employ follow-on weapons once the initial shot has timed out. If you are able to reduce TA, drive it down to 45-35 degrees, put the contact on collision, and continue to employ follow-on shots to the merge. Maintaining geometry hotter than collision will cause the fighter to fly in front of the contact, which is unacceptable. Cooling off the intercept too soon will take too long to arrive at the merge, increasing your time to kill.
- b. **Unaware group declared bogey** – The same geometry as a group declared hostile should be used, as described above. However, the fighters will not employ at all until in close and after executing a VID to ensure ROE has been met. Using these procedures will ensure that target aspect does not balloon and prevent or delay a VID from happening.
- c. If the bandit suddenly turns into the fighter, recognize the change in the situation and aggressively transition back to the aware gameplan.

2. **Aware Bandit Scenario**

Any TA < 30 degrees or Spiked with TA < 60 degrees means the bandit is aware and the fighters should consider themselves targeted. If targeted, the fighter should defend post MRM timeouts but NLT 8 NM of range.

3. **Air-to-Air Radar Missile Defense**

Defend after MRM timeout but NLT defense range (8 NM), whichever is first. If the fighters are plan skate but end up inside abort range (10 NM), then they are committed to the merge and should then defend per the timeline.

Assuming a 50 degree fighter crank, the defense heading is about 40 degrees more of turn, in the direction to place the bandit in the beam. The fighter should use contact BRAA information from the attack display to get an exact beam heading prior to defending. Using the +/- 100 +/-10 method is a quick way to calculate this heading.

If the bandit is aware, the fighter needs to assume that the bandit can employ a missile. In order to defeat the enemy's radar and any fired weapon, execute a defensive maneuver. This maneuver is:

- a. Place the bandit in the beam, turning in the shortest direction using a hard as possible, level turn.
- b. Dispense chaff (PB15) while turning and reset (PB14) the radar.
- c. Select SRM and set up the radar for the pitch in (see below).
- d. Determine how long to remain in the defense.

SNFO (PRI) - "Showtime, defend South"

SNFO (ICS) - "Heading XXX"

SNFO (PRI) - "Showtime 11, timeout, single only, 18 thousand, single group."

There are three scenarios that can occur and that determine how long to remain in the defense:

- a. If the fighters are naked when they go into the defense and remain naked, there is no minimum time to remain in the defense, and the fighters may pitch in right away.
- b. If the fighters are spiked when they go into the defense and once established become naked, the fighters should turn in immediately. The spike going away is the indication that the fighters are no longer locked by the bandit radar.
- c. If the fighters are spiked when they go into the defense and remain spiked, the fighters should wait for 10 seconds. This means that the fighters were unable to defeat the bandits' radar and that the bandits know where the fighters are located. The fighters should turn in and force the merge because remaining in the defense would allow an opportunity for the bandits to stern convert on the fighters.

Expect to pitch in to the bandit with 110-130 degrees of turn, out of the defense. Use the HSI/SA display to determine this heading by looking just past 90 left or right.

- a. If outside of 5 NM select the following:
 - i. SRM set (RWS/20NM/140/6B)
 - ii. Cursor 8-10 NM
 - iii. Bracket bandit's last known altitude
- b. If inside 5 NM, select
 - i. WACQ
 - ii. SRM

SNFO (PRI) - "Showtime, in left"

SNFO (ICS) - "Heading XXX"

When the radar picks up the contact, bring it to the nose by calling "Bricks" on ICS, employ an SRM, and prepare to merge.

"Bricks" is both a directive and a descriptive call. The call directs the pilot to disregard the assigned in heading and place the contact on the nose. The call describes that radar SA has been regained.

4. Follow-on Employment

Once pitching in from the defense, gaining radar SA, and calling "Bricks," the fighter should take an STT on the contact. With a hostile declaration and valid shot criteria, an SRM should be employed. If there is no IN LAR for the SRM, the fighter can select MRM, employ with an IN LAR, and go back to SRM selected.

With a valid SRM employment opportunity:

SNFO (PRI) – "Showtime 11, Fox-2"

Or with no SRM and employing a MRM:

SNFO (PRI) – "Showtime 11, Fox-3"

SNFO (PRI) – "Showtime 11, timeout, locked, twenty thousand, single group, Fox-2"

With assess impact and a tally:

Pilot (PRI) – "Showtime 11, kill, locked, twenty thousand, single group"

5. Transition to WVR

The transition from BVR employment to WVR employment is a critical moment in the intercept. After the crank and timeout of the missile, followed by the decision to "banzai," the fighter crew must adjust their mindset for the employment of WVR tactics. The fighter crew must quickly determine:

- a. Does the bandit have SA?
- b. Will this be an offensive, defensive or neutral merge or possibly a stern conversion?
- c. What is the best first move option post merge?
- d. When will I be in LAR for the SRM?

1310. WVR/MERGES

When approaching a merge situation, it is imperative for the fighter aircrew to gain a “tally” as soon as possible and to be able to effectively communicate the position of the hostile aircraft to pilot and wingman. “Tally” communication takes practice and experience. Comm brevity and format standardization are key to successful WVR communication. Clock codes are used for directing the pilot and/or wingman’s eyes on the bandit(s) using a modified BRAA format. For example:

SNFO (AUX) - “*Showtime 11, tally 2, left 10 o’clock, 2 miles, 5 high*”

Approaching the merge, the fighter aircrew transitions to the High Aspect BFM game plan. Ideally, the fighter arrives at the merge with a positional advantage (angles, turning room) an energy advantage (airspeed) and low to high merge geometry for advantage in gaining tally and for weapon optimization (blue sky background). Figure 13-6 depicts the various aspects a fighter aircrew will see at the merge to determine if they have an advantage, are neutral, or at a disadvantage with a bandit.

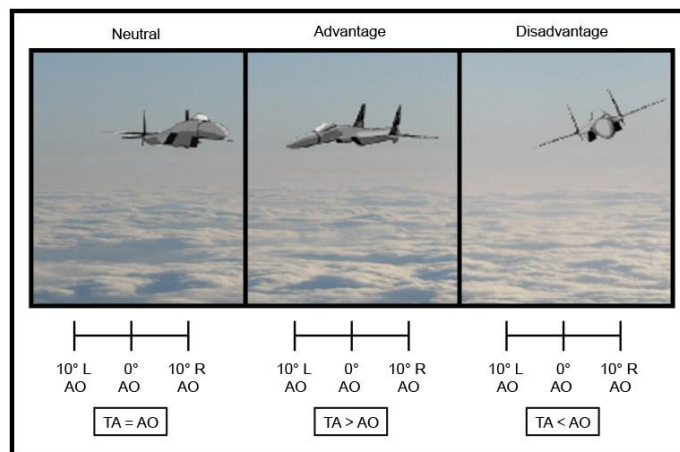


Figure 13-6 Merge Scenarios

1311. POST-MERGE FLOW AND SHORT RANGE RADAR (SRR)

Post-Merge flow will be decided based on whether or not the fighters have high or low SA at the merge.

- High SA: Fighter tally or timeout/kill approaching the merge
 - Fighters should flow into the threat sector. The threat sector is wherever the bandits were originally coming from at the beginning of the intercept.

- Low SA: No tally or shots at the merge
 - Fighters should continue on their pre-merge heading.

SRR procedures are used anytime the fighter clears a merge or makes a kill. SRR procedure is designed to rebuild the fighter's radar SA from short to long-range as they flow into the threat sector. If performed properly, SRR mechanics should take no more than 30 seconds to accomplish. Fighters must be disciplined during SRR and ensure that there is no communication being performed. This silence is imperative so that if the fighter does gain radar SA to a contact they can immediately ask for a declaration from AIC. SRR mechanics are:

- Post merge command a flow heading on PRI and select SRM, BST and then WACQ
 - i. SNFO (PRI) – *“Showtime, flow 090”*
 - ii. This enters scan to clear flight path
 - iii. Remain in WACQ for 10 seconds once established on flow heading
- Exit ACM mode to enter the SRM set RWS/20 NM/140/6B
 - i. Cursors at 10 NM
 - ii. Set scan volume elevation centered around ownship altitude
 - iii. Hold for 10 seconds
- Select MRM and enter RWS/80 NM/140/6B
 - Cursors at 25 NM with elevation bracketing ownship altitude

Upon completion of SRR the fighter should request a picture from AIC, ensuring that they anchor themselves off the bullseye. If at any time a contact is detected during SRR the fighter needs to get a declaration and be ready to employ weapons if necessary.

SNFO (PRI) - “SABRE, Showtime 11, Denver 140, 70, flow 090, picture.”

1312. TARGET MANEUVERS

It is safe to assume that threat pilots understand basic radar/RWR indications and missile employment timelines, and that they will take action to thwart our employment plan. Their goal is to force fighters into short range, WVR maneuvering. Anytime STT is commanded on a contact, the contact may be alerted to the fighter's intentions. A target maneuver is defined as a

change in bandit heading that results in a TA greater than 50 degrees. The threat reaction could be any combination of the following:

- Turning into the fighter to close as rapidly as possible **or** turning away in an attempt to escape.
- Perform some pre-planned deceptive maneuver, such as going to the beam.
- Activate electronic countermeasures (ECM).

When a bandit maneuver is detected, the fighter should near simultaneously execute the following steps using the process of “Recognize, Communicate, Act.”

1. **Recognize**

The following indications of contact maneuvers will occur:

- a. Change in V_C . The radar detects and measures the Doppler shift of returning energy and is extremely sensitive to changes in Doppler velocities. Even a subtle change of less than 10 degrees of heading change by the bandit will be very detectable by the radar. Changes in V_C are instantaneous with bandit maneuvers.
- b. Change in Range Rate of Closure (RROC). A closely related concept to rate of closure is range rate of closure. Where V_C is displayed next to the range caret, RROC is the speed at which the contact moves down the attack display.
- c. Change in contact heading and drift rate. The contact heading display in the upper left corner of the tactical area is a near instantaneous indication of contact maneuvers. If the track crossing rate of the contact has increased, stagnated, or reversed, a change in contact heading has occurred.
- d. Change in elevation caret and differential altitude (for altitude maneuvers). The antenna elevation caret, located on the left side of the tactical region, provides the primary cue to changes in altitude. The contact's differential altitude above or below the fighter, is displayed next to the elevation caret. As the bandit changes altitude these numbers change accordingly.
- e. Change in target aspect vector orientation. The target aspect vector will change with contact changes in heading. The aspect vector is the last maneuver indication the radar will display and will lag relative to actual bandit nose position.
- f. The MEM cue may appear at the bottom of the radar display indicating aged contact information that is unreliable. An effective defensive maneuver will trigger this radar indication.

- g. Speed changes will be indicated only by changes in V_C , RROC, and L&S speed information.

Recognition of maneuvers will be primarily via Rate of Closure, then through bandit heading and Target Aspect Vector.

2. Communicate

Once a maneuver is recognized, the player that sees it shall first communicate to the section to put the contact on the nose with a “go pure” call on AUX. Next, that fighter will communicate on PRI to give SA to all players. Communicate by transmitting Callsign, group name, maneuver.

SNFO (AUX) “Go pure.”

SNFO (PRI) “Ripper 11, Trail Group, maneuver.”

3. Act

After putting the contact on the nose, evaluate trackfile stability. If the radar mode the fighter is currently in is providing a stable trackfile with no MEM cues, the fighter should stay in that radar mode. If the current radar mode is not keeping up with the bandit maneuver and the trackfile becomes unstable or presents a MEM cue, continue to back out radar modes (STT -> TWS -> RWS) every 10 seconds until positive radar SA is regained. With a loss of radar SA and a return to the RWS mode, MRM set should be used for bandits outside of 20 NM, SRM set for bandits between 10 and 20 NM, and WACQ for bandits inside of 10 NM. With a loss of radar SA, call “Clean” on PRI and follow the no radar SA contingency.

- a. Place the L&S on the nose and evaluate the stability of the trackfile.
- b. Select appropriate radar mode to expand the scan volume and maintain contact.

With a stable trackfile and when the bandit turns back toward hot, the SNFO should employ another MRM shot once IN LAR, crank, and resume the timeline. This should occur even if a previous shot does not go LOST due to the severe reduction in P_k , assuming timeline and loadout allows.

If the bandit’s TA stabilizes in a beam or drag aspect, evaluate whether the TA and range meet the reset criteria and execute reset procedures if required.

1313. RESET AND RECOMMIT

In situations where the bandit maneuvers and is no longer a threat, and pursuing it wastes fuel or the fighters cannot rapidly achieve a kill, it makes most sense to turn away and return to the CAP. When the fighter terminates an intercept to return to its cap, it is resetting.

Resetting is a tactical procedure that should be executed in order to ensure the proper disengagement of the targeted group and hand over monitoring of that group to AIC. Note that with TA > 60 degrees, the fighter will not be on the bandit's radar, so the threat is significantly reduced. By maintaining radar contact, the fighter has a tactical advantage. However, the bandit could maneuver again and become a factor very quickly.

- Outside of 20 NM, if TA > 60 degrees the fighters should reset, as pursuing a contact at range burns fuel.
- Inside of 20 NM, the bandit would have to maneuver to > 95 degrees TA to trigger a reset. The fighters should look for a much larger TA because the bandit is a serious threat inside of 20 NM.
- Inside of 10 NM, reset is not an option as this is considered within visual range (WVR). Use geometry to decrease range as rapidly as possible and employ valid Fox-3s or Fox-2s to kill the bandit.

1. **Reset Procedures**

To ensure safe separation from the bandit at reset, use the following procedure:

- a. Turn hard to place the bandit 50 degrees ATA on the cold side of the attack display. This is also called "stiff-arming."
- b. Monitor for 5 seconds if outside of 20 NM; 10 seconds if inside 20 NM.
- c. Execute a hard as possible turn in the shortest direction to put the bandit at the fighter's six o'clock (about 130 degrees of turn); then unload and extend while commanding a buster to set a 0.1 IMN speed advantage. Execute SRR mechanics.
- d. Direct AIC to monitor the group. (PRI) - "*Showtime reset west, SABRE monitor single group.*"

If the bandits reach commit criteria again, the fighters will once again commit.

2. **No Radar SA by 15 NM**

If the fighters have not gained radar SA by 15 NM, they should execute the last two steps of the reset procedures based on BRAA information. If the fighter had radar SA but it was then subsequently lost due to student error, the timeline shall be followed and the defense should occur NLT 8NM.

1314. **GAME PLAN**

All of the previous sections within this chapter detail the game plan for Advanced Intercepts. The timeline in Figure 13-7 is generic and represents a basic training tool used to teach correct

and timely decision making. Because it is specific to your training at VT-86, realize that your employment game plan and Strike/Fighter timeline will change when you employ tactical aircraft in the Fleet. The timeline must be committed to memory.

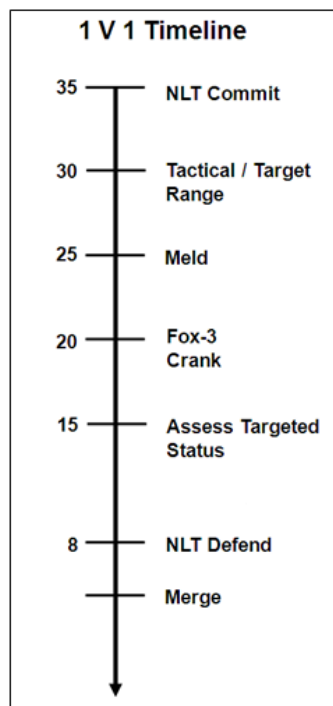


Figure 13-7 Timeline Revisited

– **Construction of the Timeline and Game Plan Considerations**

The basic considerations used to devise the timeline are outlined below. It is built using “open source” threat capabilities and adjusted to meet specific VT-86 mission and training objectives.

- a. Determine shot range from threat information.
 - Threat shot range at VT-86 is expected to be 10-15 NM.
- b. Identify relationship between Range ROC and timeline tasks based on anticipated airspeeds.
- c. Compute minimum and maximum shot range in order to maintain an advantage.
- d. Determine ranges for melding, targeting and commit, based on anticipated shot range and time required to complete each.
- e. Determine NLT Defend and Abort range based on threat shot range.

When constructing the fighter mission conduct based on timeline requirements, the following considerations are included:

- a. Coordination with AIC controllers
- b. CAP planning (position, types of orbit, orientation)
- c. Weapon employment plan and weapon inventory
- d. Fuel management plan
- e. Training rules and training resources

While there will be many more factors included in the employment of tactical Fleet aircraft, this outline provides a basic road map to A/A mission planning. The key in Advanced Intercepts, as with all air-to-air intercepts, is sound Formation (Form), Radar Mechanics (Sensor), and Communications (Comm).

1315. CONCLUSION

The 1v1 Advanced timeline is the same timeline that will be used for the rest of the AWI phase. The commit, label, name, target, meld, sort, shoot, crank, and decide sequence will apply throughout all events. This chapter included an immense amount of information that must be thoroughly understood for success at VT-86. Focus your efforts on knowing this information in great detail as it will provide you with basic skills and knowledge from which to build upon in the future.

CHAPTER FOURTEEN

INTRODUCTION TO SECTION AIR-TO-AIR EMPLOYMENT

1400. INTRODUCTION

Fighters rarely employ as a single aircraft. Fighters will typically employ in section, division or as a part of a larger Strike package or Defensive Counter Air package containing multiple elements.

Fighters throughout history have proven the benefits of having a Wingman in combat. Two aircraft, working together, are more powerful and capable than one, whether their mission is Forward Air Control (Airborne) or a Fighter Sweep. The fighters in a section or division work together to achieve and maintain tactical situational awareness and to achieve air superiority.

This chapter will discuss planning and employment considerations for two aircraft in the air-to-air environment. The information presented here is relevant to section radar attacks (SRA), 2vX intercepts, and the Self Escort Strike (SES). How a section employs in a known 2v2 scenario, such as those seen in SRA, is not different from how the same section will employ in a multi-group environment in 2vX or SES.

1401. SECTION MISSION PLANNING CONSIDERATIONS

As with any kind of military operation, the key to successful employment is robust pre-mission planning. As WWI German General von Moltke stated, “no plan survives contact with the enemy.” It is the planning of the mission that allows fighter aircrew to be adaptable to emerging situations and still accomplish the mission. Planning for an air-to-air mission must take into account all of the following:

- Mission requirements
- Enemy situation and capabilities
- Friendly assets and capabilities
- Weather, terrain and environmental effects

1. Mission Requirements

The fighter's must plan to accomplish the mission. All other considerations are secondary. Planning for an Offensive Counter Air (OCA) mission may be very different from the considerations for Defensive Counter Air (DCA) such as the area defense of a Marine landing force or Carrier Strike Group. The Mission Commander must understand the fundamental requirements of the mission and ensure that all planning supports those goals while adhering to standard operating procedures, applicable flight regulations and utilizing prescribed and proven tactics.

In the fleet, tactical and squadron SOPs for employment will serve as a template for employment in many different strike/fighter scenarios. For the 2v2, 2vX, and SES stages of training, the previously introduced 1v1 timeline is the basis for mission planning. SNFOs should know this timeline from memory and should know how to execute an intercept from any point on the timeline as the tactical situation may dictate.

Some additional considerations that often arise in planning are:

- a. Is target destruction required or desired? “Required” will lead to a higher risk mindset than “desired” and will shape the fighter’s gameplan accordingly.
- b. Are friendly losses acceptable? In operations supporting a no-fly zone or other operations other than war, friendly losses are usually not acceptable. In a military operation, fighter losses may be acceptable, especially if target destruction is required or when defending high value assets such as the Carrier Strike Group. This means that your life and that of your crew depends on your ability to plan and execute effectively and with lethality.
- c. Are merges acceptable? Often, the answer to this depends on enemy capabilities. Against a highly trained and capable adversary, skate mindset will probably be briefed as preferable to banzai.

2. Planning Radar Coverage

Normally, squadron’s tactical SOP will cover the scan volume altitudes and placement, but the aircrew is ultimately responsible for taking into account the specifics of the AOR for the mission including:

- a. Terrain elevation and significance
- b. Radar resolution and performance
- c. Probability of detection, or P_D , of the radar against the expected threat
- d. Scan volume parameters, including frame time and target aging

Tactical SOPs usually have these parameters factored in for a greater than 90% solution. However, arguments can be made for modifying scan volume, altitude responsibilities, or other parameters to maximize the chance of mission success.

For example, a narrowly defined AOR in a mountainous region where the threat is typified by low altitude strike aircraft and helicopters that may prove difficult for AIC to detect, might dictate that the fighters narrow their scan volume and that both concentrate in a low block look schedule with one fighter scanning above the fighters every few frames in coordination with AIC.

The point of this example is this: Fighter aircrew are the experts at employing their weapons systems to maximum effect. It is ultimately the Mission Commander's responsibility to ensure the mission is accomplished. A thorough understanding of capabilities and limitations is required and may lead to a deviation from "standard" employment. Any deviation from tactical recommendations must be thoroughly researched, briefed and evaluated for operational risk to mission accomplishment.

At VT-86, intercept scenarios are over open water, so standard radar employment considerations from lvl will still apply. Specifically, the fighters should use RWS/140/6B/80 NM setup with 16 second aging for their radar..

1402. BRIEFING AND DEBRIEFING

A solid section flight begins with a solid brief. The crew member briefing, usually the mission commander, should speak from notes and not simply "read" the brief. Mission commander's should speak with confidence and provide specific examples. The briefing officer should:

- Speak to flight related admin in detail including weather and NOTAMS that may affect the flight. These factors can have a tactical impact on the mission such as fuel planning for adverse weather recovery and diverts
- Procedures on and off the route or CAP, especially if they differ from normal procedures
- The communications plan
- Tactical plan and timeline in detail
- Contingencies in detail.

The briefing officer should seek to avoid:

- Reading the brief. The mission commander should know the plan.
- Using the phrase "as SA dictates."
- Briefing items that are SOP. The exceptions to this are safety of flight and mandatory briefing items.
- Using hands to represent aircraft. Use models instead and seek to represent realistic geometry.

Mission commanders should brief the most likely contingencies. However, no briefing can cover every possible contingency.

1. Conducting the Briefing

Briefs should follow the standard format of:

- a. Administration (Admin)
- b. Tactical Administration (Tac Admin)
- c. Mission Conduct
- d. Contingencies

Since the SNFO should be intimately familiar with the Admin and Tac Admin portions of the briefing, this discussion will focus primarily on the conduct and contingency portions of the briefing.

2. Briefing Mission Conduct

The mission commander will brief mission conduct in great detail. As support for this, the intercept timeline should be displayed on the board. The timeline should be briefed so that formation, sensor employment, communications and crew coordination is covered for each significant part of the intercept. The timeline for section employment is shown in Figure 14-1.

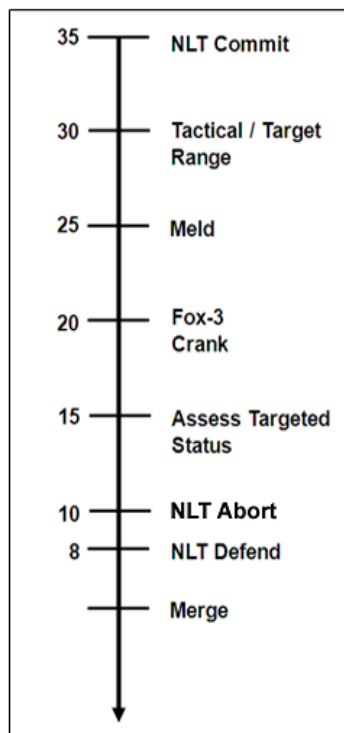


Figure 14-1 SRA and 2vX Timeline

3. Section Tasks

The fighter section must accomplish these intercept tasks any time they commit:

- a. Build and recognize the picture with AIC through active listening
- b. Identify the highest priority group in the picture
- c. Commit on timeline
- d. Identify the group which the fighters will target
- e. Set geometry to achieve an advantage on the targeted group
- f. Meld all fighter radars into the targeted group
- g. Sort the contacts in the targeted group to engage
- h. Employ on timeline
- i. Determine weapon effects and follow-on flow
- j. Flow to follow-on engagements or separate from the fight

All of those tasks must be planned for to ensure the section acts as a team in accomplishing the mission.

4. Mutual Support

Section mutual support begins with solid understanding of formation use and sensor planning. In combat, the formation used will be dependent on the tactical situation. In an offensive scenario, the fighters will likely be close to or potentially exceed visual limits using Combat Spread.

At VT-86, T-45C combat spread is defined as:

- a. 90 or 270 degree bearing line
- b. $\approx$ 1 NM lateral separation
- c. $\pm$ 1,000 – 3,000 feet

This formation is used in air-to-air missions and differs from that used on low levels in that the fighters will have at least a 1,000 foot altitude difference. This formation will be briefed for all Tac Conduct portions of the flight with other formations per the SOP for flight management to and from the working area.

Visual mutual support between formation members provides a means to maintain the best defensive posture while executing the fighter mission. Without a visual, the fighters rely solely on aircraft sensors and communications to provide mutual support. The concept of mutual support will be addressed in more detail in the FRS during the section engaged maneuvering events.

5. Section Communications

The briefing officer should use clear, correct, concise, SA enhancing communication examples for all stages of the intercept. The Lead SNFO should also brief examples of section fuel, weapon checks, and all appropriate communications brevity that will be used, where appropriate.

6. The Four Assumptions

When briefing communications, the flight should assume that that the other fighter is:

- a. Visual – meaning they have sight of the other fighter
- b. No Joy – meaning they do not have a tally on any unknown or hostile aircraft
- c. Naked – meaning the fighters are not spiked by hostile radars
- d. Clean – meaning the fighter has no radar contacts

If any of the above change, the fighter must inform the other aircraft on the applicable radio.

7. Sensor Setup

The fighters should brief a sensor setup that covers:

- a. Mode
- b. Azimuth
- c. Bar scan
- d. Range scale
- e. Aging

In written form, this can be written as: Mode/Azimuth/Bar Scan/Range. Because the selected aging is based on the selected Azimuth and Bar Scan, once the aging has been stated in the brief it will be assumed to remain that value for that Az/Bar. For 140 azimuth and 6 bar scan the aging will be set to 16 seconds. For example, range while search, 140, 6 bar, 80 NM:

RWS/140/6B/80

VMTS radar can retain sets via the SET option on the attack display.

8. Sensor Employment

The Lead SNFO is responsible for briefing sensor employment. This should include a depiction of the attack display on the board with bullseye and GeoRef information (Figure 14-2). This drawing should allow the briefing officer to update mode, range, azimuth, bar scan and cursor placement (with appropriate altitude scan volume indications) as required during the brief to demonstrate expected sensor employment. Correct details here matter.

Contacts should be displayed with the appropriate symbology for the mode being used. This means the briefer should have a well-studied scenario ready to brief.

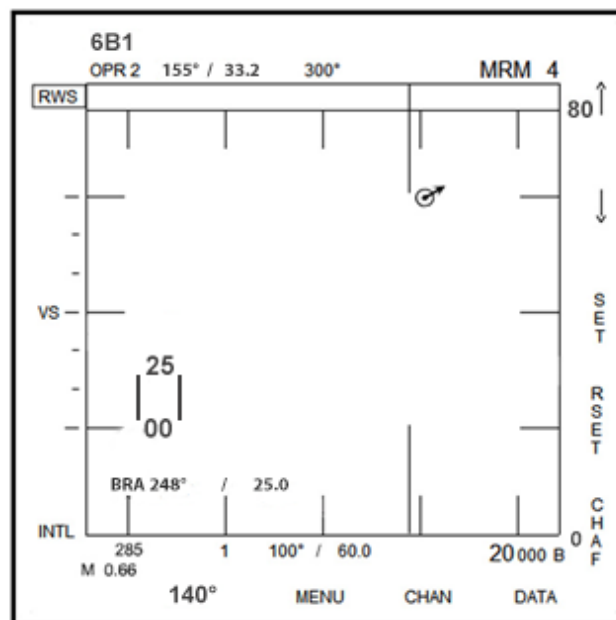


Figure 14-2 Radar Attack Display on a “White Board”

9. Crew Coordination

Effective section employment mandates that all crew members understand and can communicate about the tactical situation. These responsibilities are:

- a. Lead pilot
 - i. Maintains orientation on CAP
 - ii. Maintains altitude and airspeeds as directed by Lead SNFO
 - iii. Performs flight management comm

- iv. Overall responsibility for flight
- b. Lead SNFO
 - i. Backs up pilot on navigation, area and fuel management
 - ii. Performs mission communications
 - iii. Primary sensor operator
 - iv. Directs Lead Pilot for intercept maneuvering
 - v. Directs Wing SNFO on sensor and weapon employment, as required, if different from brief
 - vi. Responsible for mission accomplishment
- c. Wing Pilot
 - i. Keep sight of Lead
 - ii. Maintains formation as briefed
 - iii. Backs up Lead with navigation, fuel and area management
 - iv. Maneuvers as directed by Wing SNFO and Lead Pilot
- d. Wing SNFO
 - i. Primary sensor operator
 - ii. Conducts Wing specific mission communications
 - iii. Backs up Lead SNFO on sensor and weapons employment

The Lead SNFO will be responsible for mission accomplishment, but the Wing SNFO is responsible for backing up the tactical decisions of the Lead SNFO. As always, safety of flight is the responsibility of all aircrew, with the Flight Lead retaining ultimate safety of flight decision making authority.

10. Contingencies

The briefing officer must brief, at a minimum, contingencies for no AIC, adverse weather, system degradations, and any other appropriate contingency that may affect mission accomplishment.

1403. TYPES OF MISSIONS

A section may be employed in either an offensive or defensive counter air mission. The differences are briefly discussed below.

1. Offensive Counter Air

OCA missions are offensive operations to destroy, disrupt, or neutralize enemy aircraft, missiles, launch platforms, and their supporting structures and systems both before and after launch, but as close to their source as possible. These operations include attack operations, fighter sweep, escort, and suppression of enemy air defenses (SEAD). OCA missions are not confined to just air-to-air combat. Rather, any mission undertaken to disable airfields, aircraft or their supporting structures is an OCA mission. Figure 14-3 depicts a generic OCA strike package. Specific fighter missions may include:

- a. Fighter Sweep – may or may not be associated with a strike package
- b. Escort – ensure the safety of the strike package
- c. Close Escort – dedicated air-to-air fighters embedded with the strikers
- d. SEAD – suppress enemy air defenses
- e. Strike – destroy surface targets that impact the enemy's air order of battle

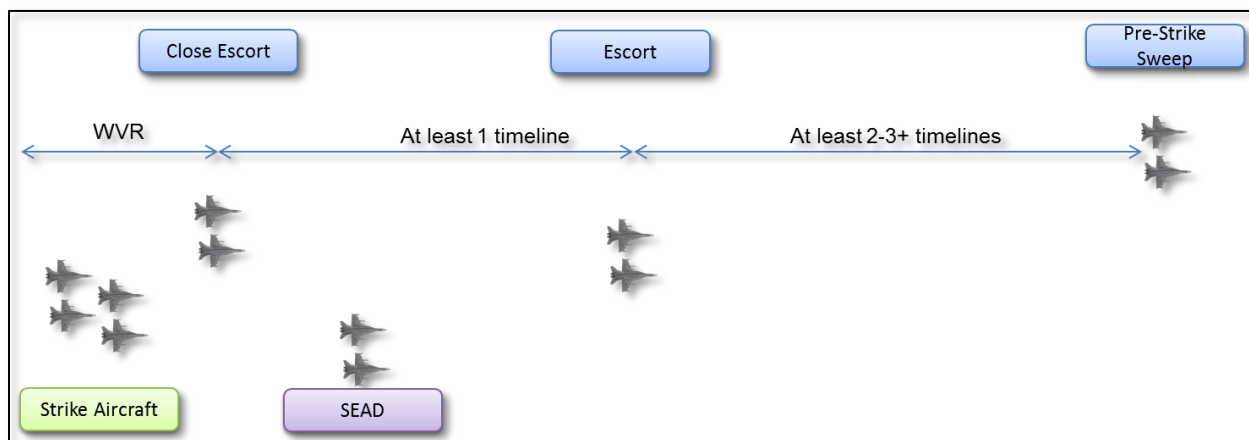


Figure 14-3 Generic OCA Strike Package

These missions may all occur simultaneously as shown in the theoretical OCA strike package example. Note that the ranges here are notional and are tied to the air-to-air timeline, allowing the actual distance to vary depending upon the threat. This allows for each dedicated air-to-air section to perform all timeline tasks without overrunning the preceding section's execution.

2. Defensive Counter Air

The DCA mission is used to detect, identify, intercept, and destroy or negate enemy forces attempting to attack or penetrate the friendly air environment. DCA is often divided into either a point or area defense. Point defense is the defense of a specific asset or location, such as a disabled ship or the Strike Group's Aircraft Carrier. Area defense is the defense of a series or group of assets across a broader geographic region, such as a country border or an assault landing area. Friendly ("blue") losses will likely be deemed "acceptable" if required to protect the asset. Typical DCA missions include combat air patrols (CAP), and:

- a. Barrier Combat Air Patrol (BARCAP) – defense of an area with the intent to not let hostile aircraft past a designated line; the proverbial line in the sand (Figure 14-4)
- b. High Value Airborne Asset Protection or CAP (HVAAP/HVAACAP) – defense of a high value asset such as the E-2, E-3, Growler or other national asset
- c. RESCAP – CAP established in to support rescue or Tactical Recovery of Aircraft or Personnel (TRAP) missions
- d. Point Defense
- e. Area Defense

An example of a DCA arrangement is shown in Figure 14-4. DCA CAP missions often must integrate with surface based air defense, such as Patriot or AEGIS systems to provide a gap-free and multi-layer defense for friendly forces.

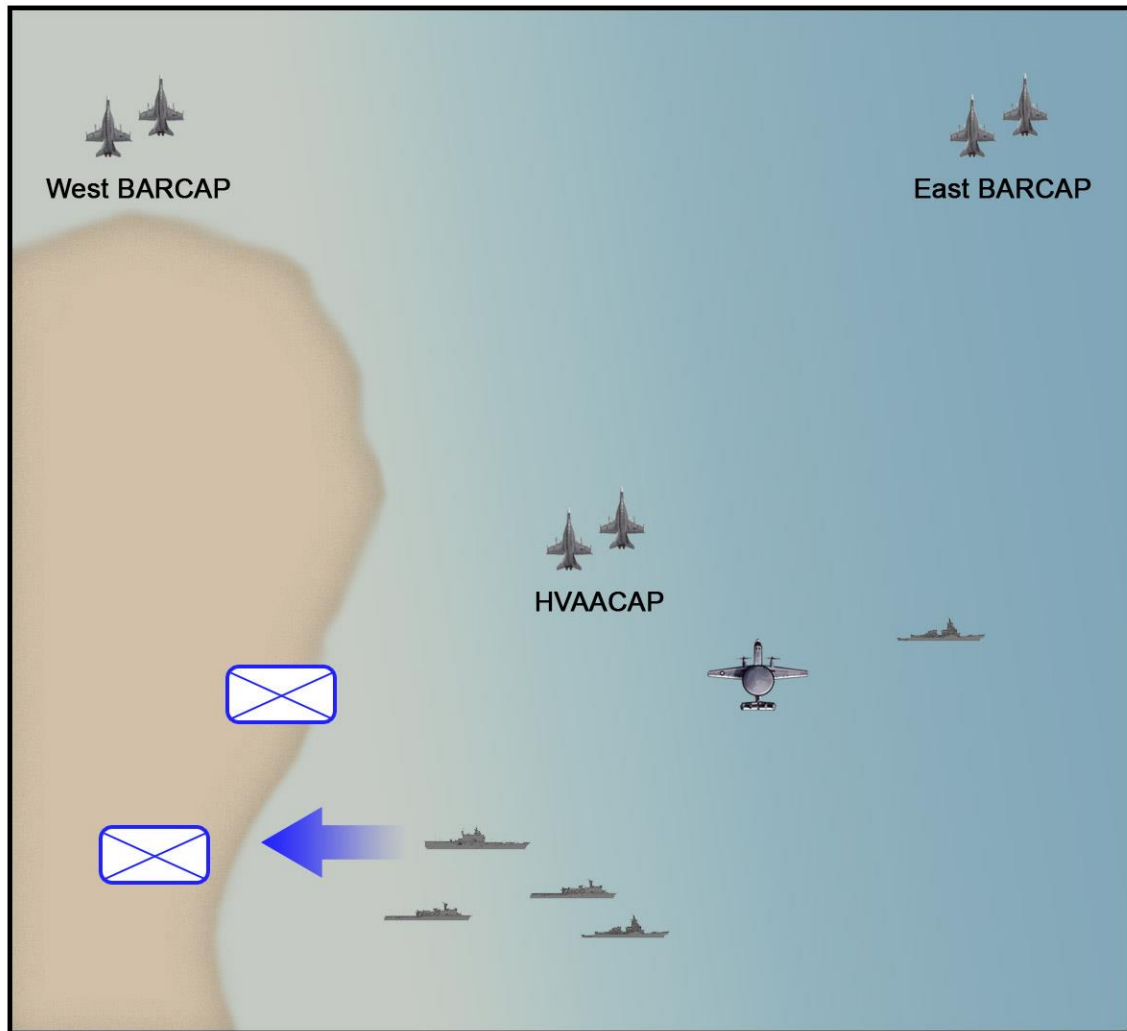


Figure 14-4 DCA BARCAPS

1404. THE ROLE OF THE FIGHTER SECTION IN AIR-TO-AIR WARFARE

Regardless of the briefed mission, the fighter section coordinates their actions to accomplish the following:

- Sanitize airspace
- Detect and Identify unknown contacts
- Destroy hostile contacts meeting ROE
- Integrate into the larger air defense picture

Whether OCA or DCA, the role of the fighter section can be summed up by the above.

1405. 2V2 VERSUS 2VX INTERCEPT EXECUTION

The remainder of this chapter will discuss section intercept procedures. SNFOs will encounter a single, sortable group against which they will practice the timeline skills inherent to a section intercept. These intercepts consist of two fighters versus one bandit (2v1) or two fighters versus a single group of two bandits (2v2) scenarios. This stage is known as section radar attacks or SRA. These scenarios provide introduction and practice of section radar employment skills.

Following SRA, the SNFOs will encounter multiple group scenarios against an unknown number of bandits (2vX). In these scenarios, the section will be tested on teamwork and decision making in addition to the skills initially encountered in SRA.

Procedurally, there is no difference between the two intercepts. The fighter section should approach every scenario as a potential 2vX scenario until AIC or fighter SA demonstrates that the scenario is not.

1406. INTERCEPT EXECUTION

Any section intercept will follow the timeline shown previously to accomplish the following intercept tasks in order:

- Pre-commit picture building
- Commit
- Label / Name
- Target
- Meld
- Sort
- Shoot (Employ)
- Decide
- Merge

Regardless of the number of hostile or bogey groups, the fighters will execute these tasks against every group meeting commit criteria.

1407. PRE-COMMIT PICTURE BUILDING

The air-to-air picture is a combination of situational awareness: AIC, and both of the fighters. Initially, the picture will be built using broadcast control anchored to bullseye. The brief should follow the “form, sensor, comm.” format as shown in Figure 14-5.

Range	Action	Form/Geometry	Sensors	Comm
	Pre-Commit	Cap – Alt as briefed/ 250 KIAS	MRM/RWS/140/ 6B/80 NM Mate 25 NM	<i>“Rage 11, tapes on, fight’s on” PRI</i> Broadcast Control anchored from Bullseye

Figure 14-5 Briefing Should Follow “Form, Sensor, Comm” Format

1. Pre-Commit Formation

Prior to the commit, the fighters will be on CAP. Flight Lead/SNFO will brief a CAP management plan including types of turns and CAP orientation. The CAP should be oriented to have sensors focus into the threat sector on “Hot” legs. In place turns will be used unless otherwise briefed or called in flight. “*Hot R/L*” and “*Cold R/L*” imply in place turns onto the CAP leg, and will be called by the lead pilot. “*Visual*” and “*Six Clear*” calls are not required while on cap.

The fighters will CAP in tac wing at 250 KIAS at briefed CAP altitude.

2. Pre-Commit Sensor Employment

The fighters should sanitize airspace with MRM selected in RWS/140/6B/80NM. The fighters should mate their radars at 25 NM. This is done by placing the cursor at 25 NM, and adjusting the elevation so that no altitude gap in fighter coverage exists outside of 25 NM. In a 6 bar scan, this corresponds to 25,000 feet of altitude coverage at 25NM. The lead SNFO will normally bracket 40,000 to 15,000 feet, while the wing SNFO will bracket surface to 25,000 feet. This is shown in Figure 14-6.

As you can see, this provides for an overlap in radar coverage between the two radars. Each fighter will sanitize their Area of Responsibility (AOR) according to the briefed plan. The boundary between Lead’s AOR and Wing’s AOR is called the “Razor,” and in this case would be 20,000 feet.

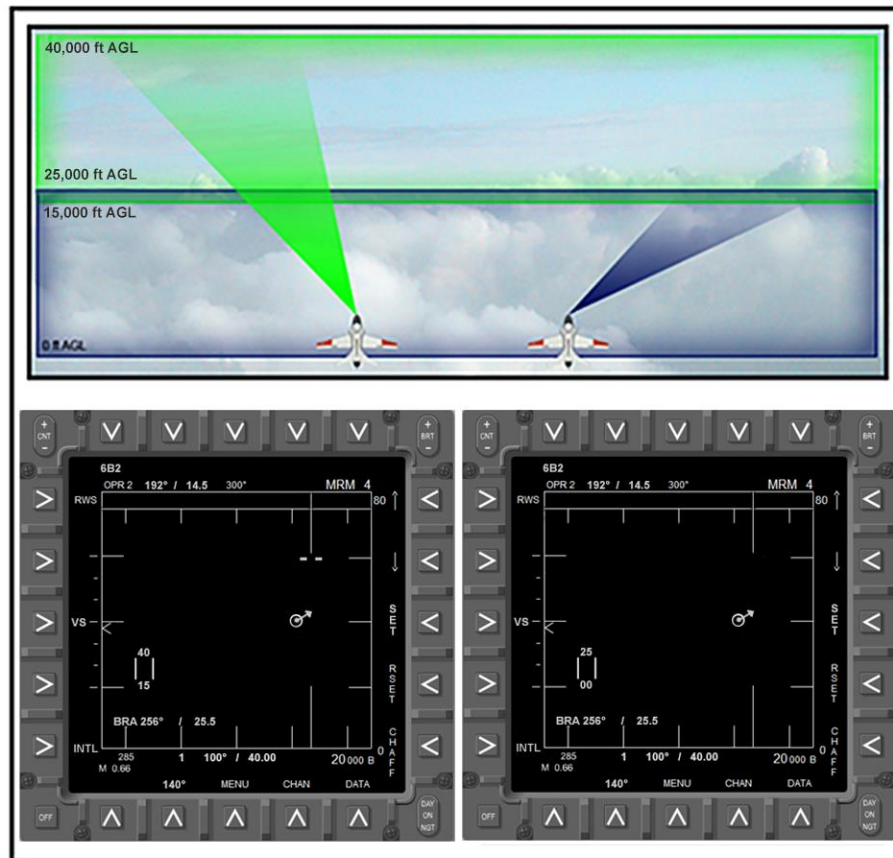


Figure 14-6 Properly Mated Radars at 25 NM

3. Pre-Commit Communications

The picture will continue in broadcast control, anchored to bullseye until the fighter's commit. The Comm priority will be AIC, Lead, and then Wing by exception. A typical call will be:

LEAD (PRI) - "SABRE, Rage 11, picture"

AIC (PRI) - "SABRE, three groups; group Vegas 215, 25, 19 thousand, track south, hostile; group Vegas 270, 50, 22 thousand, track east, hostile; group Vegas 250, 60, 22 thousand, track east, hostile."

Once a group meets commit criteria, the fighters will commit and control will change to tactical control, anchored to bullseye.

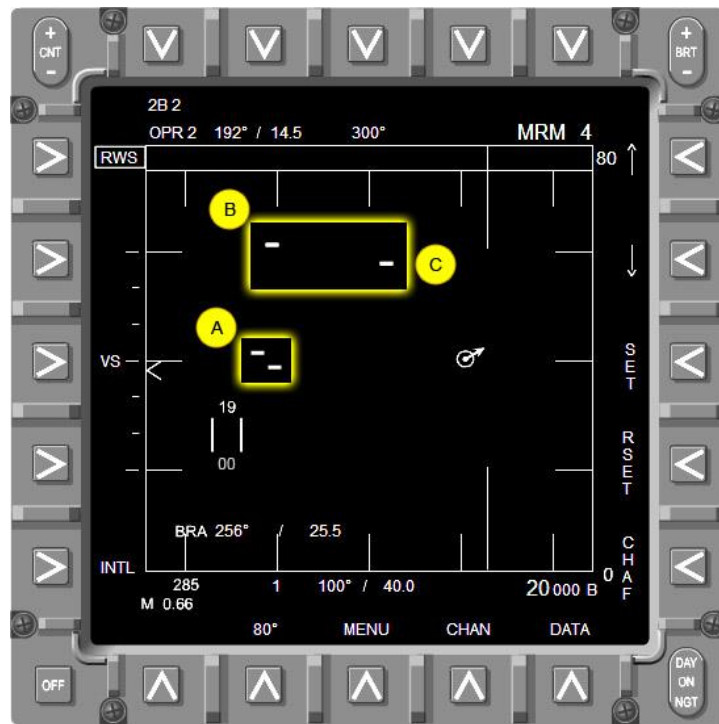


Figure 14-7 Group Definition

A group is defined as two or more contacts within 5 NM range, 5 NM azimuth and within 5,000 feet in elevation. The rule of 60 or the cursor BRAA information can be used to determine azimuth requirements for a group. In the Figure 14-7, group A has two contacts at 40 NM. Those contacts are less than 5 NM in range and are 4 degrees in azimuth. At 40 NM 4 degrees is about 3 NM in azimuth, so the contacts in A are labeled a “group.” Groups B and C are 35 degree apart in azimuth at 60 NM. By the rule of 60, they are 35 miles apart making them separate groups. The radar picture in Figure 14-7 contains three groups and could be described in a similar manner as the picture call from the example above.

As a review of AIC communication control and formats:

- Broadcast control will identify groups by their location in a way relevant to all listeners and is normally anchored to bullseye.
- Tactical control adds a target aspect to each group, relevant to the fighter under tactical control with the addition of the picture label and group names.
- Format anchored to bullseye references group positions from a designated geographical reference.
- The BRAA format anchors group locations to the fighter’s nose.

Communication formats and priority talkers are shown in the Figure 14-8:

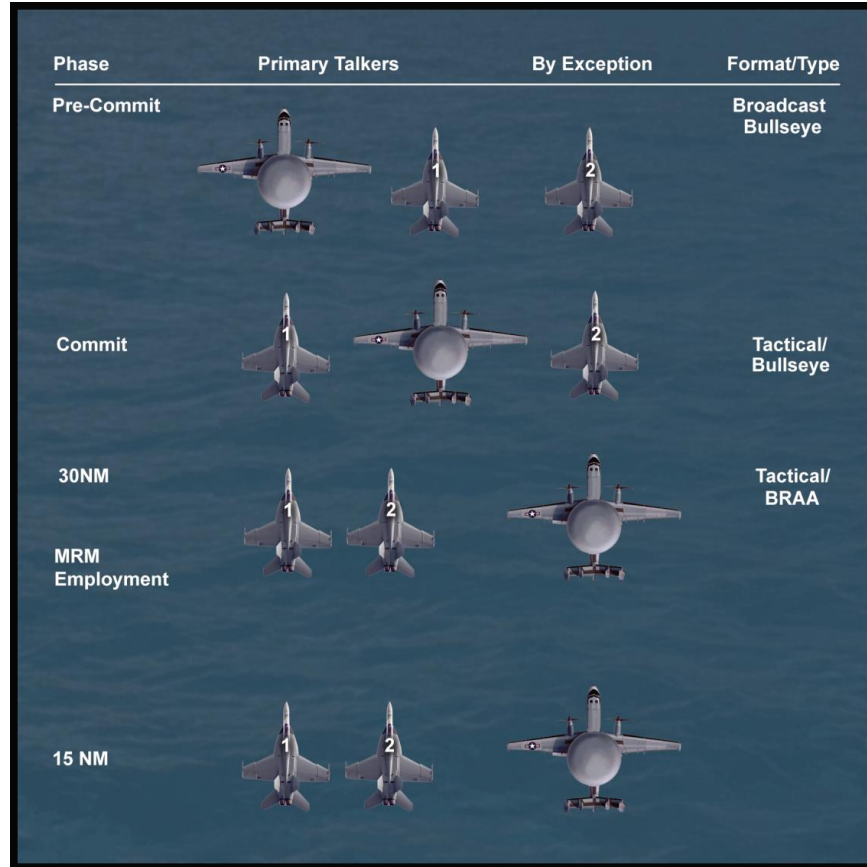


Figure 14-8 Comm Format and Priority Talkers to MRM Employment

1408. COMMIT

1. Commit Criteria

Commit criteria for 2v2 or 2vX intercepts will be slightly different:

- SRA – commit when the single group is flank or less aspect but no closer than 35 NM
- 2vX – commit when any group in the picture meets briefed commit criteria. At a minimum this will be a group with flank or less aspect but no closer than 35 NM

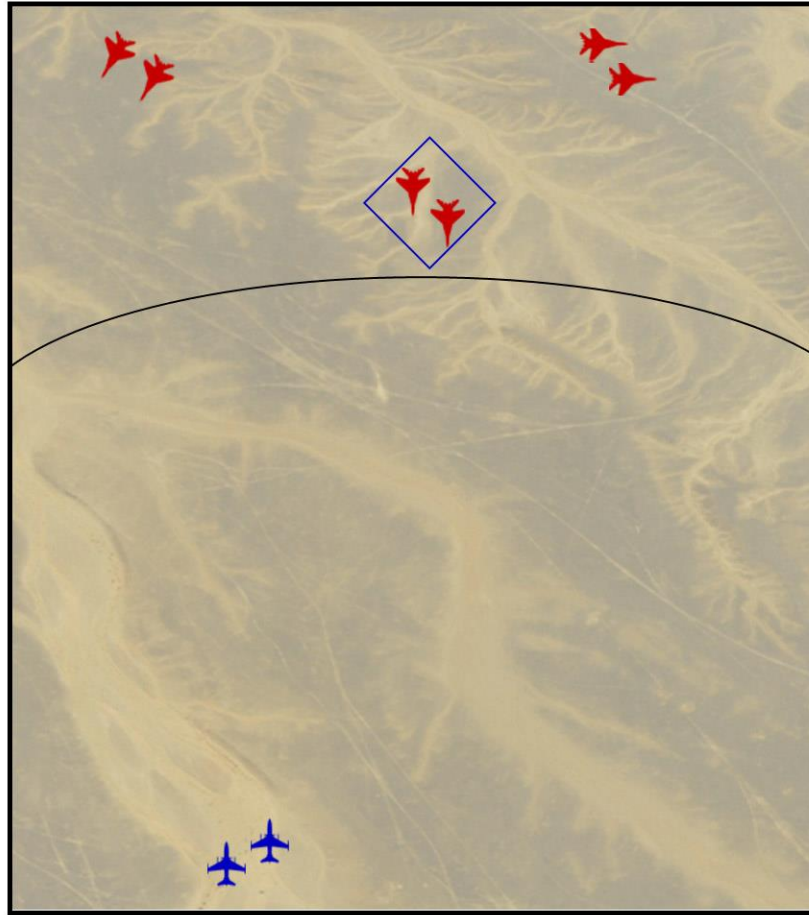


Figure 14-9 Group Meeting Commit Criteria

The Lead SNFO must realize that the group that meets commit criteria may not be in the lead's AOR. Wing should be prepared to recommend commit if Lead does not recognize that a group in the low AOR meets commit criteria (Figure 14-9).

2. Formation at Commit

At the commit, the fighter's formation will transition to combat spread. The fighters should set geometry by placing the group which meets commit criteria on the nose initially until radar detection occurs and then set the appropriate geometry.

3. Sensor Employment at Commit

The fighters should continue to sanitize in their AOR using the sanitization set, mated at 25 NM.

4. Commit Communications

Communication for the commit should use the section's callsign to avoid any confusion. At commit, AIC will automatically transition from broadcast to tactical control. For example:

Lead SNFO (PRI) - *“Rage, commit”*

(AUX) - *“Check Tapes, Master Arm”*

Figure 14-10 shows formation, sensor employment, and communications at the commit.

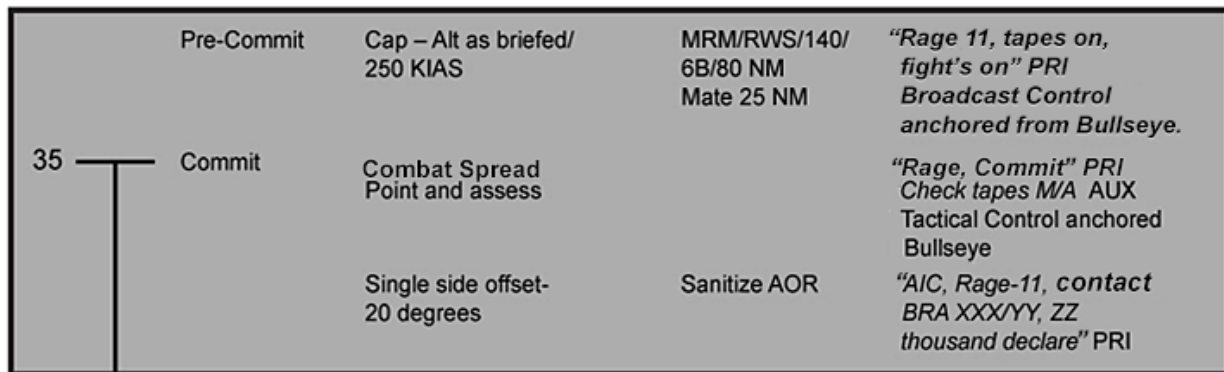


Figure 14-10 Commit Actions Form, Sensors and Comm

5. When should Wing speak up?

Outside of 30 NM, Comm priority is Lead, AIC, then Wing by exception. Wing is obligated to speak whenever the following occurs:

- Any of the four assumptions violated.
- Wing sees a group or possesses information not being talked about by AIC or Lead.
- As Wing and outside of 30 NM, speak only when your comm enhances overall SA.

As Wing, when transmitting on the control frequency (PRI), use clear, concise, SA building comm with appropriate brevity. One bad call can trash SA for all who hear it. Wing's comm must be as clear and concise as Lead's and/or AIC.

Wingmen should not hesitate to contribute SA building information, but should ensure they broadcast information that is not already known by the other blue players.

Inside 30 NM, comm priority is Lead, Wing, then AIC by exception. Wingmen should understand when they are required to speak, what information to pass and when it is appropriate in accordance with the briefed plan and this FTI. Examples of when Wing could speak up include (but are not limited to):

- a. “*Ripper 22, clean (group name)*” is required if AIC is calling a group in Wing’s AOR that is not seen by Wing (no radar SA to a group in your AOR post-Tactical Range call).
- b. Radar SA to a group inside 25 NM at any altitude when a meld hasn’t been called.
- c. Any SA to move the section to maintain timeline in order to survive (abort, defense, etc.).

1409. LABELING AND NAMING

Once the fighter’s commit, AIC will label the picture and include group names if possible.

1. Labeling

The picture will be labeled only if it fits one of the pre-briefed labels agreed upon between the fighters and AIC (shown in Figure14-11).

In SRA stage, the fighters will encounter only a single group. Single group labels and definitions include:

- a. Single group
 - i. Two contacts (strength 2)
 - ii. This is both the picture’s label and the group’s name

Possible fill-ins:

- b. “Heavy” - more than strength 2
- c. “Stack” - contacts have significant altitude difference (> 5K feet)

In 2vX, the fighters can expect to encounter any number of groups in any configuration.

2. Naming

Groups will be named appropriately based on their location. When the picture fits a prebriefed label, the groups will be named accordingly. In SRAs, only single groups are presented and named “Single Group.” A single group scenario is the only time the label and name will be the same. If the single group contains more than two contacts, AIC will add the modifier “heavy,” if AIC has that SA. Group names for labeled pictures are shown in Figure 14-11.

Two groups in echelon will be labeled as either range or azimuth first, and then echelon will be used as a descriptive modifier. For example: “Two groups azimuth 20, echelon northeast” describes two groups with 20 NM azimuth separation with the east group offset in range to the

northeast (Figure 14-11, middle right). The only configuration of two groups that does not fit into one of these pre-briefed labels is “two groups stack” (not shown) in which the two groups are directly above/below each other in altitude.

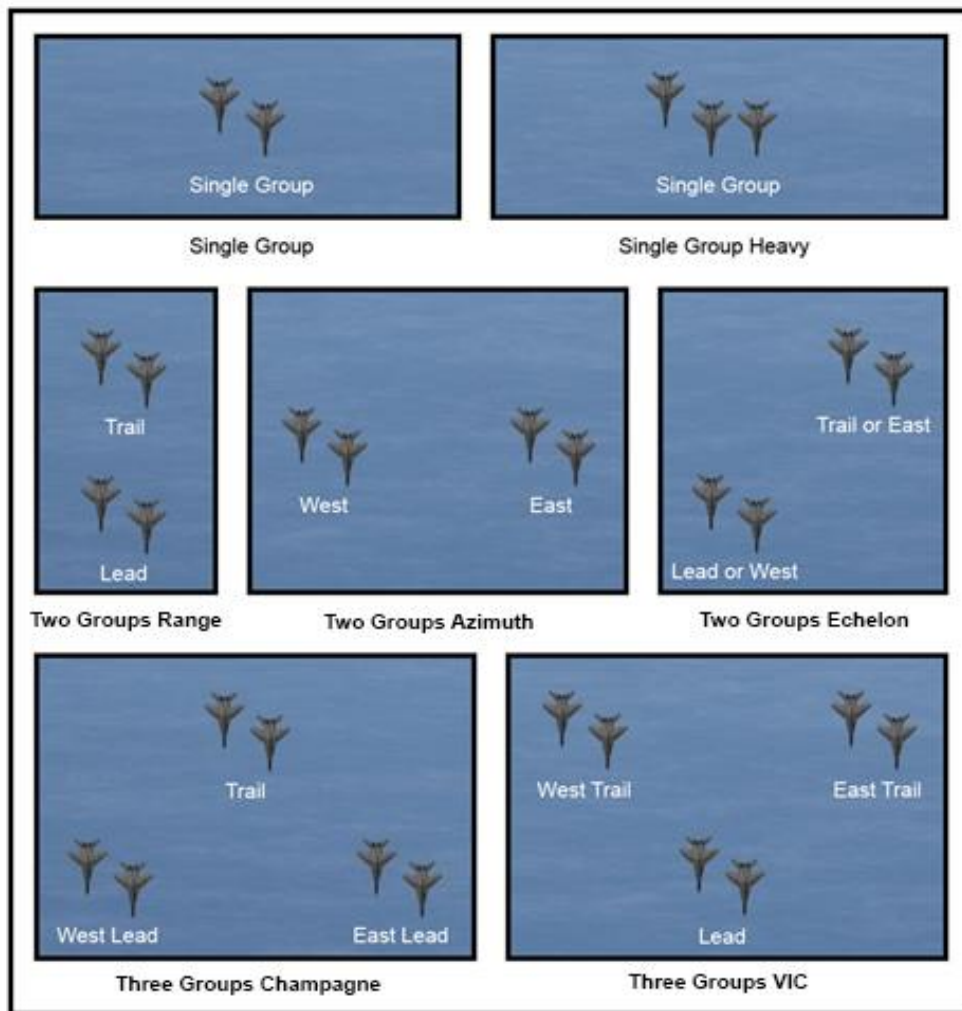


Figure 14-11 Picture Labels and Group Names

3. Labeling and Naming Communications

Unless in extremis or without any AIC available, the fighters will not label the picture or name the groups. Lead fighter can recommend a label and group names or ask for a new picture if the tactical situation dictates. AIC will communicate the label and names in tactical control anchored to bullseye as shown in the following examples:

4. Single Group

Single group scenarios are the only case where the picture label and name are the same.

- a. AIC - *“Sabre, single group, Vegas 270, 35, 25 thousand, flank West, hostile”*
- b. AIC - *“Sabre, single group, Vegas 270, 35, 25 thousand, hot, heavy, hostile”*

5. Two Groups Range or Azimuth

In two group pictures, the picture can be defined by the location of the first group identified; AIC will not provide a bullseye location to the second group in a range presentation. In other words, *“two groups range 30”* implies that the lead group will be identified and the trail group is 30 NM behind the lead group. AIC is required to transmit the altitude, if known, aspect and declaration to the other group. For groups in azimuth, AIC will anchor both groups. For example:

- a. AIC - *“Sabre, two groups range 30. Lead group Vegas 250, 40, 26 thousand, hot, hostile. Trail group 30 thousand, flank southwest, hostile”*
- b. AIC - *“Sabre, two groups azimuth 30. West group Vegas 140, 35, 30 thousand, hot, hostile. East group Vegas 152, 47, 20 thousand, beam east, hostile”*

6. Two Groups Echelon (+ sub-cardinal direction)

Two groups in an echelon will be named according to the range or azimuth convention, depending on the depth or width of the echelon. For example, an echelon with more range than azimuth will be labeled as *“two groups range”* with the modifier *“echelon”* and a subcardinal direction. For example:

- a. AIC - *“Sabre, two groups, range 30, echelon northeast. Lead group Vegas 180, 40, 19 thousand hot hostile; Trail group 22 thousand, flank southwest, hostile.”*
- b. AIC - *“Sabre, two groups, azimuth 30, echelon northeast. West group Vegas 205, 40, 19 thousand hot, hostile; East group Vegas 190, 35, 15 thousand, flank southeast, hostile.”*

7. Naming with More than Two Groups

There are too many possible arrangements for three or more groups of bandits to all be covered here. AIC will label the picture and name the groups in a logical fashion using the basic guidelines presented here. The SNFO should be able to recognize the picture by being familiar with the labels and names presented here.

1410. TARGETING

The presence of multiple groups does not mean all groups will be factor groups, or have any impact on the tactical picture. The true intent for any enemy formation is often unknown. The fighters may receive information from *“off board”* sources that aid in determining which groups of bandits are *“fighters”* and which are *“strikers.”* As a general planning principle:

- On OCA sweep missions:
 - i. Commit will be “liberal”
 - ii. “Sweep” mentality; kill anything flying declared hostile
 - iii. More tactical employment options
- On OCA escort missions:
 - i. Escort’s purpose is to defend the strikers
 - ii. Groups not a threat to strikers are not targeted
 - iii. Limited tactical employment options
- DCA point or area defense:
 - i. Defend friendly assets
 - ii. Kill only threat groups
 - iii. Priority target enemy strikers
 - iv. Very limited tactical employment options

1. Targeting Priorities

The next phase of the intercept after labeling and naming is targeting. The fighters will target the group that is the highest threat in the tactical picture. The highest threat group is called the factor group.

A 35 NM NLT commit provides time to identify the threat group based on the following criteria:

- a. Elevation
- b. Range
- c. Azimuth

Other, mission dependent considerations:

- d. Bogey over bandit or hostile (DCA)
- e. Strikers over escort or screen (DCA)

The type of fighter mission will determine what criteria will be used to determine the factor group. On a DCA mission, for example, the enemy's strike aircraft will have a higher priority than their escort or screen fighters.

At VT-86, fighters should determine the highest factor group based on which groups is the:

- a. Closest
- b. Hottest
- c. Highest
- d. Known Threat

It should be noted that this is not necessarily a list of priorities and that each of these factors need to be taken into consideration real-time to determine the group that is the largest threat to the fighters.

2. Flow Range

Flow range is the minimum distance needed between two groups that allow the fighters to perform all timeline tasks versus the targeted group, take that intercept to a logical conclusion and then execute all timeline tasks against the follow on group. Usually this range is not less than meld range. Other considerations for flow range include, but are not limited to:

- a. Allows for destruction/merge with targeted group
- b. Immediate meld to far group for BVR weapon employment
- c. Time-to-kill (TTK) considerations with the first group using WVR ACM (always strive to minimize TTK)
- d. Acquire and meld into the follow-on group on timeline

At VT-86 flow range will be 30 NM.

- a. Flow range is not a consideration for:
 - i. Single group
 - ii. Single group heavy
 - iii. Single group stack

- b. Flow range is always a consideration for:
 - i. Two groups range
 - ii. Two groups azimuth
 - iii. Two groups with an echelon component

3. Formation and Geometry at Targeting

The fighters will remain in defensive combat spread throughout the intercept. The fighters must take into account their area of responsibility (working area in training), hostile surface threats, engagement zones and follow-on group flow when determining which direction and how much to offset. All things being equal, the fighters should establish a minimum of 20 degree single side offset by placing the targeted group at least 20 ATA to affect intercept geometry and create a positional advantage. SSO priorities are:

- a. Away from untargeted groups
- b. Shortest direction

The Lead SNFO must realize that the targeted group may not be in the high AOR and should be prepared to not have radar contact with the targeted group until meld.

4. Groups in Azimuth or with Echelon Component

In a multi-group environment, the fighters should choose an offset that provides the best opportunity to avoid untargeted groups and possibly turn any echelon component into a range problem.

The fighters should utilize an aggressive SSO 40-50 ATA with the previously discussed priorities. By using an aggressive SSO and a crank away from untargeted groups, the fighters are able to manage the WEZ to follow-on groups and influence flow to the max extent possible. SNFOs must understand that while they may flow from group to group during range problems, they should always abort with groups in azimuth at decision range. To determine if the picture is flowable, ask AIC for separation call. If AIC re-labels the picture as range greater than flow range, the fighters can update their flow decision.

The reason for this is simple: The timeline is designed for sequential, not simultaneous, engagements. In future training, when more than one fighter section is present or when the threat level dictates, the fighters may split into two sections or target as singles. At VT-86, the section will always stay together and target one group at a time.

5. Targeting Sensors

The fighters should continue to sanitize using the sanitization set. Fighters may designate an appropriate L&S at any time radar SA is available, and should attempt to target as close to tactical range as possible but NLT 30 nm.

6. Targeting Communications

Once the fighters have determined which group in the picture to target and at tactical range, this is communicated to AIC using the following call:

- a. In SRA - Lead SNFO (PRI) - *“Rage, target single group”*
- b. In 2vX - Lead SNFO (PRI) - *“Rage, target lead group”*

In 2vX it is implied that AIC will monitor the untargeted groups and provide SA if the picture drastically changes.

7. Stack Targeting

Whenever a stack is called or recognized, the Lead SNFO must assign or re-issue targeting prior to meld based on High and Low contacts. This recognition is especially important coming from the wing aircraft, who is sanitizing the low AOR. Outside of 30 NM, the fighter with SA to the uncalled contact should give a separate contact call to AIC for correlation.

Wing SNFO (PRI) – *“Sabre, Ripper 12, contact BRAA 110, 32, ten thousand, declare”*

If wing has achieved correlation to the low contact with previous information passed via a picture or contact call, and are inside 30 NM, they can give a shortened call to lead on AUX.

Wing SNFO (AUX) – *“Ripper 12, single group stack”*

Targeting should match each aircraft's AOR. Each fighter will then adopt an owning mindset. Meld comm should also be modified so as to not include altitude. If the meld comm includes altitude, it shall be considered a directive call and the out of AOR aircraft should meld based on the call. Stacks should not be confused with an elevation sort. The fighters should consider to group a stack when two contacts are outside the same block.

Lead SNFO (PRI) - *“Ripper 11, Target High Contact, Ripper 12, Target Low Contact”*

Lead SNFO (AUX) - *“Ripper, meld, BRAA 110, 25”*

1411. 30 NM TACTICAL RANGE AND FOLLOW-ON FLOW

At 30 NM, communications will transition to tactical control in the BRAA format for the remainder of the intercept.

The 30 NM tactical range call is a reminder to all players that meld is in 30 seconds and serves to update everyone's timeline SA.

1. 30 NM Range Formation and Geometry

The fighters will remain in combat spread, and should be in a climb or descent to get into a block that will allow for the best possible merge. If the fighters recognize that the initial geometry was incorrect, the Lead can direct, and Wing can recommend, a geometry change up to this range. After 30 NM the fighters focus is on employing weapons on the targeted group and not correcting intercept geometry.

2. 30 NM Range Sensors

At 30 NM, the fighters are 5 NM or 30 seconds away from meld. They have two more frames with which to sanitize their AOR and correlate any contacts found. The fighters must remain patient and let the radar sanitize their AOR so as not to miss any undetected groups.

3. 30 NM Communications

At 30 NM, the fighter with the targeted group in AOR will call "*Callsign, group name, 30 miles.*" For example, in SRA with a single group at 13,000 (the low AOR):

Wing SNFO (PRI) - "*Rage 12, single group, 30 miles*"

Or, in 2vX in a range presentation with the targeted group at 28,000

Lead SNFO (PRI) - "*Rage 11, lead group, 30 miles*"

1412. MELD AND SORT

Melding is a directive call to cease sanitization and focus both fighters' radar into the targeted group for sorting and weapons employment. Melding was introduced in Advanced 1v1 AWI and will be revisited here in the context of section employment. Sorting is determining which contact in the targeted group that each fighter will employ weapons against.

Melding and sorting is the foundation for air-to-air success. Fighters will rarely employ as a single ship. Many threat nations use formations that are large in number and closely spaced. Often an unobserved bandit entry into the visual arena is directly traceable to:

- Poor/lazy/inappropriate sanitization at range

- Poor meld mechanics
- Not having a sort
- Poor visual lookout

Disciplined radar employment is key to BVR air-to-air success. A WSO's reputation is often built on their ability to execute disciplined intercept procedures in a timely and accurate fashion.

1. **Meld**

The melding and sorting procedures should commence no earlier than 25 NM and are executed with the same discipline and procedural knowledge as an immediate action "boldface" procedure.

2. **Meld Formation**

The fighters remain in combat spread during meld and sort

3. **Meld Sensor Employment**

Meld mechanics were introduced in Advanced 1v1. There is now one additional step at the end of the meld that all fighters must perform. The meld procedure is now:

- a. Range to 40 NM
- b. Reduce to 80 degree azimuth
- c. Center scan volume on meld call with full action trigger
- d. Place cursor 1-2 NM in front of meld called range
- e. Center elevation scan volume to bracket meld called altitude
- f. Command STT on contact
- g. Enter TWS via the TWS push button at PB10

SNFOs are encouraged to use HOTAS for bump functionality to adjust range and azimuth. While this is taught here as "technique," it will be the preferred method in the F/A-18. Execution of the correct procedure is more important than the technique used.

The purpose in entering STT prior to TWS in sort mechanics is to force the radar into AUTO scan centering. SNFOs should not deviate from the meld mechanics by going directly to TWS from RWS at PB5. The meld procedure ensures the TWS scan volume is centered in azimuth and altitude on the targeted group.

4. **Meld Communication**

The meld communication will be in the BRAA format.

The purpose of meld is to bring all the fighter radars into the targeted group. Since the fighters are sanitizing the same amount of azimuth volume, but different altitudes, the most important part of the meld call is altitude.

The fighter with targeted group and radar contact in AOR should make the meld call. Flight callsign is used, not individual aircraft callsign. The meld call format is “*Callsign, meld, BRAA XXX, YY, ZZ Thousand.*”

For example, with the group at 25 thousand feet in SRA:

Lead SNFO (AUX) - “*Rage, meld, BRAA 320, 25, 25 thousand*”

For contact at 15 thousand in Wing’s AOR:

Wing SNFO (AUX) - “*Rage, meld, BRAA 320, 25, 15 thousand*”

As a reminder, when melding into a group that is a stack, the altitude should be removed from the meld call.

5. **Sort**

Meld is followed immediately by the sort. When sorting, the fighters will designate the contact within the targeted group against which they will employ weapons.

6. **Sort priorities, assumptions and assignments**

The sort “contract” within the section is briefed using the sort priorities in the order of:

- a. Azimuth
- b. Range
- c. Elevation

When sorting, the following assumptions are made:

- a. Group is strength 2
- b. Group formation is sortable in azimuth
- c. No comm is required if assumptions are met

For prioritizing the sort, the following assignments are made:

- a. Lead will sort
 - i. Left
 - ii. Lead
 - iii. High
- b. Wing will sort
 - i. Right
 - ii. Trail
 - iii. Low

7. Sort Formation

The fighters remain in combat spread while sorting.

8. Sort Sensor Employment

The most important part of sensor employment is sorting the correct contact. The briefed sorting gameplan is known as the “sort contract.” The standard sort contract used at VT-86 is shown below in Figure 14-12.

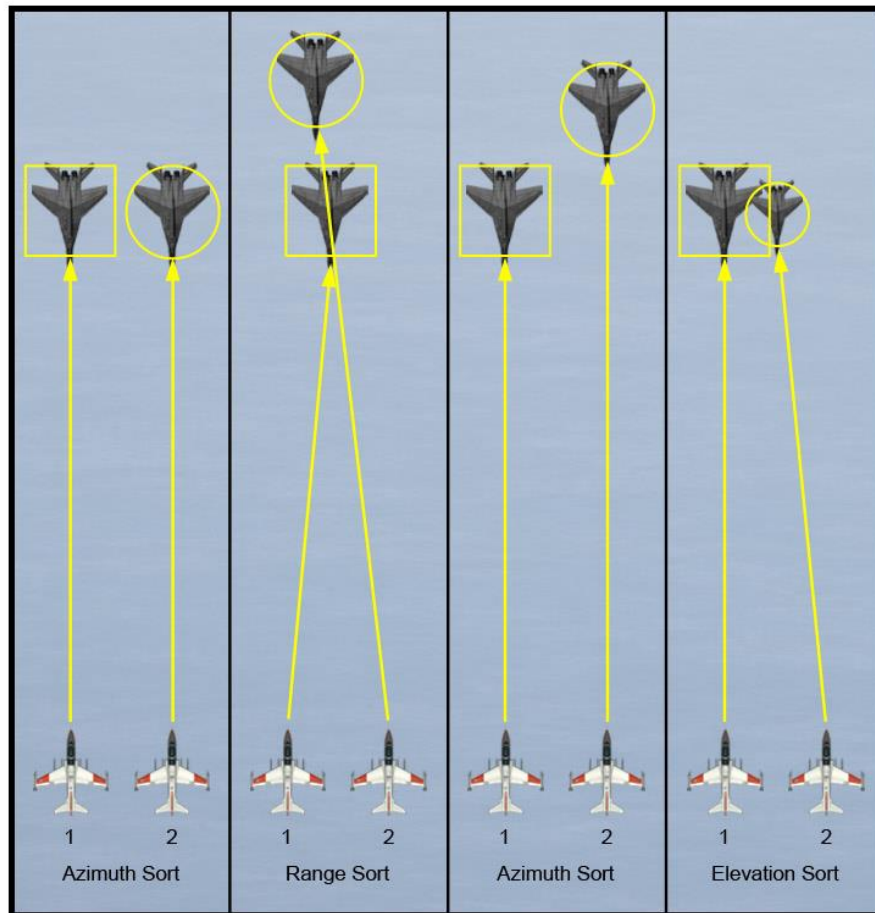


Figure 14-12 Sorting Plan

If all assumptions are met, the section will use an azimuth sort with Lead sorting left and Wing sorting right. If a “B-sweep” can fit between the two contacts then the group is sortable in azimuth. Notice that in panel 3 in Figure 14-12 that a B sweep can fit between the contacts so the fighters can prioritize sorting the contacts in azimuth.

9. Sort Mechanics

The sort procedure is:

- a. Lead
 - i. Make sort contract the L&S
 - ii. Designate the other contact as a DT2
- b. Wing
 - i. Make sort contract the L&S

- ii. Command STT on sort contract

10. Post Sort Radar

Figure 14-13 shows a Wing sort on the left screen (STT), and a Lead sort on the right screen (TWS AUTO).

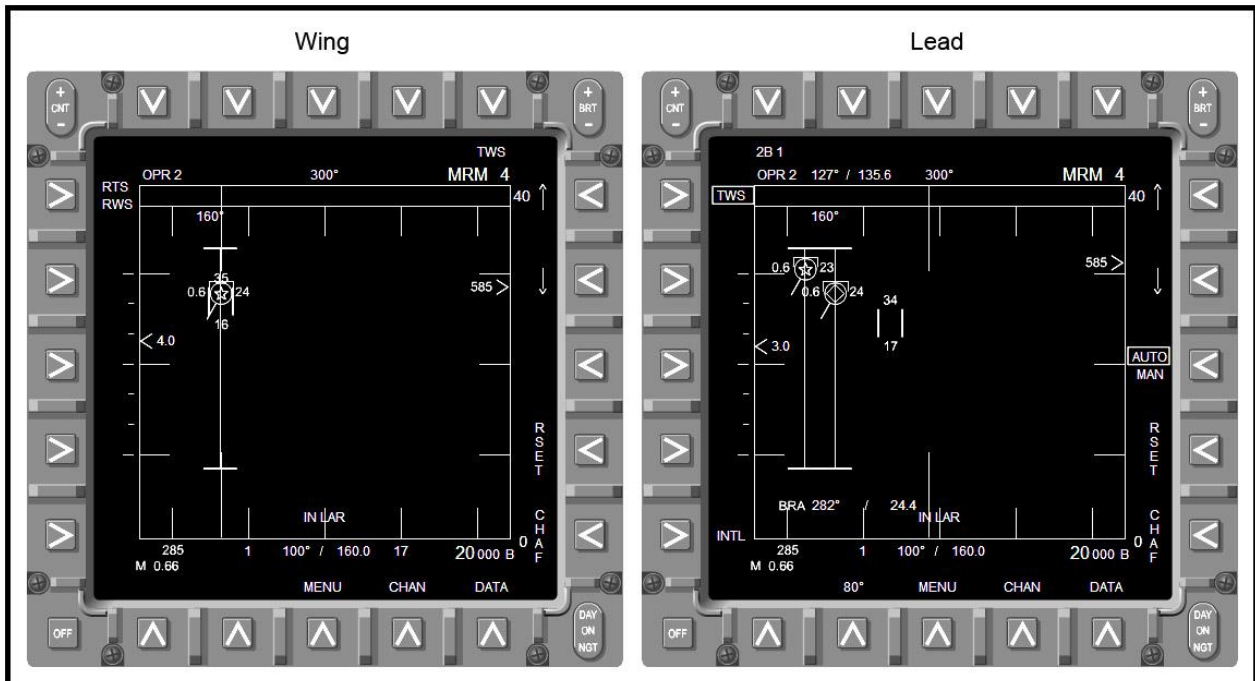


Figure 14-13 Post Sort Radars for Wing (Left) and Lead (Right)

11. Groups Not Sortable in Azimuth

A group is considered not sortable in azimuth if a B-sweep will not fit between the contacts. This is shown in Figure 14-14. This situation should be recognized by both fighters.

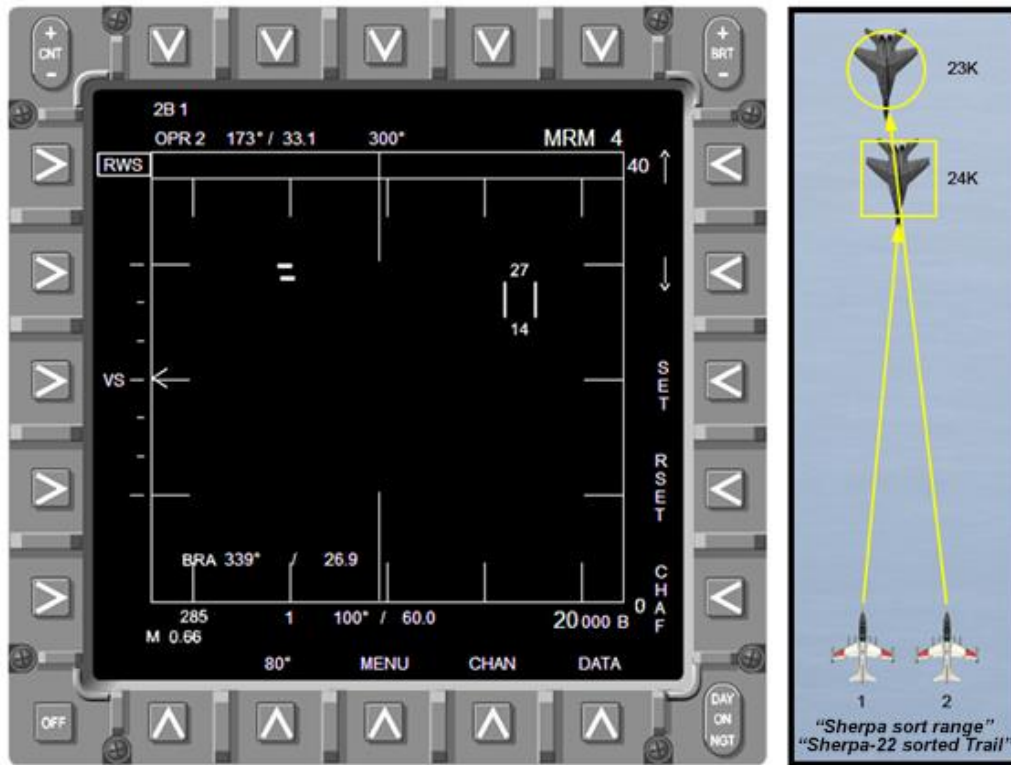


Figure 14-14 Groups Not Sortable in Azimuth are Sorted in Range

If this case is recognized by Lead they will be directive. Lead directs a range sort with the following call:

Lead SNFO (AUX) – *“Range, Sort Range”*

If recognized by Wing, or if directed by Lead, the wingman will be descriptive. The descriptive call sounds like:

Wing SNFO (AUX) – *“Range 12, sorted trail man”*

If the contacts are not able to be sorted in azimuth or in range but there is an altitude differential, the contacts can be sorted by elevation. SNFOs should be careful to not confuse an elevation sort with stack targeting; an elevation sort would be contacts within the same block, whereas a stack would be contacts in different blocks.

12. Single Contact in a Group

A group with a single contact will be identified by the fighter with SA on AUX as *“Call Sign, group name, single only, XX thousand,”* with the other fighter responding on AUX, *“Callsign, same.”* If both fighters agree, Wing will employ on the single contact from STT and Lead will hold the shot and sanitize around the contact from TWS.

If the section disagrees, the fighter with two contacts will own the group and employ on both out of TWS. The section should follow the mantra “He who sees 2 shoots 2, he who sees 1 shoots none.”

13. “Sorted” vs. “Locked”

If a fighter has an STT but was unsure of sort, then the fighter is “locked, XX thousand.” The other fighter will attempt to sort around this call using altitude as the discriminant. Disciplined sort mechanics will prevent a “locked” status.

14. More Than Two Contacts In A Group

If a group contains more than two contacts there needs to be a discussion on AUX. The first player to recognize should make the call “*Callsign, Group Name, Strength X.*” If both fighters agree, employ based on the employment tree in Figure 14-15 or as directed by Lead.

The gameplan for any 2v1 strength disagreement is “he who sees two, shoots two; he who sees one shoots none.” The fighter who only sees one should hold their shot, and the fighter that sees two will employ out of TWS on both contacts. For any other strength disagreements (3v2, 3v1), the fighters should employ via their respective line on the employment tree that corresponds to the strength composition that they see.

15. Directive Sort

If the assumptions for an azimuth sort are not met, then Lead will direct the sort as a range or elevation sort as depicted in Figure 14-12.

16. Meld and Sort Summary

A successful intercept in either SRA or 2vX depends on disciplined execution of meld and sort mechanics prior to weapon employment. It is the SNFO’s primary responsibility during the intercept to ensure that the correct sort is made, every time. The entire intercept can fall apart inside 20 NM without a good sort. The SNFO should never give up trying to get a sort or ensuring that the section has the maximum amount of SA prior to weapon employment.

1413. SECTION WEAPONS EMPLOYMENT AND CRANK

Weapon employment is the culmination of all intercept tasks to 20 NM. At weapon employment, the fighters will shoot and crank as a section. Recall from previous chapters that the primary consideration for a valid MRM employment is an “IN LAR” condition at launch. The fighter crew must continue to assess shot quality throughout missile time of flight. The fighters must employ IN LAR and on timeline.

The employment tree (Figure 14-15) graphically depicts the sorting and employment plan for bandit formations from a single aircraft to a three-ship. It assumes a group sortable in azimuth. If the bandits are in a formation other than azimuth, Lead needs to be directive on how to sort the

group prior to shots. An agreed upon single bandit will be engaged by Wing from STT and Lead will hold shots and remain in TWS looking for additional contacts. For a two-ship, Lead will employ from TWS on the left bandit (or per briefed sort), Wing will employ from STT on the right bandit (or per briefed sort). In a group with strength 3, Lead will employ from TWS on the center and left contacts while Wing employs on the right contact from STT.

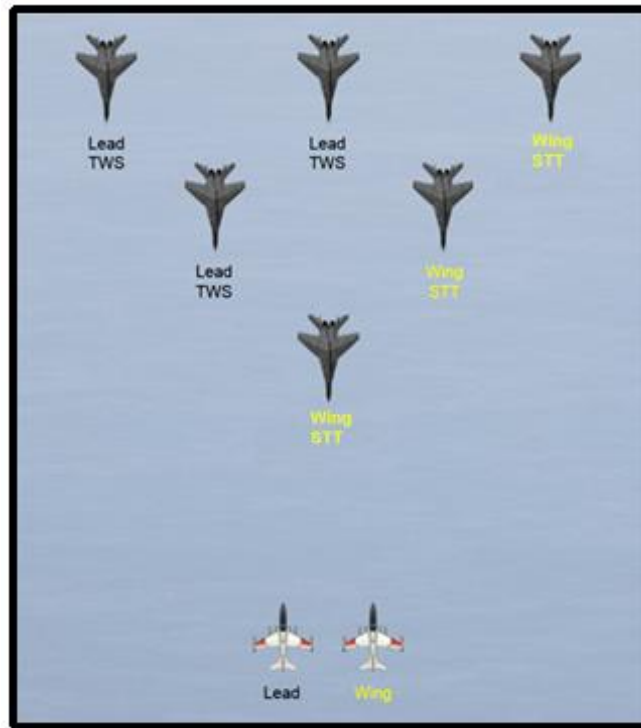


Figure 14-15 Employment "Tree"

1. Section Employment Considerations

Assuming a correct sort, the following will happen:

- a. Wing will employ a single missile from STT on sort contract.
- b. Lead will employ from TWS:
 - i. Sort contract
 - ii. Up to two missiles, one on each of the left contacts in group with strength 3

If the section is locked: Wing will employ and Lead will attempt to complete the sort around what Wing has locked.

In the event a group strength 3 is encountered, Lead will employ one missile on each of the left most contacts from TWS while Wing will employ from STT on the right bandit. This is summarized in the employment “tree” graphic in Figure 14-15.

2. Post-Employment WEZ Management – The Crank

Immediately following employment, but no later than 20NM, the fighters will turn to manage the enemy WEZ as introduced in 1v1 AWI.

3. Employment/Crank Formation

Immediately following employment, the fighters will maneuver to manage the hostile WEZ. This maneuver is called a crank. The fighters will crank whether or not they employ (such as in the case of a group declared bogey).

In SRA, the fighters will crank to place their sort contract at 50 degrees ATA in the shortest direction and adjust to maintain visual mutual support.

In 2vX the fighters will crank in the most favorable direction based on the following priorities:

- a. Away from untargeted groups
- b. In the shortest direction

Deconfliction is the responsibility of the Wing pilot. The wingman may become acute due to the crank geometry.

4. Employment Sensor Consideration

Per Figure 14-15, Lead will employ out of TWS. Wing will employ out of STT. Lead will remain in TWS and monitor for additional contacts to appear which is known as a “late breakout.” The fighters should evaluate their shots through time of flight to ensure they remain valid and determine if follow-on BVR employment is warranted. If the fighters see a LOST missile status, then the shot is likely trashed and follow-on employment should be considered. If a late breakout is encountered Lead needs to evaluate if an MRM will be able to reach timeout prior to NLT defense or NLT abort. If so, a missile should be employed on that contact and if not, the fighters should plan skate.

5. Employment and Crank Communications

Comm brevity for MRM employment will follow the BVR employment standard on PRI:

- a. “Fox 3” is used for the employment of one missile
- b. “Fox-3, two ship” means employment has occurred against more than one trackfile

- c. “*Callsign, second Fox-3*” indicates follow-on employment against the same trackfile with the previous missile still in flight (i.e., prior to timeout)

Following the employment call, the Lead fighter should call for the crank on AUX:

Lead SNFO (PRI) - “*Rage 11, Fox-3*”

Wing SNFO (PRI) - “*Rage 12, Fox-3*”

Lead SNFO (AUX) - “*Rage, crank Left (or Right)*”

If, due to geometry, Wing is at risk of losing radar contact, Wing can call on AUX “*Rage 12, gimballs.*” Lead should immediately adjust geometry to compensate while maintaining geometry and section visual support.

At missile timeout, the fighters should make the appropriate timeout call. If trashed, the fighters should make a trashed call. The going in game plan should be that Wing will make the timeout call first so that Lead can direct the section into the defense and then Lead can make their timeout call.

1414. DECISION RANGE AND TRANSITION TO WVR

After weapons have been employed and fighters are established in the crank, Lead must decide what the next action will be. This decision is based on how offensive the section is and how much of an advantage the fighters will have at the merge. If the fighters will have the advantage or the merge will be neutral they can consider themselves winning. If the section would find themselves defensive at the merge the fighters should think of themselves as losing.

With multiple groups, AIC should automatically provide a separation call to the untargeted group once the fighters have employed on the targeted group. If AIC does not automatically give the separation call, Lead must request it prior to decision range.

Lead SNFO (PRI) - “*Sabre, Ripper 11, separation trail group*”

AIC (PRI) - “*Sabre, trail group separation 33, 21 thousand, hot, hostile*”

1. Winning and Losing

As a refresher from 1v1 AWI, you are winning if the section:

- a. Sorted, and
- b. Employed/cranked on time, and
- c. Naked

Or

- d. Sorted, and
- e. Employed/cranked on time, and
- f. Spiked

Or

- g. Sorted, and
- h. Employed/cranked late, and
- i. Naked or TA >30 degrees

You are losing if:

- a. Locked/unresolved sort, or
- b. No weapon employment and Spiked

Or

- c. No contact by 15 NM, and
- d. Spiked

2. **Banzai**

“Banzai” is a directive call to timeout missiles currently in flight and fight to the merge; regardless of the affect those missiles have on the hostile forces. At VT-86, the section will banzai whenever they are winning. This is a briefed assumption and communication is required only when there may be confusion about whether or not the section will banzai. The call on AUX is “*Callsign, banzai.*”

3. **Section Execution of Defense and Banzai**

If either of the fighters are spiked, they should defend as a section post every missile timeout but no later than 8 NM. Although the defense is an individual effort, the section will both turn to place the spike azimuth in the beam and dispense chaff. Each fighter should state their specific defense headings over ICS. While in the beam, the fighters should select the SRM set (RWS/140/6B/20NM), place the cursors 8-10 NM and bracket bandits last known altitude in preparation of turning in.

Once naked, the fighters should turn in approximately 110-130 degrees from the defense heading to place the threat on the nose and prepare for SRM employment and the merge (Figure 14-16).

If at any time the bandits are assessed to be inside of 5 NM the fighters should select WACQ with SRM selected.

4. Communications During A Banzai

Banzai communications are initiated by the fighter that is spiked and will flow as follows:

Wing SNFO (AUX) - *"Rage 12, spiked"*

(PRI) - *"Rage 12, timeout, Trail man, Single group"*

Lead SNFO (PRI) - *"Rage defend east"*

(ICS) - *"Heading 075"*

Wing SNFO (ICS) - *"Heading 077"*

Lead SNFO (PRI) - *"Rage 11, timeout, Lead man, Single group"*

When Wing goes naked:

Wing SNFO(AUX) - *"Rage 12, naked"*

Lead SNFO (PRI) - *"Rage in left (right)"*

(AUX) - *"Heading XXX "*

If lead is spiked, the spiked call is not required. Lead will direct the defense and turn in as appropriate.



Figure 14-16 Banzai Geometry and Comm

5. Skate and Abort

“Plan skate” is a descriptive call to timeout missiles currently in flight and go out. “Abort” is a directive call to execute the abort mechanics. The abort is executed from the crank. This is shown in Figure 14-17.

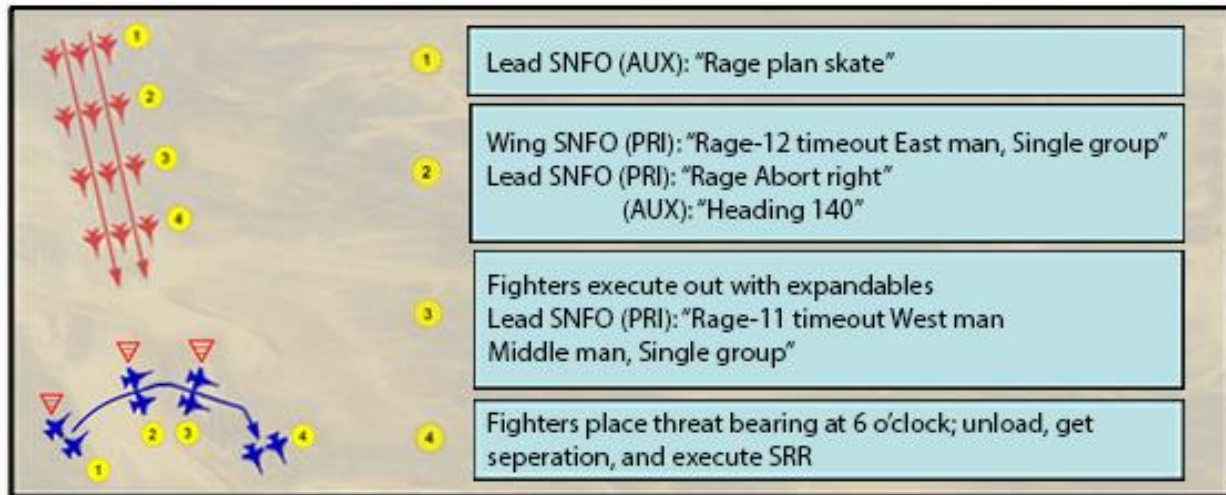


Figure 14-17 Abort Geometry and Comm

Skate communications are initiated by Wing when timeout occurs and will flow as follows:

Wing SNFO (PRI) - *"Rage 12, timeout, East man, Single group"*

Lead SNFO (PRI) - *"Rage, Abort right"*

(AUX) - *"Heading YYY"*

Lead SNFO (PRI) - *"Rage 11, timeout, West man, Middle man, single group"*

6. Minimum Defense Range

Minimum Defense Range (MDR) is the range at which the skate or banzai decision is executed, not the range at which the decision is made. In general, in SRAs the fighters should:

- Banzai against a single group when they are winning
- Abort against a single group when they are losing
- Either option is a defensible decision against a single, strength 3 group with two kills at timeout if the fighters are otherwise winning. Be prepared to defend the decision that is made.

In 2vX the fighters should:

- a. Banzai when winning against the targeted group with separation in range of 30 NM or greater from the closest, non-targeted group AND AIC has SA to the follow-on group.
- b. Skate in all other circumstances
 - i. If winning against targeted group and no SA to follow-on group(s)
 - ii. If losing against targeted group

The decision to skate or banzai may be defensible, even if executed against the guidance here. However, the SNFO choosing to deviate from this guidance must be prepared to defend the decision and discuss the consequences.

1415. SECTION SHORT-RANGE RADAR EMPLOYMENT

As with BVR radar employment, the section must coordinate short-range radar (SRR) mechanics in order to maximize their effectiveness. Section SRR procedures are used anytime the fighters clear merges or make a kill. SRR procedure is designed to rebuild the fighter's radar SA from short to long-range in preparation of flowing to the next group. If performed properly, SRR mechanics should take no more than 30 seconds to accomplish. This means that a group at flow range of 30 NM will potentially be at 25 NM by the time the fighter gets a radar into the AOR and detect the follow-on group. Section SRR mechanics are:

- Select SRM, BST and then WACQ
 - i. This enters scan to clear flight path
 - ii. Hold this for 10 seconds once established on flow heading
- Exit ACM mode to enter the SRM set RWS/20 NM/140/6B
 - i. Mate at 10 NM
 - ii. Lead sets scan volume elevation 1K below ownship altitude and above
 - iii. Wing sets scan volume elevation 1K above ownship altitude and below
 - iv. Hold for 10 seconds
- Select MRM and enter RWS/80 NM/140/6B
 - Mate at 25 NM with altitudes set according to the briefed sanitization set

Once the fighters have mated at 25 NM, follow-on targeting can occur. When flowing group to group, the fighters should be prepared to enter the timeline at the 30 NM tactical range call or less. With a group at flow range when the fighters merge with the first group, the follow-on group may be as close as meld when the fighters get radar SA to them.

The key to short range employment as a section is coordination with AIC and strict adherence to the timeline on the follow-on group. If SA is high, the fighters can immediately meld, sort and employ on timeline to the follow-on group. Banzai may be an option.

In the event the fighters have low SA to the follow-on group or are employing behind timeline, they should consider the abort.

The fighters need to be ready to target, meld, sort, shoot and crank on the follow-on group all in accordance with the timeline. If no targeting calls have been made by the fighters at meld range, AIC will directively target the fighters into a follow-on group using BRAA format, which the fighters should use to meld. Flowing group to group, fighters may climb or descend to an appropriate block based on AIC's altitude information.

AIC (PRI): "Rage, target trail group BRAA 260, 25, 14 thousand, hot, hostile"

SNFO (PRI): "Rage 11"

1416. WVR SECTION EMPLOYMENT

Inside 10 NM, the fighters are within visual range (WVR). The fighter's priorities switch from radar mechanics, sanitization and employment to attriting the enemy threat. The fighter's priorities are:

- Maintain mutual support
- Get tallies
- Maintain SA
- Kill, or blow through, together
- Minimize time-to-kill

1. Mutual Support

There are three types of mutual support: Visual, Sensor and Communication.

2. Visual Mutual Support

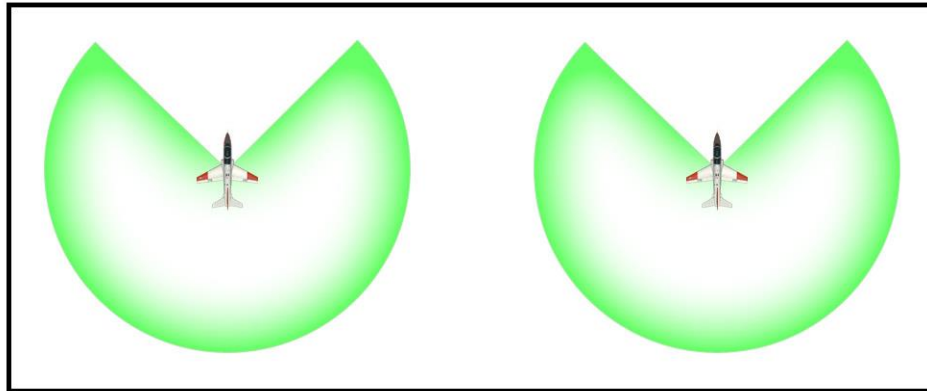


Figure 14-18 SNFO/WSO Visual Lookout AOR

When supporting each other visually, the pilots will visually clear their own aircraft's 10 to 2 o'clock as well as clear through the section while maintaining sight of the other aircraft.

The SNFO responsibility is to visually clear their own aircraft's 2 to 10 o'clock and outside the section. This is shown in Figure 14-18.

3. Sensor Mutual Support

Sensor mutual support (Figure 14-19) exists when one fighter has radar contact with the other, but is not visual. This may occur during or after a turning fight, or if the fighters become separated due to intercept geometry, such as in a VID scenario. The ability to discriminate which contact is friendly and which is the bandit in a turning engagement requires effective communication.

The fighter without a "visual" must use good sensor selection, be descriptive about own SA and work to obtain tally and visual. This should be done with clear, concise, SA building comm.

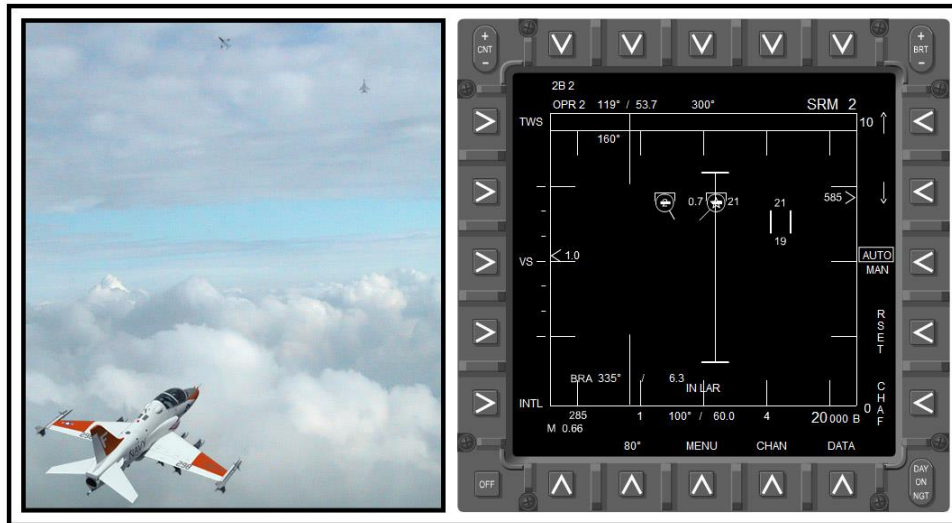


Figure 14-19 Sensor Mutual Support

4. Support through Comm

The goal of support through comm should be to establish sensor or visual support as soon as possible. Lead should describe their position and flow direction to Wing until sensor or visual support is regained. At a minimum, Lead should provide Wing with a direction and altitude or a destination to which flow should be initiated.

1417. VISUAL IDENTIFICATION PROCEDURES

A “bogey” declaration presents the fighters with a very dangerous situation. If the identity of the group remains bogey throughout the intercept, the fighters will be forced to visually identify the aircraft and then likely enter a visual turning fight if the group is visually identified as hostile. In order to survive when faced with this situation, the fighters must rely on sound execution.

Groups declared bogey may receive priority targeting above bandit or hostile groups if they meet commit criteria based on the type of mission and its objectives. A group declared bogey in the battlespace is a true unknown, and every friendly asset will be working to identify that group’s identity.

1. Visual ID Execution

Visual Identification (VID), if required for mission accomplishment, is directed if there is a bogey declaration, or no declaration, by 10 NM. Without a hostile declaration the fighters should continue to execute the gameplan in accordance with their timeline ensuring they crank at 20 NM. At 15 NM the fighters should make an aware/unaware bandit decision.

If the bandits are unaware the fighters need to execute the briefed plan for an unaware bandit, either stern convert or best possible merge. Once fighters decide bandits are unaware they need to immediately transition to hot geometry to prevent target aspect from ballooning out of control.

Lead will perform best geometry per the game plan, and Wing should prioritize good visual mutual support, accepting some error in geometry, as long as they maintain an IN LAR. Wing should also, approaching the VID, take their own separation 3-5 NM with a slight heading away from their Lead.

With aware bandits the fighters will still need to honor that threat and defend as a section. Because no MRMs were employed there will be no timeouts, but it is not advantageous to wait until 8 NM to defend. The fighters should then execute the standard section defense between 10-12 NM and then execute a section high aspect VID. The fighters should establish “eyeball” and “shooter” roles during the pitch-in. The eyeball will make the VID while the shooter will employ immediately on the correct contact if the group is visually identified and declared hostile.

Although either aircraft may assume either role, all things being equal, Lead will make the ID and Wing will employ. If it is advantageous for Wing to take the eyeball roll, Lead will be directive prior to the pitch-in with a call of “*Rage 12, you’re eyeball*” on AUX. Examples of when this could happen are due to section geometry or for loadout considerations (i.e., Wing has zero SRMs available).

With fighters in the defense and when pitch-in criteria are met, in this example Lead will establish the role of eyeball and Wing will establish the role of shooter. The shooter will delay their pitch-in by 5-10 seconds to gain adequate shot separation (3-5 NM) from the eyeball aircraft. The eyeball should make a high aspect pass with the bandits, VID them from the Forward Quarter, and the shooter will employ an appropriate missile when ROE has been satisfied. Any sort on a section VID shall be ignored due to system limitations and the group should be terminated once section VID training objectives have been met. The section high aspect VID comm flow is:

Lead SNFO (PRI) - “*Rage, defend south*”

Lead SNFO (PRI) - “*Rage 11, In left, Eyeball*”

Wing SNFO (PRI) - “*Rage 12, In left, Shooter*”

Lead SNFO (PRI) - “*Rage 11, Standby (Group name)*” (5-10 seconds before VID)

Lead SNFO (PRI) - “*Shoot, shoot, (NATO Identifier), (Group name) hostile*”

Wing SNFO (PRI) - “*Rage 12, Fox-2/3*”

Wing’s responsibility is to be tally, visual when Lead merges and makes the ID. If hostile, Wing should immediately employ to kill prior to the merge. The fighters should minimize TTK, and then flow to the next engagement.

2. VID Comm

Recall that when making the ID, the eyeball uses the directive comm: “*shoot, shoot,*” “*skip it,*” or “*unable ID*” and identify the aircraft by its NATO identifier. For example:

Lead SNFO (PRI) - “*C/S, standby (Group name)*”

Lead SNFO (PRI) - “*Shoot, Shoot, FULCRUM, (Group name) hostile*”

Wing SNFO (PRI) - “*Fox-2*”

Or

Lead SNFO (PRI) - “*Friendlylies, friendlylies, skip it*”

When properly executed, a VID against a group visually identified as a hostile should result in a quick kill at the merge by the shooter.

1418. CONTINGENCIES IN THE SRA AND 2VX ENVIRONMENT

1. Abort

If the fighters have no SA to the targeted group by 15 NM the fighters should abort. Abort is a “tactical emergency procedure.”

Abort is maximum performance, 135 degree overbank, nose slicing turn to put the threat at the 6 o'clock position and accelerating to .7 IMN. Abort should not be confused with “skate,” which is the descriptive call to all fighters that Lead has recognized that it would not be tactically sound to merge with the targeted group and has made the decision to go out.

Abort criteria must always be tailored to specific events. One event may elect to abort for very different reasons than another event. For 2v2 events the section must abort whenever they are at a strength disadvantage (heavy), because they cannot make simultaneous high aspect merges happen (stack), or do not have the missiles to clean up a merge. For 2vX the fighters must also abort on a group when other factor groups are able to affect that merge, such as groups in range less than FR or groups in azimuth. The fighters must assume that follow-on groups maintain hot geometry. Suggested abort criteria are:

- a. A group that is heavy
- b. A group that is a stack
- c. Groups in range < FR
- d. Groups in azimuth

- e. Less than 1 missile per fighter
- f. No/Bad radar SA by 15 NM
- g. Low SA
- h. VID (brief dependent)

Comm for the abort is:

SNFO (PRI) - *“Nikel, Abort left”*

SNFO (AUX) - *“Heading 180”*

2. **Flowing Cold Checklist**

Post–abort, or whenever the fighters are flowing cold, the PABST checklist shall be conducted. This is:

P - Put the threat at your six o'clock

- Calculate the 180 out heading (as discussed earlier) and execute Abort
- Complete SRR

A - Assess targeted status

- Request a picture
 - SNFO (PRI) - *“SABRE, Nikel 11, picture”*

B - Blue force composition

- Execute section powder check (loadout and fuel state)
 - SNFO (AUX) - *“Nikel 11, three (MRM), oh (N/A FI), two (SRM), 1.5”*
 - SNFO (AUX) - *“Nikel 12, three, oh, two, 1.4”*
- Address items for follow-on targeting, minimum cold targeting range, and flow

S System setup

- MRM sanitization set with appropriate AOR

T Target / Turn in

- Target follow-on group (if required and with minimum cold targeting range)
 - o SNFO (PRI) - *“Nikel, target pop-up group”*
- Turn hot and get back on timeline
 - o SNFO (PRI) - *“Nikel, in left”*
 - o SNFO (AUX) - *“Heading 300”*

3. Minimum Cold Targeting Range (MCTR)

Should the fighters abort and have live bandits behind them, they will need a minimum amount of range to target follow on groups in accordance with the flowing cold checklist. In order to sort and take shots on timeline, a baseline of 30 NM should be used for MCTR (2 NM for the turn in and 3 NM to gain radar SA before meld range). MCTR may be briefed differently based on mission planning factors. Re-committing is not required to target follow-on groups, as long as AIC is in TCBE or BRAA format. If follow on groups are inside MCTR, fighters should continue to flow cold and the group should be terminated if no further training objectives remain.

4. Loss of Radar SA

“Clean” is not called outside of MELD range. If the in-AOR player does not have radar SA approaching 25 NM that fighter must broadcast that lack of radar SA in order to get enough information to MELD on time. Initiate the clean comm approximately 20 seconds after the 30 NM call to stay on timeline. The first fighter to recognize they are clean should call (AUX) “C/S, clean.” If the other fighter has radar SA they should respond in BRAA format or initiate MELD based on timeline. The second fighter to recognize they have lost radar SA should broadcast (PRI) “C/S, clean, group name.” AIC will then respond. Once the fighters have gained SA to the group’s location, they need to assess where they are on timeline. For example, approaching MELD, call the MELD; approaching 20 NM, own the targeted group; approaching 15 NM, abort. If timeline allows, the fighters should adjust their radar scan volume and proceed with the intercept. Use the steps of MELD mech to adjust the radar into the appropriate location.

5. Bent Gadget

In the event one fighter's radar suffers a failure during the intercept, the fighters should maintain visual support and plan to skate. The fighter with a working radar should employ one MRM against each trackfile in the targeted group from TWS.

If neither radar is working, the fighters should abort no later than 15 NM.

1419. SECTION EMPLOYMENT SUMMARY

The fighter section is the smallest unit expected to employ autonomously in the air-to-air environment. The timeline is the same for SRA as 2vX. Keys to success as a section start with proper planning and briefing. Once airborne, success depends on proper formation management, effective, coordinated sensor employment, knowing and adhering to the timeline, executing tasks on time and effective communications.

Section tactical employment is a very dynamic environment. The principles in this chapter serve as a foundation for training at VT-86 and building blocks for future training in the fleet.

CHAPTER FIFTEEN

THE SELF ESCORT STRIKE ROUTE

1500. INTRODUCTION

A self-escort strike is a mission in which the strike aircraft are expected to provide their own defense and response to air-to-air threats during ingress to and egress from the target area. The “strike route” more closely simulates the multitasking demand of modern fighters. SES is a culmination of skills from Strike (A/S) and AWI (A/A) stages. SNFOs are expected to:

- Build a mental picture of the spatial relationships between the fighter, planned strike route, target and multiple threat groups by using active listening
- Effectively transition from the air-to-air to air-to-surface mindset and vice versa during mission execution
- Make sound decisions with regard to targeting, threat avoidance and target attack timing

1501. ROLE AND OBJECTIVES

1. Role

The role of a strike fighter on a self-escort mission is complex. SES missions are characterized by the fighter having a limited air-to-air load out based on the necessity to carry air-to-surface weapons to destroy the target.

The key to SES is recognizing that the primary mission is to strike the target, on time, with the correct amount of ordnance. The air-to-air portion of the mission is secondary and executed as required only for self-protection. That being said, the fighters cannot strike the target if they get killed or if they are forced to jettison their ordnance in order to survive.

2. Objectives

Specifically stated, the objectives of the SES at VT-86 are:

- a. Bombs on target, on time (BOTOT)
- b. Target all factor groups
- c. No blue losses

1502. SCENARIOS/THREAT

1. Scenario Generation

The SNFO will be required to generate a scenario in which the SES mission will occur. The scenario may be based on current events. The scenario briefed should include all of the information listed below. The scenario should build a framework of geographical and political factors that are behind the military's involvement in the situation. An example of a simple scenario:

"The country of Escambia has been providing a safe haven for pirate activity in the Gulf of Mexico. NATO has responded with an increased naval presence centered on Carrier Strike Group 2 (CARSTRKGRU 2, or CSG-2) based on the USS GEORGE H.W. BUSH (CVN-77). The coalition task force is conducting interdiction operations from port of Panama City to Gulfport to destroy or disrupt potential pirate activity. The Escambian military has responded to the presence of the task force with an increased defensive posture, including active air patrols inside their air space."

The scenario should include a description of threat nation, reasons for military action and how that action is being accomplished.

2. Mission Statement

The mission statement is a statement that clearly identifies the purpose for the mission. The briefing officer needs to explain how the fighter's mission fits into the bigger picture of a larger campaign without spending an inordinate amount of time on the subject.

The mission statement should serve as a foundation by which mission success or failure can be measured. This can be done by giving a brief explanation of the "Commander's intent" for the outcome of the mission. Blue losses will not be acceptable. A typical mission statement may be:

"Our mission is to destroy the ATC radar site at NAS Pensacola in order to degrade the enemy's ability to coordinate their air defense assets. Target destruction is desired; blue losses are not acceptable"

3. Threat

The threat for self-escort strike will be comprised of both surface and air threats. The air threats will be a MiG-29 or Su-27 series aircraft with a AA-10 SAR missiles with a maximum range of 15 NM and a AA-11 IR missiles with 3 NM FQ and 1.5 NM RQ WEZ.

The surface threat should include at least one medium range, surface-to-air (SAM) system. Anti-aircraft artillery (AAA) is expected below 15,000 feet in all scenarios, as are MANPADS below 10,000 feet.

All threat information can be summed up by the enemy's order of battle (EOB). The Air Order of Battle (AOB) and Ground Order of Battle (GOB), which includes SAMs, should be briefed by the briefing officer.

1503. ROUTES

SNFOs should refer to the latest STAN notes and the latest edition of the briefing guide for the most current list of routes, waypoints, GeoRefs and other pertinent information. The WHODAT will be the primary SES area. If any other area is briefed (i.e., W-155A, EAGLE-A, etc.) route information and restrictions need to be thoroughly briefed.

SNFOs should produce a route diagram (Figure 15-1) during the brief that include at a minimum:

- SNFO names and route number
- Bullseye
- Route points and labels, including bullseye location of route points and target
- ETAs for route points and distances
- Point coordinates

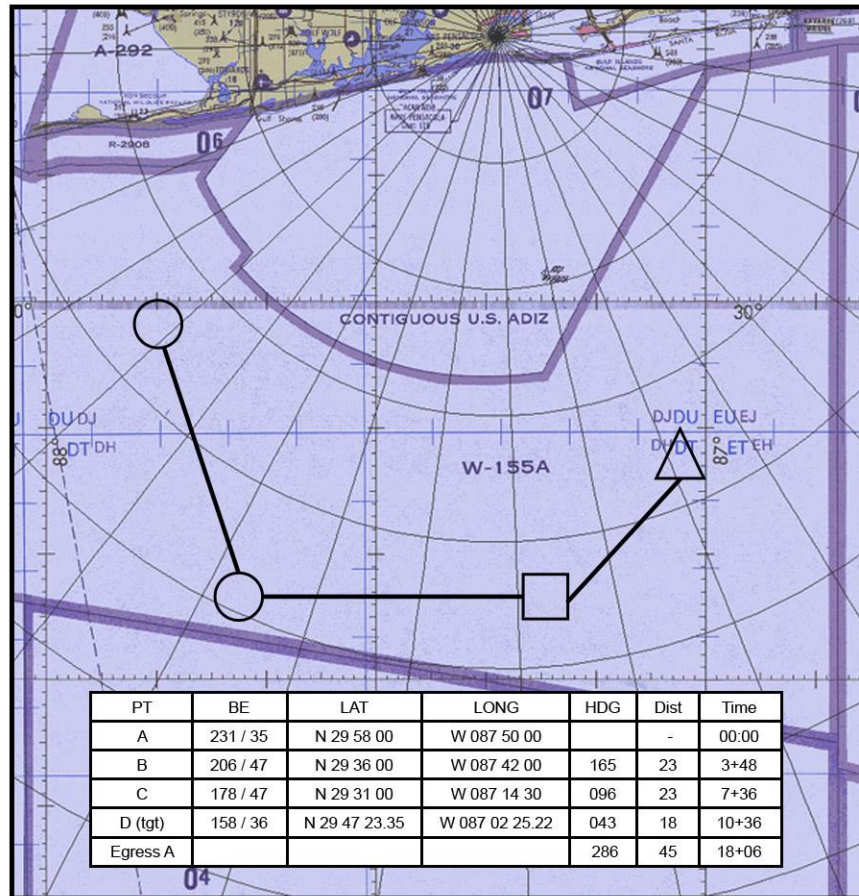


Figure 15-1 Example Route Card for a Notional W-155A Route

1504. THREAT PRESENTATIONS

SNFOs should expect multiple group presentations on both the ingress and egress portions of the route. These groups may or may not be a factor to the mission. A solid set of commit criteria and a targeting plan should be briefed to accommodate up to two groups in the picture, with one affecting the route at any given time.

Both friendlies and hostile aircraft will be present. At a minimum, the brief should cover:

- Suppression of Enemy Air Defense (SEAD) plan for EA-18G/EA-6B; location and time on station
- Expect jamming platform to turn "music" on at your pre-planned IP time
- Controlling agency (E-2) callsign and location

15-4 THE SELF ESCORT STRIKE ROUTE

Multiple friendly and bandit/unknown groups will clutter the picture. Clear, concise and SA building communications are key to success. Correlation will be important to avoid potential fratricide (blue on blue) or blue on white (commercial or private air traffic).

1505. COMMUNICATIONS

The SNFO will initiate the fight with *“Hammer 11, Tapes on, Fight’s on”* on PRI two minutes prior to the start of the route. At that point, AIC will start building the picture with broadcast control, anchored from bullseye.

At push time, the fighters will call, *“Hammer, push”* on PRI to inform all players that the fighters are starting the route.

At the IP, the fighters will call *“Hammer, IP inbound”* on PRI to inform all players that they are beginning the target attack.

Lead will direct the section to transition from the A/A to A/S midset by using the *“Attack”* call. This call will be on AUX and be made as part of the target attack.

If the fighters are off the route due to an intercept and need to proceed directly to the target rather than return to the route, the call *“Hammer, target direct”* on PRI can be used to inform all players of the fighter’s intent. This call is made to ensure that AIC, the SEAD asset and all fighters understand the intent of the fighters is to proceed with the target attack from the fighters’ current location, and is made in place of an *“IP inbound”* call..

Both the *“IP inbound”* and *“target direct”* calls are used as SA building calls to ensure all strike assets are aware of the striker’s intentions. SNFOs must realize that these two calls are mutually exclusive, the section is either *“IP inbound”* (heading to the target along the prebriefed route) or *“Target direct”* (heading from their current location toward the target not on the prebriefed heading).

Finally, once the last striker has delivered ordnance on the target, the call *“Hammer, Miller time”* will be made on PRI to inform all assets that the strikers are egressing the target area.

1506. TIMING PROBLEMS AND CORRECTIONS

The real challenge of SES is integrating a required air-air intercept into the strike timing problem using a preplanned route. The SES route is planned at 360 KGS, with the exception of the target leg, which is planned for 420 KGS for Surface-to-Air threat considerations. Route planning considerations, taught in earlier stages of training, provide useful tools when faced with an air threat delaying the fighters along the route. Specifically, a nonlinear path from the push point to the IP affords the strikers the opportunity to use time and distance geometry to correct for delays along the route. Without this preplanned geometry, there is no way the fighter can make up time proceeding directly to the IP or target after prosecuting an airborne threat.

1. Setting TOT

Time on target (TOT) will be determined and briefed by the Lead SNFO using whole minutes. For strike execution a 1 minute TOT window is acceptable using TOT +/- 30 seconds. For example, a TOT of 10:17 means bombs on target no earlier than 10:16:30 and no later than 10:17:30. To calculate the TOT airborne:

- a. Take the current real world time and add the time it will take to run the route
- b. Add 2 minutes for the time from fight's on to the Push
- c. Round that result to the next whole minute

Once a TOT has been calculated, Lead SNFO will transmit the TOT to all players over PRI

SNFO (PRI) - *"Ripper, set TOT 1-7"*

IGS (PRI) - *"Bandits copy TOT 1-7"*

2. Correcting Timing

The SNFO should use all available cues (SA page, HSI, spider card) to maintain positional awareness to the target. A 35 NM intercept from commit through merge may take 3+ minutes and large deviations off the route may create timing problems. Understanding the geometry of the route is the key to fixing timing problems that develop and making the TOT.

Prioritize using route geometry over speed increases for timing corrections. This is a real world factor since strike fighter aircraft are always fuel limited, as are T-45's. Set ground speed to meet timing gates at either 300 KGS=5 miles/minute, 360 KGS=6 miles/minute. The route will be run

- a. Below 20k feet: Minimum airspeed post-fight's on shall be 250 KIAS; once established on the target leg or whenever within IP distance of the target (defensive considerations), minimum airspeed shall be 300 KIAS.
- b. Above 20k feet: Minimum airspeed post-fight's on shall be 0.5 IMN; once established on the target leg or whenever within the IP distance of the target, minimum airspeed shall be 0.6 IMN.

If early, the fighter should use an offset prior to the IP to delay if the tactical situation allows it. If late, turn direct to the IP or target to make up the difference. If the fighters determine that geometry will not be enough to get the timing result they desire then speed may be adjusted according to the guidance above. Fighters shall not turn more than 30 degrees away from the target to correct timing once established on the target leg. "Established" is defined as IP distance inbound and within +/- 30 degrees of planned attack heading.

The air-to-air intercept can draw a fighter far from the planned route. SNFOs need to be cognizant of their location in relation to the planned route and where they are on timeline based on the set TOT.

1507. WEAPONS DELIVERY

Weapons delivery for SES will be in accordance with Strike Stage Standards as delineated in the current Strike FTI for CCIP bombing. SNFOs should plan on a radar-aided PGM delivery on a pre-briefed target that has been designated on the radar.

The valid drop criteria for SES are:

- Fighters must be in the 15-19 (25-29 for Eagle-A) blue block
- Correct system setup (i.e., A/G Master Mode, Armed, BOMB/CCIP/TGT HT)
- Target Captured
- Minimum airspeed or greater (as defined above)
- Inside the attack cone of +/- 30 degrees of planned run in heading with the target on the nose
- Release must be within 0.2 NM of planned release point (i.e. 3.2-2.8 NM)

1508. EVENT BRIEFING AND EXECUTION

1. Brief

As with 2vX, a successful Self Escort Strike flight begins with a professional brief. The briefing should include an in-depth and detailed A/S portion covering items such as the route overview, target acquisition and attack considerations as well as an A/A portion covering all 2vX considerations. See the STAN notes for the latest SES board layout.

2. Execution

- a. System Setup: Radar - The fighters should call “Hammer 11, Tapes on, Fight’s on” when established and ready to proceed with the event. MRM sanitization set, A/A waypoints, GeoRefs on the HSI/SA and sequences for the route and the area should all be briefed, selected and displayed prior to fight’s on. In the MRM sanitization set, the fighters are covering an enormous amount of area (Figure 15-2). The SNFO needs to recognize what a contact on the edge of the attack display means in relation to the strike route.

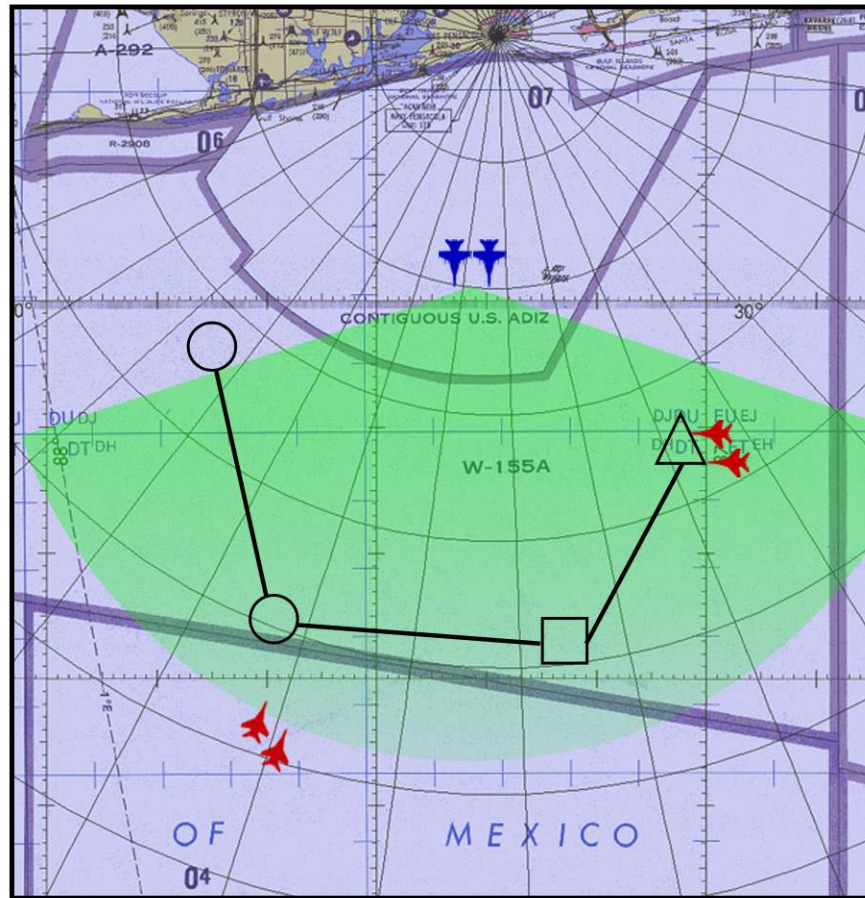


Figure 15-2 Radar Coverage

- b. System Setup: Navigation - In the OFT, the HSI should be set up per SOP for strike flights. This may include:
 - i. Route sequenced
 - ii. Area Sequenced
 - iii. GeoRefs per SOP
 - iv. TACAN to Bullseye
 - v. Waypoint - manual
- c. In the VMTS T-45, the HSI and SA page should be set up to display the following
 - i. Route sequenced
 - ii. Area Sequenced

- iii. GeoRefs per SOP
- iv. A/A Waypoint – to briefed Bullseye
- v. AIR and GROUND Boxed
- vi. Waypoint - manual

3. Event Flow

The SES event sequence will flow as one scenario beginning with the fight's on call and ending with a KIO. The fighters will begin the route, build the picture, commit, target, complete the intercept, decide whether to flow group to group, perform a PGM attack on the target, and egress without breaks or reloads. The route should be run twice, with a new TOT set and fight's on call following the first "*knock it off*" and before the next run.

The SNFO should apply all concepts from 2vX and consummate A/A engagement if required for self-protection on the ingress and for survival on the egress. The fighters should use geometry to stiff arm hostile groups if possible. The fighters may be required to VID on the egress and should be prepared for any air-to-air situation they have previously encountered in training, to include deceptive tactics on the part of the enemy.

Specific training objectives for the SES events are:

- a. Target factor groups
- b. Bombs on target, on time
- c. Section mutual support to kill/survive

The priority is to strike the target and survive to return home. This should drive all of the fighter's decision making.

4. Commit Criteria

The fighters should adhere to the 2vX timeline for both ingress and egress, with a 35 NM NLT commit on groups showing flank or less aspect. The fighters should assume the any contact with less than 60 TA has radar SA to the fight. Also, the fighters should commit on any picture that includes groups that are or will be a factor on the route.

5. Post Commit Geometry

The fighters must consider post commit intercept geometry as it will affect the route, follow-on group to group flow and route timing. Applying aggressive offsets to turn echelon or azimuth presentations into range problems can severely impact timing on the route. If the fighters adhere

only to the planned route, they will likely find themselves in tactically disadvantageous positions.

The fighters should always target factor groups while closely monitoring reset/drop criteria to avoid getting pulled too far off the route. Follow-on group flow and route geometry will determine how the fighters will set geometry after the commit. SSO priorities are now:

- a. Away from untargeted groups/threats (follow-on A/A groups and SAM rings)
- b. For timing (either toward the route or toward the target)
- c. Hot
- d. Shortest direction (for $TA < 10$)

6. 30 NM, Meld and Sort

As in 2vX, Flow Range will remain 30 NM. The tactical range call at 30 NM will be given to remind all fighters and AIC that the section is 30 seconds from meld on the targeted group.

Meld and sort should be on timeline as performed in 2vX. Lead will employ from TWS and Wing from STT (or as per the brief). The flight should maintain section visual support throughout the intercept and should use AIC to maintain SA to untargeted groups.

7. MRM Employment, Crank, and Decision Range

The fighters should employ on timeline in accordance with the 2vX employment tree. The fighters should then immediately crank, NLT 20 NM, with the following priorities:

- a. Away from untargeted groups/threats
- b. For timing
- c. Cold
- d. Shortest direction (for $TA < 10$)

The fighters should plan to banzai unless a tactical situation is presented that dictates they skate.

8. Timeout and Flow

Fighters should timeout their weapons and execute the standard 2vX timeline procedures, assessing bandit awareness at 15 NM and minimizing the time-to-kill with surviving bandits. Timely Forward Quarter MRM/SRM employment and expeditious tallies are keys to minimizing TTK. SNFOs should adhere to a missile conservation mindset. This means that on the ingress,

one fighter will prioritize MRM post-defense, and the other will prioritize SRM. On the egress, the reverse will apply.

Group to group flow should begin with disciplined execution of SRR mechanics once section visual support has been regained. Anticipate that the fighters may be faced with an immediate meld to a follow-on group and be prepared to execute on timeline without delay. Flow decisions into follow-on groups must be made using the same considerations of geometry, threat locations, and strike route. SNFOs are responsible for managing follow-on group geometry and applying these considerations.

If there is a group in the vicinity of the target, or between the target and IP, that is hot to the fighters, then making the planned TOT is unlikely. The fighters should flow into the next engagement as in 2vX, prior to executing the target attack, and keep in mind strike abort criteria, which is discussed later in the chapter.

A group that is in the fighters' beam or flank position (60 degrees ATA or greater on the attack display) away from the route or a path directly to the target will not be a factor during the target attack. However, this group may close the distance and become an immediate factor to the fighters once they are off target, but should not impact the fighter's target attack decision or execution.

Finally, any group that is not a factor on the ingress will still be alive and may be a factor on egress. Target only factor groups. Fighters on the way home shouldn't be looking for a fight, especially when they are low on fuel and air-to-air weapons.

9. Target Area Commit and Target Attack

The decision to attack a target or commit on an airborne threat in the vicinity of the target is called target area commit criteria. At VT-86, SNFOs should make a picture call at the IP. If the Lead SNFO assesses that the fighters can flow to the target, release ordinance, and have flow range to the called group at weapon impact, the fighters will attack the target. If range to the follow-on group is expected to be less than flow range at weapon impact, the fighters must target the follow-on group and take that intercept to a logical conclusion at the expense of striking the target. Additionally, a group that remains in the vicinity of the target will make it impossible for the fighters to transition to A/S and release their ordinance. These are just a few examples of what would meet conditions for a Strike Abort. The fighters shall Strike Abort for the following on the ingress:

- a. Less than one missile per fighter
- b. A group declared bogey
- c. If the fighters determine they will be more than a minute early or late for their TOT
- d. Any non-flowable air-to-air pictures (i.e. anything causing a section abort)

The fighters should also remain cognizant of the surface to air threats in the area, and whether the bandits are attempting to drag the fighters into a SAM MEZ by their actions.

10. Target Attack

The target attack is the same PGM attack used in Strike and CAS Stages. It is shown in Figure 15-3 as a reminder.

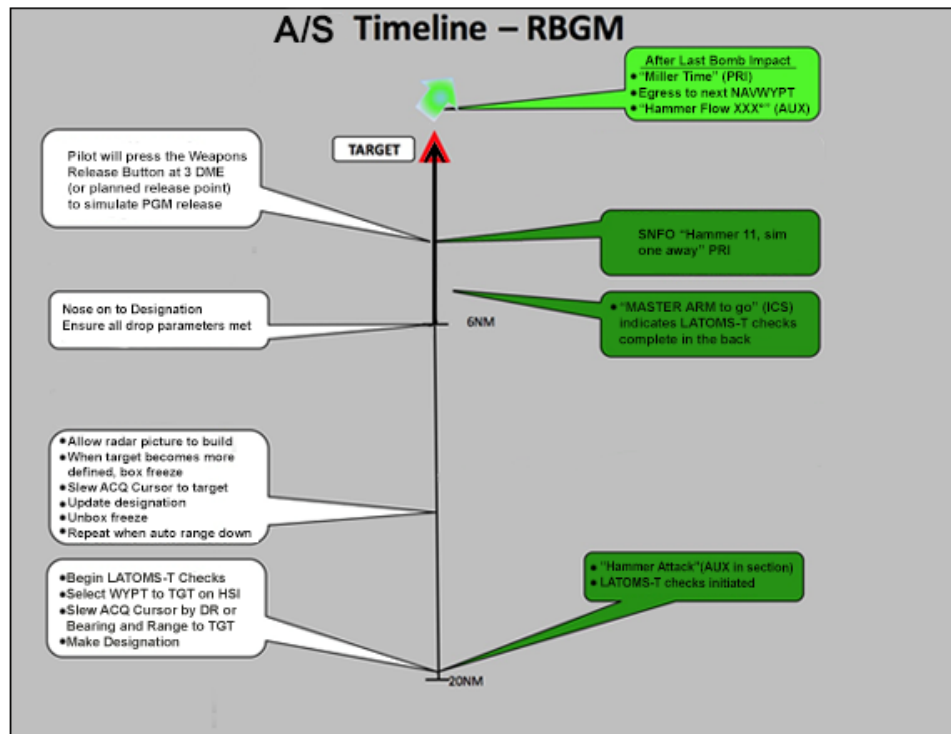


Figure 15-3 VT-86 Sensor-Aided PGM Attack

11. Egress

At "Miller time" the fighters should execute short range radar procedures and begin to rebuild the picture. The SNFO should select steering to the Egress Flow Point (EFP). The Lead SNFO will direct the flow of the flight.

Lead SNFO (PRI) - *"Ripper, flow 270"*

(After completing SRR) Lead SNFO (PRI) - *"SABRE, Ripper 31, picture"*

AIC will assist the fighters in their egress by providing tactical control. The fighter should sanitize their AOR during egress to cover gaps in AIC coverage.

The fighters must remember that they have to survive and RTB. A speed increase can be used, fuel permitting. Stiff-arming groups is preferred to being pulled off the egress route. Be alert,

15-12 THE SELF ESCORT STRIKE ROUTE

separate and avoid being dragged back into the SAM threat ring. Mission commanders should be decisive and disciplined. Lazy execution against an aggressive bandit will get the fighters shot down over hostile territory after they have released their bombs.

1509. SUMMARY

SES is the culmination of all of the skills the SNFO has acquired during flight training from start to finish. SNFOs are expected to demonstrate airmanship, leadership and sound headwork in a tactical environment to achieve mission success. The air-to-air employment of the section must be sound and flow seamlessly into air-to-ground employment.

SES at VT-86 is the closest thing to a “real world” Strike/Fighter mission in the TRACOM. Keep in mind that there are often multiple solutions to any tactical presentation. SNFOs should rely on their training, good headwork and application of sound tactics to accomplish the mission.

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APPENDIX A
MULTI-SERVICE TACTICAL BREVITY CODES

Abort(ing)(ed)	Directive/informative to cease action/attack/event/mission
Action	Directive to initiate a brief attack sequence or maneuver
Active	An emitter is radiating
Alpha Check	Request for/confirmation of ownship bearing and range to described point
Anchor(ed)	1. Orbit about a specific point; reference point flown by tanker 2. Informative to indicate a turning engagement about a specific location
Angels	Altitude of friendly aircraft in thousands of feet MSL
Azimuth	Two or more groups separated laterally in bearing
Bandit	A positively identified enemy aircraft. The term is a function of identification and does not necessarily imply direction or authority to engage.
Beam(ing)	Target maneuvering stabilized within 61 to 120 degrees aspect; generally given with cardinal directions; east, west, north, or south
Bent	System indicated is inoperative
Bingo	Prebriefed fuel state which is needed for recovery using prebriefed parameters
Bird	Friendly surface-to-air missile (SAM)
Bittersweet	Notification of possible BLUE ON BLUE situation relative to a designated track or friendly aircraft
Blind	No visual contact with friendly aircraft/ground position. Opposite of Visual.
Blow Through	Directive/informative call that indicates aircraft will continue straight ahead at the merge and not turn with target/targets
Bogey	An unidentified air contact
Bogey Dope	Request for target information as requested or for the closest group in BRAA

BRAA	Format of tactical control providing target bearing, range, altitude, and aspect from fighter
Bracket	Indicates geometry where friendly aircraft will maneuver to a position on opposing sides, either laterally or vertically from the target
Break (direction)	Directive to perform an immediate maximum performance turn in the direction indicated. Assume a defensive situation.
Broke Lock	Loss of radar/IR lock (advisory)
Buddy Lock	Locked to a known friendly aircraft
Bugout (direction)	Separation from that particular engagement/attack/operation; no intent to re-engage/return
Bullseye	An established point from which the position of an object can be referenced by bearing and range.
Buster	Directive call to fly a max continuous speed (military power)
CAP/Capping	1. Directive call to establish an orbit at a specified location 2. An orbit at a specified location
Capture	Aircrew has identified and is able to track a specified A/S target with an onboard sensor
Cease Engagement	In air defense, break the engagement on the target specified and prepare to engage another target. Missiles in flight will continue to intercept.
Cease Fire	Do not open fire, or discontinue firing; missiles in flight are allowed to continue to intercept; continue to track
Chattermark	Begin using briefed radio procedures to counter communications jamming
Check (Left/Right)	Turn degrees left or right and maintain new heading
Cherubs	Altitude of a friendly aircraft in hundreds of feet above surface (usually refers to helicopter traffic)
Chicks	Admin term used to denote friendly aircraft
Clean	1. No radar contacts on bandits, bogies or aircraft of interest 2. No visible battle damage
Cleared	Requested action is authorized (no engaged/support role are assumed)

Cleared Hot	Ordnance release is authorized
Closing	Decreasing in range/separation
Cold	1. Attack geometry will result in a pass or roll out behind the target 2. On a leg of the CAP pointed away from the anticipated threats 3. Threat group heading away from fighters
Come off	(Left-Right/Low-High) Directive to maneuver as indicated to either regain mutual support or to deconflict flight paths for an exchange of engaged and supporting roles. Implies both “visual” and “tally.”
Commit(ted)	Fighter intent to engage/intercept groups of interest
Cons/Conning	Unknown/non-friendly aircraft leaving contrails
Contact	1. Sensor contact at the stated position 2. Acknowledges sighting of a specified reference point
Continue	Continue present maneuver, does not imply clearance to engage or expend ordnance
Crank	Maneuver to put target at gimbal limits to manage target weapons envelope
Cutoff	Request for, or directive to intercept using cutoff geometry
Dash (#)	Aircraft position within a flight. Use if specific callsign is unknown.
Deadeye	Informative call by the laser designator indicating the laser/IR system is inoperative
Declare	Inquiry as to the ID of a correlated group, a specific track, or target
Defensive	(Spiked/Missile/SAM/Mud/AAA) Aircraft is in a defensive position and maneuvering with reference to the stated condition and unable to ensure deconfliction of mutual support.
Divert	Proceed to alternate base
Drag(ing)	1. (AF) Target maneuvering to 0-60 degree aspect 2. (NAVAL) Target maneuvering to 121-180 degree aspect
Drop(ing)	Directive/informative to stop monitoring a specified emitter/target and resume search responsibilities

Echelon	Fill-in to a picture label describing groups aligned behind and to the side of the closet group.
Engaged	Maneuvering with the intent to kill. If no additional information is provided (bearing, range, etc.) this implies visual/radar acquisition of target
Estimate	Estimate of size, range, height or other parameter of a specified contact; implies degradation
Extend	Short term maneuver to gain energy, distance or separation; normally with the intent of re-engaging
Eyeball	Fighter with primary visual identification responsibility
Faded	Radar contact is lost on unknown/non-friendly contact
Father	TACAN station
Feet Wet/Dry	Flying over water/land
Fence (In/Out)	Set cockpit switches as appropriate prior to entering/exiting the combat area
Fireball	Visual confirmation an air-to-air contact has been destroyed
Flank(ing)	1. (USAF) Target with a stable AA of 120-150 degree 2. (NAVAL) Target with stable aspect of 31-60 degree
Flash	Temporarily turn on prebriefed IFF mode or system
Float	Directive/informative to expand the formation laterally within visual limits to maintain a radar contact or prepare for defensive response
Fox (number)	Simulated/actual launch of air-to-air weapons ONE- semi-active radar-guided missile TWO- Infrared-guided missile THREE- active radar-guided missile
Friendly	A positively identified friendly contact
Furball	A turning fight involving multiple aircraft; non-friendly and friendly aircraft are in close proximity
Gadget	Radar or emitter equipment

Gimbal	Radar target is approaching azimuth or elevation limits
Gorilla	Large force of indeterminate numbers and formation
Green	A descriptive term referring to the direction the fighter must head in order to move away from threats
Group	Radar targets within 3 NM in azimuth and range of each other
Guns	An air-to-air or air-to-surface gunshot
Hard (Direction)	High-G, energy sustaining turn in direction called
Head/Head On	1. (USAF) Target with an aspect of 160-180 degrees 2. (NAVAL) target with an aspect of 0-20 degrees
Heads Up	Alert of an activity of interest
Heavy	A group known to contain three or more contacts
Hit(s)	1. (A/A) Radar return in search with altitude reference 2. (A/S) Weapons impact within lethal distance
Holding Hands	Aircraft in visual formation
Hold Fire	An emergency fire control order used to stop firing on a designated target, to include destruction of any missiles in flight
Home Plate	Home airfield or carrier
Hostile	A contact positively identified as enemy in accordance with theater rules of engagement and may be engaged with a clear field of fire
Hot	1. Attack geometry will result in roll out in front of the target 2. On a leg of the CAP pointing towards the anticipated threats 3. Threat group heading towards fighters. Opposite of Cold Navy: 0 - 30 Aspect 4. Ordnance employment intended or completed 5. Defined area is expected to receive fire
Hotdog	Informative/directive call that an aircraft is approaching or at a specified stand-off distance from the sovereign airspace of a nation (as defined by national boundaries or territorial sea and airspace) (COLOR may indicate additional standoff distance) Follow briefed procedures

I.D.	1. Directive to identify the target 2. I.D. accomplished followed by type
In	(Direction) Informative indication a turn to a hot aspect relative to a threat/target
India	Mode IV IFF
Jink	Unpredictable maneuvers to negate a gun tracking solution
Joker	Fuel state above Bingo at which separation/bugout/event termination should begin
Judy	Aircrew has radar/visual contact on the correct target, has taken control of the intercept, and only requires situation awareness information. Controller will minimize radio transmissions.
Kill	1. Clearance to fire 2. In training, a fighter call to indicate kill criteria have been fulfilled
Knock It Off	Directive to cease all phases of air combat maneuvers/attack/exercise activities
Ladder	Three or more groups/contacts in range
Laser On	Start/acknowledge laser designation
Lead/Trail	A descriptive term referring to a single group in a multi-group range or echelon presentation
Leaker(s)	Airborne threat has passed through a defensive layer. Call should include amplifying information.
Line Abreast	Two contacts within a group side-by-side
Locked	(BRAA/Direction) Final radar lock-on; sort is not assumed
Magnum	Launch of friendly anti-radiation missile
Mapping	Multi-function radar in an A/S mode
Marking	Friendly aircraft leaving contrails
Marshal(ing)	Establish(ed) at a specific point

Merge (merged plot)	1. Information that friendlies and targets arrived in the same visual arena 2. Call indication radar returns have come together and bearing and range information will be unavailable until aircraft separation is achieved
Monitor	Maintain radar awareness on or assume responsibility for specific group
Mother	Parent ship
New Picture	Used by controller or aircrew when tactical picture has changed. Supersedes all previous calls and reestablishes picture for all players
No Factor	Not a threat
No Joy	Aircrew does not have visual contact with the target/bandit/landmark. Opposite of Tally
Off (Direction)	Informative call indication attack is terminated and maneuvering to the indicated direction
Offset	Informative call indicating maneuver in a specified direction with reference to the target
On Station	Informative unit/aircraft has reached assigned station
Opening	Increasing separation
Out	(Direction) Informative indicating a turn to a cold aspect relative to the threat. Opposite of In.
Package	Geographically isolated collection of groups/contacts/formations
Padlocked	Informative call indicating aircrew cannot take eyes off an aircraft or surface position without risk of losing tally/visual
Parrot	IFF transponder
Picture	Request of provide air information pertinent to the mission in a digital bullseye format
Pig's away	JSOW launch
Pigeon	Magnetic bearing and range to Homeplate (or specified destination)
Pince/Pincer	Threat maneuvering for a bracket attack

Playmate	Cooperating aircraft
Playtime	Amount of time aircraft can remain on station
Pop	Starting climb for air-to-surface attack
Popeye	Flying into IMC (i.e., clouds or area of reduced visibility)
Popup	Informative that a contact has suddenly appeared inside a prebriefed range
Posit	Request for position; response in terms of a geographic landmark, or off a common reference point
Post Attack	Directive transmission to indicate direction after completion of intercept/engagement
Post Hole	Rapid descending spiral
Press	Directive to continue the attack; mutual support will be maintained. Supportive role will be assumed
Pump	A briefed maneuver to low aspect to stop closure on the threat or geographical boundary with the intent to re-engage
Pure	Informative indicating pure pursuit is being used or directive to go pure pursuit
Push	(Channel) Go to designated frequency
Pushing	Departing designated point
Range	Two or more groups separated in range
Reference (Heading)	Directive to assume stated heading
Reset	Proceed to a prebriefed position or area of operation
Resume	Resume last formation/station/mission ordered
Rider	A bogey that is conforming with safe passage routing/airspeed/altitude procedures
Ripple	Two or more munitions released or fired in close succession
Rolex (Time +/-)	Timeline adjustment in minutes from preplanned mission execution time

SAM	Visual acquisition of a SAM or SAM launch, should include clock code position
Scram	Emergency directive to egress for defensive or survival reasons
Separate	Leave a specific engagement; may or may not reenter
Shackle	A crossing weave/scissors maneuver to adjust/regain formation parameters that contains a single flight path crossing between lead and wing
Shooter	Aircraft/unit designated to employ weapons
Shotgun	Prebriefed weapons loadout at which separation/bugout should begin
Skate	Informative/directive to execute launch and leave tactics
Skip It	Veto of fighter commit, usually followed with further directions
Skosh	Aircraft is out of/or unable to employ active radar missiles
Snap (Direction)	Immediate vector to the group described
Sort	Directive to assign responsibility within a group; criteria can be met visually, electronically (radar) or both
Sorted	Sort responsibility has been met
Sour	1. Equipment indicated is operating inefficiently 2. Invalid response to an administrative IFF check
Spitter	An aircraft that has departed from the engagement or is departing the engaged fighters targeting responsibility
Splash	1. Air target destroyed 2. Weapons impact
Split	Request to leave formation to engage a threat; visual may not be maintained
Spoofing	Information that voice deception is being employed
Spot	Acquisition of laser designation
Squawk	Operate IFF as indicated or IFF is operating as indicated

Stack	Two or more groups/contacts/formation with high/low altitude separation in relation to each other
Status	Request for tactical situation/position
Steady	Directive to stop oscillation of IR pointer
Steer	Set magnetic heading indicated
Stern	Request for, or directive to, intercept using stern geometry
Stinger	Formation of two or more aircraft with a single aircraft in trail
Stop	Stop IR illumination of a target
Stranger	Unidentified traffic that is not associated with the action in progress
Strangle	Turn off equipment indicated, usually IFF
Stripped	Informative call from Wingman/element indicating out of briefed formation/position
Sweet	1. Equipment indicated is operating efficiently 2. Valid response to an administrative IFF check
Switch	Indicates an attacker is changing from one aircraft to another
Tactical	Request/directive to switch to tactical control
Tally	Sighting of a target/bandit/landmark. Opposite of No Joy
Target	Directive to assign group responsibility to aircraft in a flight
Targeted	Group responsibility has been met
Ten Seconds	Directive to terminal controller to standby for “Laser On” call in approximately 10 seconds
Terminate	1. Stop laser illumination of a target 2. Cease local engagement without affecting the overall exercise
Threat	Untargeted HOSTILE/Bandit/Bogey within prebriefed range/aspect of a friendly
Tied	Positive radar contact with element/aircraft

Tiger	Enough fuel and ordnance to accept a commit
Tracking	1. Stabilized gun solution 2. Continuous illumination of a target 3. Contact heading
Trashed	Informative call that missile has been defeated
Unable	Cannot comply as requested/directed
Uniform	UHF radio
Victor	VHF/AM radio
Visual	Sighting of a friendly, aircraft/ground position; opposite of Blind
Warning (Color)	Hostile attack is: RED - imminent or in progress YELLOW - probable WHITE - improbable (all clear)
Weapons	(Fire only): FREE - on a target not identified as friendly in accordance with current ROE TIGHT - at target positively identified as hostile in accordance with current ROE SAFE - in self-defense or in response to a formal order
What Luck	Request for results of missions or tasks
What State	Report amount of fuel and missiles remaining. Ammunition and oxygen are reported only when specifically requested or critical ACTIVE - number of active radar missiles remaining RADAR - number of semi-active missiles remaining HEAT - number of IR missiles remaining FUEL - pounds of fuel or time remaining
Winchester	No ordnance remaining
Words	Mission-pertinent information
Yard	Directive to use A/A TACAN for ranging

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APPENDIX B

OFT AIRCRAFT RECOGNITION GUIDE

The information in this Appendix is taken from the U.S. Army Field Manual 44-80, **Visual Aircraft Recognition** dated September 1996. FM-44-80 is a document originated by the U.S. Army Air Defense Artillery School. Its purpose is to train U.S. Army ground observers with little or no aviation experience how to identify aircraft and tell friendly aircraft from hostile aircraft. Although this perspective differs slightly from those of the SNFO and fighter aircrew, the fundamentals presented in FM 44-80 are sound and a good starting point to aircraft identification.

Aircraft recognition is a required aircrew skill. Being able to visually identify aircraft is a starting point. In follow-on training you will be required to associate sensors and weapons systems with particular aircraft and sub-models in order to better prepare and brief your mission. The farther out an aircraft can be detected, recognized, and identified, the more time aircrew have available to make an engagement decision. However, in the air, identification can be difficult as many aircraft share design features.

All aircraft are built with the same basic elements, recalled as WEFT (Figure B-1):

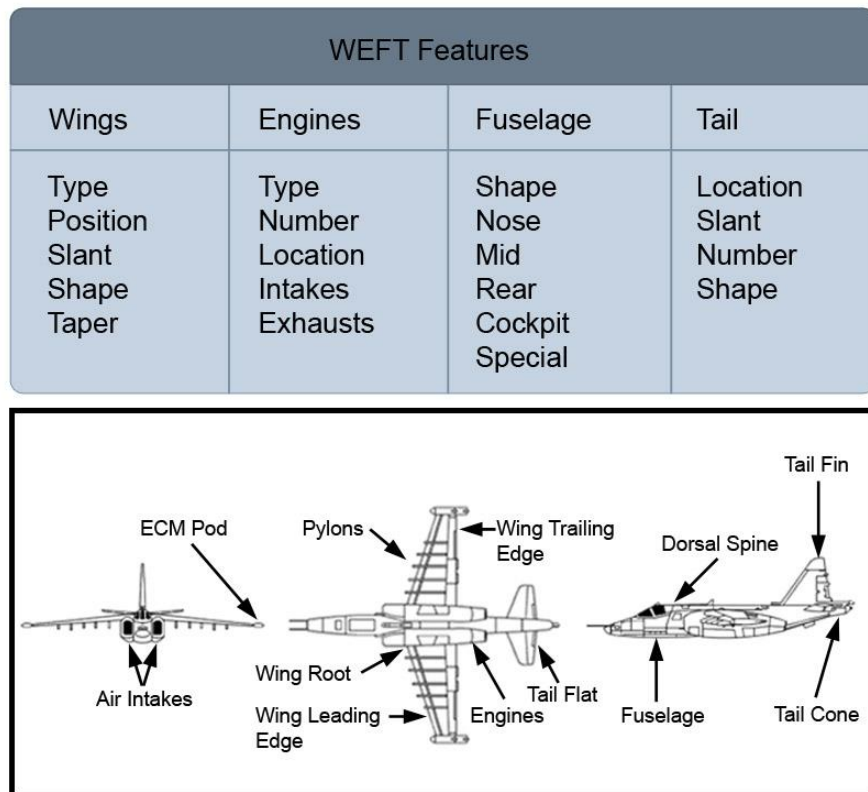


Figure B-1 WEFT Recognition

The WEFT features are unique to each aircraft, and will assist strike fighter aircrews with positive identification. The key to using these features is studying aircraft and knowing what the NATO identifications are for the aircraft being identified. Aircrew need to invest ample amounts of time on-deck studying aircraft from all corners of the globe. The Battle Group is mobile and strike fighter aircrew will operate anywhere in the world.

The information here pertains specifically to the aircraft available in the Aircrew Training Device 2F205A, also known as the Operational Flight Trainer (OFT). The complete FM 44-80 is available from multiple sources and online. It is unclassified, distribution is unlimited, and is now considered obsolete. Nevertheless, the information is still valid. Aircraft recognition is a required aircrew skill. Being able to visually identify aircraft is only a starting point. Aircrew is required to associate sensors and weapons systems with particular aircraft and sub-models in order to better prepare and execute missions.

1. **Wings**

There are many different types of aircraft wings. General attributes that can be used to identify an aircraft by its wings are:

- a. Type
- b. Position
- c. Slant
- d. Shape
- e. Taper

Wing examples are shown below in Figure B-2.

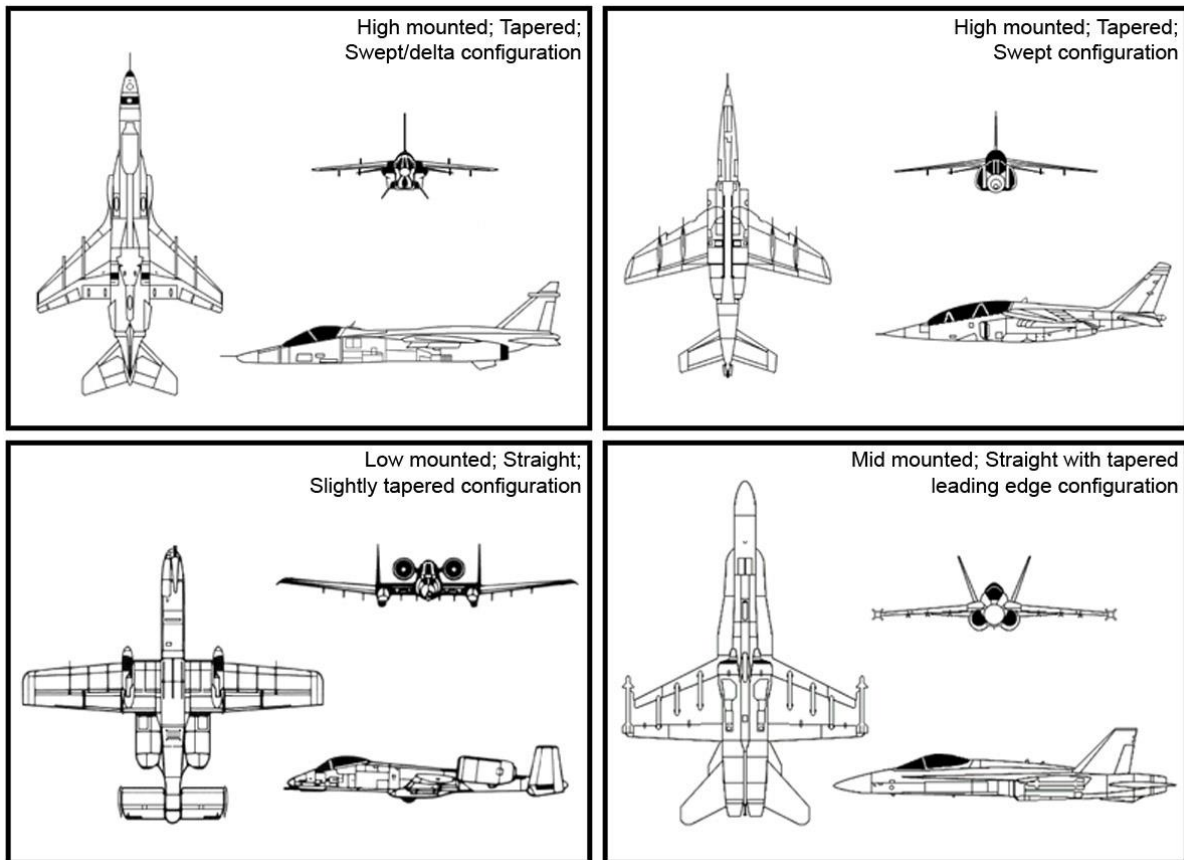


Figure B-2 Various Wing Types

2. Engines

Engine identification characteristics include:

- a. Type
- b. Number
- c. Location
- d. Intakes
- e. Exhaust

Engine examples are shown in Figure B-3.

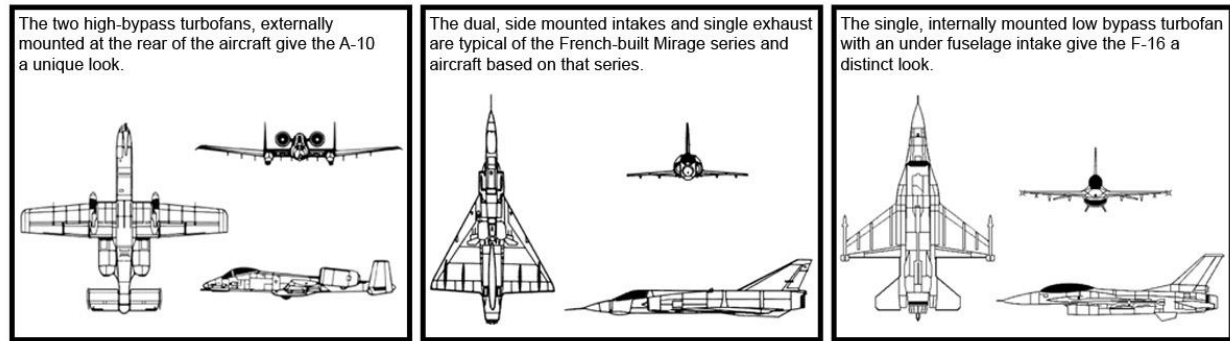


Figure B-3 Engine Number and Mounting

3. Fuselage

Intricacies of fuselage design are difficult to identify in modern fighters. However, some basic features include:

- a. Overall shape
- b. Nose, mid-section and rear shape and appearance
- c. Cockpit location
- d. Canopy

Fuselage examples are shown in Figure B-4.

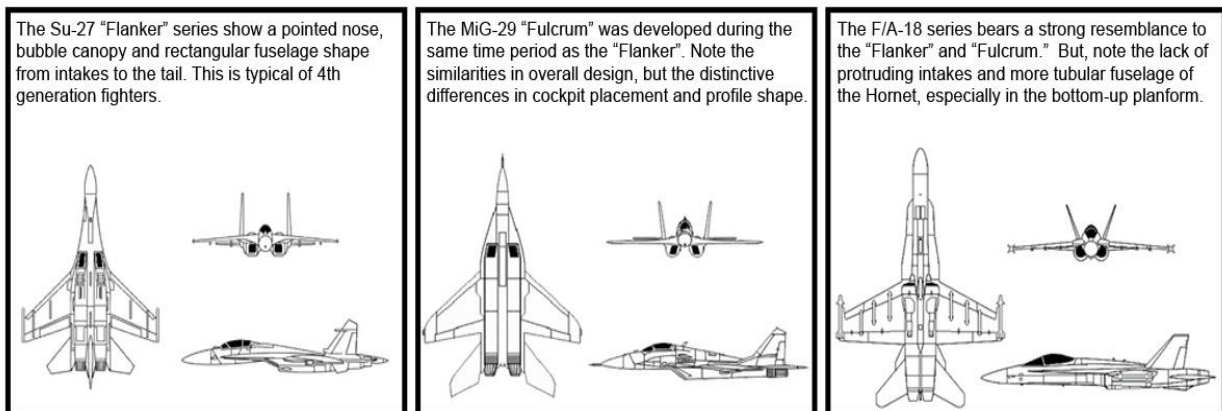


Figure B-4 Fuselage Configurations

4. Tail

The tail of an aircraft is often as distinctive as the wing shape and even more than the fuselage. Some basic features include:

- a. Location of vertical and horizontal stabilizers
- b. Number of vertical stabilizers
- c. Slant
- d. Shape

Tail examples are shown in Figure B-5.

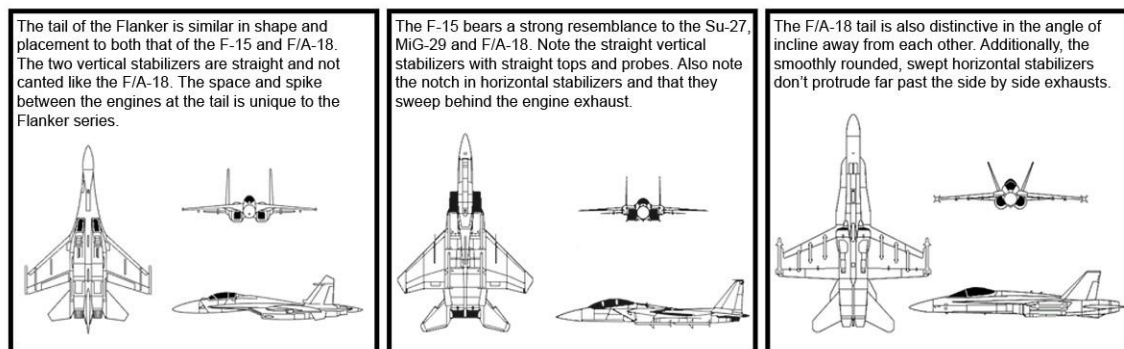


Figure B-5 Tail Configurations of Three Similar Aircraft

The remainder of this Appendix is devoted to aircraft found in the OFT. The SNFO should be able to recognize any of these when prompted by the scenario or instructor.

- a. F-15E Strike Eagle
- b. F-16C Fighting Falcon
- c. F/A-18A+/C/D Hornet
- d. F/A-18E/F/G Super Hornet
- e. F-35 Lightning II
- f. E-2C/D Hawkeye
- g. E-3 Sentry
- h. KC-10 Extender

- i. F-4 Phantom II
- j. F-5 Tiger
- k. F-14B
- l. Mirage 2000
- m. MiG-21
- n. MiG-29
- o. Su-27/30

5. F-15E Strike Eagle

The F-15E Strike Eagle (Figure B-6) is the multi-role version of the F-15 Eagle. It is optimized for strike missions and carries a crew of two, a pilot and weapon systems officer (WSO).

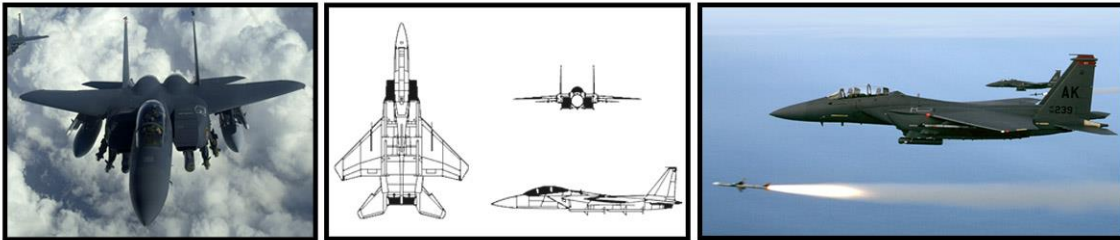


Figure B-6 F-15E Strike Eagle

Length	63 feet 9 inches
Wingspan	42 feet, 9 inches
Wings	High, semi-delta, blunt, angled tips;
Engines	Two mounted in the rear. Diagonally-shaped, box-like air intakes alongside the fuselage. Dual exhausts.
Fuselage	Long, pointed nose and a bubble canopy. Large, box-like center section that tapers slightly to the front and rear.
Tail	Two fins; tapered leading edges, straight trailing edges, and square tips; mid-mounted, swept, tapered with saw tooth horizontals
Similar Aircraft	F-14 Tomcat, Su-24 FENCER, Tornado, MiG-29 FULCRUM, Su-27 FLANKER.

6. F-16 Fighting Falcon

The F-16 Fighting Falcon (Figure B-7), is a small, agile, single engine multi-role aircraft. The F-16 is the most numerous U.S. jet fighter produced with over 4,500 produced for 25 countries.

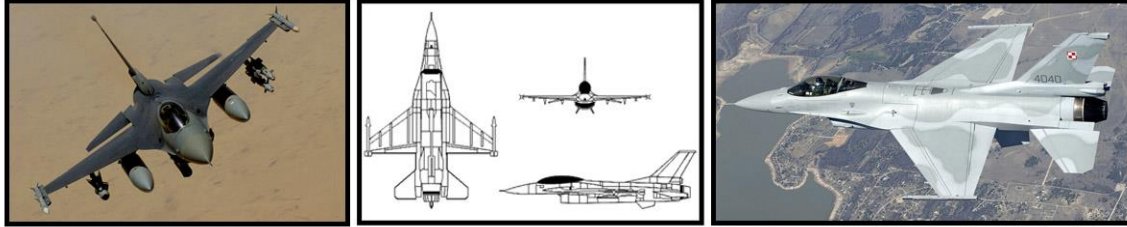


Figure B-7 F-16 Fighting Falcon

Length	47 feet, 8 inches
Wingspan	31 feet
Wings	Mid-mounted, delta-shaped. Missiles are normally mounted at the wing tips.
Engine	One in body. Oval air intake under the center of the fuselage. Single exhaust.
Fuselage	Long and slender with widening at the intake; bubble canopy with one or two aircrew (dependent on model)
Tail	Swept-back, tapered fin with square tip. Flats mid-mounted on the fuselage, delta-shaped with square tips, and a slight negative slant. Two belly fins.
Similar Aircraft	F/A-18 Hornet, MiG-29 FULCRUM, Mirage F1

7. F/A-18A+/C/D Hornet

The F/A-18A+/C/D Hornet (Figure B-8) is the primary fighter of the Navy and Marine Corps. All A models in service have been upgraded to the A+ standard. The USMC is the only U.S. service to use the D operationally.

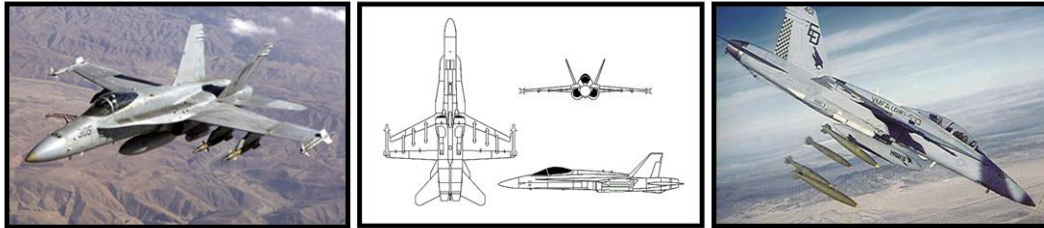


Figure B-8 F/A-18A+/C/D Hornet

Length	56 feet
Wingspan	37 feet, 6 inches
Wings	Mid-mounted, semi-delta with prominent leading edge root extension on sides of fuselage from the wing to the front of the cockpit. Missiles are usually on square tips.
Engines	Two turbofans mounted in the aircraft rear section. Oval air intakes under the wings.
Fuselage	Barrel-shaped with solid, pointed nose. Aircraft widens at the air intakes and tapers to the rear. Bubble canopy for one (A+/C) or two aircrew (D model)
Tail	Swept-back and tapered tail flats mid-mounted on the body. Twin, swept-back, and tapered tail fins mounted forward on the fuselage. Fins have an outward tilt.
Similar Aircraft	F-16 Fighting Falcon, MiG-29 FULCRUM, Su-27 FLANKER, F-15 Eagle

8. F/A-18E/F Super Hornet

The F/A-18E/F Super Hornet (Figure B-9) is about 25% larger than the C/D model and incorporates a new radar and new mission computers. It also serves as the basis for the EA-18G Growler.

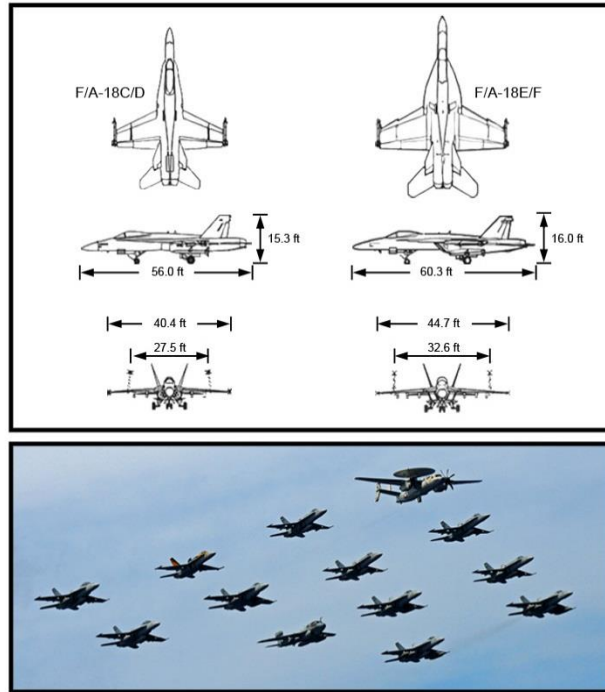


Figure B-9 F/A-18E/F Super Hornet

Length	60 feet, 4 inches
Wingspan	44 feet, 7 inches
Wings	Mid-mounted, semi-delta with large LEX on sides of fuselage
Engines	Two mounted on aircraft rear section with rectangular air intakes and oval exhaust
Fuselage	Barrel-shaped with widening at the intakes and a bubble canopy for one (E model) or two aircrew (F model)
Tail	Twin vertical stabilizers canted outward
Similar Aircraft	F-16 Fighting Falcon, MiG-29 FULCRUM, Su-27 FLANKER, F-15 Eagle

9. F-35 Lightning II (Joint Strike Fighter)

The F-35 Lightning II (Figure B-10), also known as the Joint Strike Fighter, is the U.S. Military's next generation tactical aircraft. It is being built in three versions, one each for the Navy (F-35C), Marine Corps (F-35B) and Air Force (F-35A).



Figure B-10 F-35 Lightning II

Length	51 feet, 2 inches (A&B) 51 feet , 4 inches (C)
Wingspan	35 feet (A & B); 43 feet (C)
Wings	Mid-mounted, semi-delta with prominent leading edge root extension
Engines	Two rectangular intakes at cockpit; one exhaust between tails
Fuselage	Barrel-shaped with solid, pointed nose; aircraft widens at the air intakes and tapers to the rear
Tail	Swept-back, and tapered tail; twin, swept-back, tapered tail fins with an outward tilt
Similar Aircraft	F-16 Fighting Falcon, MiG-29 FULCRUM, Su-27 FLANKER, F-15 Eagle, F-22 Raptor

10. E-2C/D Hawkeye

The only carrier based, turboprop aircraft, the E-2 (Figure B-11) family has provided Airborne Warning and Control (AWACS) functions for Carrier Strike Groups (CSG) since Vietnam.

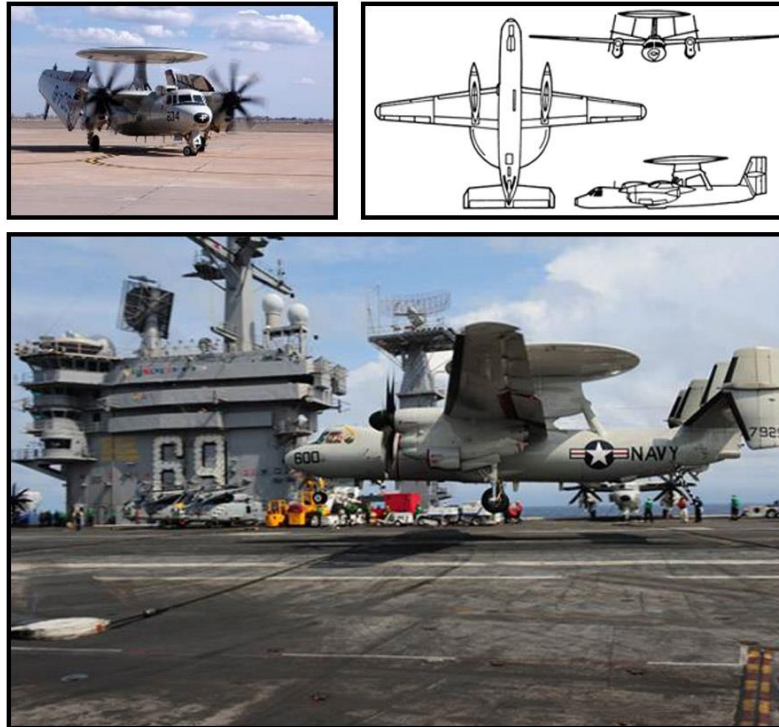


Figure B-11 E-2C/D Hawkeye

Length	57 feet, 6 inches
Wingspan	80 feet, 6 inches
Wings	High-mounted and equally tapered with blunt tips
Engines	Two turboprops mounted under the wings; engines extend well beyond the wings' leading edges
Fuselage	Oval that tapers to the rear; rounded nose; stepped-up cockpit; large radome
Tail	Four fins, horizontal stabilizers that are high-mounted on the fuselage with an upward slant
Similar Aircraft	C-160 Transall, G.222, An-24, An-26; none have distinct radome. C-2 is similar design, used for carrier on-board delivery (COD).

11. E-3 Sentry

The E-3 Sentry (Figure B-12) is an Airborne Early Warning and Control System, or AWACS aircraft used by the USAF and other U.S. allies. It is based on the Boeing 707 airframe.



Figure B-12 E-3 Sentry

Length	152 feet, 9 inches
Wingspan	154 feet, 7 inches
Wings	Low-mounted, swept-back, and tapered
Engines	Four mounted on pylons under the wings
Fuselage	Round, cigar-shaped, tapers to the rear; large radome on top of the body between the wings and tail section
Tail	Swept-back, tapered fin with a square tip; horizontal stabilizers are swept-back, tapered, and mid-mounted on the fuselage
Similar Aircraft	An-50 Mainstay (AEW aircraft based on the Il-76 transport) KC-135 based on same 707 airframe

12. KC-10 Extender

Based on the DC-10, the KC-10 Extender (Figure B-13) is the primary USAF tanker. It was designed to refuel aircraft using both USAF and USN/USMC methods during the same mission.

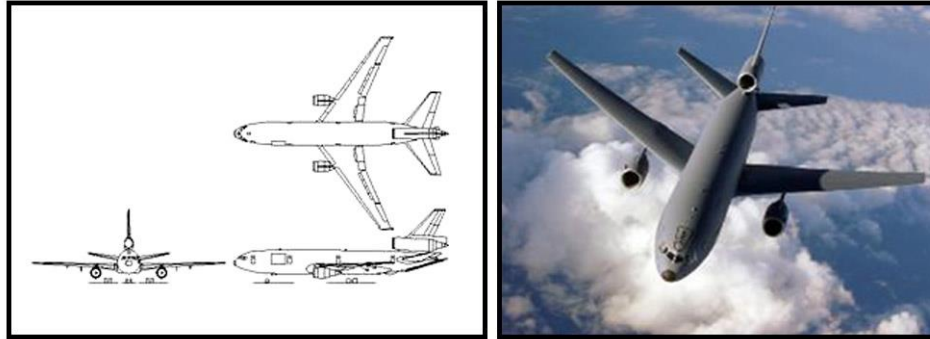


Figure B-13 KC-10 Extender

Length	181 feet, 7 inches
Wingspan	165 feet, 4.5 inches
Wings	Low-mounted, swept-back, and tapered
Engines	Three; one in each under wing pylon, one in the vertical tail
Fuselage	Cylindrical, tapered at each end; a refueling boom is beneath the aft end
Tail	Swept-back, tapered fin with a square tip; engine-mounted, horizontal stabilizers are swept-back, tapered, and mid-mounted on the fuselage
Similar Aircraft	KC-135 in role; DC-10, MD-11, L-1011 similar looking, 3 engine airliners

13. F-4E Phantom II

The F-4E Phantom II (Figure B-14) is a two seat, twin engine, all weather, long range supersonic fighter-bomber. The Islamic Republic of Iran Air Force (IRIAF) has approximately 200 F-4D and E models in service.

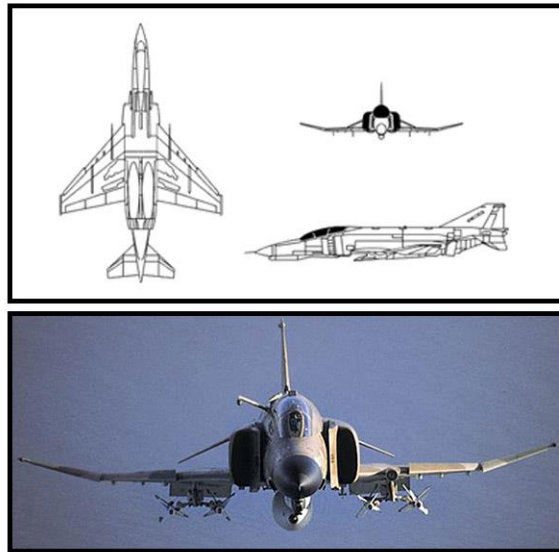


Figure B-14 F-4E Phantom II

Length	63 feet
Wingspan	38 feet, 5 inches
Wings	Low-mounted, swept-back, semi-delta with square tips; positive slanted wing tips and a saw tooth in leading edges of the wings
Engines	Two engines; rectangular air intakes both sides rear cockpit; two exhausts beneath an overhanging rear section
Fuselage	Rectangular midsection, pointed droopy nose
Tail	Horizontal stabilizers are mid-mounted on body; delta-shaped with a negative slant; sharply back-tapered fin with a square tip
Similar Aircraft	Jaguar, A-4 Skyhawk, Super Etendard.

14. F-5E Tiger II

The F-5 (Figure B-15) was exported to many countries, including Iran, during the 1970s. The IRIAF maintains 75 in their inventory. It is also used extensively by the US military as an aggressor, due to its similarity to the MiG-21, and many US allies.

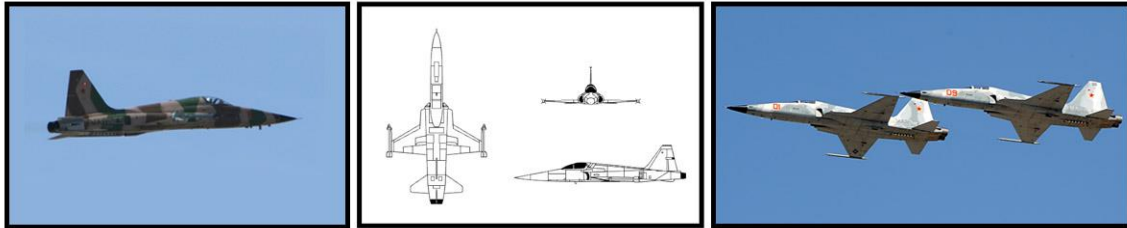


Figure B-15 F-5E Tiger II

Length	48 feet
Wingspan	26 feet, 8 inches
Wings	Low-mounted, stubby, and unequally tapered; missile rails on tips for AIM-9 or equivalent
Engine	Two engines inside the body, semicircular air intakes forward of the wing roots
Fuselage	Bullet-shaped, long, drooping nose; bottom is flat from the air intakes to the dual exhausts
Tail	Horizontal stabilizers are low mounted and equally tapered; fin is large and equally tapered with a square tip

15. F-14A Tomcat

The only air force to still fly the F-14A Tomcat (Figure B-16) is the IRIAF, with 44 in their inventory, 25 of which are believed operational.

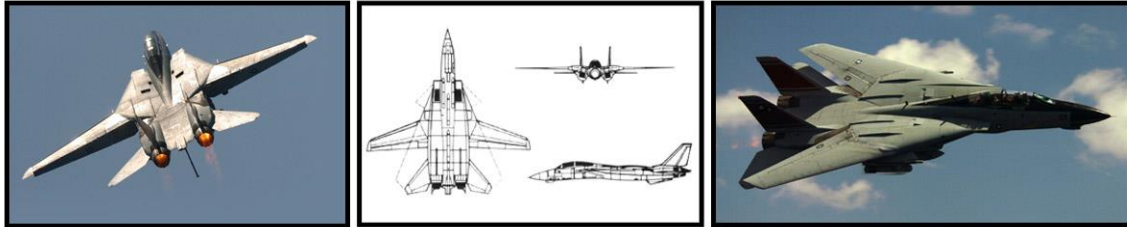


Figure B-16 F-14A Tomcat

Length	62 feet
Wingspan	64 feet
Wings	High-mounted, variable, swept-back, and tapered with curved tips
Engines	Two turbofans in fuselage; diagonally shaped, box-like air intakes alongside the fuselage; dual spaced exhausts
Fuselage	Box-like from the air intakes to the rear section; pointed nose; bubble canopy
Tail	Twin swept-back, tapered tail fins; mid-mounted horizontal stabilizers on the fuselage; swept-back, and tapered; fins beneath engines extend forward to horizontal stabilizers
Similar Aircraft	F-15 Eagle, Su-24 FENCER, Tornado, Su-27 FLANKER, MiG-29 FULCRUM

16. Mirage 2000

The Mirage 2000 (Figure B-17) is a single engine, multi-role aircraft sold in many variants, including a two-seat strike version. With the exception of the Mirage F1, the Mirage series, and aircraft based on them are very similar in physical appearance.

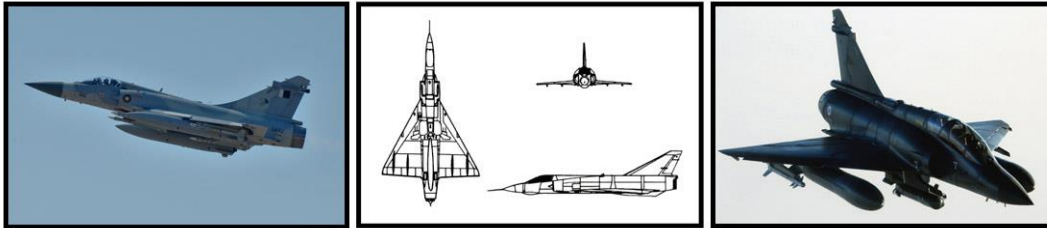


Figure B-17 Mirage 2000

Length	50 feet, 3 inches
Wingspan	29 feet, 5 inches
Wings	Low-mounted, delta with clipped tips
Engines	One turbofan mounted in the fuselage; semicircular air intakes alongside the fuselage forward of wings; large, single exhaust protrudes past the tail
Fuselage	Tube-shaped with pointed nose and bubble canopy
Tail	No horizontal stabilizers; fin is swept-back and tapered with a clipped tip
Similar Aircraft	Mirage III/5 (difficult to distinguish); Kfir, Viggen

17. MiG-21 (NATO IDENTIFIER FISHBED)

The MiG-21 (Figure B-18) (NATO identifier “FISHBED”) is the most widely produced jet fighter aircraft in the world. First produced in 1959, the aircraft has had continuous capability updates. The newest versions possess BVR missiles and Helmet Mounted Sights (HMS). Production ended in 1989 at 11,496.

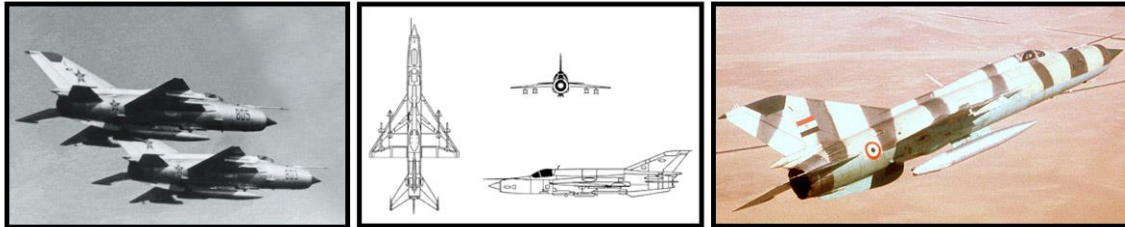


Figure B-18 MiG-21

Length	51 feet, 8 inches
Wingspan	23 feet, 5 inches
Wing	Mid-mounted, delta wing with small square tips
Engine	One, with round air intake in the nose; single exhaust
Fuselage	Long, tubular body with blunt nose and cone intake; dorsal spine flush with the canopy
Tail	Fin swept-back and tapered with square tip; horizontal stabilizers mid-mounted on the body, swept-back, and tapered with square tips; belly fin under tail
Similar Aircraft	Su 17/22 Fitter, Mirage III/5, A-4 Skyhawk; F-5 Tiger II

18. MiG 29 Series (NATO IDENTIFIER “FULCRUM”)

The MiG-29 (Figure B-19) (NATO identifier “FULCRUM”) is a single or two seat fighter. It has evolved into a true multi-role fighter comparable to the F/A-18C/D.

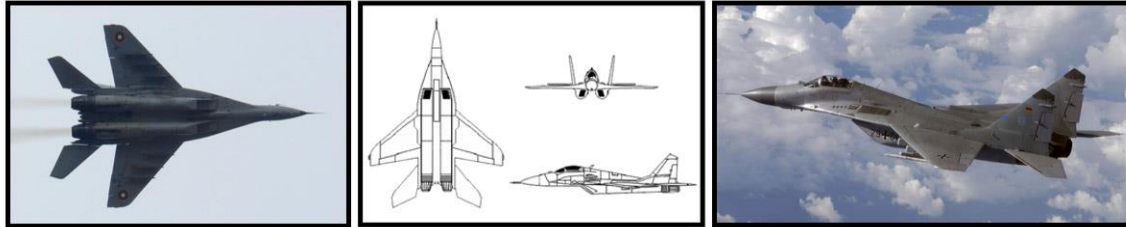


Figure B-19 MiG-29

Length	50 ft 10 in (15.6 m)
Wingspan	33 ft 7 in (10.26 m)
Wings	Swept-backed, tapered with square tips; LEXs are wide and curved down to the front; LEX begins even with front of canopy
Engines	Two; two slanted, rectangular intakes being at cockpit; two widely spaced exhausts
Fuselage	Tapered, wedge shaped body, widens at air intakes and tapers to exhausts; pointed nose; bubble canopy
Tail	Two swept-back, outward canted, tapered fins with slanted tips; stabilators extend well past exhausts
Similar Aircraft	F/A-18 Hornet, F-16 Fighting Falcon, F-15 Eagle, Su-27 FLANKER

19. Su-27/30 Series (NATO IDENTIFIER FLANKER)

With 12 air-to-air hard points, the Su-27 series (Figure B-20) (NATO identifier “FLANKER”) is the premier, Chinese and Russian-built multi-role fighter. The single seat Su-27 fighter evolved into the two seat Su-30 strike aircraft.

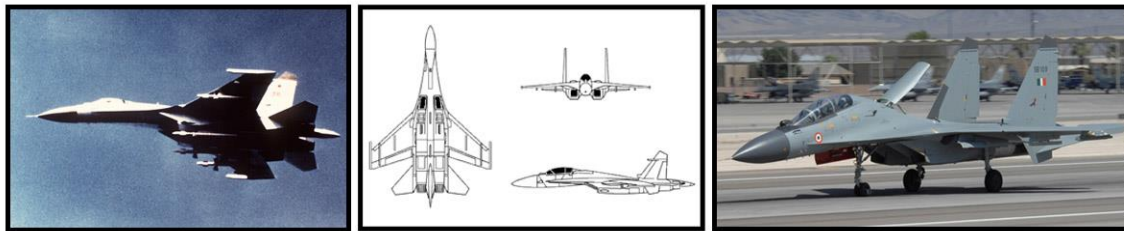


Figure B-20 Su-27/30 Series

Length	69 feet
Wingspan	47 feet, 6 inches
Wings	Mid-mounted, semi-delta with square tips; LEX extends to cockpit; missile rails on tips
Engines	Two with square, diagonally-cut air intakes mounted under the wings well aft of canopy
Fuselage	Rectangular from air intakes to the tail; pointed nose and bubble canopy; spike between exhausts
Tail	Fins are swept-back, tapered with slanted tips, no cant, mounted outboard engines; stabilators are mid-mounted, swept, tapered and do not extend far past exhausts
Similar Aircraft	F-15 Eagle, F-14 Tomcat, MiG-29 FULCRUM

APPENDIX C: COMM EXAMPLES

A section of fighters, tactical callsign Showtime 11-12 are executing an AWI event with Sabre as AIC and 'Denver' as bullseye.

1. Starting the fight

- a. Bandits PRI: *"Showtime 11, your bandits are set"*
 - i. If the fighters are ready PRI: *"Showtime 11, Tapes on, fight's on"*
 - ii. If the fighters need more than 30 seconds PRI: *"Fighters need (time)"*

2. Precommit

- a. PRI: *"Sabre, Showtime 11, picture"*
 - i. Response to AIC's picture PRI: *"Showtime 11"*

3. Commit

- a. PRI: *"Showtime, commit"*
- b. AUX: *"check tapes, master arm"*

4. Contact/Declaration

- a. PRI: *"Sabre, Showtime 11, contact BRAA XXX/YY, (alt), declare"*
- b. PRI: *"Sabre, Showtime 11, contact Denver XXX/YY, (alt), declare"*
- c. If looking for declaration but already correlated. PRI: *"Sabre, Showtime 11, declare (group name)"*

5. Targeting

- a. PRI: *"Showtime, target, (group name)"*
- b. PRI: *"Showtime 11 target high contact, Showtime 12 target low contact"*

6. 30 NM comm

- a. PRI: *"Showtime 11, (group name), 30 miles"*

7. Meld

a. AUX: *"Showtime, MELD BRAA, XXX/YY, (alt)"*

i. Leave off altitude if targeting a stack

8. Sort

a. AUX: *"Showtime, sort range"*

b. AUX: *"Showtime 12, sorted trail man"*

c. AUX: *"Showtime, sort elevation"*

i. AUX: *"Showtime 12, sorted low man, (alt)"*

d. AUX: *"Showtime 11, (group name), single only, (alt)"*

i. AUX: *"Showtime 12, same"*

ii. Only if different than the altitude previously called. AUX: *"Showtime 12, (group name), single only, (alt)"*

9. Shot/Crank

a. PRI: *"Showtime 11, fox-3"*

i. *"Showtime 11, fox-3, two ship"* indicates missiles were employed on both contacts in a group

ii. *"Showtime 11, second fox-3"* indicates two missiles were employed on the same contact

b. AUX: *"Showtime, crank (right/left)"*

10. Missile Status comm (always on PRI)

a. Missile status comm always follows the same format: *"C/S, (missile status), (sort), (group name)"*

b. MRM

i. *"Showtime 11, timeout, single only, (alt), single group"*

1. Used to indicate a missile timed out on a group assessed strength single only

- ii. *“Showtime 12, timeout, trail man, lead group”*
 - 1. Used to indicate a missile timed out on the trail man of the lead group
- iii. *“Showtime 11, timeout, two ship, high contact”*
 - 1. Used to indicate a missile timed out on a group stack and the high contact was assessed strength two
- iv. *“Showtime 12, timeout, single only, (alt), low contact”*
 - 1. Used to indicate a missile timed out on a group stack and the low contact was assessed strength single only
- v. *“Showtime 12, kill, locked, (alt), west group”*
 - 1. Used to indicate a fighter was tally at missile timeout, and had not sorted
- vi. *“Showtime 11, timeout, south man, middle man, trail group”*
 - 1. Used to indicate missiles timed out on two of the three contacts within a group
- vii. *“Showtime 12, shot trashed”*
 - 1. Used to indicate a missile was trashed
- c. SRM
 - i. *“Showtime 11, kill, locked, (alt), single group”*

11. **Fighter Request for SA**

- a. PRI: *“Sabre, Showtime 11, separation (group name)”*
 - i. Used to get separation between groups, for example groups in range:
 - 1. *“Sabre, Showtime 11, separation Trail group”*
- b. PRI: *“Sabre, Showtime 11, status (group name)”*
 - i. Used to ask for an update on a group, for example groups in range and the lead group maneuvers
 - 1. *“Sabre, Showtime 11, status Trail group”*
- c. PRI: *“Sabre, Showtime 11, BRAA (group name)”*

12. Defense

- a. PRI: *"Showtime, defend (cardinal direction)"*
- b. ICS: *"Heading XXX"*

13. In

- a. PRI: *"Showtime, in L/R"*
- b. AUX: *"Heading XXX"*
- c. PRI: *"Showtime 11, fox-3"*
- d. PRI: *"Showtime 11, timeout, locked, (alt), (group name), fox-2"*
- e. PRI: *"Showtime 11, kill, locked, (alt), (group name)"*

14. Post Merge

- a. Immediately post-merge. PRI: *"Showtime, flow 270"*
- b. Post SRR. PRI: *"Sabre, Showtime 11, Denver XXX/YY, flow 270, picture"*
 - i. Flow is called with a heading, not a direction.

15. Abort

- a. PRI: *"Showtime, abort L/R"*
- b. AUX: *"Heading XXX"*
- c. Post-SRR complete. PRI: *"Sabre, Showtime 11, picture"*

16. Clean

- a. First fighter to lose radar SA. AUX: *"Showtime 11, clean"*
 - i. Second fighter has radar SA. AUX: *"Showtime, MELD BRAA, XXX/YY, (alt)"*
 - ii. Second fighter does not have radar SA. PRI: *"Sabre, Showtime 11, clean (group name)"*

17. Maneuvering Targets

- a. Wingman has first radar SA to maneuver. PRI: *“Showtime 12, (group name) maneuver”*
- b. AUX (Showtime 11): *“go pure”*

18. Reset

- a. PRI: *“Showtime, reset (cardinal direction), Sabre monitor (group name)”*
- b. AUX: *“Heading XXX”*

19. Buddy Lock / Buddy Spike

- a. AUX: *“Showtime 11, buddy spike”*
 - i. Used by the fighter that assess they are being spiked by their wingman
- b. AUX: *“Showtime 11, buddy lock”*
 - i. Used by the fighter that realizes they have locked their wingman

20. Section VID

If lead is assuming eyeball role:

- a. PRI: *“Showtime, defend (cardinal direction)”*
- b. ICS: *“Heading XXX”*
- c. PRI: *“Showtime 11, in L/R, eyeball”*
- d. (5-10 seconds later) PRI: *“Showtime 12, in L/R, shooter”*
- e. (5-10 seconds prior VID) PRI: *“Showtime 11, standby (group name)”*
- f. PRI: *“Shoot, Shoot, NATO ID, (group name), hostile”* or *“Friendlies, Friendlies skip it”* or *“Unable ID”*
- g. PRI: *“Showtime 12, fox-2/3”*

If directing wingman to be the eyeball:

- a. PRI: *“Showtime, defend (cardinal direction)”*
- b. ICS: *“Heading XXX”*
- c. Lead SNFO AUX: *“Showtime 12, you’re eyeball”*
- d. PRI: *“Showtime 12, in L/R, eyeball”*
- e. (5-10 seconds later) PRI: *“Showtime 11, in L/R, shooter”*
- f. (5-10 seconds prior VID) PRI: *“Showtime 12, standby (group name)”*
- g. PRI: *“Shoot, Shoot, NATO ID, (group name), hostile”* or *“Friendlies, Friendlies skip it”* or *“Unable ID”*
- h. PRI: *“Showtime 11, fox-2/3”*